How Data Visualisation Works

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Preface

Visualization is essentially a mapping process from computer representations to perceptual representations, choosing encoding techniques to maximize human understanding and communication.[1]

This is a book about data visualisation.

Like many books, it scratches an author's itch.

This particular itch began a long time ago, when first introduced to the works of Edward Tufte. Books like *The Visual Display of Quantitative Information* and *Envisioning Information*[2] inspired an interest in how to produce effective graphics, but they were also frustrating.

Reading a Tufte book to get better at data visualisation is a bit like watching Roger Federer play to get better at tennis. It is possible to glean fragments of insight into the master's skill and technique, but it is difficult to come away with a clear set of rules for how to succeed on our own.

At the same time, encountering the work of William S. Cleveland presented a different problem.[3] Here were hard experimental results with clear implications for effective data visualisations: use length not area; use bars not pies. But they only told us about a small subset of the types of data visualisations that we could create and they only told us about a small subset of the types of questions that we could answer with those data visualisations.

The next decade or so were spent distracted by building tools for *creating* data visualisations[4] before returning again to the question of *designing* data visualisations. In those intervening decades, a lot has happened.

Data visualisation is an area of research for psychologists, for computer scientists, for geographers, and even for statisticians. The literature is wide and deep and complex.

This has significantly reduced the size of the second problem. We know a lot more about a wider range of data visualisation types and data visualisation tasks. Unfortunately, this has introduced a new problem. There are comprehensive and detailed overviews of current knowledge,[5] but it is now difficult to consume and make sense of so much knowledge.

At the same time, there are more accessible and practical guides for creating effective data visualisations.[6] These eschew much of the gory detail in favour of simple, applicable advice. The problem here is that, without the underlying explanations, the guidelines can become just a long laundry list to commit to memory. It is useful to know that bars should always start at zero, but is more satisfying to know why.

For some of us there is a need for a middle path. We would like to understand why we are making certain data visualisation choices, but we have brains that can only hold so many ideas at once. For us there is utility in a framework that contains a small number of core concepts, that covers a wide range of data visualisations, but connects them all with a relatively simple explanation.

This book attempts to provide such a framework. It is a synthesis of a large number of research articles and books and it is a simplification of a large amount of knowledge. It is an attempt to prune the complexity so that we can see the forest through the trees, while at the same time leaving enough foliage that we can still recognise a tree when we see one.

I hope that it helps.

How to Read this Book

Although the goal is to provide a relatively simple conceptual framework for thinking about data visualisation, this is not a short book. The Executive Summary provides the simplest possible statement of the framework and it may be useful to resurface and revisit this summary if you become lost in the depths of one of the chapters.

A slightly longer summary of the framework is provided in the Review. The Review also provides some guidance on how the framework can be applied to the creation of a new data visualisation and the criticism of existing an one.

Another major issue with a subject this complex and a subject that is studied from so many different angles is *terminology*. There is no consistency across all of the different sources of information from which this book was drawn, so all that we can do is attempt to be consistent with ourselves. Some of the most important terms that will be reused relentlessly throughout this book are highlighted in the Executive Summary. The meaning of these terms is hopefully approximately what you already think, but they will be made clearer as the book progresses.

All section and figure cross-references are hyperlinks, so clicking them will navigate to the relevant section or figure. Even more helpfully, hovering the mouse over a hyperlink will show a preview of the relevant section or figure. This is particularly effective for figures because it obviates the need to reproduce the same image multiple times; just hover over the figure link to see the image that the text is referring to.

Acknowledgements

This book is based upon a large number of research articles and books, but in order to make the text appear more accessible and friendly, there are no formal citations in the main text. There are, however, a large number of footnotes, which provide the links between what is being described in the main text and the underlying research, and those footnotes contain proper citations. The real work is not being ignored, it is just being hidden from view.

This book has been generated using the Quarto publishing system. If any parts of the book look nice to you it is probably thanks to Quarto.

Notes

[1] This quote is from an early web resource on data visualisation that was produced by the ACM SIGGRAPH Education Committee (Owen et al. 1999).

[2] *The Visual Display of Quantitative Information* (Tufte 1983) and *Envisioning Information* (Tufte 1990) are just two of a series of influential and visually attractive books that also includes *Visual Explanations* (Tufte 1997) and *Beautiful Evidence* (Tufte 2006).

[3] There were several studies by Cleveland, some in collaboration with Robert McGill, which were summarised and extended in two books: *The Elements of Graphing Data* (William S. Cleveland 1985b) and *Visualizing Data* (William S. Cleveland 1993).

[4] Some of that work is described in Murrell (2019) and some more of it is described here.

[5] *Information Visualization: Perception for Design* (Ware 2021) reviews a very wide range of research that has relevance to data visualisation in one way or another and *Visualization Analysis and Design* (Munzner 2014) provides a comprehensive and theoretical coverage of the complete process of creating effective data visualisations.

[6] For example, *Creating More Effective Graphs* (Robbins 2005), *Data Visualization: A Practical Introduction* (Healy 2018), and *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures* (Wilke 2019).

License

Executive Summary

- Creating a data visualisation involves an **encoding** of **data values** as **visual features**.

  For example, numeric **counts** are **encoded** as the **heights of bars** to create a bar plot.

- Reading a data visualisation involves a **decoding** of **visual features** to obtain **data values**.

  For example, the **heights** of bars in a bar plot are **decoded** to obtain **counts**.

- The **effectiveness** of a data visualisation depends on the effectiveness of the **decoding**.

  For example, a bar plot is effective for comparing **counts** because the visual system is effective at comparing the **heights** of bars.

- More **complex encodings** produce more **complex visual features**.

  For example, a scatter plot **encodes two numeric variables** as the x- and y-locations of points, which produces the **position in space** of points.

- More **complex visual features** lead to more **complex decodings**.

  For example, the **position in space** of points in a scatter plot is **decoded** as the **correlation** between two numeric variables.

- An **effective data visualisation encodes** data values to visual features that facilitate an effective **decoding** from visual features back to information about the data.

1 Introduction

A data visualisation presents data values in a visual format. In an effective data visualisation, this allows properties of the data to be perceived very rapidly and without conscious effort. For example, Table 1.1 shows data on the crime rate for youth offenders in New Zealand, for the ages 14 to 16, for each year from 2011 to 2021. With this purely text-based, tabular presentation of the data, it requires considerable time and effort to determine any trends in the crime rates.

Table 1.1: A table of the number of distinct youth offenders, per 10,000 of popuplation, aged 14 to 16 from 2011 to 2021. It is difficult to perceive trends in the crime rates from this purely text-based, tabular presentation of the data.

age	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
14	583	524	446	378	318	305	280	281	270	244	219
15	712	619	528	443	369	339	315	304	272	262	246
16	810	723	615	482	426	365	328	334	314	302	257

By contrast, with a simple line plot data visualisation like Figure 1.1 we can perceive the trends in crime rates easily and immediately. It is also easy to perceive more detailed properties like the fact that the trend in crime rates for 16-year-olds is not completely monotonic and the fact that the crime rate for 15-year-olds (the blue line) is always greater than the crime rate for 14-year-olds (the red line) and less than the crime rate for 16-year-olds (the green line).

Figure 1.2 shows a different data visualisation of the same data, this time a heatmap. If we use a data visualisation like Figure 1.2, while very basic trends are apparent, perceiving detailed changes in the crime rates is very much harder compared to Figure 1.1. In other words, not all data visualisations are equally effective.

Although Figure 1.1 is very effective for the questions that we have considered so far, it is far less effective if we are interested in a different question, for example, whether the *total* crime rate is completely monotonic. Does the *sum* of the red, blue, and green lines always decrease year on year? In other words, a single data visualisation cannot be effective for all possible questions.

Suppose that we are interested in what the *exact* crime rate was for 16-year-olds in 2011. Neither Figure 1.1 nor Figure 1.2 is the best option for answering this quesion. The best option in this case is actually Table 1.1.

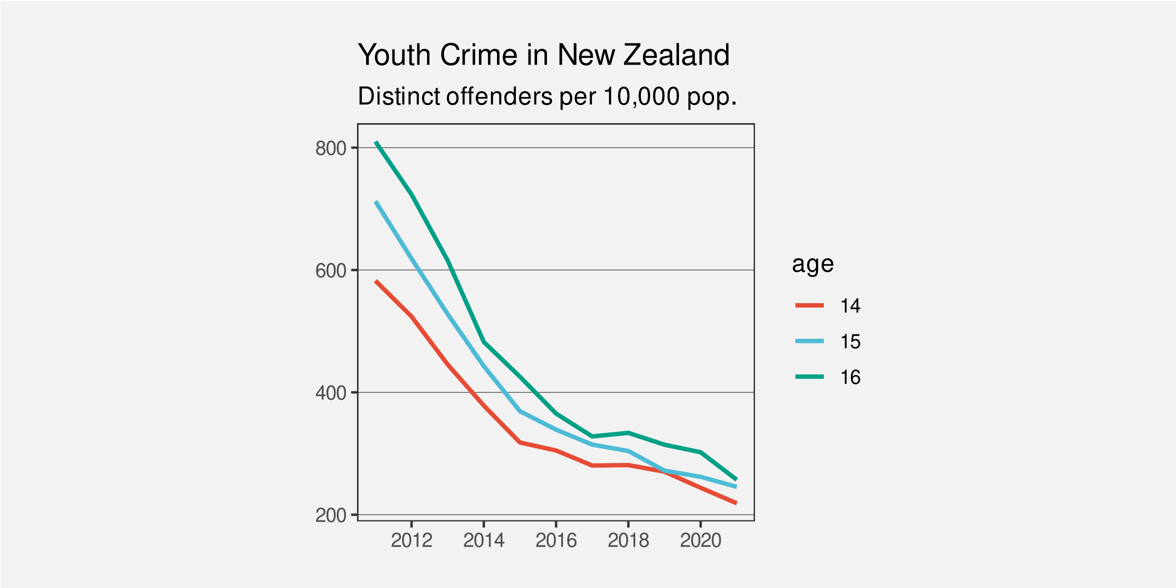


Figure 1.1: A line plot of the crime rate amongst youth offenders aged 14 to 16 from 2011 to 2021. Detecting trends in the crime rates is very fast and easy with this data visualisation.

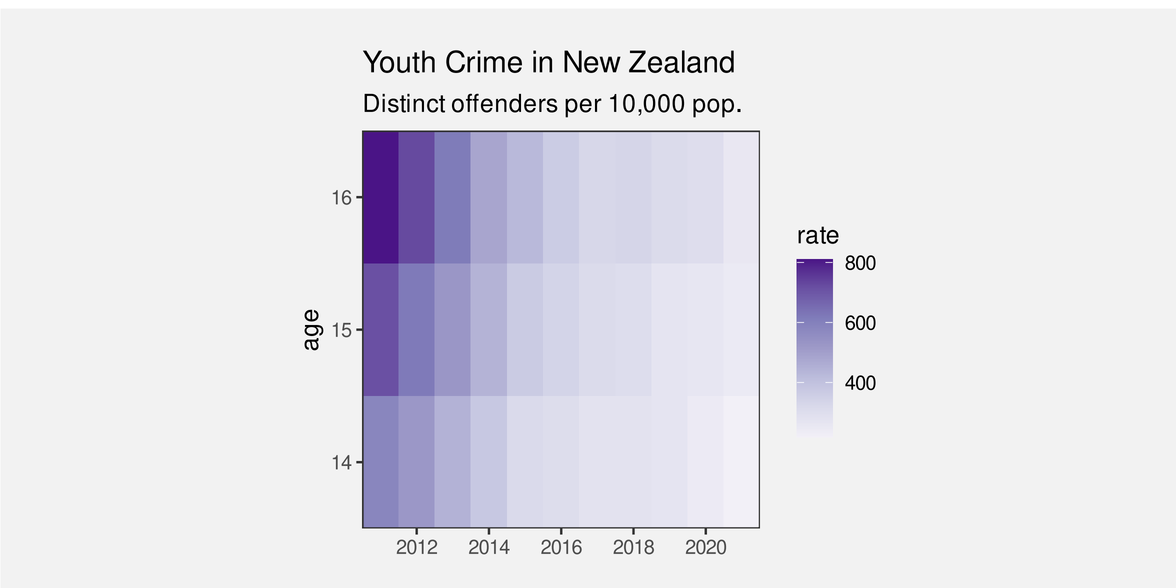


Figure 1.2: A heatmap of the crime rate amongst youth offenders aged 14 to 16 from 2011 to 2021. Some basic trends are visible, but details are harder to perceive compared to Figure 1.1.

What makes a data visualisation effective? Why are some data visualisations more effective than others? The purpose of this book is to provide explanations for why, when, and how data visualisation works.

1.1 One big idea: Mappings

Figure 1.1 and Figure 1.2 are examples of different types of data visualisation, a line plot and a heatmap respectively. We will explore many different types of data visualisations throughout the book, but rather than address each type of data visualisation on its own, we want to identify underlying concepts that help to explain how all data visualisations work.

One concept that we will lean on very heavily is the idea of **mappings**: how is information being converted into a visual representation?[7]

The most obvious mapping in a data visualisation involves the mapping of **data values** to **data symbols** (Figure 1.3). For example, in Figure 1.1, all of the data values for 14-year-olds, the number of offenders for each year, have been mapped to an orange line. The data values for 15-year-olds have been mapped to a blue line and the data values for 16-year-olds have been mapped to a green line. In other words, each row of Table 1.1 is mapped to a separate coloured line.

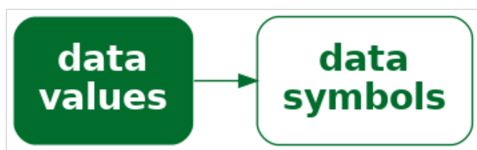


Figure 1.3: A data visualisation can be described by the mappings that it employs to represent data values as data symbols.

The heatmap of the same data (Figure 1.2) provides a different visual representation because it employs a different mapping. In this case, the data symbol is a rectangle, with one data symbol for each combination of age and year, and with the colour of the rectangle reflecting the number of offenders of that age for that year. In other words, each cell of Table 1.1 is mapped to its own coloured rectangle.

The journey that we take throughout this book will involve exploring a variety of different ways that information can be mapped to a visual representation. As much as possible, we will attempt to reduce any data visualisation to a set of mappings from different pieces of information to visual representations of that information. By understanding the different mappings that we can use, we will come to understand why some data visualisations are more effective than others.

1.2 How will it help?

You may have heard that statisticians hate pie charts.[8] But do you know why? Do pie charts have any redeeming features?

Conversely, bar plots are extremely popular, but why are they so good? Knowing more about why different plots work for different situations will help us to make better decisions about when to use what sort of data visualisation.

There is a lot of useful information and understanding about how data visualisations work, but it is distributed throughout and embedded within a very broad, deep, and detailed data visualisation literature. This book aims to produce a simple, coherent explanation that is accessible to a less expert audience.[9]

Understanding how data visualisation works will enable us to:

- Critically analyse data visualisations created by others.
- Decide which data visualisations we should use.
- Justify our choice of data visualisation.
- Reason about how to improve the effectiveness of a data visualisation.
- Reason about whether a novel visualisation will be effective.

1.3 Scope

The focus of this document is **static** data visualisations for **presentation**. The assumption is that our data values have an interesting property and we want to produce a display that makes it easy and efficient for viewers to perceive that property. Some of the information that we cover will also apply to interactive graphics and to exploratory graphics, but the specific requirements of those types of data visualisation will not be addressed.[10]

This document is also focused on what **vision science** has to tell us about effective data visualisation. We will consider factors that influence how rapidly and how accurately information can be extracted from a data visualisation. This will include findings from a broad range of research, encompassing Statistics, Computer Science, Psychology, and Cartography.

However, this document does not cover all fields of research that impact on data visualisation. We will not, for example, address the ethical use of data visualisation, beyond a basic assumption that we want to faithfully represent the properties of the data.[11]

This document is also aimed at a numerate audience, so we assume an appreciation of the importance of collecting the right data in the right way for the questions that we hope to answer.[12]

It is also important to acknowledge that vision science cannot yet tell us everything about how data visualisation works. However, what is known will help us to have some understanding of why some data visualisations are so effective and others are less effective.

As well as being a synthesis of a large range of data visualisation research, this document is also a simplification of that research. Some details are smoothed over and some details are just left out. The goal is to provide a framework that is easy to understand by non-experts and easy to apply to a broad range of data visualisations. References to the relevant literature are provided in footnotes so that the interested, or sceptical, reader can dig deeper.

1.4 Software

This book is about the choices that go into *designing* a data visualisation. It does not cover how to use software to implement those design choices and *create* a data visualisation.

All data visualisations in this book were created with R.[13] The code for all data visualisations is available in the github repository for this book.

1.5 Summary

Data visualisations can be incredibly effective for presenting important properties of data values. However, any one data visualisation may only be effective for certain data values and for certain data properties.

The goal of this document is to explain why some data visualisations are more effective than others; how data visualisation works.

We will focus on how information is **mapped** to a visual representation. We can characterise a data visualisation in terms of the mappings that it uses to **encode** data values as data symbols.

Notes

[7] The idea of characterising a data visualisation as a set of *mappings* from data values to visual representations aligns with ideas from the *Grammar of Graphics* (Wilkinson 2005).

[8] In fact, pie charts have received a lot of hate from all quarters for over a century:

"In a sense, it might be construed as an insult to a man's intelligence to show him a pie chart" Karsten (1925, p98).

"the only worse design than a pie chart is several of them … pie charts should never be used." Tufte (1983, p178).

For some counter-arguments, see Section 7.2.

[9] Besides scratching the author's itch, this book addresses an identified need. For example, Sönning (2022) deconstructs how a line plot works in great detail, but there is a market for something that is more accessible to and applicable by a less-expert audience.

"Going forward, we must do a better job of translating the academic research into practice, making it easier for academics and nonacademics alike to create useful, well-designed graphics." (Vanderplas, Cook, and Hofmann 2020)

The aim is to provide an overview that is similar in scope to more expert frameworks like (Kosslyn 1989), but more consumable by a less expert audience. The goal of this document is to provide a description that is sufficiently straightforward for the non-expert to consume, hopefully without making too many experts roll in their graves or send them to an early one. In most cases, it will be easy to see for ourselves that a visual effect is real through a simple diagram or data visualisation, but references to the relevant theories and experimental results in the literature will be provided in footnotes like this one.

[10] Some examples of books that include a discussion of interactive graphics are Cook and Swayne (2007), Theus and Urbanek (2008), and Unwin, Theus, and Hofmann (2006).

[11] Alberto Cairo's writing often includes the ethical dimensions of data visualisation, e.g., Cairo (2014) and Cairo (2016b).

The *Data by Design* project is a striking example of work with a focus on ethical data visualisation (Emory Digital Humanities Lab 2021).

[12] We are assuming that the data we have is sufficient for the question(s) of interest and contains no *substantive* problems, in the sense of Healy (2018).

Examples of books that provide a more comprehensive treatment of the entire data visualisation process are Cairo (2019) and Munzner (2014).

An example of a much shorter, but still broader view of the data visualisation process is Bertini (2025).

[13] Many of the diagrams in this book were also created with R (R Core Team 2025), but some of diagrams were created using graphviz (Ellson et al. 2002).

2 The Visual System

Understanding why some mappings are more effective than others will depend upon an understanding of how the human visual system works. In order to understand the effectiveness of a mapping from information to a visual representation, we need to know how we get from visual representations back to the original information.

We will talk about the transformation of data values into data symbols as an **encoding** of the data values. Later chapters will explore many different encodings. In this chapter, we will consider the reverse operation: the **decoding** of data symbols back to data values (Figure 2.1). The effectiveness of an encoding from data values to data symbols will depend on how well we can decode from the data symbols back to data values.

In this chapter we will establish a very simple model of the visual system, which will help us to understand the effectiveness of the different mappings that we consider throughout the book.[14]

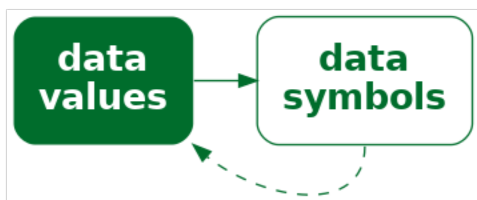


Figure 2.1: The effectiveness of a data visualisation will depend on how well our visual system can **decode** from the data symbols to the properties of the data.

2.1 The eye

In order for us to perceive a data visualisation, light must emit from a computer screen or reflect off a page and enter the eye. Light passes through the pupil, is focused by the lens, and is projected onto the retina at the back of the eye. The retina contains around 100 million light-sensitive nerve cells and signals from those are combined into about 1 million nerve fibres in the optic nerve, which passes signals on to the brain (see Figure 2.2). A very large amount of visual information is gathered by the eye and passed on to the brain.

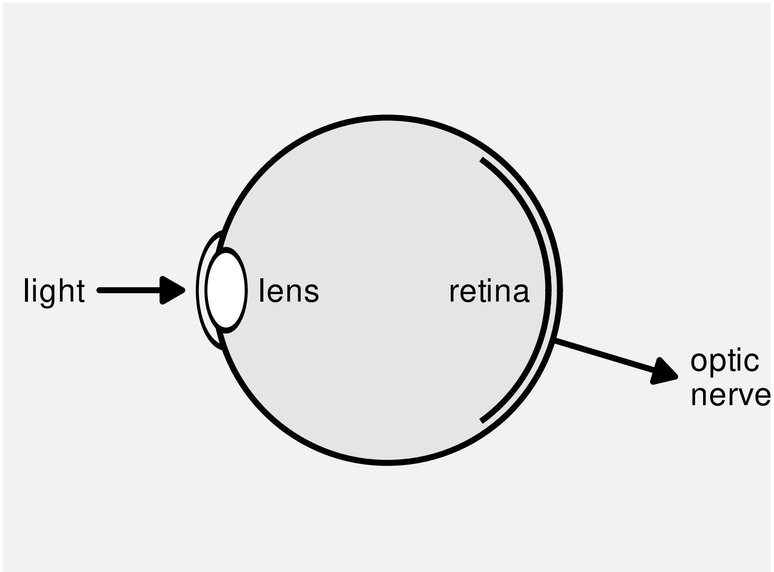


Figure 2.2: A very simple diagram of the human eye. Light enters via the pupil and is focused by the lens onto retina. Light-sensitive neurons in the retina pass signals via the optic nerve to the brain.

> One reason why data visualisation is effective is because the visual system gathers an enormous amount of information.

2.2 Visual processing

Processing of the signals from the optic nerve occurs in several areas of the brain.[15] Very basic **visual features** are extracted first, such as **position**, **angle**, **colour**, and **patterns**.[16] This low-level information is passed on and built upon to produce higher-level visual elements, such as regions and **shapes**. Information is then also integrated from prior knowledge to assist with the identification of **objects**—things like trees, animals, and faces.[17]

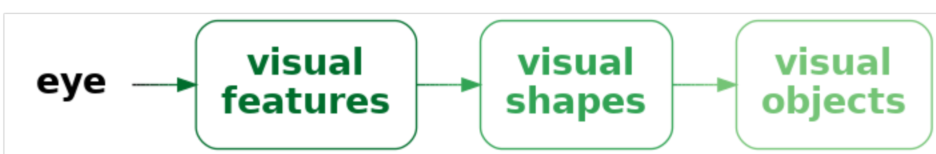


Figure 2.3: A very simple model of visual processing.

Lower-level processing occurs subconsciously and more rapidly than higher-level processing, which can require conscious effort. Furthermore, lower-level processing occurs in parallel and can involve many instances of visual features at once, whereas higher-level processing will only focus on a small number of items or a single item at once. On the other hand, lower-level visual features are rapidly discarded as our attention shifts, while higher-level visual objects are more persistent and may even become part of long-term memory (see Figure 2.4).

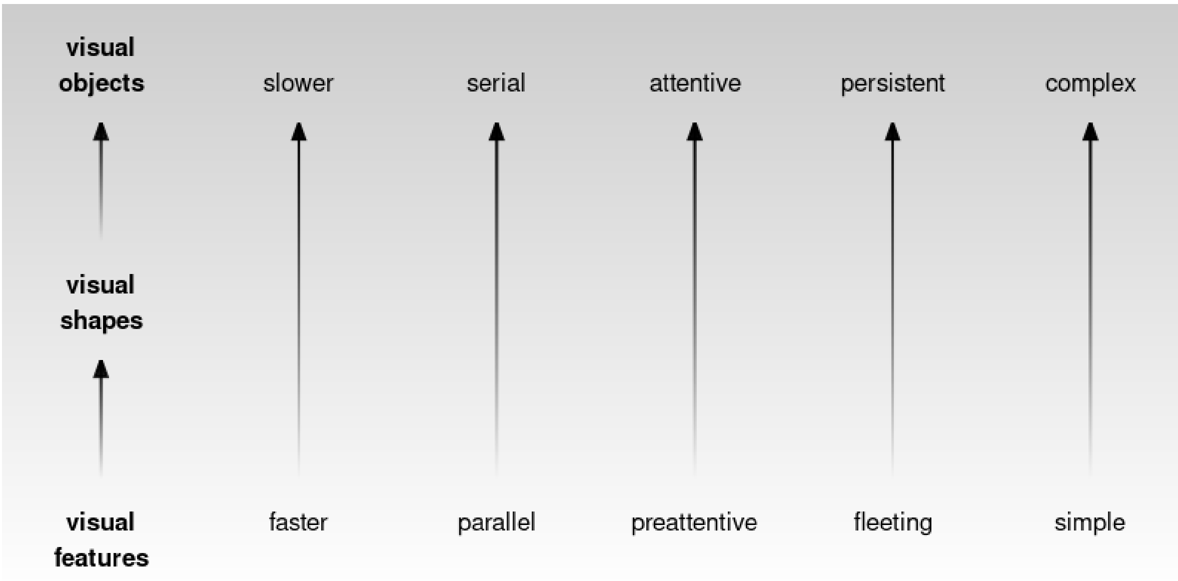


Figure 2.4: Some of the properties of visual processing.

Figure 2.5 shows an image that is made up of many small items: tiny triangles with different **positions**, **lengths**, **angles**, and **colours**. The angles of the triangles represents wind direction and the length of the triangles represents wind strength. Blue triangles are over the sea and green triangles are over land.[18] The visual system can very quickly perceive many small **visual features** at once. There is no need to concentrate separately on every individual triangle in the image. At a higher level of processing, darker and lighter **shapes** are identified as well as regions of different colours. For example, a band of lighter wind forms a line from the top-left corner at a roughly 45-degree angle. At a higher level of processing again, at least for the denizens of Oceania, the green regions are recognised as a very familiar **object**: the outline of New Zealand.

The small lines in Figure 2.5 convey a lot of simple pieces of information—what is the wind strength and direction at each location—though we have to focus on a particular spot in order to extract a single piece of information. The shapes of darker and lighter regions convey several more complex ideas like areas of stronger and lighter winds. The green regions combine to convey the single complex and abstract concept of New Zealand.

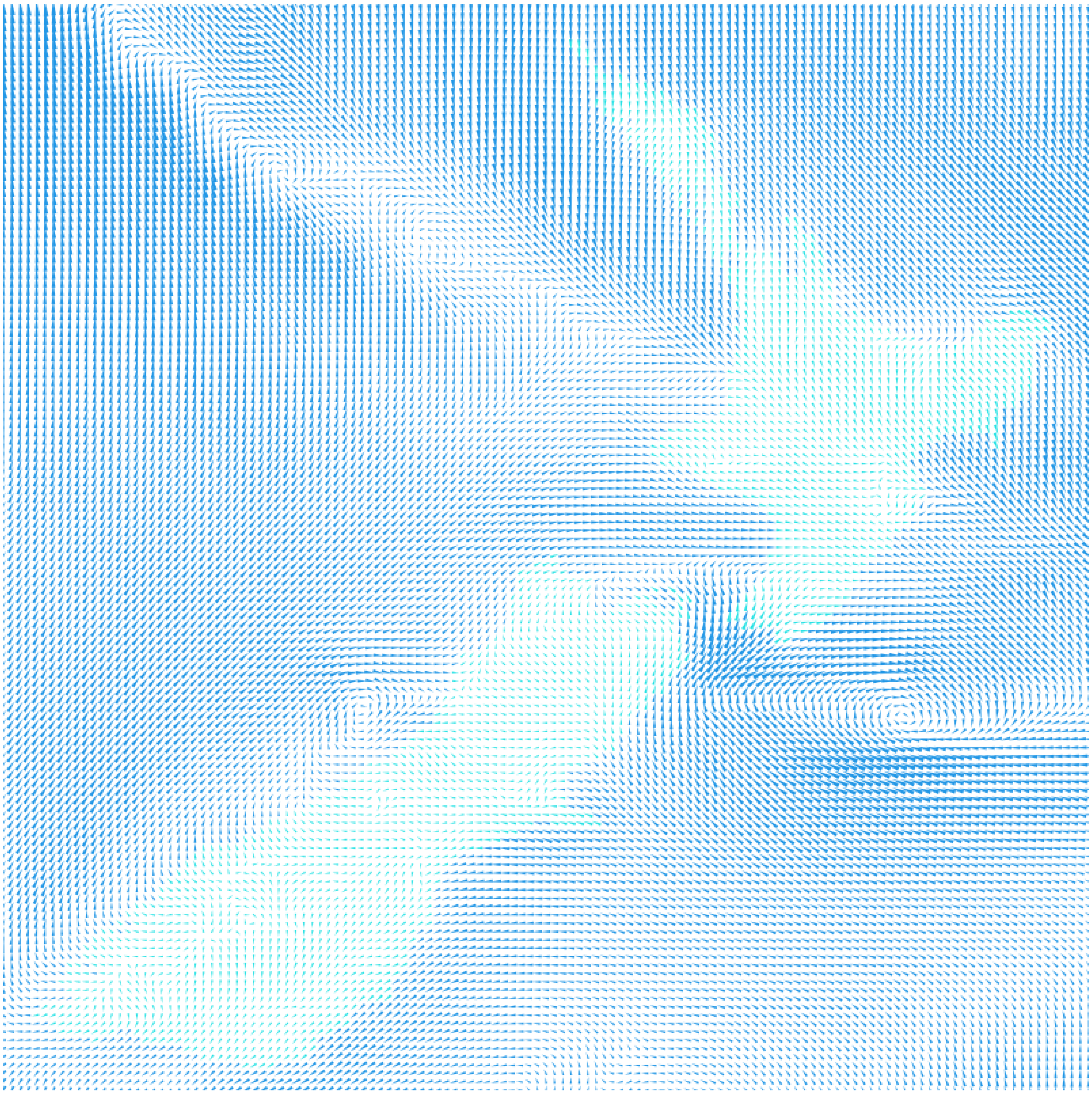


Figure 2.5: A visualisation of wind strength and wind direction that is made up of lots of small lines of different lengths and different angles and different colours.

> Encoding data values as simple visual features is effective at conveying large amounts of simple information because the visual system decodes simple visual features rapidly and all at once.

> Encoding data values as more complex visual shapes or visual objects is effective because more complex information can be decoded from visual shapes and visual objects.

2.3 Visual focus

Figure 2.6 shows a slightly more detailed view of the structure of the eye (compared to Figure 2.2). The retina contains two types of light-sensitive cells: **cones** and **rods**. The cones are sensitive to colour and are concentrated very densely at the centre of the retina, in an area called the **fovea**. Rods only detect the amount of light (monochrome vision) and are spread more sparsely towards the periphery of the retina.

This arrangement means that, despite the experience of a complete, detailed field of view, we only get detailed and colourful information from a small foveal region (less than 5 degrees of visual angle). The information from our peripheral vision is less colourful and lower-resolution.[19]

Furthermore, when we view an image, we do not simply stare at the centre and see everything. Instead, we perform a series of **fixations**, which are brief periods of focus on one area of the image, with **saccades** in between, which are rapid changes in the location of our focus. Figure 2.7 demonstrates that where we look within an image makes a difference.[20]

Figure 2.8 shows eye-tracking data from one person viewing an image for 10 seconds.[21] The original image is shown on the left and eye-tracking data is shown on the right, with hollow black circles showing the position of the coloured circles in the original image. In the eye-tracking image, the fixations are shown as purple circles and the saccades are shown as lines between the circles. For example, we can see that the viewer has fixated multiple times upon the large, colourful circles in the original image and also text at the top and bottom of the image.

> A data visualisation may fail if it requires the viewer to constantly switch focus.

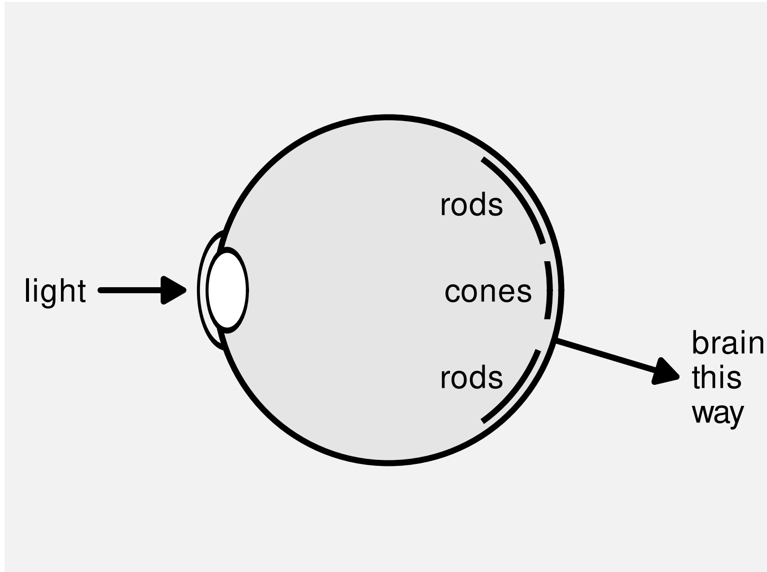


Figure 2.6: A slightly less simple diagram of the human eye. There are two kinds of light-sensitive cells in the retina. Cones are colour-sensitive and focused very densely at the centre of the retina. Rods are only light-sensitive and are spread less densely towards the periphery of the retina.

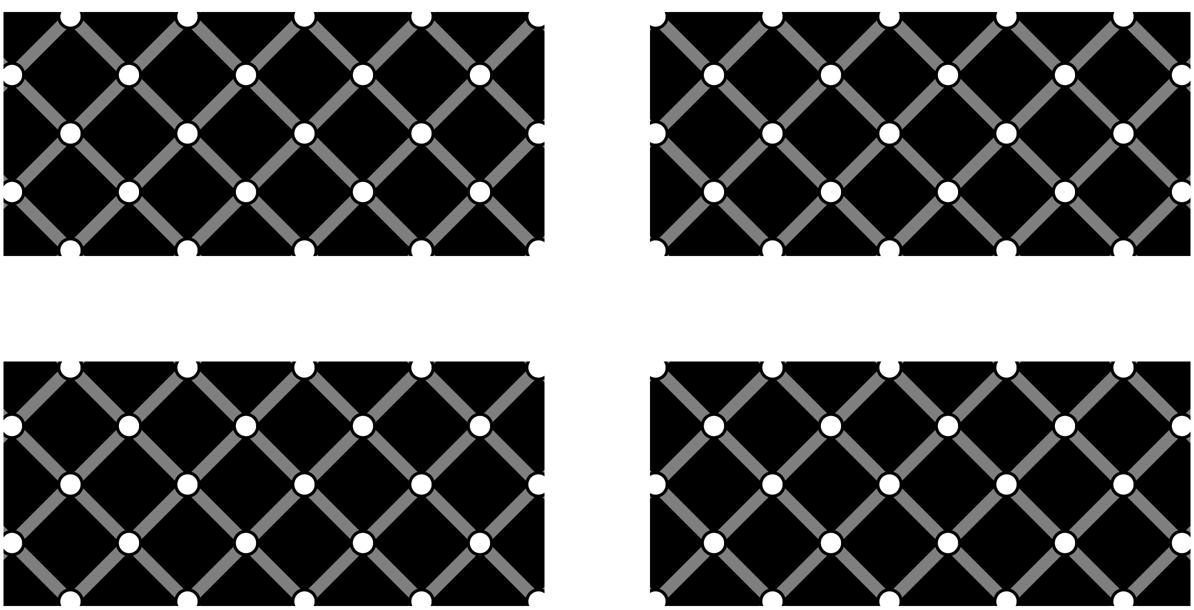
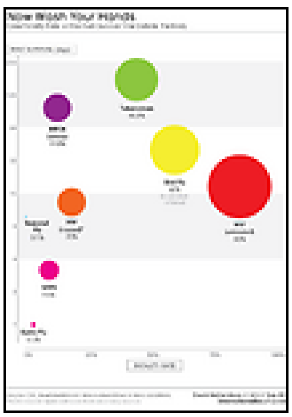
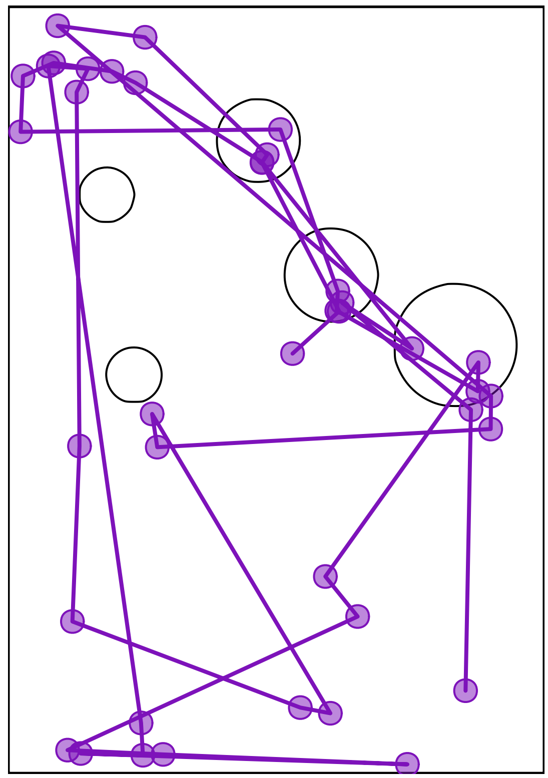


Figure 2.7: The extinction illusion. Look at one corner of the image and then switch focus to a different corner of the image. At a certain viewing distance, the white dots should disappear when we are not focused on them. We may also see black dots within the white dots in our peripheral vision.



(a) A thumbnail of the original image.



(b) Fixations and saccades from eye-tracking data.

Figure 2.8: Eye tracking data (on the right) from the MASSVIS project that shows fixations (circles) and saccades (lines between circles) for one person viewing an image for 10 seconds. A thumbnail of the original image is shown on the left.

2.4 Visual memory

When we switch focus to different areas of an image, we are likely to need to remember information from a previous area of focus. How well can we hold visual information in memory? The most important feature of visual memory is that it is very limited. We can only hold in memory a small number of items at once.[22] Figure 2.9 provides a simple demonstration of the limitations of visual memory.

Figure 2.9

It is easier and faster to decode information from visual elements if they lie within the scope of a single visual focus. If we have to switch focus, it not only takes longer to decode information, but we may hit limits on our ability to remember visual elements from previous areas of focus.[23]

The limits on visual memory reflect well-known limits on working memory and those limits can also be worked around via familiar mechanisms such as chunking.[24] For example, we may only be able to hold in memory a small number of basic visual features such as the colours of circles (Figure 2.9), but we can effectively hold many more basic visual features if we memorise more complex visual elements that are composed of many basic visual features. In other words, can remember several **visual shapes** or **visual objects** even if we cannot remember all of the individual visual features within those visual shapes or visual objects (Section 2.2). For example, can hold in memory multiple country flags even though we cannot necessarily hold in memory all of the invidual colours on those flags.

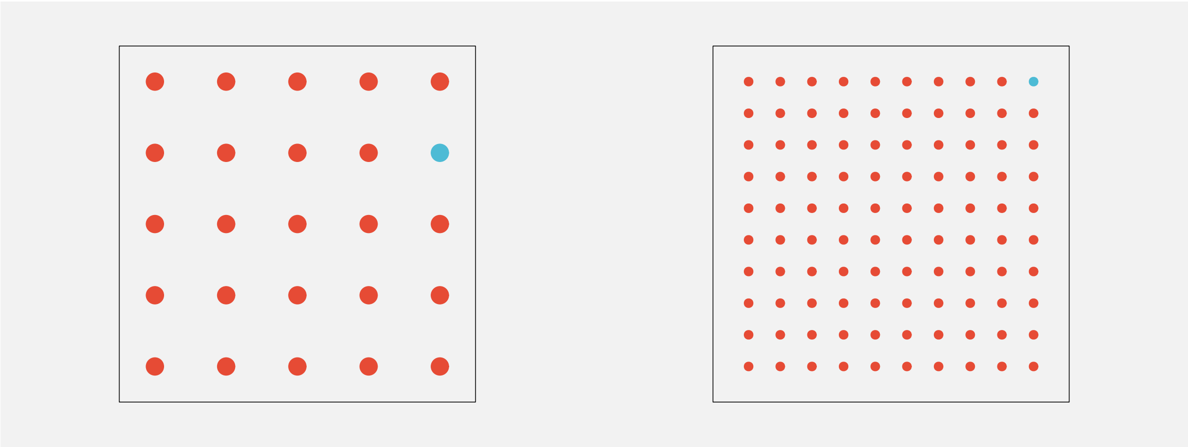
> A data visualisation will require more effort to decode if the viewer is required to switch focus and remember visual information across multiple areas of visual focus.

2.5 Visual differences

The ability to gather a large amount of basic features from an image very quickly (Section 2.2) is supplemented by rapid visual processing that combines and summarises the information. These subconscious processes extract some properties from an image without any mental effort.

For example, in Figure 2.10a we can detect the one teal dot amongst the 24 orange dots without effort. Figure 2.10b shows that this task remains effortless even if the number of dots is much larger.[25]

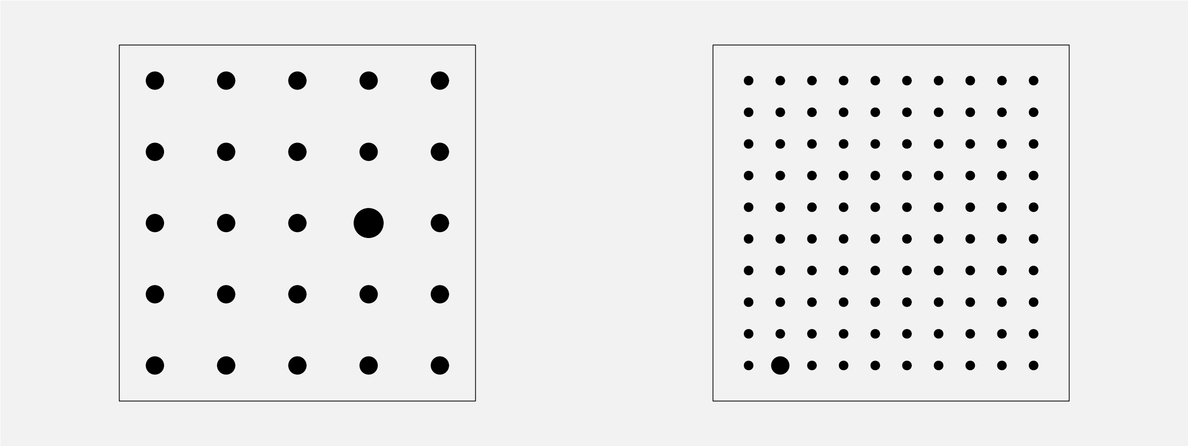
This ability to identify an item that is different from all other items within an image also applies to other basic features like size, as shown in Figure 2.11.



(a) One teal dot amongst 24 orange dots.

(b) One teal dot amongst 99 orange dots.

Figure 2.10: Amongst a collection of dots, we effortlessly perceive one dot that has a different colour.



(a) One larger dot amongst 24 smaller dots.

(b) One larger dot amongst 99 smaller dots.

Figure 2.11: Amongst a collection of dots, we effortlessly perceive one dot that differs only by size.

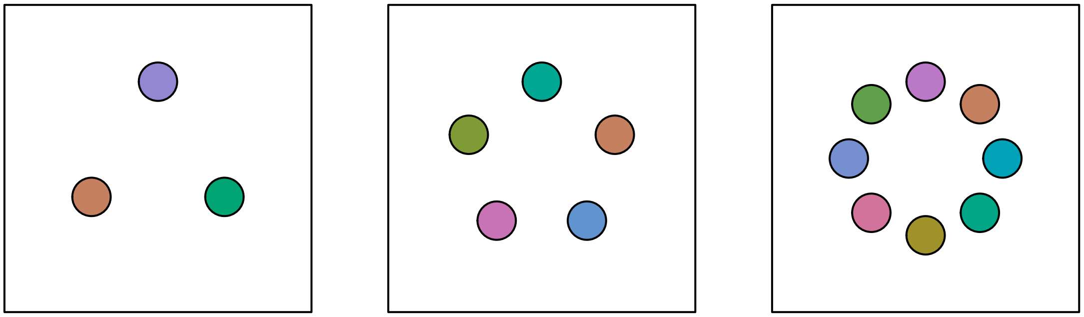


Figure 2.12: A memory task. Focus on the circles in one of the three squares then turn the page and try to tell how many of the circles have changed colour in the corresponding square in @fig-memory-static-2. The task should be obviously harder when there are more circles to remember, which reflects the limits on visual memory.

If an item differs on more than one visual feature, the effect is even stronger (Figure 2.14). This means that the visual system is very sensitive to items that are **different** in terms of basic **visual features**.

On the other hand, if the distractor items are similar on one visual feature, but different on another, and if the arrangement of the items is less structured, then finding a particular item of interest becomes much harder. For example, if we look for the single larger teal dot in Figure 2.15, it is much harder to find than it is in Figure 2.14. The visual system's sensitivity to **different** items is much greater when all other items are the **same**.[26]

> A data visualisation will be effective if important differences in the data are encoded using different colours or sizes, especially if everything else remains the same.

2.6 Visual attention

Where we focus within an image is not random. The figures from Section 2.5 show that our attention is automatically drawn to the **larger**, **brighter**, and more **colourful** items within an image. This happens due to very early visual processing and without conscious thought.[27]

> A data visualisation will be effective if it uses basic visual features like colour to draw the viewer's attention to important features.

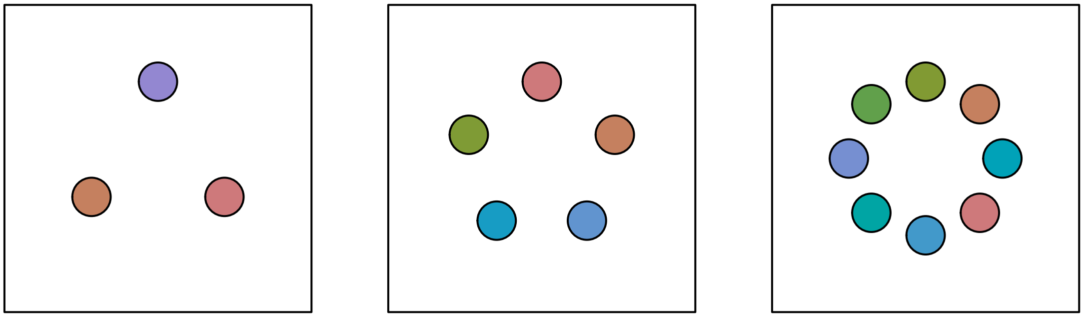


Figure 2.13: A memory task. Look at the circles in one of the squares in @fig-memory-static-1 on the previous page and then come back here and try to tell how many circles have changed colour in the corresponding square. The task should be obviously harder when there are more circles to remember, which reflects the limits on visual memory.

Where we focus within an image, and what information we extract, is also affected by conscious goals—what are we looking for?[28] For example, when you were looking for the larger teal dot in Figure 2.15, did you notice that there was also a single small orange dot? Now that your attention has been drawn to that task, it will be easier to find the smaller orange dot.[29]

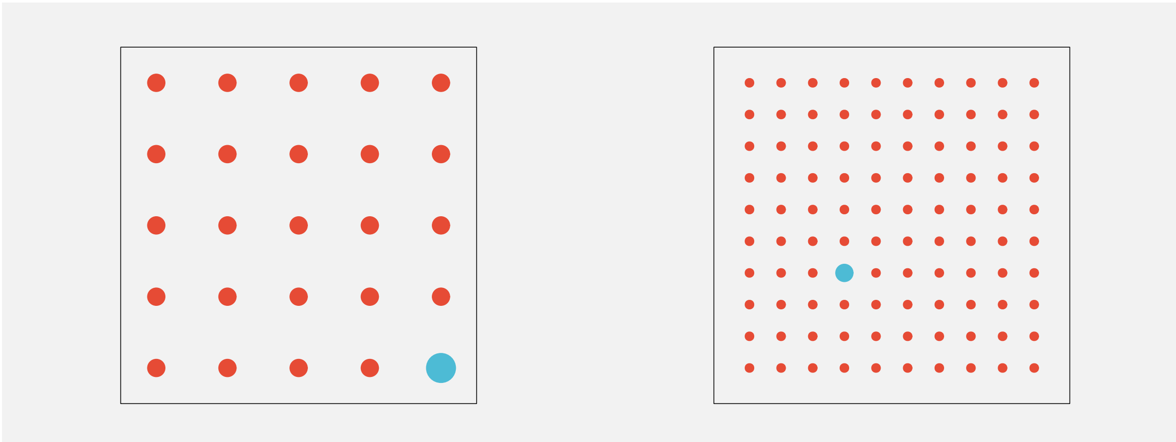
> A data visualisation will be effective if it includes instructions that direct the reader's attention to important features.

2.7 Visual similarities

In Figure 2.15, we can see several groups of items: orange dots, teal dots, large dots, and small dots. These separate groups of items are identified based on their **similarity**—each group of items have the same size and the same colour. The flip side of being able to easily identify items that are different (Section 2.5) is that we can easily identify items that are the same.

The grouping of similar items within an image also happens automatically and without conscious effort. Figure 2.16 shows pairs of dot matrices that are very similar, but in one matrix of each pair we see rows and in the other matrix we see columns. For example, in the middle pair, items that are the same colour produce groups that are either rows or columns.

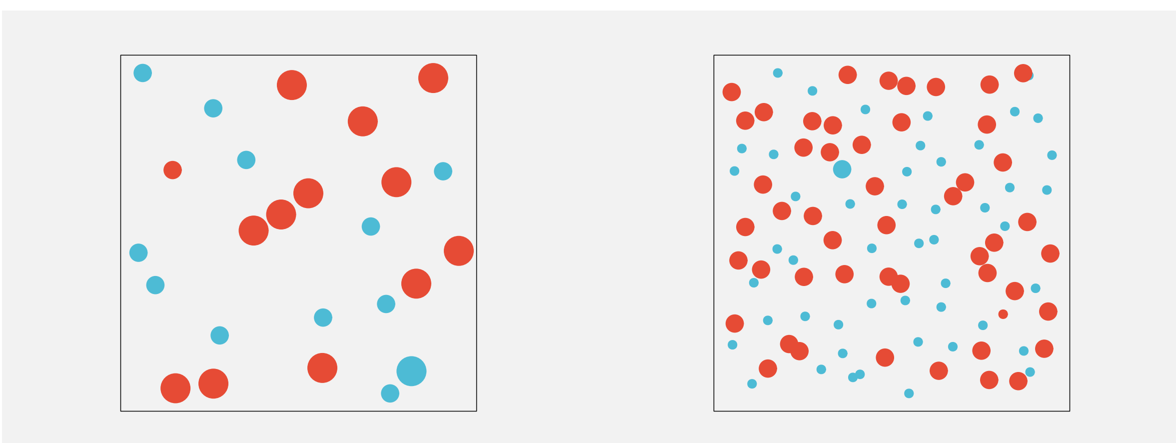
Furthermore, there are other visual properties that mean we automatically perceive items as groups: items that are close together (items that have similar **positions**) are seen as a group so we see either rows of black dots or columns of black dots on the left of Figure 2.16; items that



(a) One larger teal dot amongst 24 smaller orange dots.

(b) One larger teal dot amongst 99 smaller orange dots.

Figure 2.14: Amongst a collection of dots, we effortlessly perceive one dot that differs by both colour and size.



(a) One larger teal dot amongst 24 dots that differ by colour and/or size.

(b) One larger teal dot amongst 99 dots that differ by colour and/or size.

Figure 2.15: We have to work much harder when there is more going on.

are connected by lines are also seen as a group so we see either connected rows or connected columns on the right of Figure 2.16.[30]

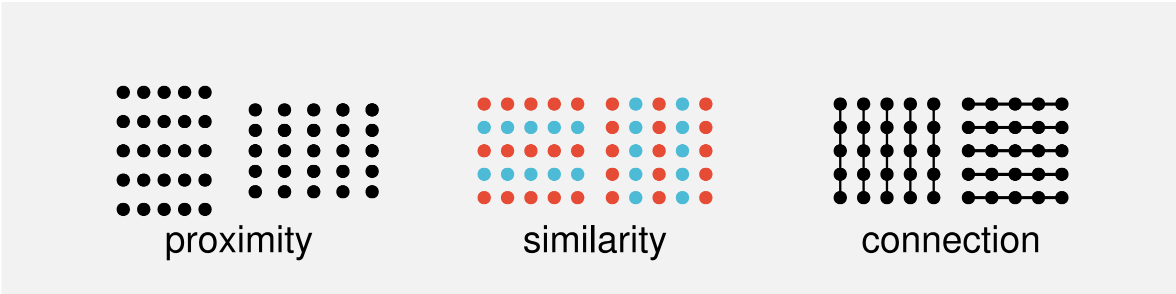


Figure 2.16: In each matrix of dots, we clearly see either rows or columns. In the pair of matrices on the left, making the dots slightly closer together horizontally produces rows, and making the dots slightly closer vertically produces columns. In the middle pair of matrices, using the same colour on each row produces rows, and using the same colour on each column produces columns. In the pair of matrices on the right, drawing vertical lines produces columns and drawing horizontal lines produces rows.

These grouping effects are very powerful and can even be used to override each other in some combinations. Figure 2.17 shows that an enclosing border can be stronger than a connecting line, connecting lines can be stronger than proximity, and proximity can be stronger than similarity.

Another type of automatic grouping is demonstrated in Figure 2.18. On the left is a collection of dots. We naturally perceive a horizontal line and a circle from these dots, as shown in the centre of Figure 2.18. We automatically complete lines, particularly smooth curves, even when the curve is broken, as in the central set of dots. We do not tend to complete lines with sharp corners, as indicated on the right of Figure 2.18.[31]

When we visualise qualitative data, we will be able to rely on the automatic identification of different groups of data symbols based on visual similarities.

> A data visualisation of qualitative data may be effective if it relies on the automatic identification of visual groups (proximity, similarity, etc).

2.8 Visual simplicity

Figure 2.19 shows an arrangement of dots, similar to Figure 2.18. The collection of dots on the left has an ambiguous interpretation. Is this a single shape, or two overlapping shapes (as

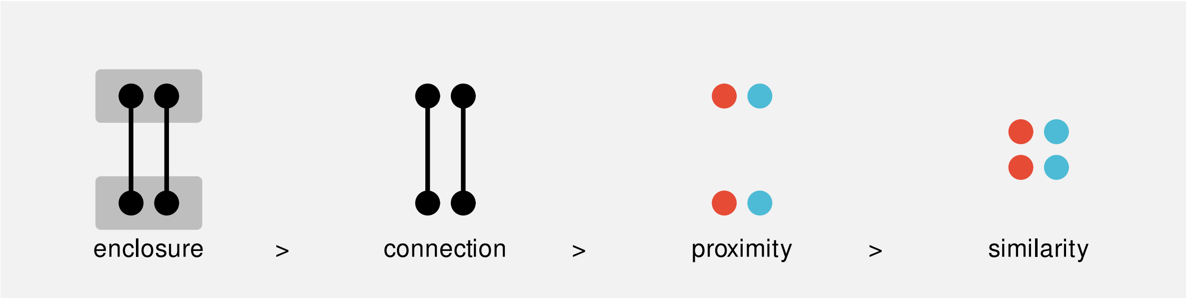


Figure 2.17: For each set of four dots, we automatically perceive two pairs of dots. On the left, the enclosing grey rectangles produce an upper pair and a lower pair, despite the fact that the left and right pair of dots are connected by lines. Second from left, the connecting lines produce a left pair and a right pair, despite the fact that the top pair and the bottom pair are closer to each other. Second from right, the positions of the dots (their proximity) produces an upper pair and a lower pair, despite the fact that the left and right pairs are coloured the same. On the right, the same colours produce a left pair and a right pair.

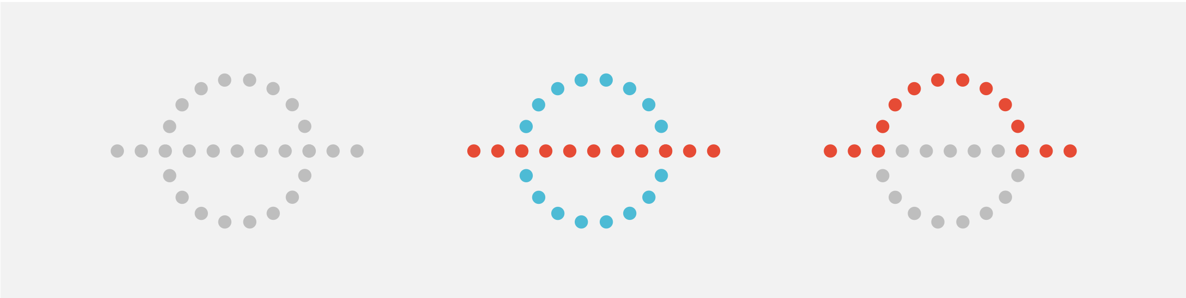


Figure 2.18: We automatically complete curves.

indicated in the centre of Figure 2.19), or some complex combination of multiple shapes (as indicated on the right of Figure 2.19)?

Our visual system has a preference for the simple interpretation. Given an ambiguous arrangement of visual elements, we will tend to perceive whatever is most orderly, regular, and coherent.[32]

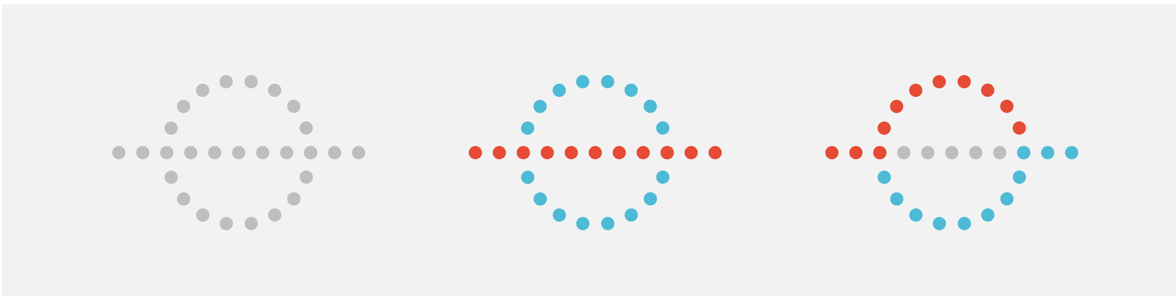


Figure 2.19: We automatically perceive simple shapes over more complex interpretations.

Figure 2.20 shows that we also perceive groups when items have a symmetric arrangement.[33]

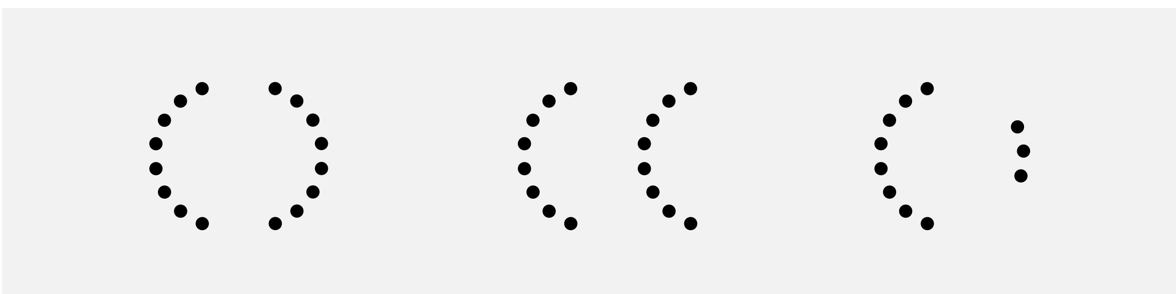


Figure 2.20: We automatically perceive groups from symmetric arrangements.

These results suggest that a data visualisations will be more effective if it is simple and orderly. We should lead the viewer where they already want to go rather than trying to force the viewer to perceive a more complex interpretation.

> A data visualisation may be effective if it is orderly and lends itself to a simple interpretation.

2.9 Visual tasks

Identifying whether visual elements are the same or different (Section 2.5) is an example of a very simple visual task. Figure 2.21 (on the left) shows another simple visual task: judging the ratio of two different lengths. However, on the right of Figure 2.21 is an example of a task that is much more difficult. In this task, we have to identify which of the four shapes in the bottom row is *not* a rotated version of the shape at the top.

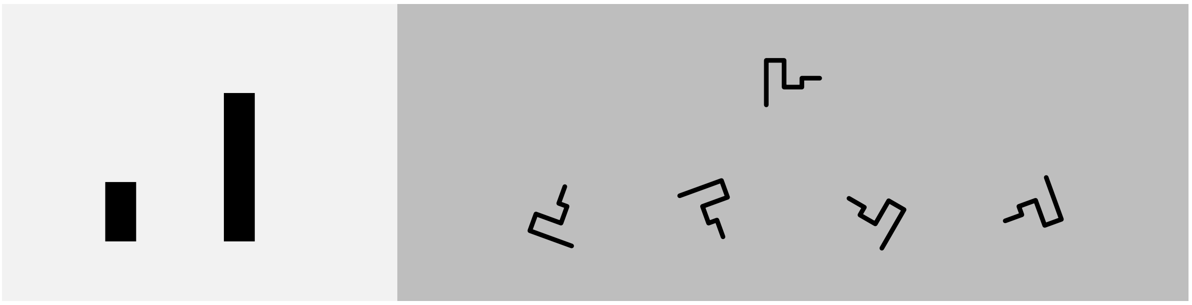


Figure 2.21: On the left is a simple visual task, to estimate the relative length of the two lines. On the right is a more complicated visual task, to decide which of the four bottom shapes is *not* a rotated version of the top shape.

The left-hand task in Figure 2.21 is effortless and can be performed immediately. The right-hand task in Figure 2.21 requires much more deliberate effort and time to perform.

> A data visualisation will be effective if it relies on automatic and rapid visual processing rather than requiring conscious mental effort.[34]

2.10 Dangers of the visual system

The value of data visualisation comes from the visual system's ability to rapidly gather and process a large amount of visual information. However, while data visualisations usually consist of artificial and simple images presented on a two-dimensional page or screen, the visual system has evolved for the unstructured, messy, and three-dimensional natural environment. This means that the visual system will sometimes generate a more sophisticated, but erroneous interpretation of an image from a very simple arrangement of basic shapes. In other words, we are vulnerable to visual illusions.

For example, Figure 2.22 shows the Muller-Lyer Illusion and the Ebbinghaus Illusion. In the former, two horizontal lines of the same length appear to have different lengths and, in the latter, two orange circles of the same size appear to have different sizes.[35]

Figure 2.23 shows another illusions involving simple lines. Every line segment in this image is the same physical length, but we perceive some segments as longer than others.[36]

Figure 2.24 shows two images that are composed of three simple parallelograms. The three parallelograms are arranged separately on the left, which means that we decode them as three

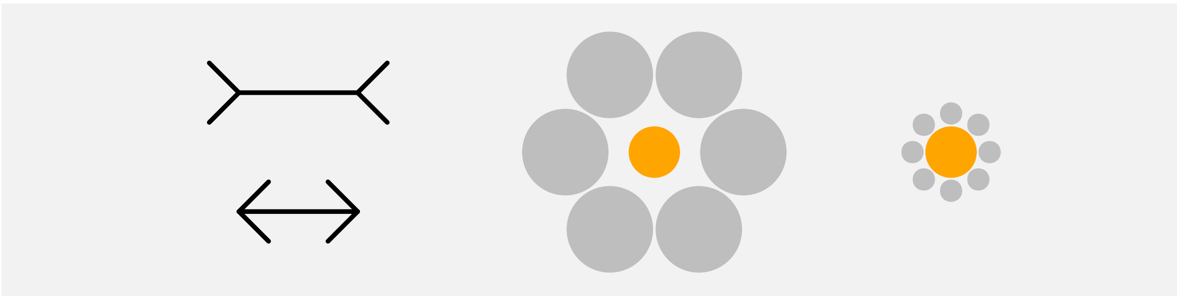


Figure 2.22: The two horizonal lines in the Muller-Lyer illusion on the left have exactly the same physical length, but the top line appears longer. The two inner circles in the Ebbinghaus illusion on the right have exactly the same physical radius, but they appear to be different sizes.

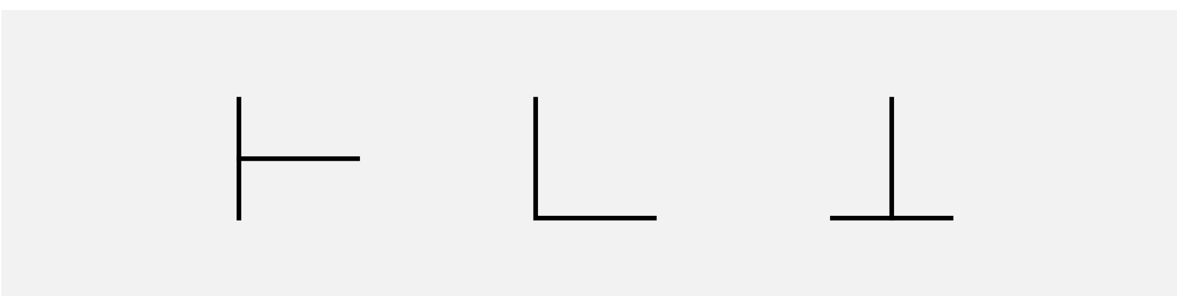


Figure 2.23: All of the horizontal and vertical line segments in this image are exactly the same physical length, but our visual system perceives some segments as longer than others.

simple shapes. However, for the arrangement of the parallelograms on the right, the natural decoding is a 3-dimensional box, even though the image is clearly 2-dimensional.

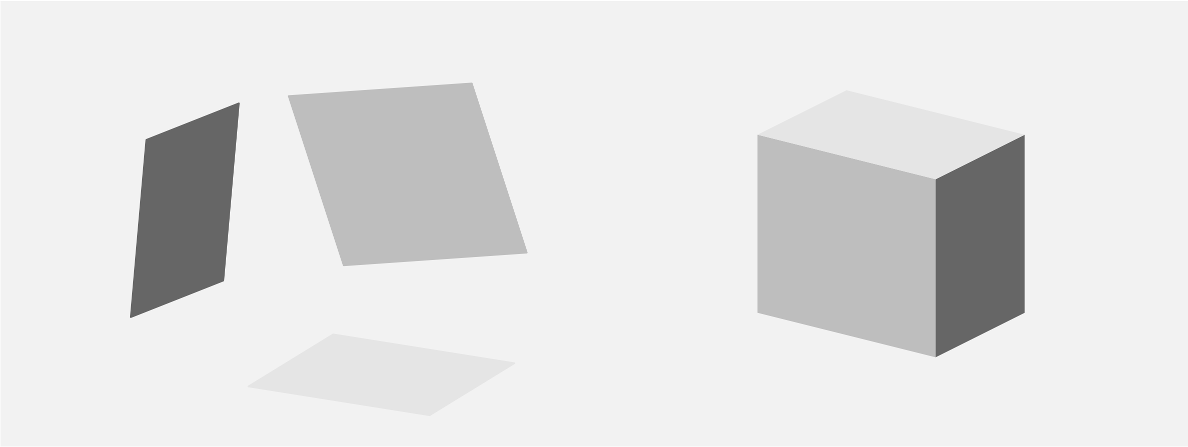


Figure 2.24: The image on the right is made up from the three simple parallelograms shown on the left. Because of their arrangement and their shading, we are tempted to decode the three parallelograms on the right as a 3-dimensional object.

These examples demonstrate that the visual system can produce incorrect or unintended results for even very simple visual tasks based on very simple visual elements.

> The effectiveness of a data visualisation will be compromised if it (even accidentally) generates a visual illusion.

2.11 Summary

The human visual system captures a very large amount of information from the visual field.

Detailed information is only available at the centre of the visual field so multiple fixations are required to capture detailed information from multiple locations within an image.

Large numbers of simple features are rapidly extracted, while more complex visual items take longer and require more congnitive effort.

Large, bright, colourful items automatically stand out.

Items are automatically differentiated and grouped based on basic visual features.

The main point is that the visual system is *very* good at perceiving certain visual properties.

An effective data visualisation will make use of the advantageous properties of the visual system and it will avoid the undesirable properties of the visual system.

Notes

[14] Simple descriptions of the human visual system can be found in many articles and books on data visualisation, for example, Sönning (2023), Franconeri et al. (2021a), and Cairo (2012). See Ware (2021) for a more in-depth, but still very accessible description. Wikipedia also has some very helpful content (Wikipedia contributors 2025).

[15] A rough estimate is that more than 50% of the brain is involved in visual processing (Van Essen, Anderson, and Felleman 1992).

[16] Low-level visual processing involves the creation of *feature maps* that identify colours, edges, and orientations, along with the spatial arrangement of those basic features within the field of view.(Ware 2021, 146)

Patterns, or *textures*, are perceived at a slightly higher level, based on low-level perception of spatial frequencies (fine detail versus coarse detail).

[17] A slightly more detailed overview is provided in Ware (2021) (page 20).

[18] Figure 2.5 is based on wind data from the National Oceanic and Atmospheric Administration's Operational Model Archive and Distribution System (NOMADS). The data were accessed on September 14 2024 and processed using the {rNOMADS} package (Bowman and Lees 2015), with raw data interpolated to a finer grid.

[19] This is not to say that peripheral vision is useless. Peripheral vision contributes significantly to our field of view (Stewart, Valsecchi, and Schütz 2020; Rosenholtz 2016), and is as capable as foveal vision at detecting motion (Mckee and Nakayama 1984). However, our ability to perceive fine details is limited to a relatively small area of focus.

[20] The extinction illusion is related to the Hermann grid illusion (Ninio and Stevens 2000).

This illusion is not explained simply by differences in foveal versus peripheral perception—there is currently no complete explanation for the effect—but it serves to show that we are getting different information from different regions of the field of view.

[21] The plot was originally from The Economist, but this image was taken from the resources provided on the MASSVIS project web site. The eye tracking data were taken from the same web site. Cite MASSVIS article. Only a thumbnail version of the original image is shown in Figure 2.8 in order to comply with terms of use of the MASSVIS data set.

[22] Some limitations of visual memory are briefly mentioned in Ware (2021) (pages 397-405 and pages 423-425).

"A … severely limited capacity is the restricted number of stimuli that can be held in VSTM [visual short-term memory]" (Marois and Ivanoff 2005)

And footnote that acknowledges some controversy (different models and number of items limit vs continuous resource limit), with ref to e.g., "Psychophysical scaling reveals a unified theory of visual memory strength" Schurgin et al (2020) and the binary map model

[23] Tufte refers to this as drawing "within the eyespan" (Tufte 1990, 67)

"In understanding human visual perception, an important component consists of what people can perceive at a glance. If that glance provides the observer with sufficient task-relevant information, this affords efficient processing. If not, one must move one's eyes and integrate information across glances and over time, which is necessarily slower and limited by both working memory and the ability to integrate that information." (Rosenholtz 2023).

[24] Miller's famous paper "The Magical Number Seven, Plus or Minus Two: Some Limits on Our Capacity for Processing Information" (Miller 1956) talks about limits on working memory, but also how to "escape" those limits, for example, by chunking information.

[25] The ability to detect features without conscious effort is called *pre-attentive* processing or *pop out* (Treisman 1985).

[26] For example, from Wolfe and Horowitz (2017): "Salience of a target increases with difference from the distractors (target–distractor heterogeneity) and with the homogeneity of the distractors (distractor–distractor homogeneity) along basic feature dimensions."

[27] Saliency maps (e.g., Itti, Koch, and Niebur 1998; Matzen et al. 2018), which attempt to predict where the viewer will focus attention, are based on the idea that attention is drawn to basic visual features like colour, angle, and size.

[28] Having a conscious target to search for not only directs our visual focus (Section 2.3), but also filters the processing of visual input to exclude visual features that are not of interest (Knudsen 2020). In other words, to some extent, we see what we are looking for!

[29] Wolfe and Horowitz (2017) describe "Bottom-up, stimulus-driven guidance" versus "Top-down, user-driven guidance".

[30] These visual effects are known as the Gestalt Grouping Principles (wagemans2012gestalt). The Proximity Principle says that items that are close to each other form a group. The Similarity Principle says that items that have the same colour or size form a group. The Connectedness Principle says that items that are connected to each other form a group. The Common Region Principle says that items within the same bounded region form a group. Todorovic (2008) provides a very accessible description of these principles and several others.

[31] This is the Gestalt Continuity Principle (Todorovic 2008): Items that lie along a smooth curve form a group.

[32] This is the Gestalt Prägnanz Principle (Todorovic 2008): The tendency to seek simplifying structure and to find order within the chaos of an image.

[33] This is the Gestalt Symmetry Principle (Todorovic 2008): Items that exhibit symmetry will form a group (in preference to items that do not exhibit symmetry).

[34] "the comprehension of visual displays tends to be easiest when simple pattern identification processes are substituted for complex cognitive processes" (Carpenter and Shah 1998, 98)

[35] The original publication of the Muller-Lyer illusion (in German) is Müller-Lyer (1889), though in that, the stimulus described is a single line with three sets of arrows, as shown below.

The Ebbinghaus Illusion is also known as the Titchener Illusion after its English-language populariser (Titchener 1916).

Various evolutionary explanations of these illusions have been proposed, e.g., (Gregory 1963) and (Howe and Purves 2005).

[36] An early in-depth exploration of the Vertical-Horizontal Illusion was provided by Künnapas (1955).

3 Visual Features

In Chapter 1, we described a data visualisation as a set of **mappings** from data values to data symbols (Figure 1.3). In Chapter 2, we established some basic properties of the visual system, which will help to explain how well the **encoding** of data values to data symbols can be reversed to **decode** from data symbols back to data values (Figure 2.1). We also identified three levels of visual processing: visual features, visual shapes, and visual objects (Figure 2.3).

There are many possible mappings between data values and data symbols. For example, in Section 1.1, we saw that the same data values could be mapped in different ways to produce different data visualisations (e.g., Figure 1.1 and Figure 1.2). The remainder of this book explores different sorts of mappings and why they are more or less effective for different purposes.

In this chapter, we start with a very simple sort of mapping. We will begin to explore data visualisations where each data value is **encoded** as a basic **visual feature** of a data symbol. Furthermore, in these data visualisations, **one** data value is mapped to **one** data symbol (Figure 3.1).

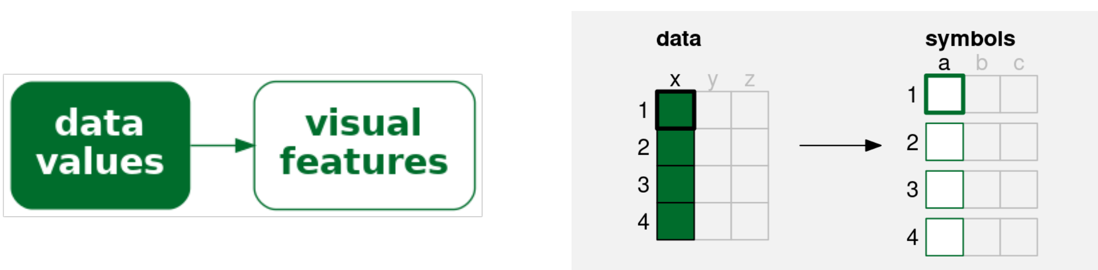


Figure 3.1: A simple data visualisation encodes each data value as a **visual feature** of a data symbol. Each data value (`x`) is mapped to a visual feature (`a`) of a single data symbol.

3.1 Bar plots

A good example of this sort of simple mapping is a bar plot. In order to demonstrate bar plots, we will work with a data set containing the total number of offenders aged 14 to 16 in

New Zealand between 2011 and 2021, for different ethnic groups. The data values are shown in Table 3.1. A bar plot of these data is shown in Figure 3.2.

Table 3.1: A table of the total number of offenders aged 14 to 16 from 2011 to 2021 for different ethnic groups.

group	total
European/Other	31274
Māori	41787
Pasifika	6993
Unknown	5632

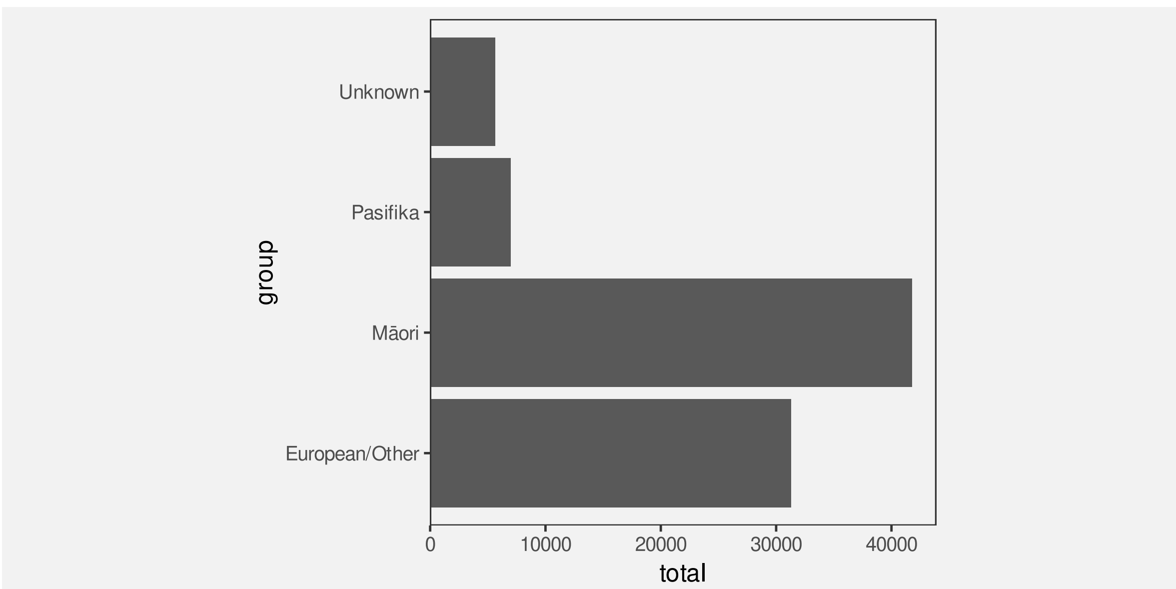


Figure 3.2: A bar plot of the total number of offenders for different ethnic groups. The data symbols in this plot are the horizontal bars.

We can see that, for each data value in Table 3.1, there is a single bar in the bar plot. For example, each total number of offenders is mapped to a single bar.

In terms of **decoding** (Figure 2.1), this means that we can decode individual data values from each data symbol. For example, we can decode a single total number of offenders from each bar.[37]

This allows us to answer simple questions about individual data values. For example, what is the largest total number of offenders?

3.2 Visual features

The data symbols in Figure 3.2 are a set of bars or, put more simply, rectangles. We have seen that each data value from Table 3.1 is mapped to one of these rectangles. For example, each total number of offenders is mapped to a rectangle.

However, we can look more closely at the mapping and consider how individual data values are being **encoded** as a data symbol. For example, the total number of offenders for each ethnic group is encoded as the **length** of a rectangle (Figure 3.3). Each data value is mapped to the length of one rectangle.[38]

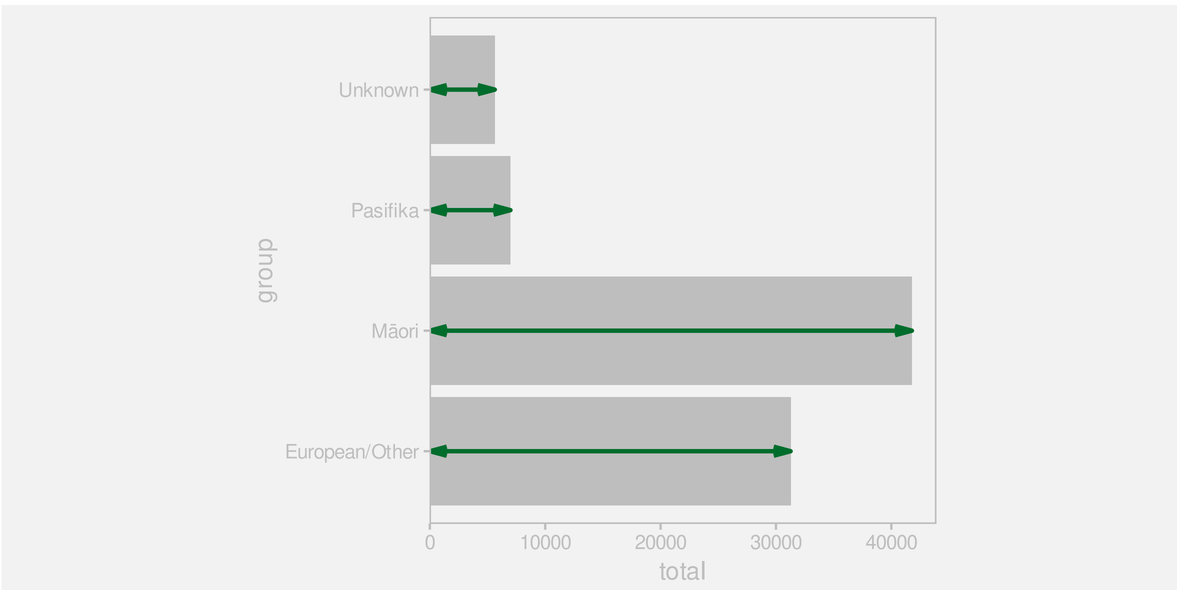


Figure 3.3: A bar plot of number of crimes for different ethnic groups. The data values, the total number of crimes, have been mapped to the **lengths** of the bars (the purple lines).

Each ethnic group data value in Table 3.1 is also mapped to a rectangle in Figure 3.2. In terms of **encodings**, each ethnic group is encoded as the vertical **position** of one rectangle (Figure 3.4).

Length and **position** are examples of basic visual features (Section 2.2). One way to make an effective data visualisation is to encode data values as the basic visual features of a data symbol. This is effective because basic visual features are identified rapidly and without conscious effort by our visual system.

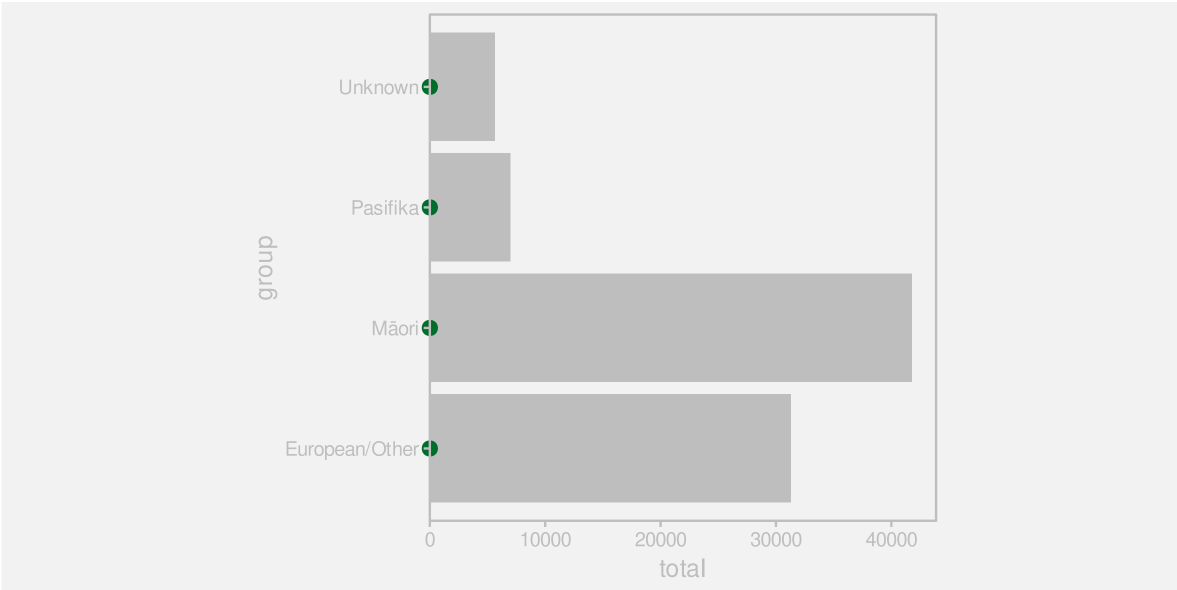


Figure 3.4: A bar plot of number of crimes for different ethnic groups. The data values, the ethnic groups, have been mapped to the **positions** of the bars (the purple dots).

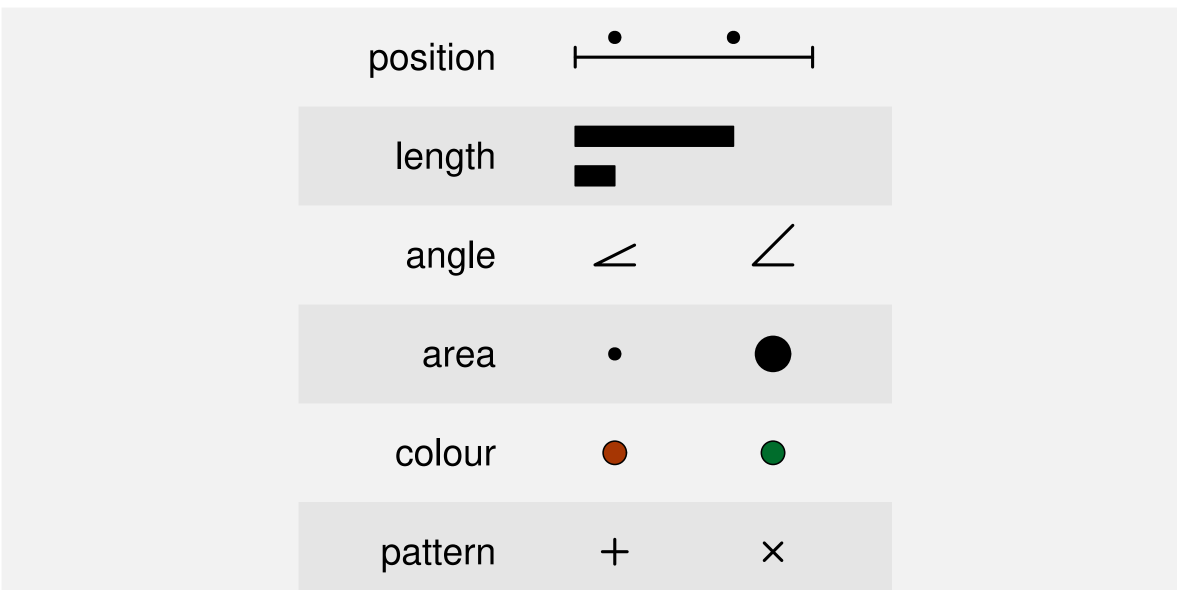


Figure 3.5: Some examples of basic visual features are **position**, **length**, **angle**, **area**, **colour**, and **pattern**.

> One reason why a bar plot is an effective data visualisation is because it involves a mapping from data values to the basic visual features of data symbols—the positions and lengths of bars—and basic visual features are processed rapidly and subconsciously by our visual system.

Figure 3.5 shows some other examples of basic visual features that are commonly used to encode data values in a data visualisation.[39] We will explore each of these in more detail in the following sections.

3.3 Quantitative features

As we discussed in Chapter 2, the real purpose of encoding data values as data symbols is to make use of the visual system to **decode** back to data values from the data symbols. In this section, we are considering mappings from data values to basic visual features, so we need to consider how well we are able to **decode** from **visual features** to **data values** (Figure 3.6).

Figure 3.6: When we encode data values as visual features, we want to map to a visual feature that allows us to decode the data values from the visual fearure.

One issue to consider is whether a visual feature is able to represent all of the information in the data values.[40] Some types of data contain more information than others:

- **Quantitative** data consists of numeric values, which naturally possess not only an ordering, but information about the size of differences between values. For example, the year 2020 comes after the year 2010 and there is a gap of 10 years in between. This is also referred to as **interval** data. **Ratio** data is quantitative data that also contains a "zero" reference value, so there is also information about absolute size and ratios of differences. For example, 200 offenders is twice as many as 100 offenders.
- **Qualitative** data consists of categorical values. **Ordinal** data consists of categories that are ordered, but there is no information about the size of differences. For example, a "high" crime level is more severe than a "medium" crime level, which is more severe than a "low" crime level, but we cannot measure the size of the differences. **Nominal**

data consists of just group labels, e.g., ethnic groups, and the only information we have is that the categories are different from each other.

Figure 3.7 shows that **position**, **length**, **angle**, and **area** are visual features that are capable of representing **quantitative** data values. For example, a bar that is twice as long as another bar comfortably represents a data value that is twice as much as another data value. All but **position** also possess a natural representation of zero.

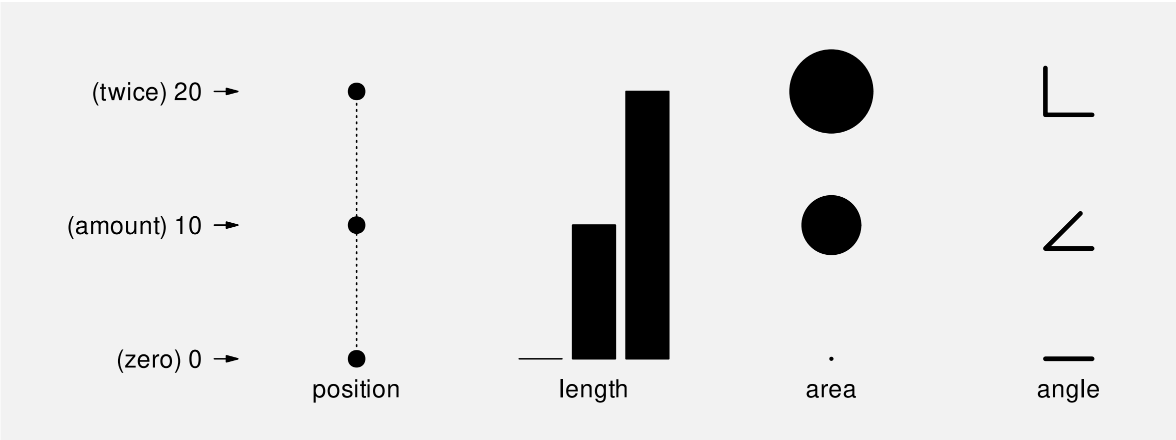


Figure 3.7: The visual features position, length, area, and angle are all appropriate visual features for visualising quantitative data values.

By contrast, Figure 3.8 shows that **colour** and **pattern** are *not* capable of representing quantitative values. For example, the colour blue is not twice the colour green and a plus sign is not 10 more than a triangle. We can tell that there are three *different* values, but in effect, these visual representations *lose* information. It is not possible to decode the data values from the visual represenations.

The bar plot in Figure 3.2 visualises the total number of offenders, which are quantitative data values. These quantitative values are encoded as the **lengths** of bars, which is a visual feature that is capable of conveying all of the information in quantitative values.

> One reason why a bar plot is effective for visualising **quantitative data values** is because the quantitative data values, the counts, are mapped to a **quantitative visual feature**, the lengths of the bars.

3.4 Qualitative features

If we have **qualitative** data, then we are primarily concerned with conveying whether data values are the same or different. Figure 3.9 shows that **position**, **colour**, and **pattern** are

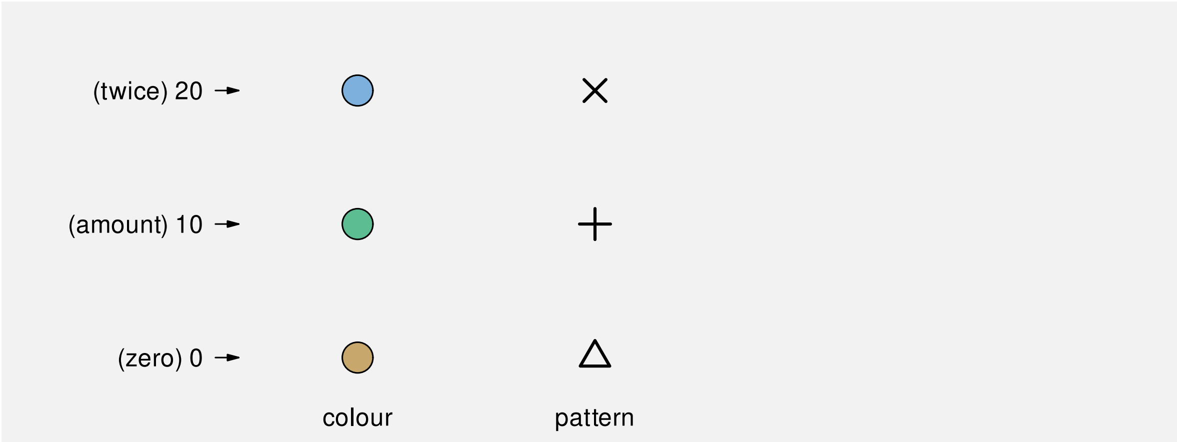


Figure 3.8: The visual features **colour** and **pattern** are not capable of visualising quantitative data values.

capable of representing qualitative data values. For example, blue is clearly different from green, which is clearly different from brown. In Section 2.7 we saw that we also easily identify visual objects that have the same visual features.

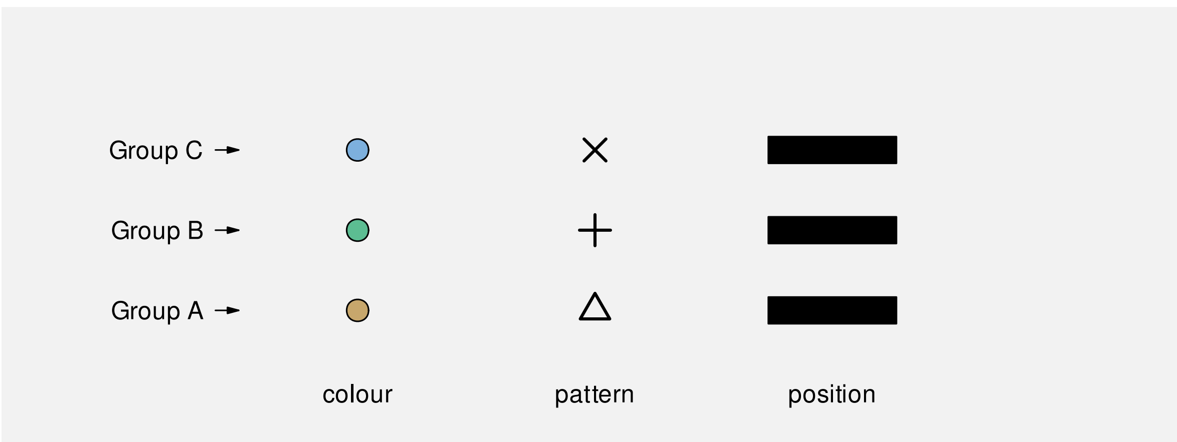


Figure 3.9: The visual features **colour**, **pattern**, and **position** are capable of representing different groups.

By contrast, Figure 3.10 shows that **length**, **area**, and **angle** are not appropriate for representing qualitative data values. While we can identify whether lengths or areas are different from each other, these visual features imply an inherent *order*, which is misleading because that *adds* information that is not present in the data values.

The bar plot in Figure 3.2 encodes the ethnic groups, which are qualitative data values, as the **position** of bars, which is a qualitative visual feature.

Figure 3.10: The visual features **length**, **area**, and **angle** are not appropriate for representing qualitative data values because they imply an ordering of data values that have no inherent order.

> One reason why a bar plot is effective for visualising **qualitative data values** is because the qualitative data values, the groups, are mapped to a **qualitative visual feature**, the positions of the bars.

3.5 Accuracy

Another issue that impacts on how well we can **decode** the data values from a visual feature (Figure 3.6) is the **accuracy** with which we can perceive quantitative values from visual features. For example, how well can we perceive the size of the difference between the length of two bars?[41]

It is easy to demonstrate that judgements of this sort are harder for some visual features than others. For example, Figure 3.11 presents sets of three bars, which differ by **length**, circles, which differ by **area**, and stars, which differ by **pattern** (number of spikes). Judging the relative sizes of the bars is easier than judging the relative sizes of the circles and decoding the stars is extremely difficult.

This issue has been studied experimentally and Figure 3.12 shows the established ordering of the accuracy of basic visual features, with more accurate visual features at the top.[42] A data visualisation that encodes data values as visual features near the top of this ranking will be more effective than one that encodes data values as visual features near the bottom.

The bar plot in Figure 3.2 encodes the total number of offenders, which are quantitative values, as the **lengths** of bars, which is an **accurate** visual feature. This means that viewers can

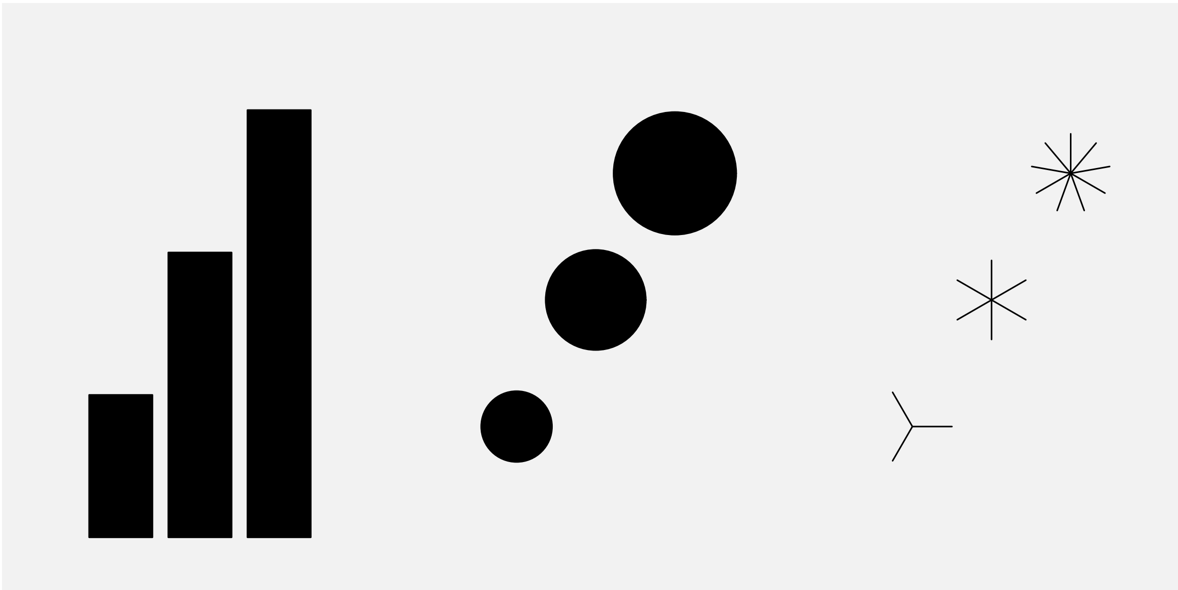


Figure 3.11: A demonstration of differences in accuracy between basic visual features. For each set of symbols, how much “larger” are the “larger” symbols?

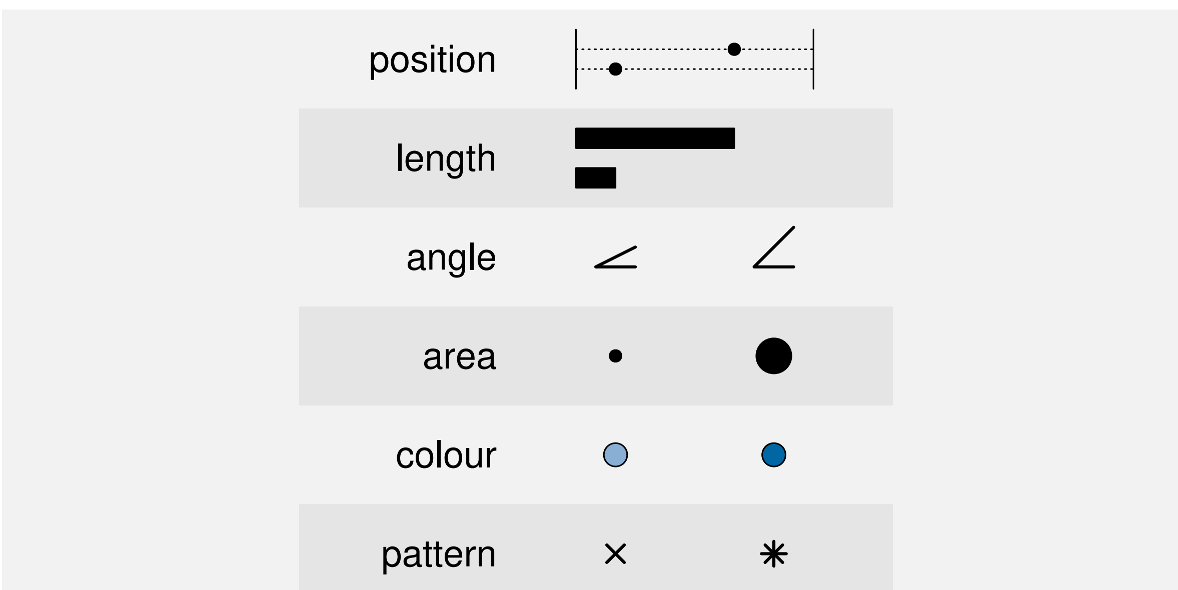


Figure 3.12: An ordering of basic visual features in terms of their accuracy, with more accurate at the top.

make accurate comparisons between the lengths of the bars.

> One reason why a bar plot is effective for visualising **quantitative data values** is because the quantitative data values, counts or proportions, are encoded as a very **accurate visual feature**, the lengths of the bars.

3.6 Capacity

When we are visualising **qualitative** data values, we are no longer concerned with accurately perceiving the size of differences; all we are concerned with is perceiving whether two values are different or the same. To be slightly more precise, we are concerned with being able to *identify* different values, for example, which colour, out of a set of known colours, is this one?[43]

The issue in this scenario is the **capacity** of a visual feature: how many different categories can be represented? In particular, how many different categories can be represented with the viewer able to reliably identify each separate category?

Figure 3.13 shows that **colour** has a limited capacity. A small number of different colours are very easy to differentiate, but the differences become harder to perceive with a greater number of colours.[44] This difficulty only increases with irregular arrangements.

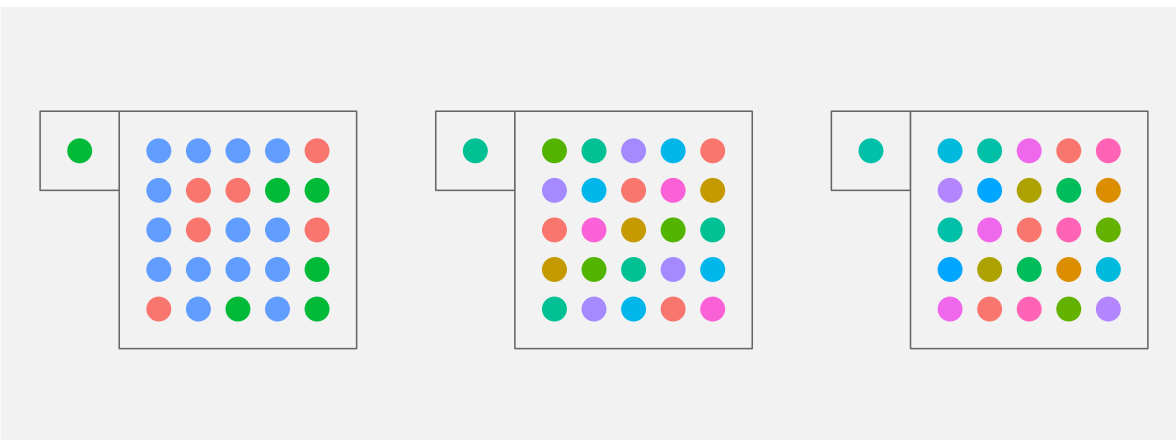


Figure 3.13: Mapping qualitative data values to colour works well for a small number of colours, but less well for a large number of colours. It is quite easy to identify which green the single dot corresponds to amongst three colours on the left, but it is much harder with the seven colours in the middle, harder still for the eleven colours on the right.

Pattern also has a very limited capacity (Figure 3.14), again exacerbated by less regular arrangements.[45]

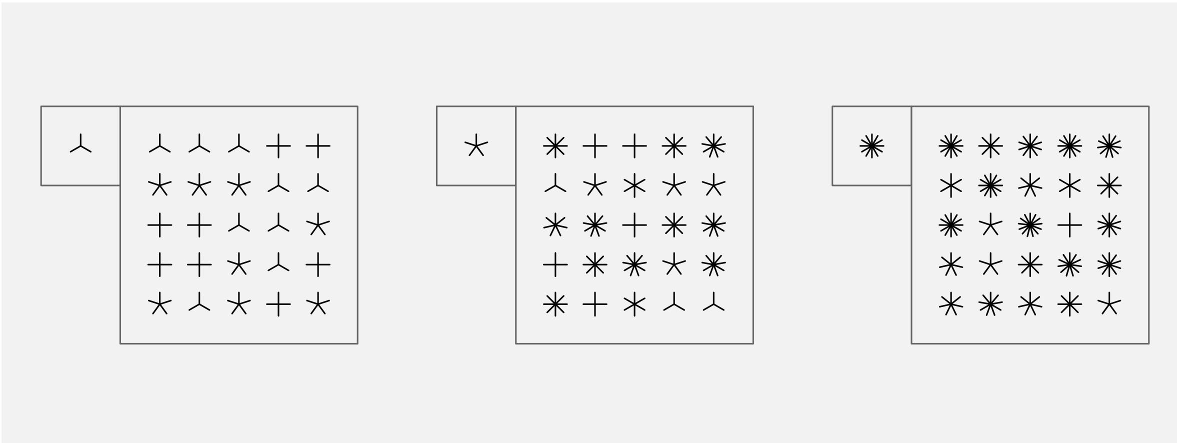


Figure 3.14: Mapping qualitative data values to pattern works well for a small number of patterns, but less well for a large number of patterns.

By contrast, Figure 3.15 shows that **position** has a greater capacity. We can easily identify the location of the single dot even as the number of alternative locations becomes large.

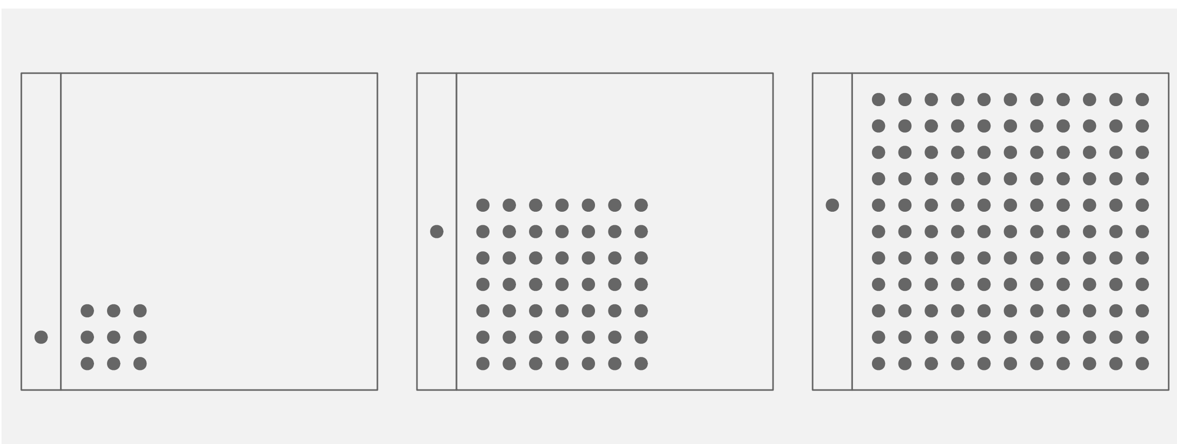


Figure 3.15: Mapping qualitative data values to position works well for both a small number of positions and for a large number of positions.

The bar plot in Figure 3.2 encodes the ethnic group as the vertical **position** of bars, which makes it easy to identify the different ethnic groups.

> One reason why a bar plot is effective for visualising **qualitative data values** is because the qualitative data values, the groups, are encoded as a visual feature that has excellent **capacity**, the positions of the bars.

3.7 Case study: Dot plots

We have seen that, for several reasons, Figure 3.2 is an effective way to visualise the total number of offenders for different ethnic groups. Figure 3.16 shows an alternative representation of these data in the form of a dot plot[46]

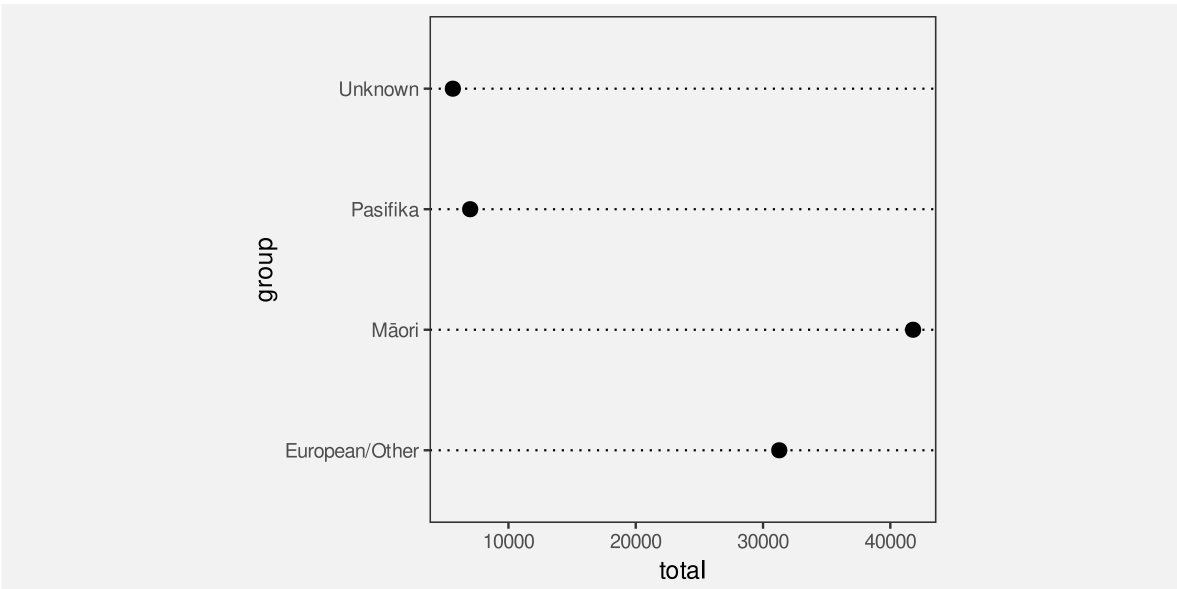


Figure 3.16: A dot plot of the number of crimes for different ethnic groups. The data symbols in this plot are the data points.

Like a bar plot, a dot plot is an effective data visualisation, and for very similar reasons. Dot plots uses a different data symbol—data points instead of bars—but **each data value** is **encoded** as the basic **visual feature** of **one data symbol**. This means that we can **decode** individual data values from the data symbols and answer questions like: what is the difference in the number of offenders between Māori and European/Other?

Furthermore, the quantitative data values (the totals) are encoded as the horizontal **position** of the points, which is a visual feature that is not only capable of representing numeric values, it is also a very accurate visual feature. The qualitative data values (the ethnic groups) are encoded as the vertical **position** of the points, which is a visual feature that is capable of representing qualitive data and is a visual feature with a high capacity.

It could be argued that the dot plot in Figure 3.16 is even *more* effective than the bar plot in Figure 3.2 for perceiving the differences between totals because **position** ranks higher than **length** in Figure 3.12. On the other hand, if we want to compare totals in terms of their ratios, Figure 3.2 is superior because it includes a representation of zero.

3.8 Case study: Pie charts

Figure 3.17 shows yet another visualisation of the total number of offenders for different ethnic groups, this time a pie chart. This is a less effective representation of the data than the bar plot in Figure 3.2 or the dot plot in Figure 3.16 and we can use familiar reasoning to see why.

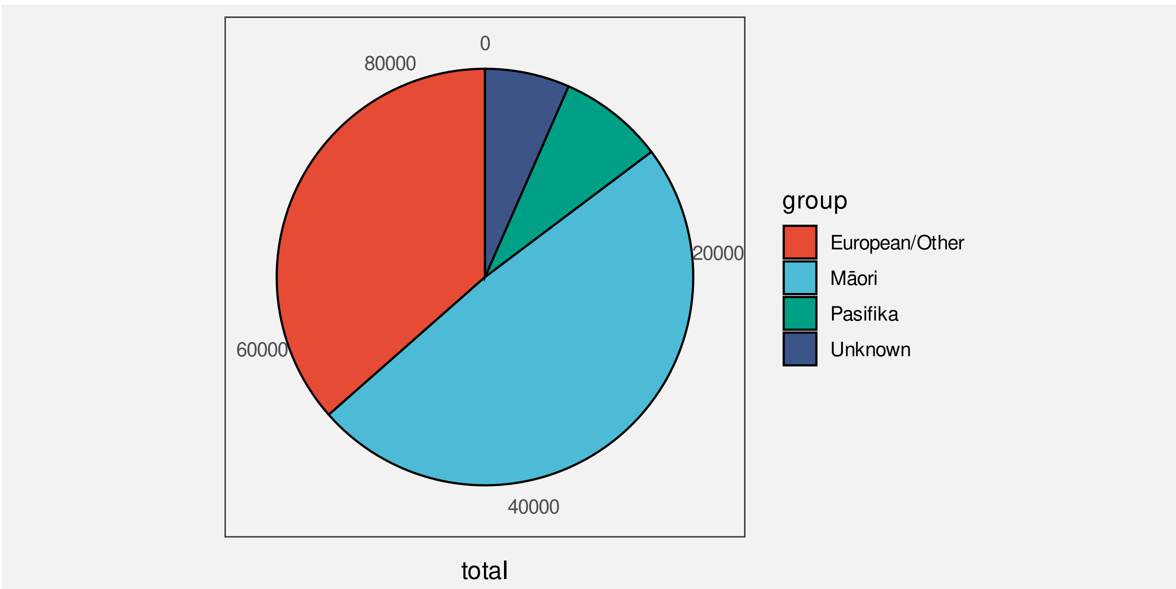


Figure 3.17: A pie chart of the number of crimes for different ethnic groups. The data symbols in this plot are the pie wedges.

The data symbols in Figure 3.17 are wedges or segments of a circle and again we have each data value mapping to a single data symbol. For example, each ethnic group maps to a separate wedge of the pie. As with bar plots and dot plots, in a pie chart the data values are encoded as basic visual features of these symbols. However, a pie chart makes use of different visual features: The total values are encoded as the **angle** and/or **area** of the wedges and the group values are encoded as the **colour** of the wedges.[47] Although **colour** is an appropriate visual feature to represent **qualitative** data, and it has sufficient capacity to differentiate between only four groups, the encoding of quantitative values as **angle** and/or **area** leads to less accurate decoding of the total number of offenders. For example, it is much harder to decode the number of offenders with an unknown ethinicity from the pie chart in Figure 3.17 compared to the bar plot in Figure 3.2 or the dot plot in Figure 3.16.

Although the situation seems bleak for pie charts, we have not yet seen all of the ways that we can assess different mappings of data values to data symbols. We will return to pie charts in Section 7.2 to show that all is not lost.

3.9 Summary

A simple mapping of data values to data symbols involves mapping **each data value to a separate data symbol**. This allows the viewer to **decode** individual data values from the data symbols.

A simple mapping of data values to data symbols also involves **encoding** each data value as a basic **visual feature** of the data symbol: position, length, area, angle, colour, and pattern. This is effective because visual features are decoded rapidly and without conscious effort.

Position, length, area, and angle are appropriate for encoding quantitative data because we can decode amounts from these visual features. We can decode position and length more accurately than area and angle.

Position, colour, and pattern are appropriate for encoding qualitative data because we can decode differences from these visual features. We can represent a large number of categories using position, but only a few categories using colours and patterns.

Bar plots, dot plots, and pie charts all provide examples of simple mappings from data values to data symbol. We can decode bar plots and dot plots accurately because they involve encodings of data values to position and length. We cannot decode pie charts as accurately because they involve encoding data values to angle and area.

Notes

[37] This one-to-one mapping between data values and data symbols is called a *bijective* mapping (Ziemkiewicz and Kosara 2009).

[38] Some authors prefer to say that the total number of offenders is mapped to the **position** of the "top" of the rectangle (e.g., William S. Cleveland and McGill 1984), but the difference is not significant because both length and position are excellent visual features in terms of decoding (Section 3.5).

It is also worth noting that there can be a discrepancy between the encoding that the creator of a data visualisation employs and the decoding that the viewer chooses to use (Hill and Imokhai 2025).

[39] In this book, we use *visual feature* to describe a basic visual quality of a data symbol, in place of more established terms like *visual variable* (Bertin 1983), *aesthetic* (Wilkinson 2005), or *visual channel* (Munzner 2014).

This choice of terminology is deliberate so that we can elide the basic visual quality of a data symbol with the early stages of visual processing (Section 2.2). There is not a direct correspondence between these two concepts, but it is a useful simplification.

There are also differences in terminology for individual visual features. For example, we use angle where others use *tilt* or *slope* and the underlying visual feature is typically referred to as *orientation*.

The use of pattern for data point *shape* is the most tenuous connection. Data point shape is commonly included as a visual channel, so we provide a connection to basic visual features via the early perception of patterns, even though there is more involved in the perception of data point shape (Chen 1982; J. Li, Wijk, and Martens 2009).

40 The idea of matching the type of data—nominal, ordinal, interval, or ratio—as well as the categorisation of visual feature types is described in Zhang (1996). Munzner (2014), following Mackinlay (1986), refers to this idea as *expressiveness*.

41 The pioneering work on the accuracy of different visual features for statistical graphics was carried out by William S. Cleveland and McGill (1984). Their results have been replicated several times in subsequent studies, including Heer and Bostock (2010). The rankings were extended to a larger set of visual features by Mackinlay (1986), although somewhat only by inference from other psychophyscial results, rather than by direct experiment.

Munzner (2014), following Mackinlay (1986), refers to this idea as *effectiveness*.

An argument for ranking the accuracy of visual features can also be made from Steven's Law (Stevens 1957), which states that the perception of the magnitude of different stimuli follows a power law. For example, the perception of length has a power of approximately one (linear), which suggests that we accurately perceive the magnitude of lengths, while area has a power less than one, which suggests that we underestimate the magnitude of areas.

42 In William S. Cleveland and McGill (1984), comparisons of bars were treated as **position** judgements rather than **length** judgements, unless the bars were unaligned (see Section 5.2), so this table is a slightly fuzzy representation of their results. However, the general ordering of visual features is basically consistent with William S. Cleveland and McGill (1984).

These rankings are also based on a relative judgement (what proportion of A is B?), which is just one possible decoding task, albeit a common one, particularly for bar plots. The ranking of visual features may differ for other sorts of decoding (McColeman et al. 2022). There is also some evidence that the rankings may differ between individuals even for simple relative judgements (Davis et al. 2024).

43 The question "Which one is this?" is different from the question "Are these two different?". The former involves "absolute judgements" (Miller 1956) to identify a stimulus from a known set of stimuli. The latter involves relative judgements and just-noticeable differences (JND), which are considerably easier and something we have a much higher sensitivity to. For example, we can differentiate between several million different colours (Pointer and Attridge, n.d.), but we can only make absolute judgements with up to 10 different colours.

44 The recommended maximum number of colours varies between sources. However, the consensus is that it is a quite small number.

"Do not use more than 10 colors for coding symbols if reliable identification is required, especially if the symbols are to be used against a variety of backgrounds." (Ware 2021, 126).

Healey (1996) says the effective maximum is 7.

45 The research evidence for the limited capacity of pattern is suggestive rather than conclusive. The recommendations of experts only extend to very small sets of patterns.

For example, William S. Cleveland (1985b) recommends using the following set of patterns, in this order: `o`, `+`, `<`, `s`, and `w`.

Tremmel (1995) only provides guidance for three different patterns and suggests using separate plots to accommodate a larger number of categories (when relying on pattern).

[46] Dot plots (William S. Cleveland 1985a) are different from scatter plots because they plot one quantitative variable against (at least) one qualitative variable, rather than plotting two quantitative variables against each other.

[47] We are assuming here, as is common, that the viewer will decode from the **angle** of pie wedges. That is not necessarily the case, for example Eells (1926b) found evidence that viewers may decode from the area, arc length, or even chord length of pie wedges. Fortunately, the larger point holds, because most of these alternatives are relatively poor visual features in terms of decoding accuracy.

4 Position

We saw in Chapter 3 that **position** is a very effective visual feature. It is appropriate for mapping quantitative data values (Figure 3.7) and it is appropriate for mapping qualitative data values (Figure 3.9). **Position** is a visual feature with high accuracy (Figure 3.12) and with high capacity (Figure 3.15).[48]

This means that, if we do not encode data values as the **position** of data symbols, it can be more difficult to decode data values from the data symbols.[49] For example, Figure 4.1 shows another data visualisation of the total number of offenders in New Zealand for different ethnic groups (Table 3.1). We have previously seen a bar plot (Figure 3.2), a dot plot (Figure 3.16), and a pie chart (Figure 3.17) of these data. In Figure 4.1, the data symbols are circles, the number of offenders is encoded as the **area** of the circles, and the ethnic groups are encoded as the **colour** of the circles. One reason why it is more difficult to decode data values from Figure 4.1, compared to, for example, Figure 3.2, is because none of the data values are encoded using **position**.

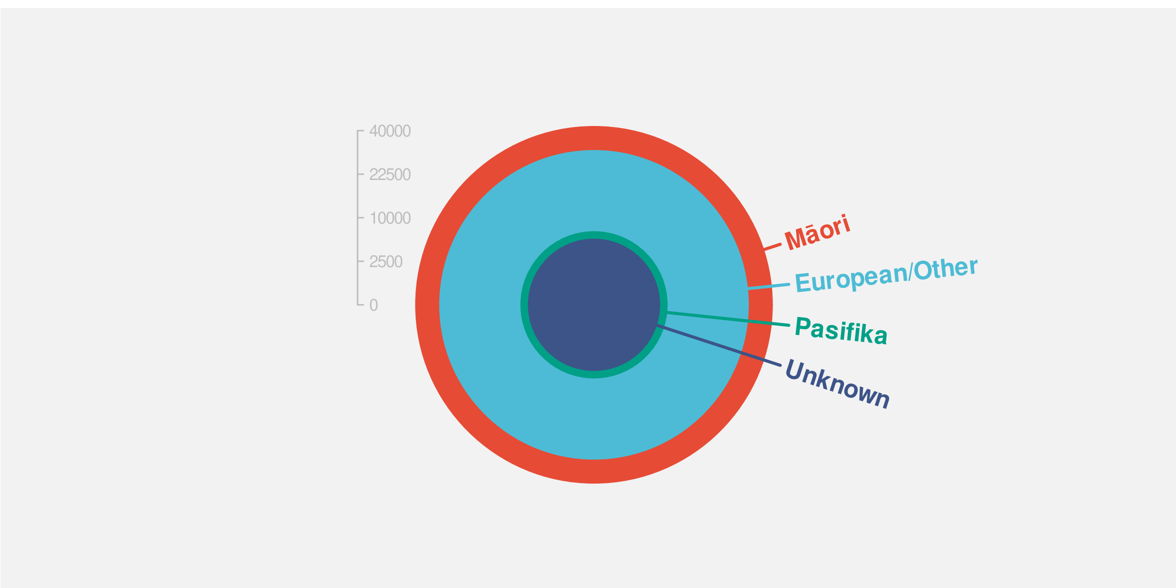


Figure 4.1: A data visualisation that does not make use of position.

However, we have already seen several data visualisations that do not map *all* data values to position (e.g., Figure 1.1 and Figure 1.2), so there must be more to learn about mapping data values to basic visual features.[50] In this chapter, we look more closely at the mapping of data values to the **position** of data symbols.

4.1 Perception is relative

In Section 3.5, we saw that **position** was the most accurate visual feature for decoding quantitative values (Figure 3.12). However, that accuracy is for *relative* judgements rather than absolute judgements. In other words, we are good at *comparisons* between positions,[51] which is one of the most important decoding tasks in data visualisation.[52]

For example, the box on the left of Figure 4.2 shows a dot at a vertical positon along a dotted line. We can decode an amount from the position of the dot, but only relative to the ends of the dotted line; the dot is three-quarters of the way from the bottom to the top of the line. We cannot decode an absolute data value from the dot.[53]

The box in the middle of Figure 4.2 shows two dots at vertical positions along two dotted lines. Again, we can only compare the positions of the two dots—the dot on the left is three times as far up its dotted line as the dot on the right.

The box on the right of Figure 4.2 shows two dots again, but this time adds an axis on the right edge of the box. It is now possible to decode an absolute value for each dot (75 and 25), but that is still only possible by making comparisons. We decode the positions of the dots relative to the positions of the tick marks on the axis.

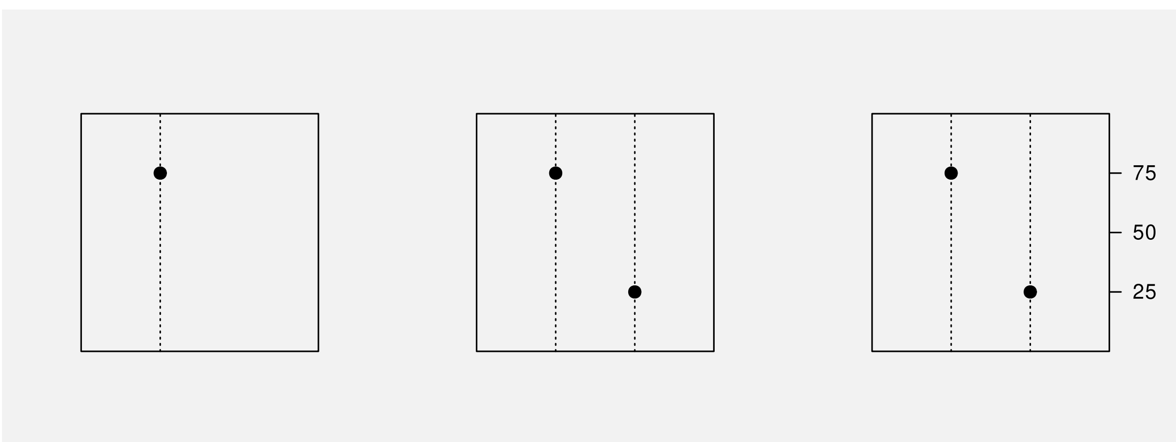


Figure 4.2: Perception of position is relative rather than absolute.

> A data visualisation that encodes a data value as the position of a data symbol is effective because we can decode the *comparisons* between the positions accurately.

4.2 Unaligned position

We have established that **position** is a very accurate visual feature (Figure 3.12), although that accuracy is for comparisons rather than for decoding individual values. Another limitation on the accuracy of **position** is that it works best when the comparison is between positions with a common baseline. For example, it is easier to compare the two positions on the left of Figure 4.3 compared to the two positions on the right.

Figure 4.3: It is easier to compare two positions on the left because they are **aligned**; they share a common baseline. The two positions on the right are harder to compare because they are **unaligned**.

Figure 4.4 shows yet another data visualisation of the total number of offenders in New Zealand for different ethnic groups (Table 3.1). This data visualisation is similar to Figure 3.16 because it encodes the number of offenders to **position** along a line. However, it is harder to compare the number of offenders between different ethnic groups in Figure 4.4 because the positions are not aligned.

> Decoding data values from positions is most effective when the positions share a common baseline.

4.3 Position is a scarce resource

In Section 3.6 we saw that **position** has a higher capacity for decoding qualitative data values compared to colour or pattern (Figure 3.15). However, position has a limitation that does not affect colour or pattern: if we use the same position for more than one data value, we end up drawing data symbols on top of each other.

For example, in Figure 4.1, we use the same position for all four circles that represent different ethnic groups, so the circles are drawn on top of each other. The total number of youth offenders for each ethnic group is encoded as the **area** of a circle. The largest number of offenders is for the Māori ethnic group–the orange circle—with European/Other the next

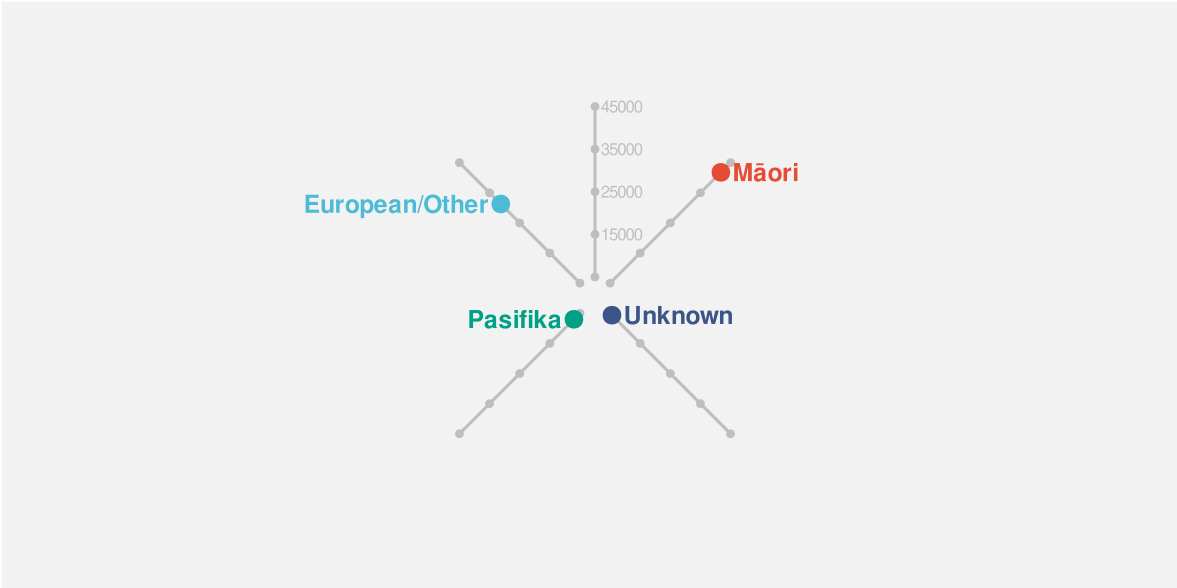


Figure 4.4: A data visualisation that makes use of *unaligned* position.

largest—the light blue circle. However, because the light blue circle is drawn on top of the orange circle, all that we can actually see is the part of the orange circle that extends beyond the light blue circle.

This represents a serious problem. It is difficult for our visual system to decode from a data symbol to a data value if we cannot see all of the data symbol.[54]

One way to express this limitation is that position is very effective for encoding *different* data values, but it is not very effective for encoding the *same* data value.

> Decoding more than one data value from two data symbols that share the same position is, at best, compromised because we cannot see all of both data symbols.

4.4 Case study: Overplotting

The Rugby World Cup is a competition that is held every four years, with the first event taking place in 1987. We have data on the the number of points that were scored in every World Cup match by Tier One nations (the top 10 rugby-playing nations). Table 4.1 shows a sample of these data.[55]

Table 4.1: A table of the number of points scored by Tier One nations in Rugby World Cup matches, along with the global hemisphere of origin. Each row represents the points scored by one team in one match. There are 294 rows in total, with just the first 6 rows shown here.

team	hemisphere	scored
Argentina	South	8
Argentina	South	18
Argentina	South	13
Argentina	South	28
Argentina	South	10
Argentina	South	7

Figure 4.5 shows a dot plot of the points scored in a match for all teams and all matches. This data visualisation has encoded the number of points scored as the horizontal **position** of data points. In theory, that allows us to decode from data points to a number of points scored. For example, we can see that one team from the southern hemisphere scored just over 100 points in one match.

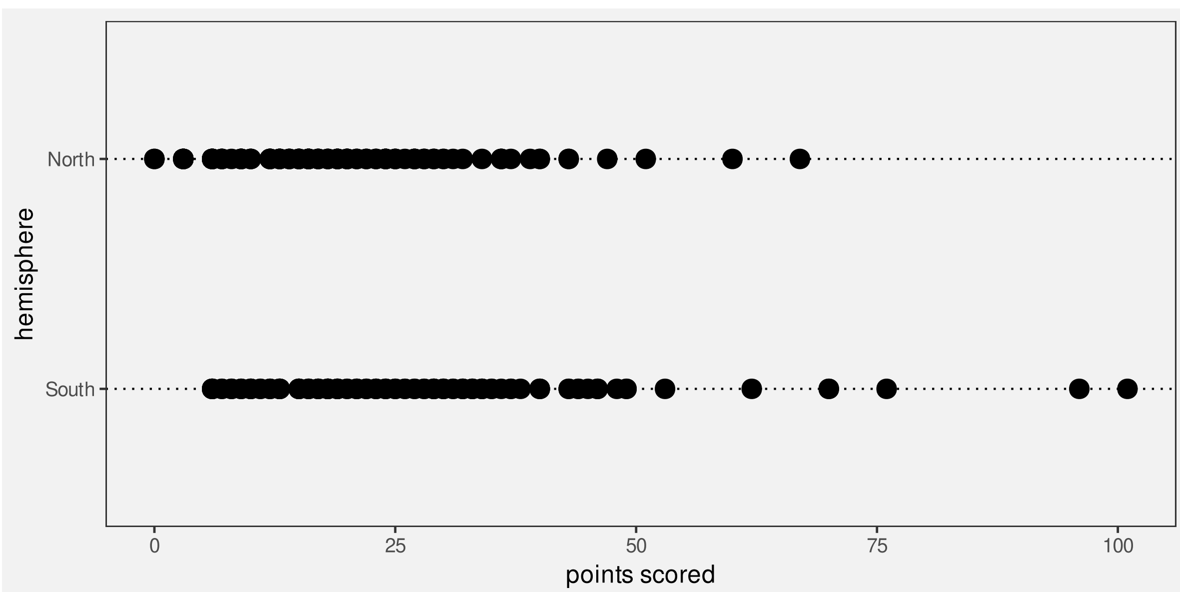


Figure 4.5: The number of points scored in Rugby World Cup games by teams from the northern and southern hemispheres.

However, as Table 4.2 shows, exactly the same number of points were scored by more than one team (from the same hemisphere) and/or in more than one match. This means that there are multiple data points at exactly the same position, which means that there are multiple data points drawn on top of each other, which in effect means that there are data points that we cannot see.

Table 4.2: A table of the number of points scored by Tier One nations in Rugby World Cup matches, along with the global hemisphere of origin. Each row represents the points scored by one team in one match. There are 294 rows in total, with just the first 6 rows shown here, ordered by the number of points scored.

team	hemisphere	scored
England	North	0
Scotland	North	0
England	North	3
Ireland	North	3
Ireland	North	3
Italy	North	3

Although it is still possible to decode that, for example, at least one team from the northern hemisphere scored zero points in at least one match, it is not possible to decode whether that was just one team or just one match. We cannot decode an individual data value from each data symbol.[56]

Figure 4.6 shows the same data as Figure 4.5, but with the data points semi-transparent so that we can see that many of the points overlap. The same number of points was scored by multiple teams in multiple matches. This demonstrates one approach to resolving the problem of overplotting; we will see several other approaches in later chapters.

4.5 Cartesian coordinates

In Figure 4.5, we have encoded *two* sets of data values using **position**. The number of points is encoded as the horizontal position of the data points and the hemisphere of origin for each team is encoded as the vertical position of the data points. This *cartesian coordinate system* is effective because the visual system is better at decoding horizontal and vertical positions compared to oblique orientations like in Figure 4.4.[57]

Because the horizontal and vertical positions are perpendicular, Cartesian coordinates also mean that we can make use of **position** twice without the position encodings interfering with each other.[58] For example, in Figure 4.5, we can decode which hemisphere that a data symbol is from separately from decoding how many points were scored. The fact that a team is from the northern hemisphere does not *by itself* imply anything about the number of points scored. Put another way, any difference between the data symbols for northern hemisphere teams and the data symbols for southern hemisphere teams must reflect something about the data.

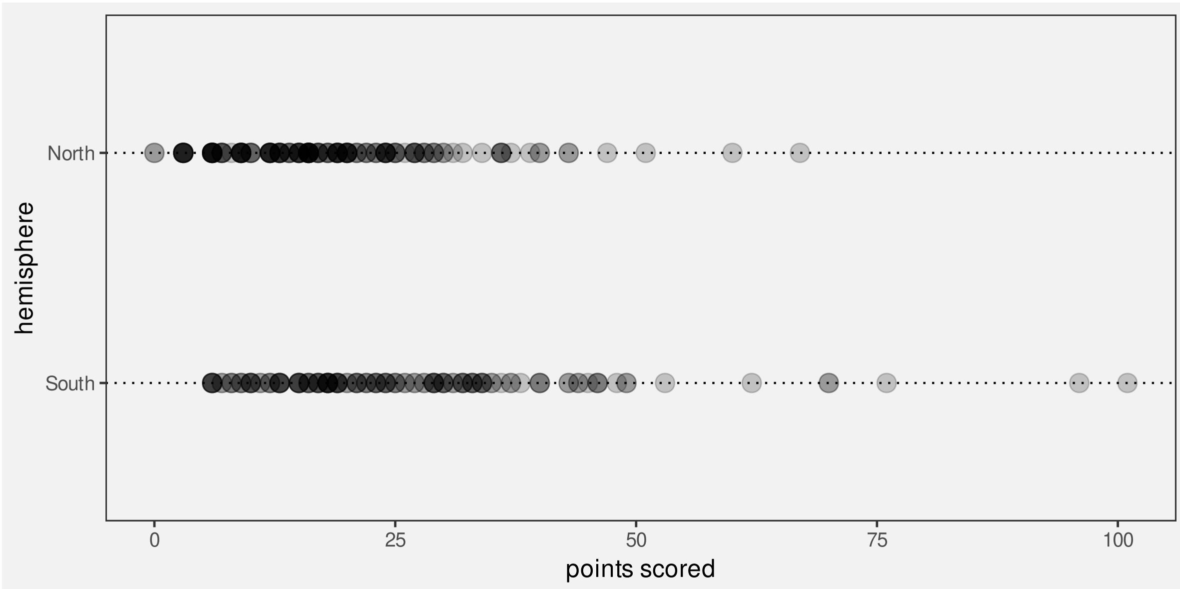


Figure 4.6: The number of points scored in Rugby World Cup games by teams from the northern and southern hemispheres.

A good example of a data visualisation that does not employ Cartesian coordinates is a pie chart, like Figure 3.17, though that is also a case where data values are *not* encoded as **position**.

> Cartesian (perpendicular) coordinates are effective because we can encode data values using position *twice*; we can decode data values from horizontal positions and we can separately decode data values from vertical positions.

4.6 Case study: Ternary plots

Table 4.3 shows data on the severity of crimes committed by youth in New Zealand. For each year, we have the proportion of criminal offences that were low, medium, or high severity.[59]

Table 4.3: A table of the proportion of crimes at different levels of severity for each year.

year	Low	Medium	High
2011	0.306	0.624	0.070
2012	0.310	0.616	0.074
2013	0.296	0.631	0.074
2014	0.293	0.638	0.069

Table 4.3: A table of the proportion of crimes at different levels of severity for each year.

year	Low	Medium	High
2015	0.254	0.656	0.090
2016	0.246	0.655	0.099

Figure 4.7 shows a ternary plot of these data.[60] In this data visualisation, there are three variables (three proportions that sum to 1), and the data values from each variable has been encoded as position. However, the positions are along axes that are not perpendicular. This means that it is not as easy to decode individual data values or compare data values because we are poorer at decoding positions that are not horizontal or vertical. Furthermore, because the three proportions sum to 1, a change in one proportion necessitates a change in at least one of the others. In other words, we cannot easily decode the difference between two data values from the difference between two data symbols.

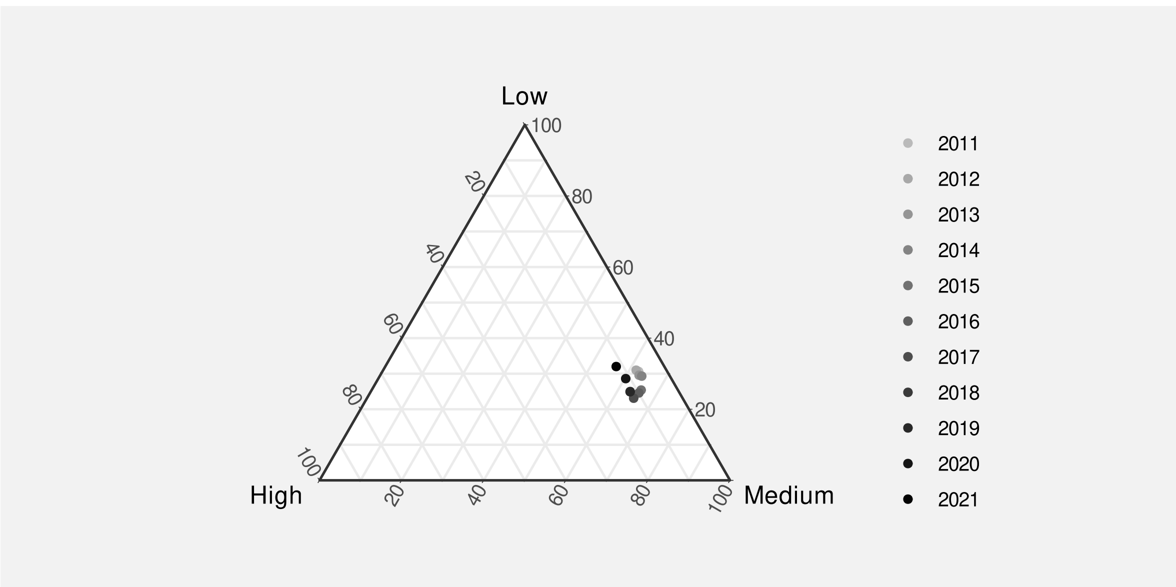


Figure 4.7: A ternary plot of the data in Table 4.3. We can see that for the entire decade around 60% of crimes have been of medium severity, with around 30% of low severity, and only 10% of high severity.

4.7 Non-linear mappings

Table 4.4 shows the total number of youth offenders broken down by type of crime. We have one qualitative variable, crime type, and one quantitative variable, total nunber of offenders.

Table 4.4: A table of the total number of youth offenders in New Zealand (2011 to 2021) broken down by type of crime. There are 16 different crime types in toral, with only the first 6 shown here.

type	total
Abductions, threats	2709
Against justice	1177
Causing injury	11326
Dangerous acts	3124
Deceptions	1198
Homicides	57

Figure 4.8 shows of dot plot of these data, with the total number of offenders encoded as the horizontal position of circles, one per type of crime. This plot is effective for comparing the total number of offenders between types of crime, both in terms of which crimes are more common than others and how much more common some crimes are than others. For example, we can see that theft is about twice as common as the next most common crime type.

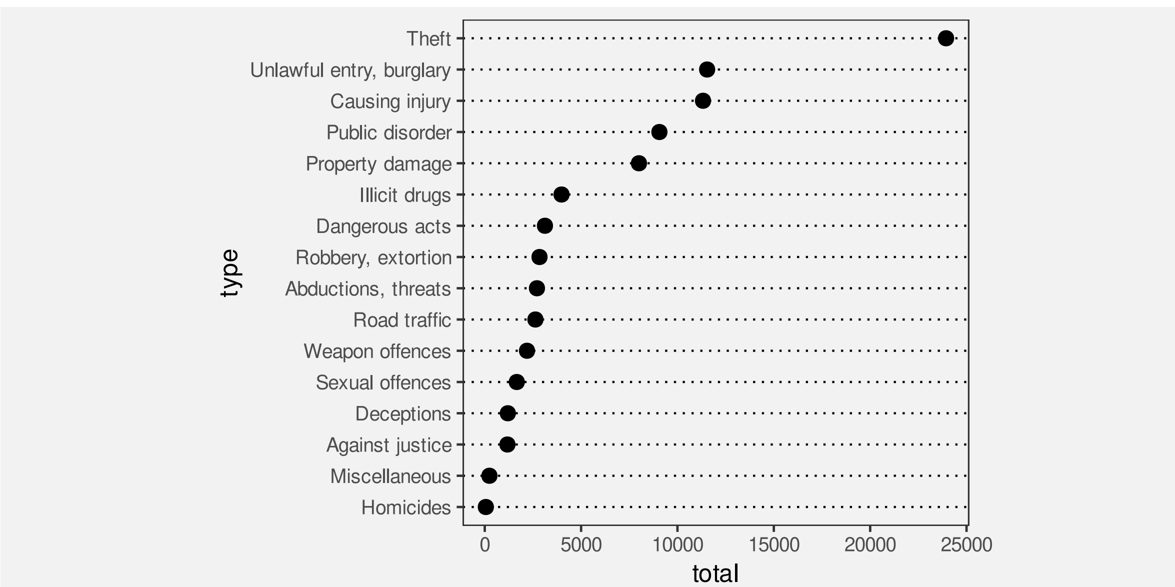


Figure 4.8: A dot plot of the number of crimes for different regions of New Zealand.

The effectiveness of Figure 4.8 comes not only from the fact that data values are encoded as the **position** of data symbols, but also from the fact that the encoding is *linear*. In Figure 4.8, the horizontal difference between 0 offenders and 5,000 offenders is exactly the same horizontal distance as the difference between 5,000 offenders and 10,000 offenders. The effective decoding of position is dependent on this sort of linear encoding.[61]

When a small number of data values are considerably larger than the majority of data values, as in Figure 4.8, a non-linear scale is sometimes employed in order to see the differences between the smaller values more easily. For example, Figure 4.9 shows the same data and the same encodings, but with a *log scale*. Rather than encoding the total number of offenders as the position of the circles, the log (base 10) of the total number of offenders is encoded as the position of the circles.[62]

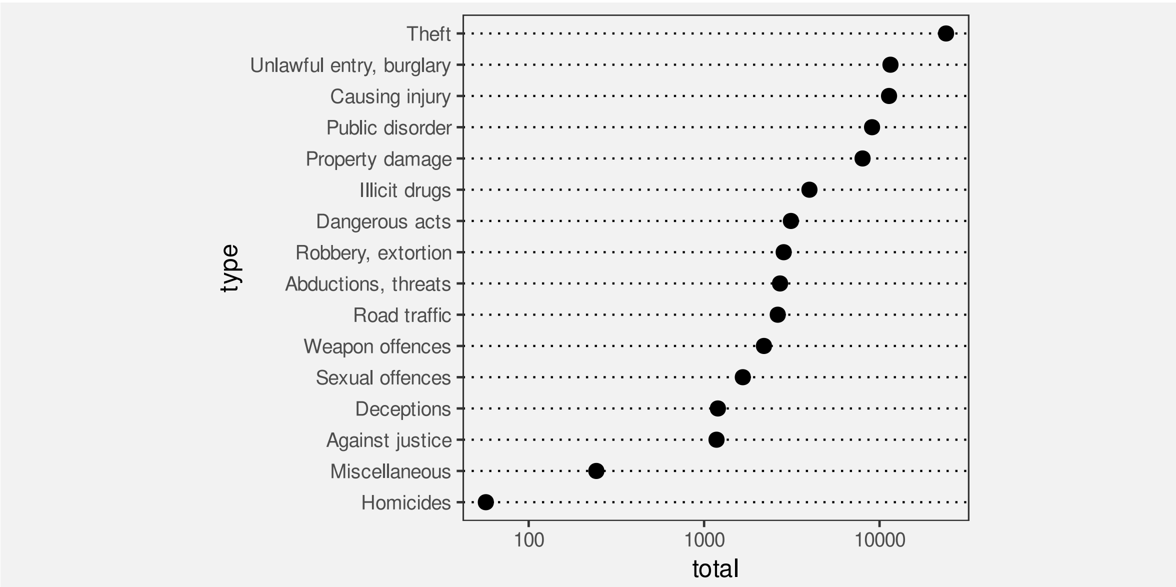


Figure 4.9: A dot plot of the number of crimes for different regions of New Zealand, with a non-linear (log) scale on the x-axis.

Unfortunately, as a result of the non-linear horizontal scale, we lose the accuracy of the position encoding. We can no longer accurately decode from position to data values because the same horizontal distance means different things in terms of data values at different locations along the x-axis. For example, it is difficult to decode the number of offenders for Miscellaneous crimes; the position of the circle data symbol is visually roughly half-way between 100 and 1000, but the data value is not half-way between 100 and 1000. It is even more difficult to decode the size of the difference between the number of offenders for Miscellaneous crimes versus Homicides. Worse still, the difference between Miscellaneous and Homicide is visually very similar to the difference between Miscellaneous and crimes Against justice, but the differences between those data values are very different.

We can still decode simple ordinal information from Figure 4.9—which type of crimes are more common than others—but we have lost the ability to decode quantitative information from the positions of the data symbols. We have essentially lost information.

Figure 4.10 shows the logged data again, but this time with labels on the x-axis that show logged values rather than the original data values. This shows that, in terms of logged data values, the positions of the data symbols are linear. We can accurately decode from the positions of the circles in Figure 4.10, but only to recover logged data values. If we want to compare logged data values, this is fine, but usually what we want is to compare the original data values—we need to decode to a logged value and then raise 10 to that power. For example, the log of the number of offenders for Deceptions is approximately 3, so the number of offenders is approximately 1000. However, this mental transformation requires much more conscious effort for most people, so we effectively lose the benefit of using a data visualisation, at least for the purpose of decoding quantitative information (Section 2.9).

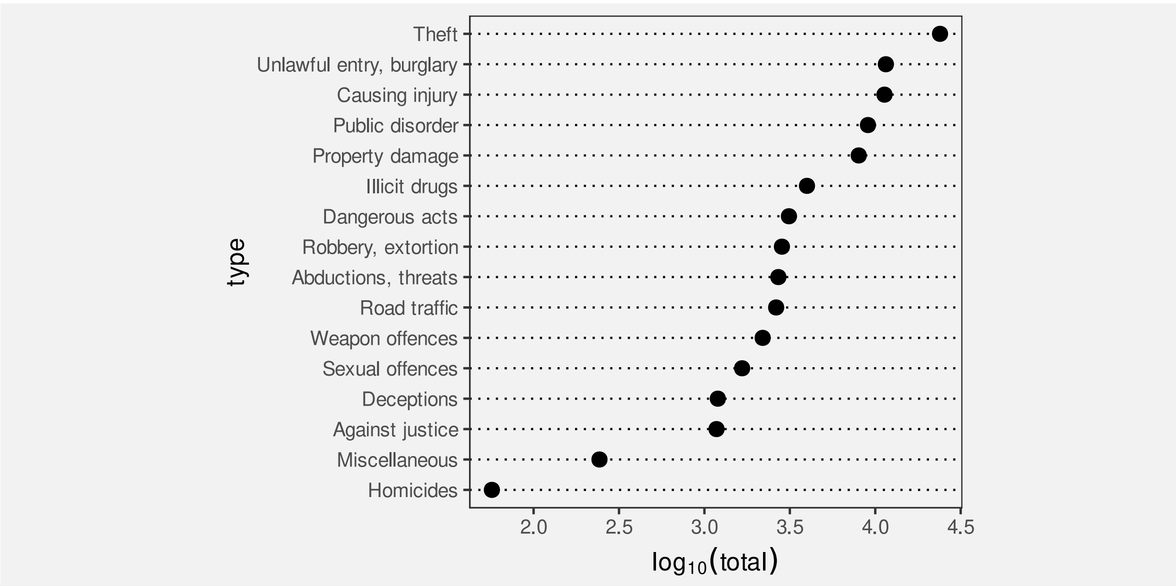


Figure 4.10: A dot plot of the number of crimes for different regions of New Zealand, with a non-linear (log) scale on the x-axis.

Although a non-linear encoding from data values to the positions of data symbols presents problems for decoding quantitative information, there are still situations where non-linear scales can be useful. For example, if we are only interested in ordinal comparisons, then non-linear positions may still be effective. We will see more examples in later sections (Section 6.8, Section 10.9).

> The encoding of quantitative data as the position of data symbols is only effective for decoding quantitative data if the encoding is linear.

4.8 Summary

Mapping data values to the position of data symbols allows for effective decoding for both quantitative and qualitative information, with the following caveats:

- For quantitative values, what we can accurately decode are *comparisons* between quantitative values, not absolute quantitative values.
- The decoding is most accurate for positions that share a common baseline.
- Encoding identical data values as the positions of data symbols means that the data symbols overlap, which compromises our ability to decode data values from the data symbols.
- We can encode one set of data values as horizontal positions and another set of data values as vertical positions because we can decode horizontal and vertical positions separately.
- Decoding quantitative data values from the positions of data symbols is only effective if the encoding is linear.

Notes

[48] It can be argued that position is a visual feature that is fundamental to our perception of any image.

"There is fairly widespread philosophical agreement, which certainly accords with commonsense, that the spatial aspects of all existence are fundamental. Before an awareness of time, there is an awareness of relations in space." (Robinson and Petchenik 1976, 14)

"The concept of spatial relatedness … is a quality without which it is difficult or impossible for the human mind to apprehend anything." (Robinson and Petchenik 1976, 123)

"According to Kubovy (1981), position in space and time have a dominant role in perceptual organization. In relation to visual information displays, the contention for space as an indispensable variable seems indisputable. For centuries, humans in all cultures have "mapped" the nonvisible to space in order to facilitate understanding." (MacEachren 1995; citing Kubovy 1981)

"We process spatial properties (position and size) separately from object properties (such as shape, color, texture, etc). Furthermore, position in space and time has a dominant role in perceptual organization, as well as in memory." (Meirelles 2013)

[49] Decoding individual data values from individual data symbols is not the only visual task that we need to perform, so it is not *always* true that we should encode data values as **position**. As we explore other sorts of mappings in later chapters, we will see reasons why **position** is not always best.

[50] As Bertini, Correll, and Franconeri (2020) put it: "Why Shouldn't All Charts Be Scatter Plots?"

[51] In the experiments described in William S. Cleveland and McGill (1984), which are the basis of the accuracy ranking in Figure 3.12, "subjects were asked to judge what percent the smaller is of the larger".

[52] Amar, Eagan, and Stasko (2005), in their taxonomy of visual tasks, consider a "low-level comparison as being a fundamental cognitive action" for reading a data visualisation.

"The fundamental task in data analysis is to make smart comparisons - we're always trying to answer the question 'Compared with what?' […] It always comes down to making and showing smart comparisons." (Tufte 2006, 127)

"Almost everything we do with data involves comparison" (Tukey 1990, 329)

"The lesson is that visualization is not good for representing precise absolute numerical values, but rather for displaying patterns of differences or changes over time, to which the eye and brain are extremely sensitive." (Ware 2021, 70)

[53] Even if we have mapped a *proportion* data value to the position of the dot, we would need to know that the bottom of the line represents 0 and the top of the line represents 1 to be able to correctly decode the proportion from the position of the dot.

[54] Wilke (2019) refers to this problem as the *principle of proportional ink*. The amount of ink used to represent a data value should be proportional to the magnitude of the data value.

Another way to decode Figure 4.1 is that there is a thin orange disc outside a light blue disk (which is outside a thin green disk, which is outside a dark blue circle). In this interpretation, we can at least see all of the data symbol, so decoding is possible, but the problem now is that the viewer is using a decoding that is different from the encoding.

The concentric-circles interpretation is perhaps the most parsimonious (Section 2.8), but just the existence of an alternative interpretation is a potential source of confusion and so a cause for concern.

[55] These data were obtained from kaggle (Begbie 2022).

[56] In effect, more than one data value is encoded as the same data symbol, which means that we cannot decode back to individual data values. This is called a *surjective* mapping (Ziemkiewicz and Kosara 2009).

[57] This preference for horizontal and vertical orientations is known as the *oblique effect* (Appelle 1972; B. Li, Peterson, and Freeman 2003)

[58] Put mathematically, horizontal and vertical positions are *orthogonal*, which means that they are *independent*.

[59] This form of data is called *compositional* data; it shows how a total is composed of its different parts.

[60] Ternary plots, also known as barycentric plots, are common in specific areas of science, for example, Earth science where the composition of rocks or soils is studied.

[61] "The representation of numbers should be directly proportional to the quantities represented." (Tufte 1983)

[62] The log (base 10) of the data values is the power to which we must raise 10 in order to get the data value. For example, if the data value is 100, the logged data value is 2.

$10^2 = 100 \iff log_{10}(100) = 2$

5 Length

Figure 5.1 shows a bar plot of the total number of offenders in each police district in New Zealand (for ages 14 to 16 and from 2011 to 2021). For the reasons outlined in Chapter 3 and Chapter 4), this data visualisation is effective for decoding the number of offenders in different police districts and making comparisons between them.

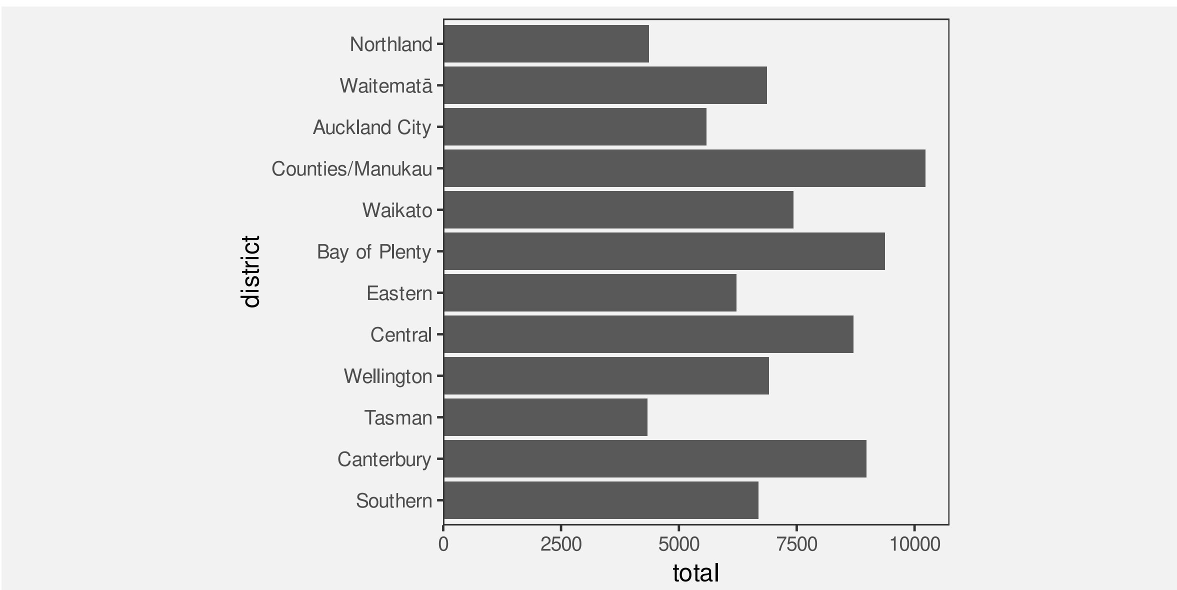


Figure 5.1: A bar plot of the number of offenders for different police districts, with the districts ordered from North to South.

Figure 5.2 shows a bar plot of the same data, but with the bars ordered from longest to shortest. This data visualisation uses the same mappings as Figure 5.1—police districts are encoded as the vertical positions of the bars and the total counts are encoded as the lengths of the bars—which means that it is also effective for comparing the total number of offenders between police districts.

However, Figure 5.2 is *more* effective than Figure 5.1 for some comparisons. For example, it is easier to compare the Northland district with the Tasman district in Figure 5.2. The fact that two data visualisations with the same mappings can have a different effectiveness suggests that there is more to know about how different visual features work.

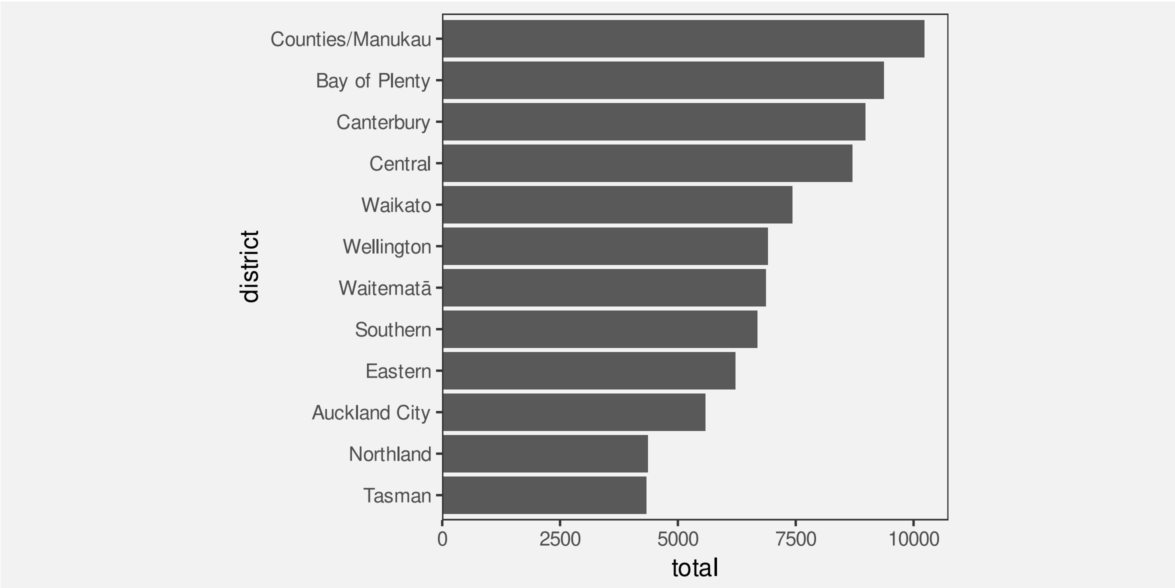


Figure 5.2: A bar plot of the number of offenders for different police districts, with the districts ordered from largest number of offenders to smallest.

This chapter covers some more details about mappings that encode data values as **length**.

5.1 The distance effect

In Section 3.5, we saw that **position** and **length** were the most accurate visual features (Figure 3.12). However, this accuracy decreases if the two visual features that are being compared are not in close proximity (Figure 5.3).[63]

This helps to explain why Figure 5.2 is more effective than Figure 5.1 for some comparisons (e.g., Northland versus Tasman).

> One reason for ordering the bars in a bar plot from longest to shortest is because, for some common comparisons, the ordering places the bars that are being compared closer together, which improves the accuracy of the comparison.

Figure 5.3: Two bars are easier to compare if they are close to each other (top row). If there is a distance between the bars and/or there are distractors (other bars) in between then comparisons are less accurate (bottom row). On both the top and bottom row, the black bar on the left is slightly higher than the black bar on the right.

5.2 Unaligned length

Figure 5.4 shows that it is harder to accurately compare the **lengths** of two bars if they are **unaligned** and do not share a common baseline (similar to the accuracy of comparing **positions**; Section 4.2).

Figure 5.5 shows an alternative visualisation of the data on the total number of offenders in different ethnic groups. The bar plot of these data that we saw in Figure 3.2 encoded ethnic groups as the **positions** of the bars so that the bars were arranged vertically with a common left-hand edge. In Figure 5.5, the ethnic groups are encoded as different **colours** and the bars are "stacked" horizontally.[64]

Although the total number of offenders has been encoded as the same visual feature in both Figure 5.5 and Figure 3.2—in both cases, the number of offenders is encoded as the **lengths** of the bars—comparing those lengths is much harder in Figure 5.5 because the lengths are **unaligned**.

> One reason why stacked bar plots are less effective for some comparisons is because we are forced to compare the unaligned lengths of the bars.

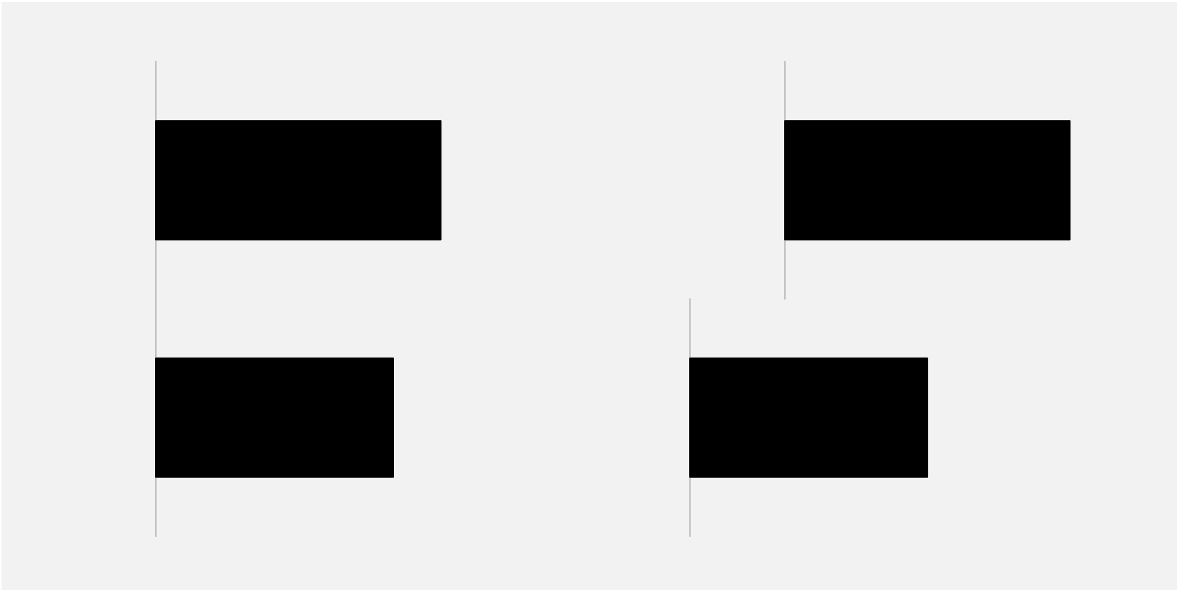


Figure 5.4: It is easier to compare the lengths of the two bars on the left because they are **aligned**; they share a common base. The two bars on the right are harder to compare because they are **unaligned**.

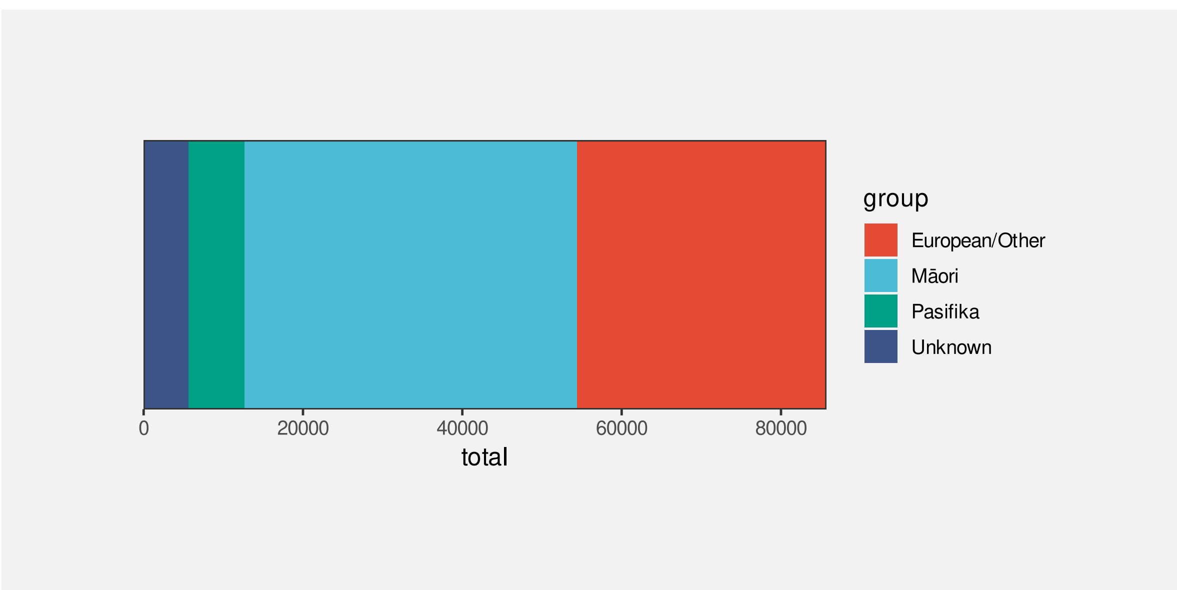


Figure 5.5: A stacked bar plot of the total number of offenders in each ethnic group.

5.3 Case study: Stacked versus side-by-side bar plots

Stacking bars, which creates unaligned lengths, is something that arises when we use a bar plot to show a quantitative variable broken down by more than one qualitative variable. For example, Table 5.1 shows data on the number of offenders in different ethnic groups for individual years (from 2011 to 2021). This is a more detailed version of the data from Table 3.1.

Table 5.1: A table of the number of offenders per year aged 14 to 16 for different ethnic groups from 2011 to 2021. There are 44 rows of data, but just the first 6 rows are shown here.

	group	count	year
1	Māori	5957	2011
2	Pasifika	1092	2011
3	European/Other	5775	2011
4	Unknown	194	2011
6	Māori	5458	2012
7	Pasifika	1040	2012

Figure 5.6 shows the number of offenders (quantitative) for each ethnic group (qualitative) and for each year from 2011 to 2021 (treated here as qualitative, or at least discrete). In this data visualisation, the number of offenders is mapped to the lengths of the bars, the years are mapped to the horizontal **position** of the bars, the ethnic groups are mapped to the colours of the bars, and the bars are stacked vertically.

The problem that we face in this situation is that there are four bars to show for each year. If we simply encode the year as the horizontal position of each bar, the bars will overlap and we will have great difficulty decoding data values from the bars (Section 4.3).

Stacking the bars is one solution, but as we have just seen, this makes some comparisons more difficult. For example, we can easily see that the number of offenders in the unknown group increased after 2016, but it is much harder to see what has happened to the European/Other group since 2016.

An alternative visualisation is shown in Figure 5.7. This has placed the four ethnic groups side-by-side for each year. It is now easier to see the changes over time for the European/Other group because all bars are now aligned. Unfortunately, as we saw in Section 5.1, some comparisons are more difficult because the bars are now further apart and have distractors in between.

Yet another variation is shown in Figure 5.8. This time we have placed all years side-by-side for each ethnic group. This provides the clearest view of the changes over time for each ethnic group because the relevant bars are close together and aligned. However, there are other

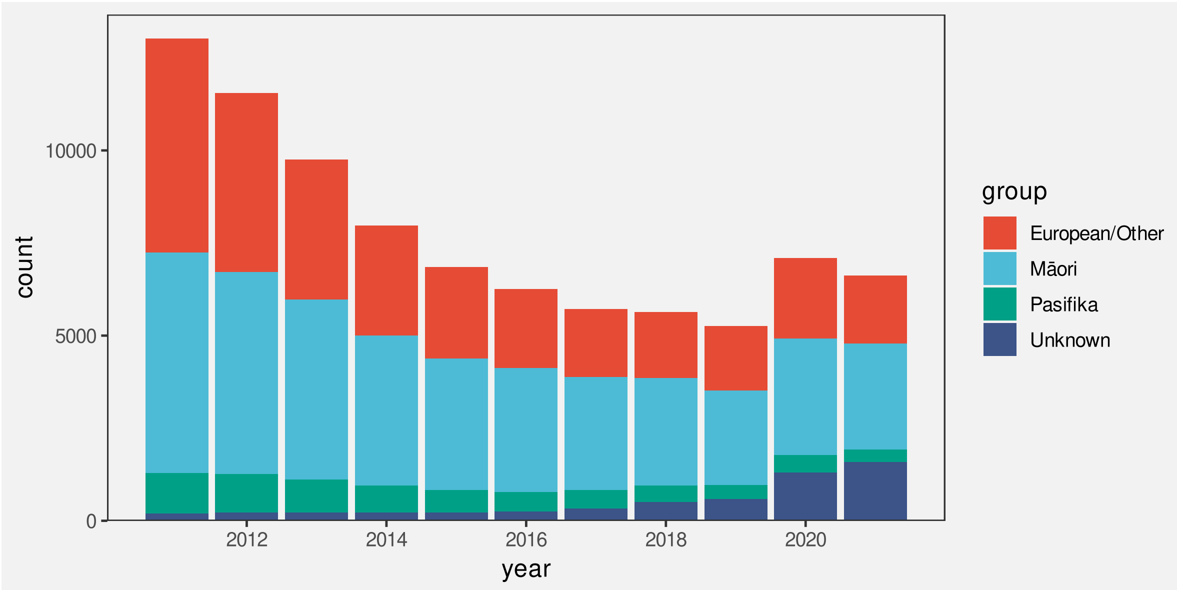


Figure 5.6: A stacked bar plot of the number of offenders in each ethnic group for each year from 2011 to 2021.

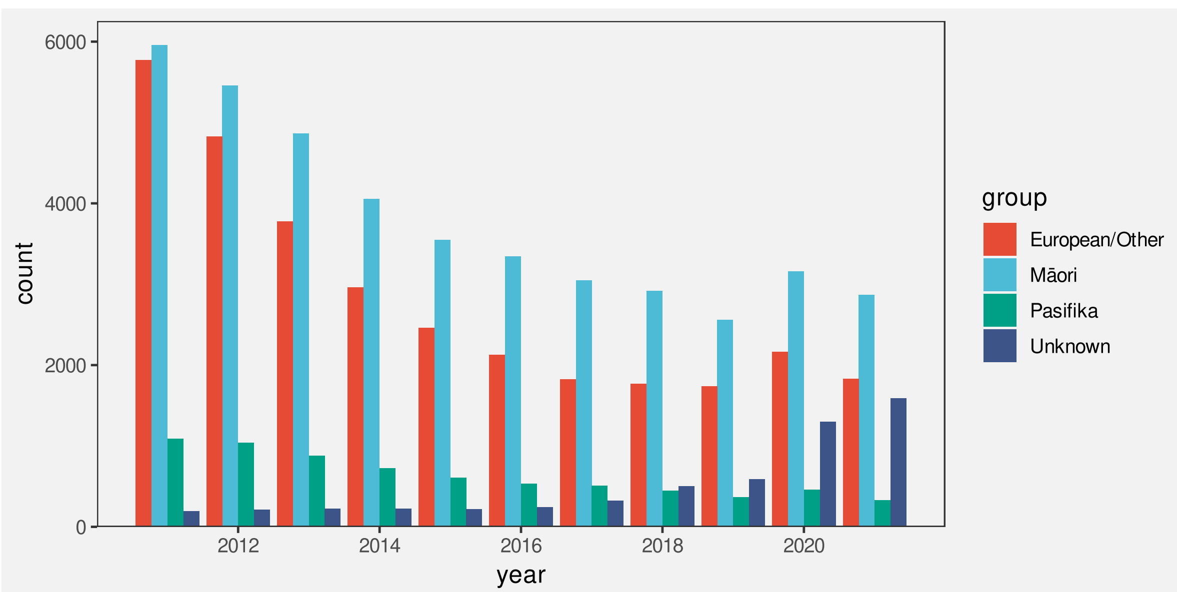


Figure 5.7: A side-by-side bar plot of the number of offenders in each ethnic group for each year from 2011 to 2021.

comparisons that are considerably more difficult because of the distance between the relevant bars, for example, comparing between ethnicities for a specific year.

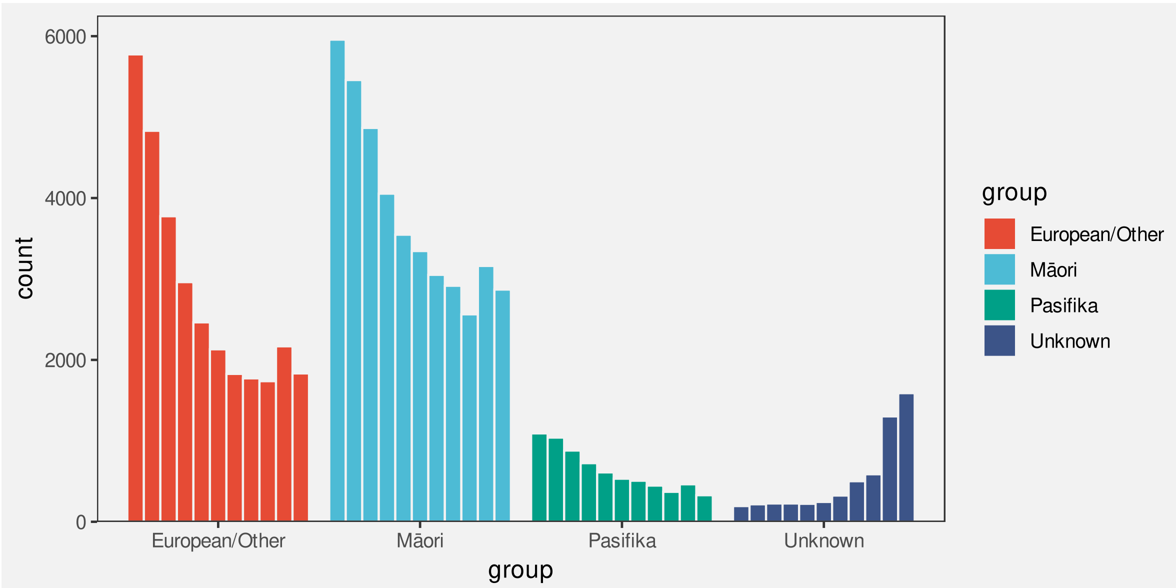


Figure 5.8: A side-by-side bar plot of the number of offenders in each ethnic group for each year from 2011 to 2021.

There is no single best visualisation in general for this situation because different visualisations each have their strengths and weaknesses. If a particular sort of comparison is to be emphasised, then one data visualisation might stand out, but there is also the option of providing more than one data visualisation in order to cover more comparisons.

5.4 Perception is relative

Figure 5.9 shows that the perception of differences between the lengths of bars is relative; it is easier to detect smaller differences when the visual features that we are comparing are smaller.[65] It is easier to see that the pair of bars on the left have different lengths and harder to see that the pair of bars on the right have different lengths. All three pairs of bars have different lengths, but the absolute size of the difference is the same, so the size of the difference relative to the bar lengths gets smaller as the bars get longer.

Figure 5.10 shows that we can make it easier to judge the differences by providing reference marks. Each pair of black bars is the same and each pair of bars differs in length by the same small amount. The difference between the bars in each pair is difficult to perceive because the difference is small relative to the lengths of the bars. The reference marks in the middle and right-hand pair of bars provide shorter lengths to compare, so we are able to more easily

Figure 5.9: A demonstration that perception of differences is relative. The pairs of bars all differ by the same absolute amount, but the difference is more easily perceived when comparing the shorter bars. In the shorter bars, the absolute difference between the bars is a larger difference relative to the length of the bars.

perceive the small differences. For example, for the pair of bars on the right, we can compare the lengths of the short white gaps rather than comparing the longer black bars.

Figure 5.11 shows a bar plot of the number of offenders in each police district (Figure 5.1) with some grid lines added. Although it is difficult to compare the bars for Northland and Tasman because they are quite far apart and there are distractors in between (Section 5.1), the addition of the grid lines makes it easier to perform the comparison because we are able to compare the distances from the ends of the bars to the nearest grid line. Those distances are much shorter than the bars so it is easier to perceive the small difference.

> One reason why grid lines are effective is because they allow comparisons of smaller lengths or distances, which means that it is easier to perceive smaller changes.

5.5 Summary

Bar plots are effective because they encode **quantitative** values as **lengths**, which are accurate visual features.

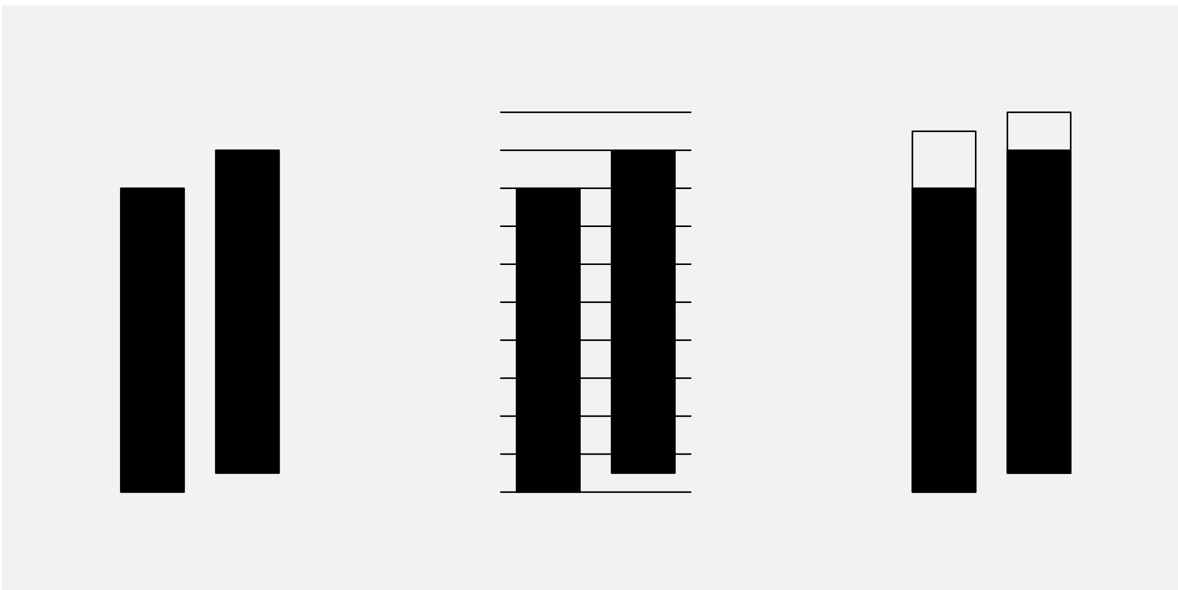


Figure 5.10: A demonstration that reference marks can make it easier to perceive small differences. Each pair of black bars is the same and each pair differs by a small amount. In the middle pair, grid lines make it easier to compare the distances from the ends of the bars to the nearest grid line and, in the right-hand pair, a border added to each bar, where the two borders are the same length, makes it easier to compare the white gaps rather than the bars themselves.

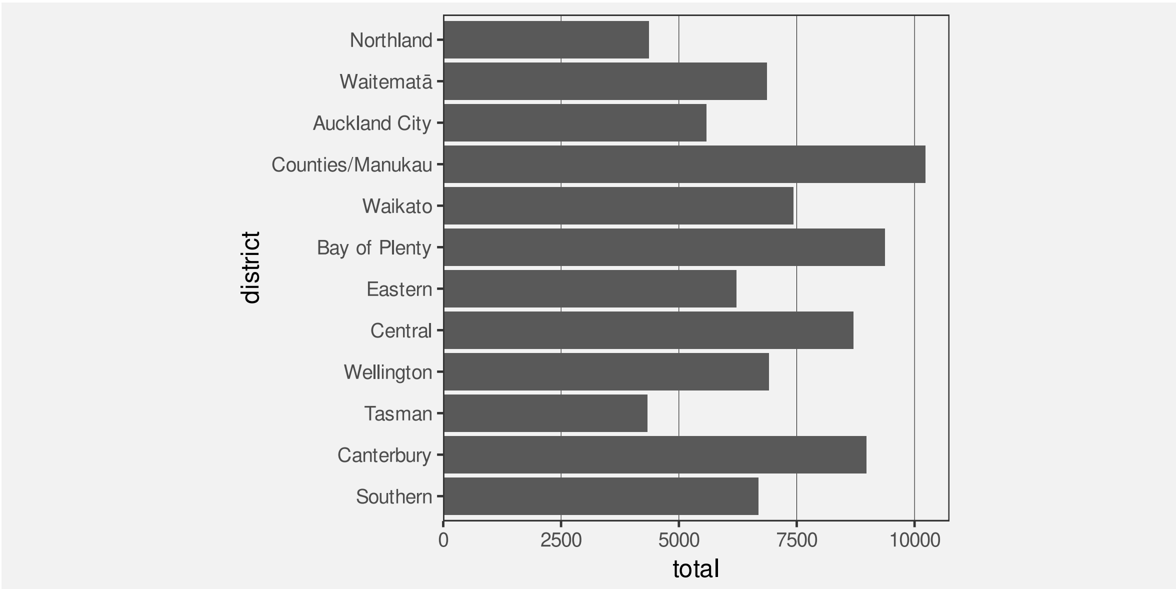


Figure 5.11: A bar plot of the number of offenders for different police districts, with the districts ordered from North to South. This is very similar to Figure 5.1, just with some grid lines added.

However, comparisons between the bars in a bar plot can be difficult if the bars are far apart, especially if there are distractors in between.

Furthermore, **stacked bars** are more difficult to compare because they require comparing **unaligned** lengths.

Finally, it is easier to perceive differences between shorter bars than between longer bars because the perception of differences is **relative**.

Grid lines can help with comparisons because they make the distances being compared smaller, so smaller differences are easier to detect.

Notes

[63] The first demonstration of the "distance effect" is attributed to William S. Cleveland and McGill (1984). It has since been replicated several times, for example, by Talbot et al. (2014) and Lu et al. (2021).

[64] In a stacked bar plot, groups are encoded as the positions of the bars and counts are encoded as the lengths of the bars, but the positions are in the same dimension as the lengths (in this case, both position and length are

horizontal). In a side-by-side bar plot, the position of the bars is in the opposite dimension to the lengths of the bars.

[65] The idea of perception of differences being relative is a very old result from Psychology known as Weber's Law (Weber 1846; Boring 1942).

6 Colour

Figure 6.1 shows a pie chart of the number of offenders for different levels of crime seriousness (between 2011 and 2021). In this data visualisation, the crime level, which is a **qualitative** variable, is encoded as **colour**, which is appropriate for representing qualitative values (Section 3.4). It is easy to decode the crime levels because it is easy to identify the colour for a particular crime level amongst the set of colours used.

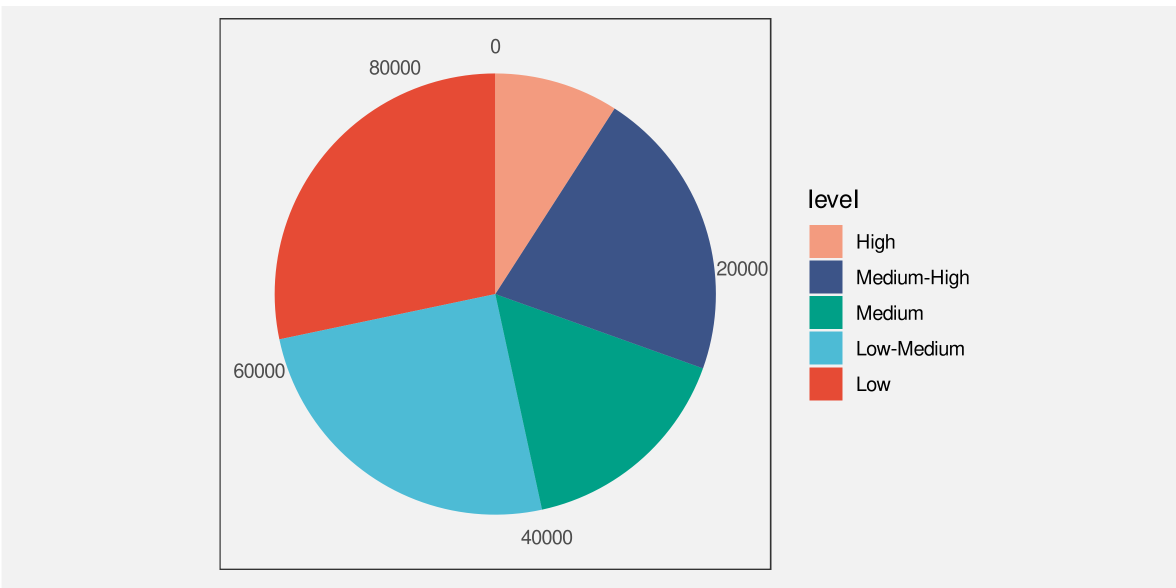


Figure 6.1: A data visualisation of the number of offenders for different levels of crime seriousness.

Figure 6.2 shows another pie chart of the same data, again encoding the **qualitative** level as **colour**. It is again easy to identify each crime level because we can easily identify each separate wedge colour. However, Figure 6.2 is *more* effective than Figure 6.1 because we can *also* perceive an ordering of the wedges from more serious crimes, which are darker wedges, to less serious crimes, which are lighter wedges.

The fact that there can be two data visualisations, Figure 6.1 and Figure 6.2, with the same mapping—the **qualitative** level of crime is encoded as the **colour** of the wedges—but the two data visualisations have different effectiveness suggests that there is still more to learn about how different visual features work.

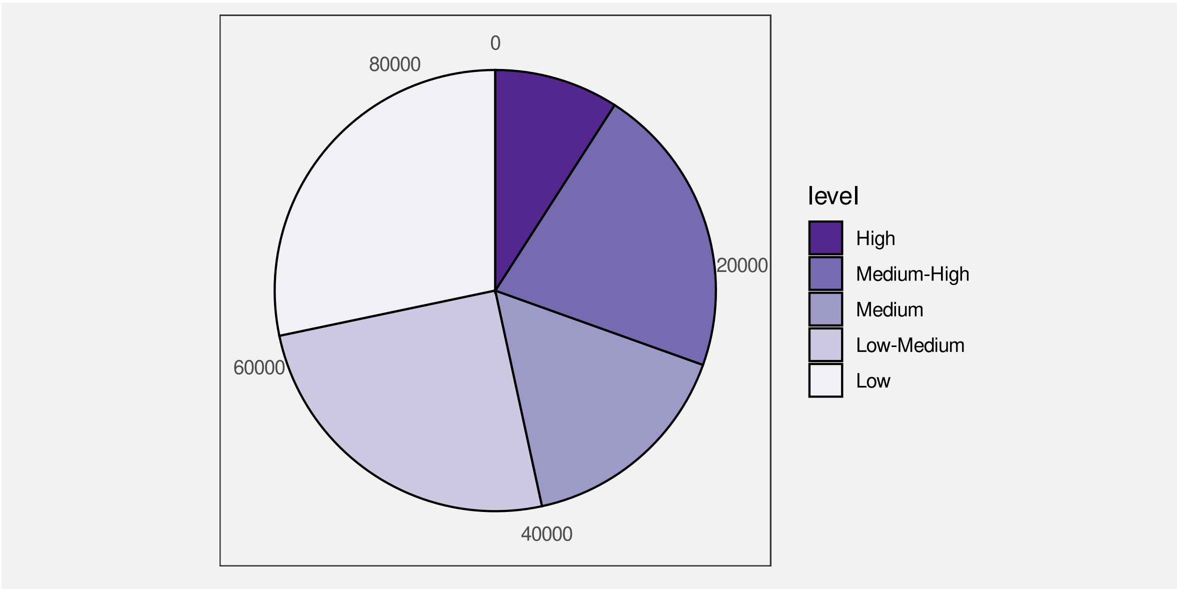


Figure 6.2: A data visualisation of the number of offenders for different levels of crime seriousness.

This chapter covers some more details about how the **colour** visual feature works.

6.1 Hue, chroma, and luminance

We have so far taken a very simplistic approach to **colour**, treating it as if it is a single **qualitative** visual feature (Section 3.4). However, the perception of colour can be divided into three separate visual features: **hue**, **chroma**, and **luminance** (Figure 6.3).[66] Hue describes what we normally refer to as the colour name: reds, oranges, yellows, greens, blues, and purples. Chroma describes the colourfulness, from bright to dull, and luminance describes the lightness, from dark to light. Our visual system is capable to perceiving changes in hue and changes in chroma and changes in luminance.

We have described colour as a **qualitative** visual feature so far, but that really strictly only applies to **hue**. For example, in Figure 5.5, we encoded different ethnic groups as different hues. We are very good at decoding qualitative data values from different hues, subject to the capacity limits described in Section 3.6.

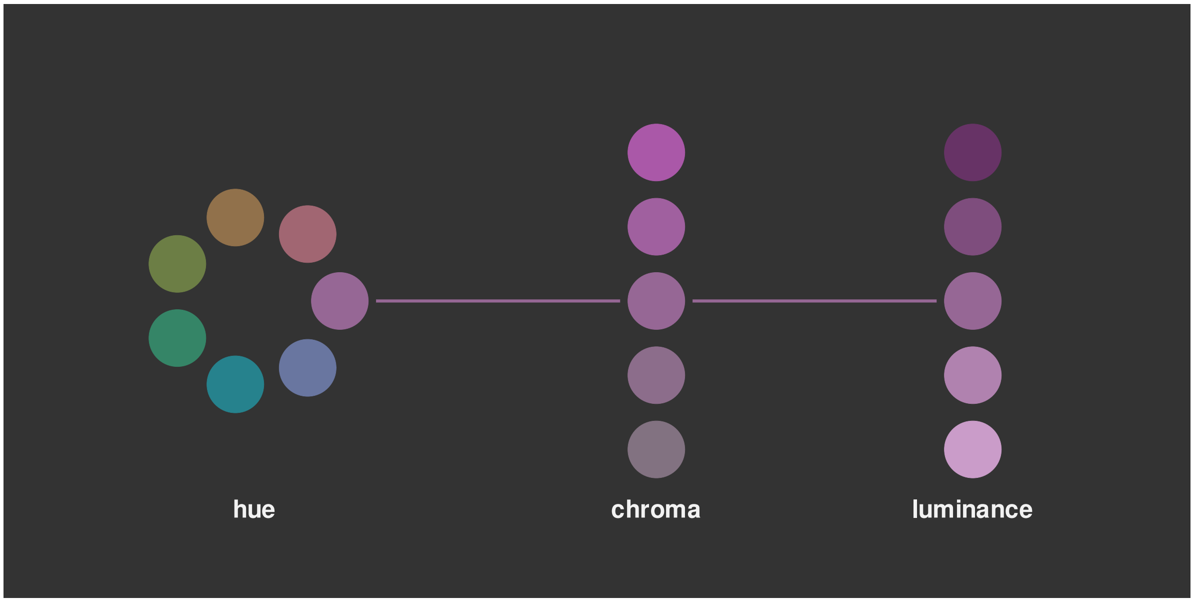


Figure 6.3: The perception of colour can be divided into three components: differences in **hue**, the colours of the rainbow; differences in **chroma**, from bright and colourful to dull and grey; and differences in **luminance**, from light to dark.

6.2 Ordinal visual features

Chroma and **luminance** are both **nominal** visual features in that they can be used to encode different categories, but they are also **ordinal** visual features because we can decode an ordering from a set of colours that differ in terms of either chroma or luminance or both. A set of colours that transitions from darker to lighter and/or brighter to duller conveys an ordering from larger to smaller.[67]

This helps to explain why Figure 6.2 is more effective than Figure 6.1. In Figure 6.2, the colours differ mostly in terms of luminance and chroma so we are able to perceive changes from larger values to smaller values because the colours transition from darker to lighter.

In this sense, **chroma** and **luminance** are able to convey more information than **hue** because they can represent **nominal** differences *and* **ordinal** differences, whereas hue can only represent nominal differences.[68]

On the other hand, Figure 6.4 shows that chroma and luminance are even more limited than hue in terms of **capacity**. We can only use chroma and/or luminance to represent **nominal** data when there are very few different groups to represent.

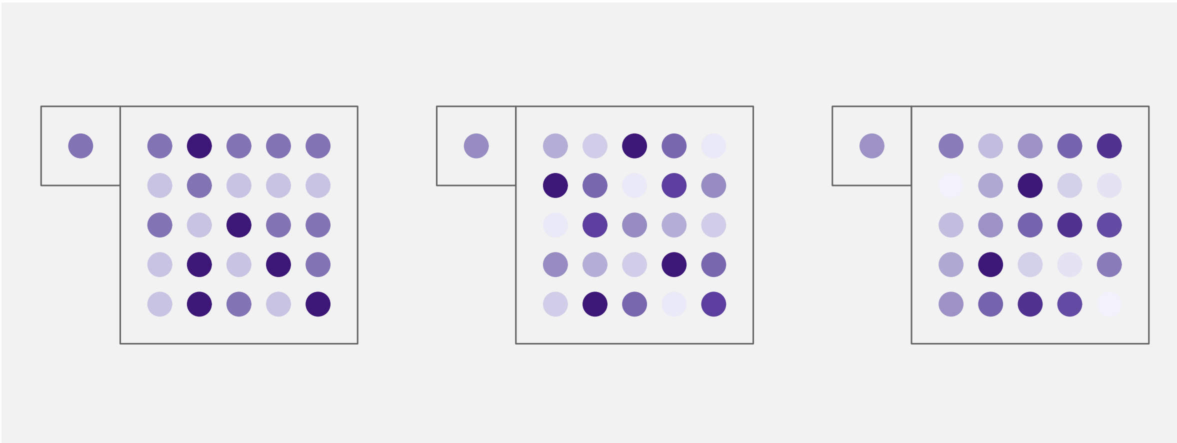


Figure 6.4: Mapping qualitative values to luminance and chroma works well for representing nominal data values, but only for a very small number of values. It is possible to identify which shade of purple the single dot corresponds to amongst the three shades on the left, but it is much harder with the seven shades in the middle, harder still for the eleven shades on the right. This figure can be compared with Figure 3.13, which demonstrates the capacity of hue for representing nominal data values.

> If we encode data values to the colour of data symbols, and the colours only differ in terms of **hue**, then we can only decode **nominal** information from the data symbols; do the data symbols represent *different* groups or the *same* group.

> If we encode data values to the colour of data symbols, and the colours differ in terms of **chroma** and/or **luminance**, it is also possible to decode **ordinal** information form the data symbols.

6.3 Perception is relative

We saw in Section 5.4 that perception of **lengths** is relative. For example, smaller differences in length are easier to see when the lengths themselves are short (Figure 5.9).

There is a similar rule for the perception of **colours**. Our perception of luminance and chroma is affected by neighbouring colours.[69] This effect is demonstrated in Figure 6.5: the same colour when surrounded by a darker background appears lighter than it does when surrounded by a lighter background.

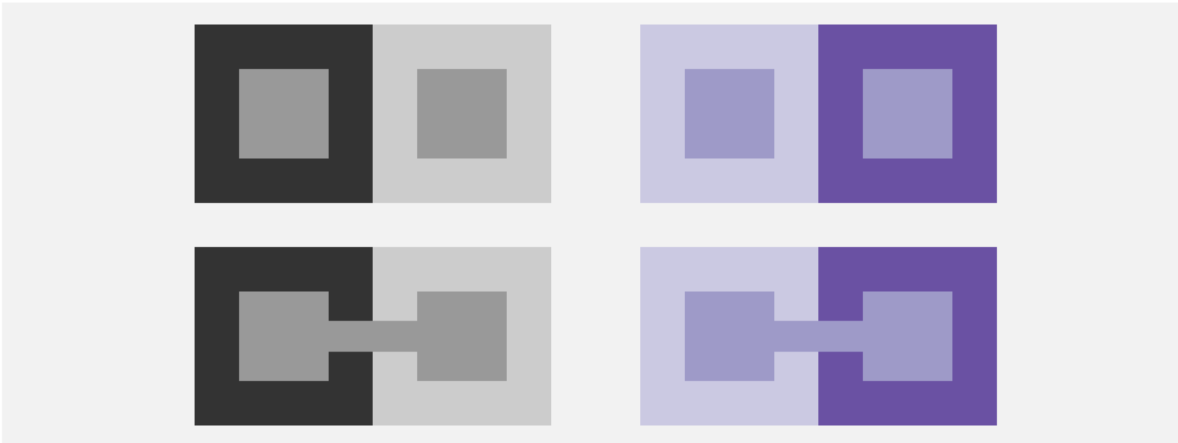


Figure 6.5: The perception of colours is relative. In the top row, the inner rectangles are exactly the same shade of grey on the left and exactly the same shade of purple on the right. The perception of the inner rectangles is affected by the surrounding colour, so the same shade appears darker against a light background and darker against a light background. The bottom row draws a line connecting the two inner rectangles to show that they really are the same shade.

This effect also applies to the perception of hue. Figure 6.6 shows a pie chart of the number of offenders in each police district, with police district mapped to the **colour** of the wedges. There are two issues with the perception of the wedge colours: each wedge is affected by its neighbour so that, for example, some of the blue wedges look more green at the boundary with their more blue neighbour and more blue at the boundary with their more green neighbour; furthermore, the colours in the legend are perceived differently from the corresponding colours in the wedges because the legend colours are surrounded by a light grey, rather than sharing a boundary with a neighbouring colour.

The pie chart in Figure 6.6 is *not* an effective data visualisation because it encodes qualitative data values with many different groups as colour hue. This exceeds the **capacity** of the **hue** visual feature; it is difficult to clearly identify which wedge corresponds to which district (Section 3.6).

Although it does not make the use of hue any more appropriate, for the purposes of demonstration, we can solve the colour perception problems in Figure 6.6 by adding a white border to all of the wedges (see Figure 6.7). Now all of the wedges and all of the squares in the legend are perceived relative to a surrounding white border. This means that, not only is each wedge perceived as a solid block of constant colour, but it is easier to match the colours within the wedges with the colours in the legend.

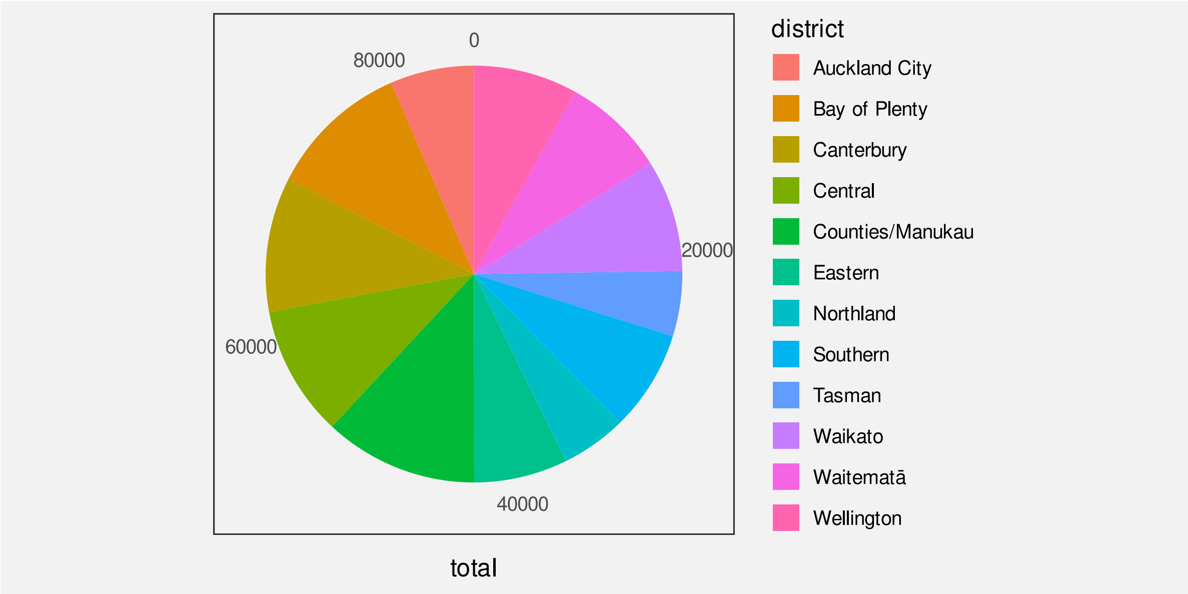


Figure 6.6: This pie chart demonstrates that the perception of hues is relative. The colours within each wedge appear to have a subtle gradient within the wedge because the perception of the wedge colour is affected differently by the neighbouring wedge colours.

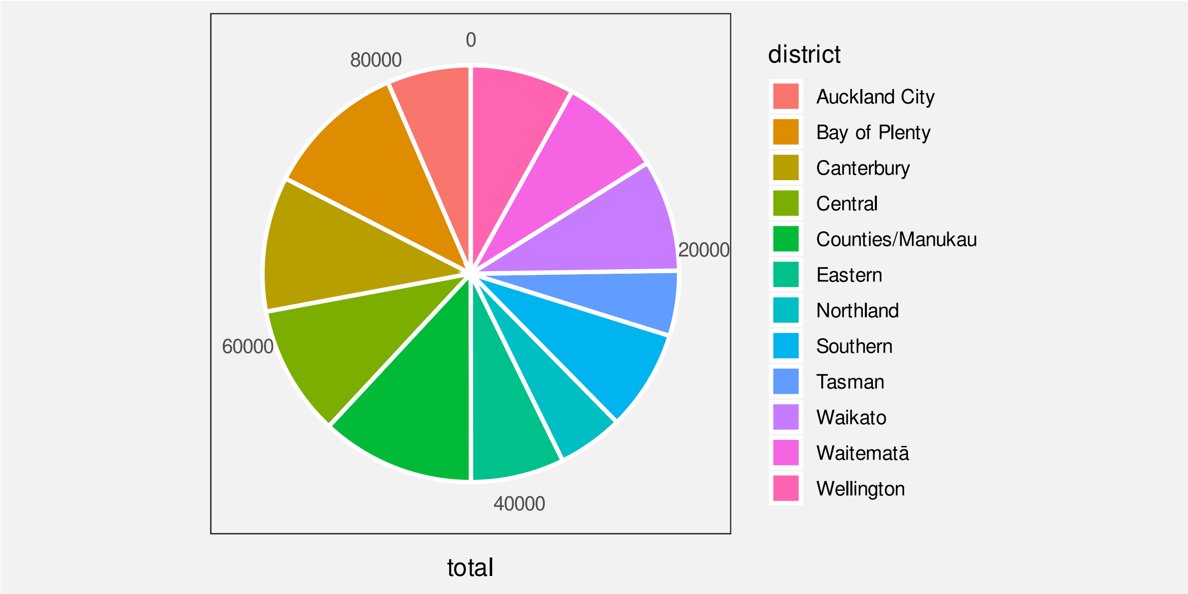


Figure 6.7: The hues in this pie chart are perceived more accurately than those in Figure 6.6 because each wedge is surrounded by a white border.

> If we encode data values as the colour of data symbols, we may have difficulty decoding the colours if the same colour is presented with different colours adjacent to it.

6.4 Size matters

The perception of colours is also affected by the size of the coloured region. In particular, small regions of colour may blend together with neighbouring colours.[70] For example, on the left of Figure 6.8, the yellow and blue stripes are, in reality, the same colours both in the top set of wide bars and in the bottom set of narrow bars, but the blue bars appear greener in the bottom set. The right side of Figure 6.8 shows that the same colours may appear different when they are used as fill colours for larger regions, like bars, compared to when they are used for thin lines. In this case, the blue line appears darker than the blue bars.

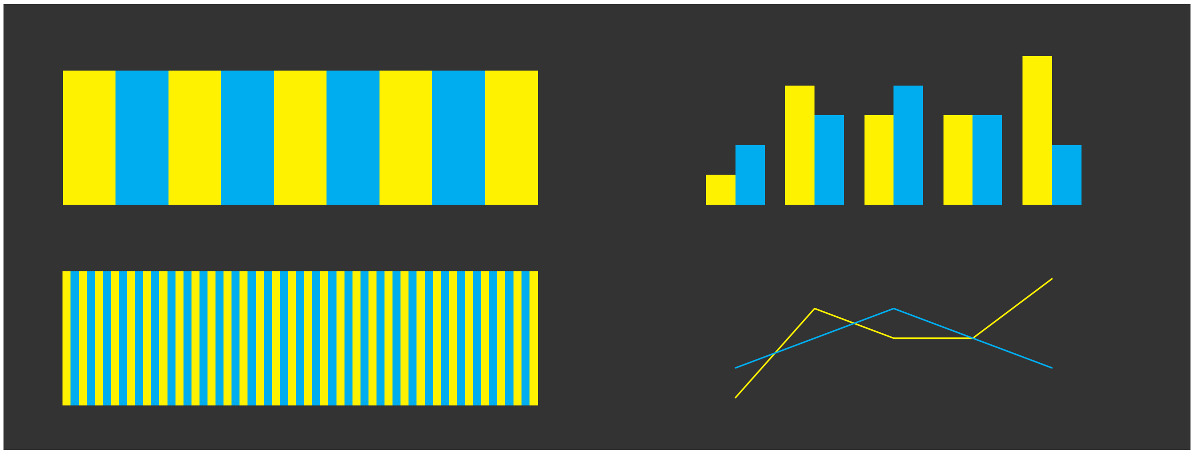


Figure 6.8: The perception of colours varies based on the size of the coloured regions.

> If we encode data values as the colour of data symbols, we may have difficulty decoding the colours if there are large differences in the sizes of data symbols.

6.5 Colours in space

Figure 6.9 shows two-dimensional slices of the set of visible colours arranged in three dimensions.[71] The centre three coloured regions (surrounded by white circles) show slices of constant luminance. Chroma increases from the centre (indicated by a white dot) and hue changes with angle. We can see the familiar rainbow of colours from red, through orange, yellow, green, blue, indigo, and violet, back around to red. The top slice is for high luminance (light) colours and the bottom slice is for low luminance (dark) colours. The coloured regions to the left and to the right (surrounded by semicircles) show slices of constant hue. Luminance increases from bottom to top and chroma increases away from the flat side of the semicircle.

Figure 6.9 provides insight into what sorts of colours and what sorts of differences between colours are possible. For example, we can see that it is not possible to produce colours that have a high chroma *and* are very dark. In other words, there is no such thing as a very colourful black. This is also true for colours that are very light.

We can also see that the range of possible chroma values depends on both the hue and the luminance of colours. For example, it is possible to produce a light yellow with either a low chroma or a high chroma, but we can only produce a dark yellow with a low chroma. Equivalently, the range of possible luminance values depends on both the hue and the chroma of colours. For example, we can produce both dark and light low-chroma yellows, but we can only produce light high-chroma yellows.

This means that, if we want to encode data values as the chroma of a data symbol, we can get a higher decoding **capacity** through careful selection of hue and luminance. For example, we can generate a wider range of light greens than dark greens and a wider range of reds that are neither too light nor too dark.

Figure 6.10 is similar to Figure 6.9 except that, instead of showing the range of colours that it is possible to perceive, it shows the range of colours that it is possible to *produce* using standard technology.[72] The simple message here is that the limits on the range of colours that we can use to encode data values is much smaller than the range of possible colours. On the positive side, the range of colours that we can produce is still very large.

A more important message is that the general points about differences between colours still apply. For example, the range of possible chroma values is dependent on the hue and luminance of colours.

> If we encode data values as the colour of data symbols, the choice of hue, luminance, and chroma affects the capacity of decoding—how many different colours can be effectively decoded.

6.6 Colour harmony

We have seen that hue is an appropriate visual feature for encoding qualitative data values because we can decode differences between hues without any ordering between hues. For example, blue is not larger than red.

However, it is not entirely true that there is no sense of *distance* between different hues. The left image in Figure 6.11 shows a colour wheel of hues for a constant luminance and chroma.

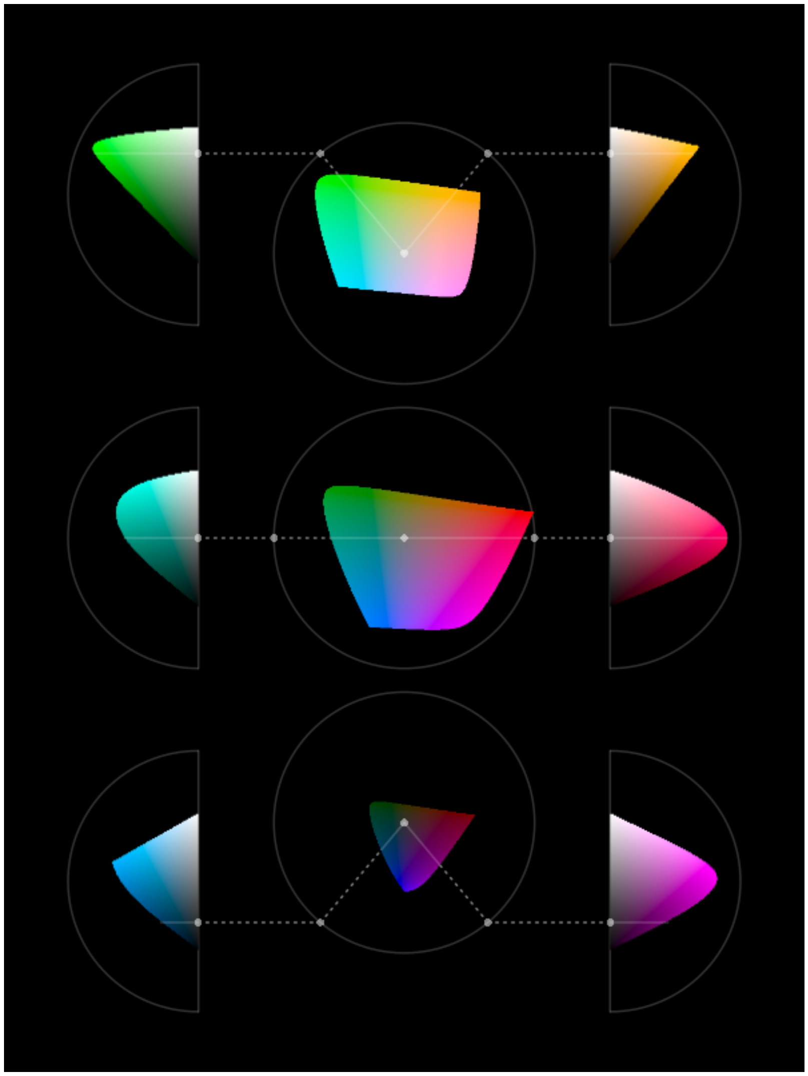


Figure 6.9: Slices through the set of colours that are visible to humans. The central images show colours that differ in terms of chroma and hue, but share a common luminance (from lighter at the top to darker at the bottom). The images to the left and right show colours that differ in terms of chroma and luminance, but share a common hue. a

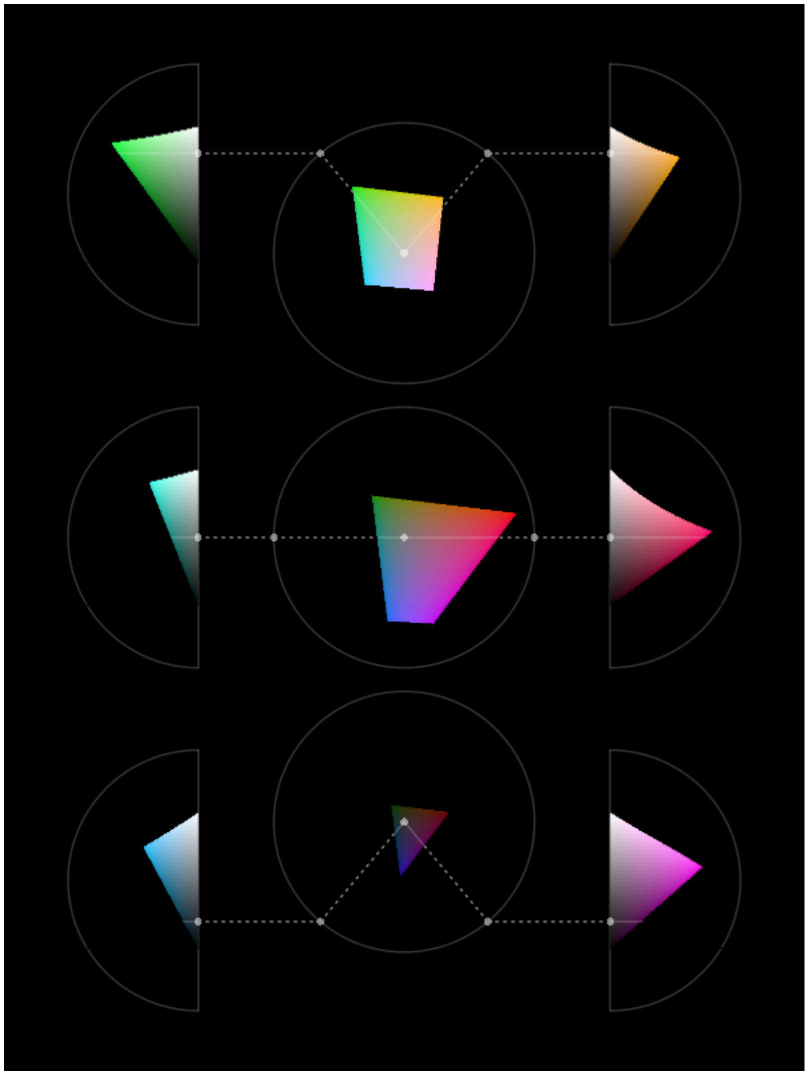


Figure 6.10: Slices through the set of colours that it is possible to produce on a typical computer monitor.

Hues that are close together on this colour wheel are decoded as more similar than hues that are on opposite sides of the wheel. For example, the top two circles on the right of Figure 6.11 have hues that are from the left side of the colour wheel so they appear similar, but the bottom two circles have hues from the left and the right side of the colour wheel so they have a greater contrast.[73]

Figure 6.11: On the left, a colour wheel that shows differet hues for a constant chroma and luminance. At top-right is a pair of hues from the left side of the colour wheel and bottom-right is a pair of hues from opposite sides of the colour wheel. There is greater contrast between the bottom pair of colours and less contrast between the top pair of colours.

6.7 Colour vision deficiency

Roughly 10% of the population, primarily men, suffer from a form of colour vision deficiency (CVD). In the most common case, the viewer will have difficulty distinguishing between red and green hues.[74] Figure 6.12 simulates how the colours in Figure 6.1 would appear to a person with severe CVD.

In order to avoid confusion for viewers with CVD, it may be necessary to vary luminance and/or chroma in addition to or instead of hue. For example, Figure 6.13 shows how the shades of purple in Figure 6.2 would appear to a viewer with severe deuteranopia. The difference in hue is of no importance in this case.

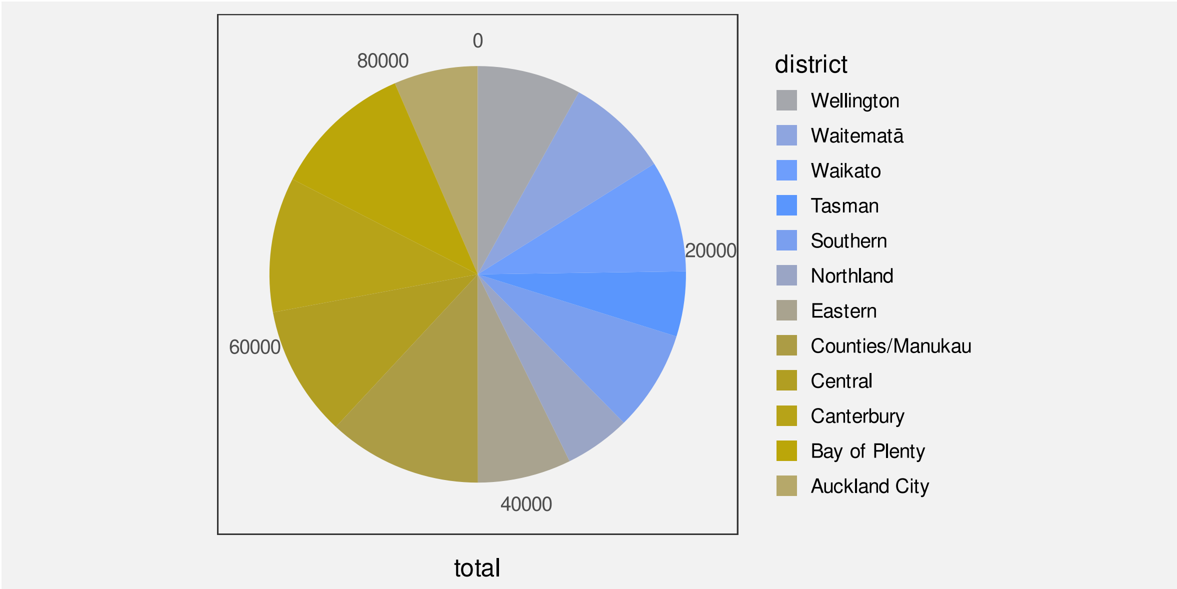


Figure 6.12: A data visualisation of the number of offenders for different police districts. This is how a person with severe deuteranopia might perceive the colours in Figure 6.6.

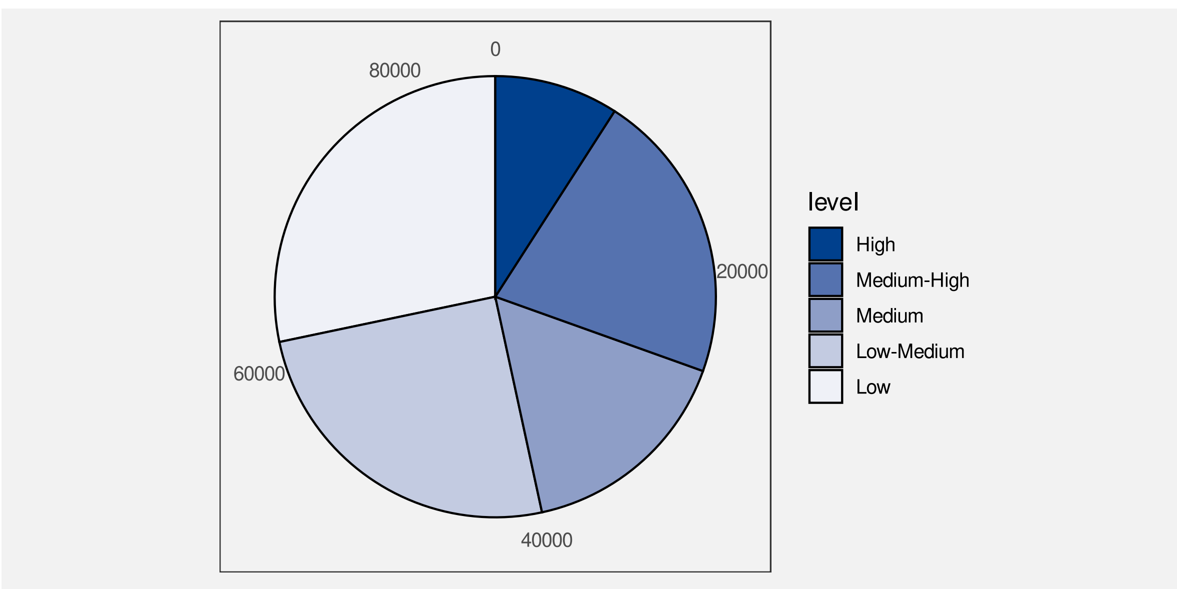


Figure 6.13: A data visualisation of the number of offenders for different levels of crime seriousness. This is how a person with severe deuteranopia might perceive the colours in Figure 6.2.

> When we encode data values as the colour of data symbols, some viewers will have difficulty decoding the colours if we do not take into account common forms of colour vision deficiency.

6.8 Non-linear mappings

The top row of Figure 6.14 shows a linear luminance mapping; we have a set of rectangles that change luminance in equal steps. The two rectangles on the left of this row appear very similar, while the two rectangles on the right of this row appear quite different. The darker rectangles appear to change more slowly and the lighter rectangles appear to change more rapidly. In other words, the *decoding* of luminance is *not* linear.

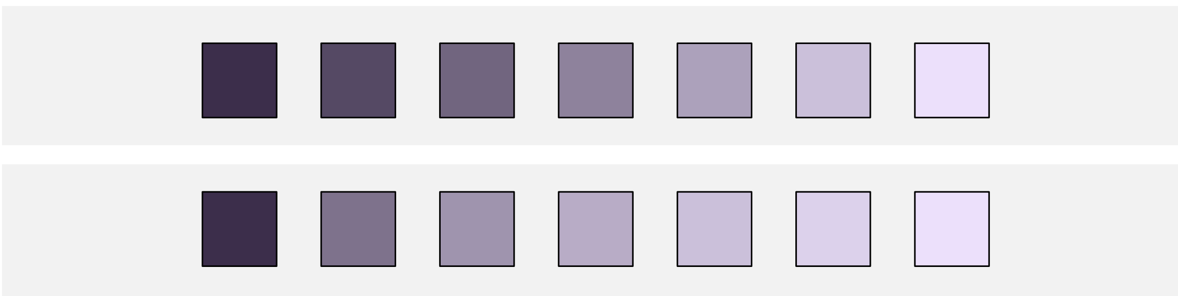


Figure 6.14: The two rows of purple values begin and end at the same level of luminance. The top row maps luminance linearly, but the bottom row uses a cube-root mapping.

We saw in Section 4.7 that accurate decoding of data values from **position** relies on using a linear encoding. That was because the decoding of position is linear. A linear encoding from data values to position, followed by a linear decoding from position to data values, leads to an accurate mapping. And that is particularly important for mapping quantitative data values.

Part of the reason why it is not appropriate to map quantitative data values to luminance is that the decoding of luminance is *not* linear. This makes it difficult to accurately decode quantitative data values from the luminance of a data symbol.

However, we can use luminance to map ordinal data values (Section 6.2). In this case, all that we need is to be able to identify that a data symbol is lighter or darker than another data symbol.

The bottom row of Figure 6.14 shows a luminance mapping that takes larger steps for darker colours and smaller steps for lighter colours. In this row, the purples appear to change more evenly. We can see differences between the darker rectangles as well as between the lighter rectangles.

In other words, a *non-linear* mapping from data values to luminance can be used to generate a larger set of luminance levels that can be differentiated from each other. A non-linear luminance mapping leads to increased **capacity** for decoding data values from luminance.

> When we are mapping nominal or ordinal data values, and we are encoding to a visual feature that has a non-linear decoding, a non-linear *encoding* can be effective for increasing the capacity of the visual feature.

6.9 Escape capacity

We have previously identified that colour has a limited capacity (Section 3.6 and Section 6.2). In general, we cannot use colour to encode more than around 7 different categories. Section 6.5 also demonstrated that this capacity varies for different combinations of hue, chroma, and luminance.

One way to increase the decoding capacity of a colour encoding is to vary more than one of hue, chroma, and luminance at once. For example, the top row of Figure 6.15 shows a set of colours that only vary in terms of luminance—hue and chroma are held constant (this is the bottom row from Figure 6.14). In the second row of Figure 6.15, in addition to increasing luminance, we have increasing chroma, so there are larger differences between colours, but still a clear ordering. Similarly, in the bottom row of Figure 6.14, the hue changes slightly as the luminance increases so that the differences are more obvious.

In effect, this is another example of a non-linear encoding. We take a non-linear path through colour space (Section 6.5).

Another way to expand the decoding capacity of a colour encoding is to reduce the number of colours that need to be directly compared. For example, if data symbols are arranged in a very regular pattern, like in a bar plot, comparisons between neighbouring data symbols are most important. This means that we can order the colours to maximise neighbour differences rather than maximising all pairwise differences.[75] Figure 6.16 demonstrates this idea with a bar plot of the crime rate in different districts for both 2011 and 2021. This bar plot uses the same set of hues as Figure 6.7, just in a different order so that adjacent bars have very different colours. It is very easy to distringuish between adjacent colours.

A similar idea is to just reuse a smaller set of hues, which means larger differences between adjacent colours, but repeat the set. The regular arrangement of the data symbols means that we can still differentiate between repeated colours because of the bar positions (Figure 6.17). For example, we can differentiate the pink for the Wellington district from the pink for the Eastern district because Wellington is at the end and Eastern is in the middle.

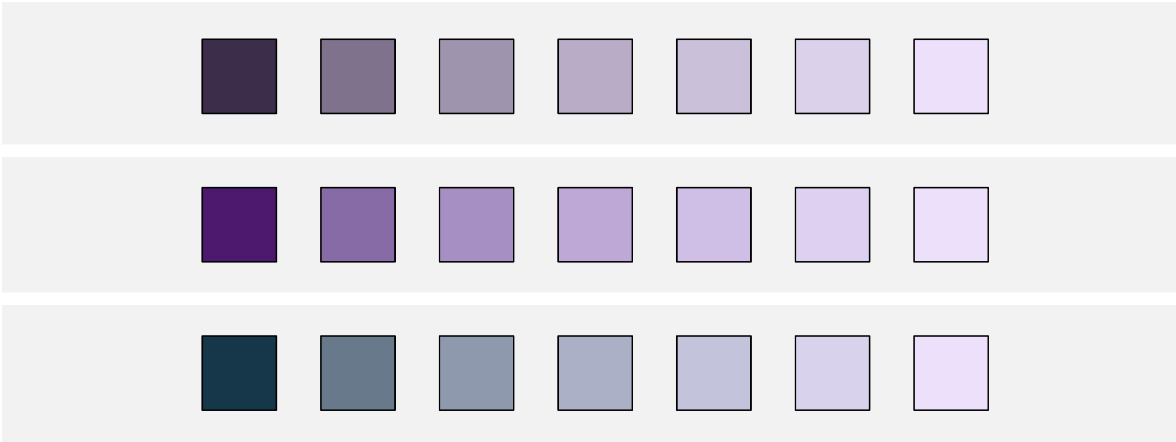


Figure 6.15: The top row holds hue and chroma constant and just varies luminance. It is limited to a very low chroma. The middle row varies chroma as well as luminance, which provides a wider range of colours compared to the top row by allowing a more colourful starting point. The bottom row varies hue and luminance, which provides a wider range of colours compared to the top row by allowing a different-hued starting point.

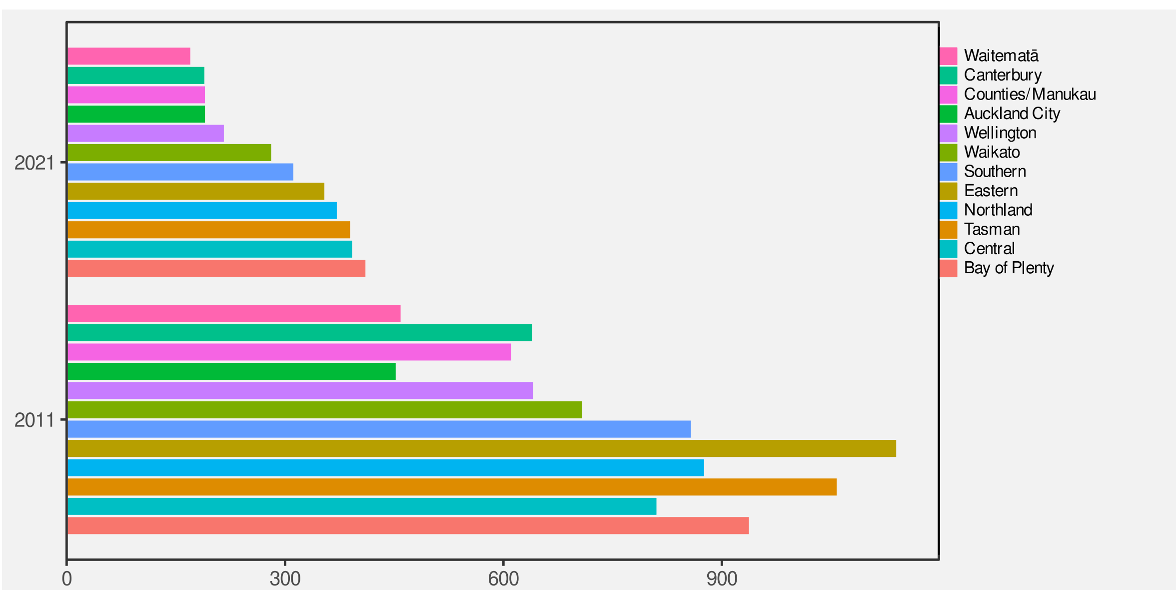


Figure 6.16: The hues in this pie chart are ordered so that adjacent hues are distant from each other.

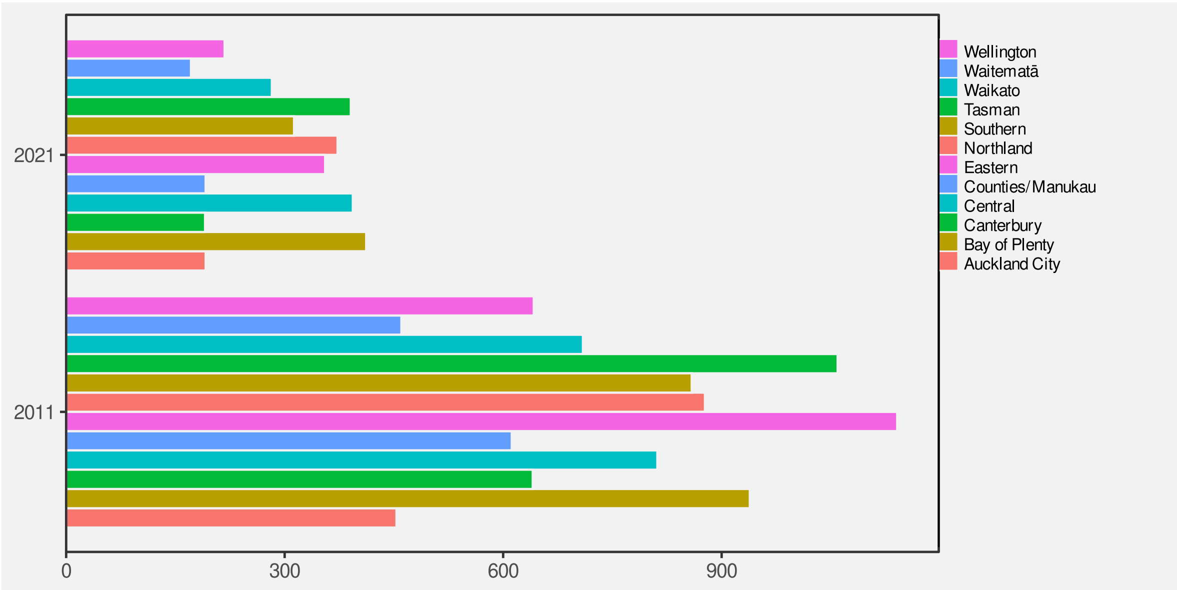


Figure 6.17: The hues in this pie chart are repeated so that adjacent hues are more distant from each other.

6.10 Colour palettes

When we encode nominal or ordinal data values as colours, we have to choose a set of colours, or a **colour palette**. For example, in Figure 6.1 we have to choose a different colour to represent each level of crime.

This choice is difficult because there are many different colours to choose from and often there is no obvious mapping from a data value to a particular colour. For example, what specific hue, chroma, and luminance should we choose to represent a "high" crime level? As we have seen in the preceding sections, there are also many, sometimes competing, criteria involved in selecting effective colours.

As an example, we can consider the problem of selecting a set of 5 different hues to represent 5 different categorical data values, like the palette that is required to represent the difference levels of crime in Figure 6.1.

We know that it is effective to encode qualitative data values as the hue of a data symbol (Section 3.4 and Section 6.1). In order to make the individual hues easy to identify, we can place them evenly around the colour circle. If we do not want the viewer to decode any ordering of the data values from the data symbols, then we should also hold the chroma and luminance of the data symbols constant. The top row of Figure 6.18 shows a palette that follows that simple approach.

The bottom row of Figure 6.18 shows one big problem with this attempt: the hues are very hard to identify for viewers with CVD (Section 6.7).

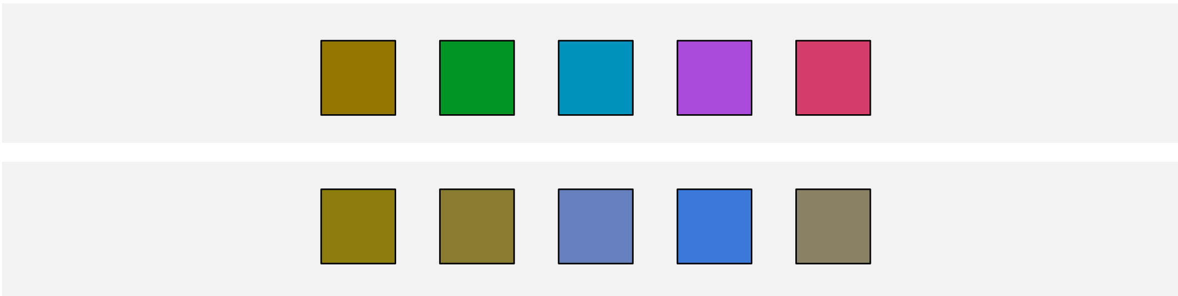


Figure 6.18: The top row shows a colour palette that attempts to vary hue evenly, in order to allow decoding of nominal data values, while holding chroma and luminance constant, in order to avoid decoding of ordinal data values. The bottom row shows how the palette appears to a viewer with red-green CVD.

Figure 6.19 shows another big problem with the palette in the top row of Figure 6.18. An attempt to hold chroma constant for a range of hues will fail on most standard computer displays because they cannot produce high chroma for all hues (Section 6.5).

Given the difficulty of selecting an effective colour palette, there is a good argument for making use of pre-existing colour palettes that have been designed by experts. For example, the ColorBrewer web site provides tools for choosing colour palettes for nominal and ordinal encodings. The Okabe-Ito palette for encoding nominal data (up to 9 categories), is specifically designed to allow effective decoding by all types of CVD (Figure 6.20). [76]

6.11 Case study: Diverging colour palettes

We have data on the total number of points that were scored and conceded in World Cup matches for Tier One nations (the top 10 rugby-playing nations) at each World Cup. Table 6.1 shows a sample of these data. This an aggregated version of the data set that we saw in Table 4.1.

Table 6.1: A table of the total number of points scored and conceded by Tier One nations in Rugby World Cup matches at each World Cup. Each row represents the total points scored by one team in one World Cup. There are 100 rows in total, with just the first 6 rows shown here.

team	year	scored	conceded	diff
Italy	1987	22	95	-73
Italy	1991	27	67	-40

Table 6.1: A table of the total number of points scored and conceded by Tier One nations in Rugby World Cup matches at each World Cup. Each row represents the total points scored by one team in one World Cup. There are 100 rows in total, with just the first 6 rows shown here.

team	year	scored	conceded	diff
Italy	1995	51	52	-1
Italy	1999	10	168	-158
Italy	2003	22	97	-75
Italy	2007	30	94	-64

Figure 6.21 shows a heatmap of the points differential (points scored for minus points scored against) for Tier One nations at each Rugby World Cup. The data symbols are rectangles, the teams are encoded as the vertical **positions** of the rectangles, and the years are encoded as the horizontal **positions** of the rectangles. These are both effective mappings and we can easily **decode** to teams or years. The points differentials are encoded as the **colour** of the rectangles, but in a slightly complicated way. Positive data values map to a shade of green, while negative values map to a shade of brown. In both cases, we have a constant **hue**, with increasing **chroma** and decreasing **luminance** as the data values get further away from zero.[77]

The use of two **hues** creates two different groups (positive and negative values), while the changes in **chroma** and **luminance** convey changes in absolute magnitude (if only for ordinal comparisons).[78]

> A diverging colour palette is effective because it creates two distinct visual features: a nominal hue to differentiate between two groups and changes in chroma and/or luminance to encode ordinal values within each group.

6.12 Case study: The rainbow

In some fields of research, for example medical imaging and geographic imaging, a common problem is to visualise numeric data values on a regular spatial grid. Figure 6.22 shows an example, where the grid is a set of longitudes and latitudes in the vicinity of New Zealand and the data values are measures of wind strength.

A common solution is to encode the numeric data values using colour[79] and it is also common to use a rainbow colour palette for that encoding (the image on the right of Figure 6.22). However, there are several problems with this approach.[80]

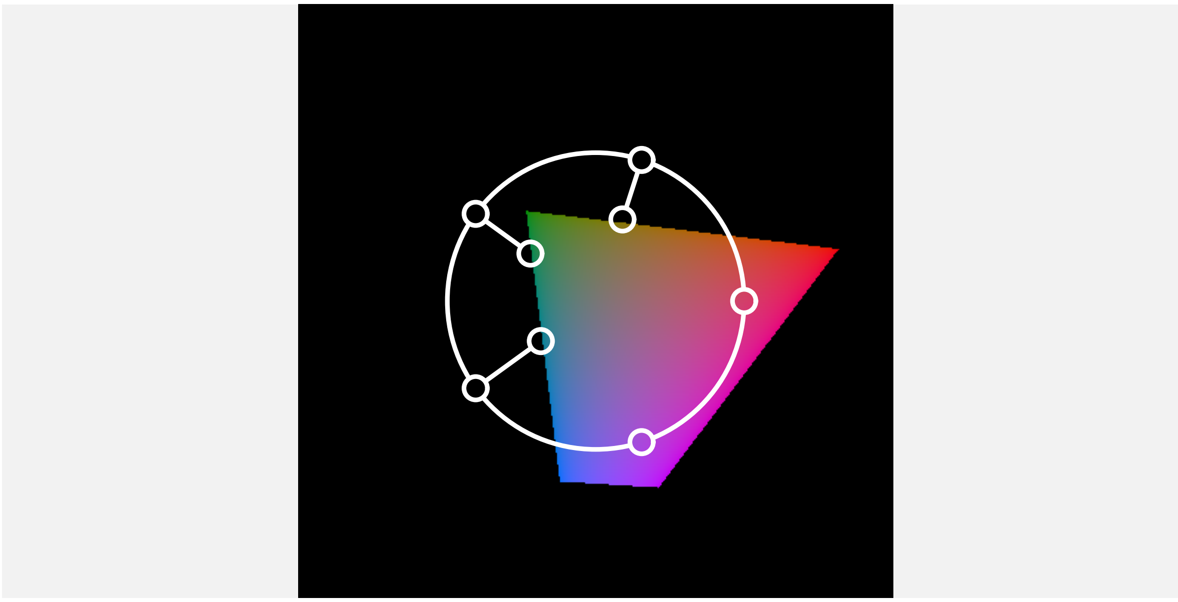


Figure 6.19: A view of the colour palette from Figure 6.18 relative to the range of colours that are possible on a standard computer monitor. The palette attempts to produce the same chroma for different hues, but only a much lower chroma can actually be achieved for some hues. The result is that the brow, green, and blue in the colour palette are less colourful than the purple and red.

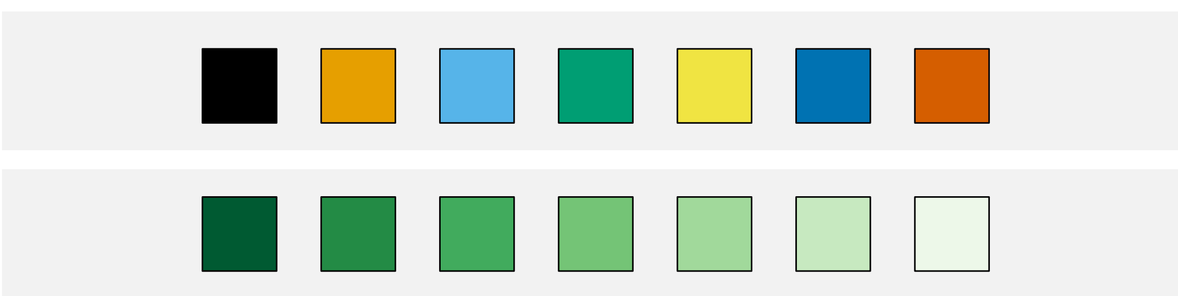


Figure 6.20: The top row shows the first 7 colours from Okabe-Ito. The bottom row is 7 sequential “Greens” from ColorBrewer.

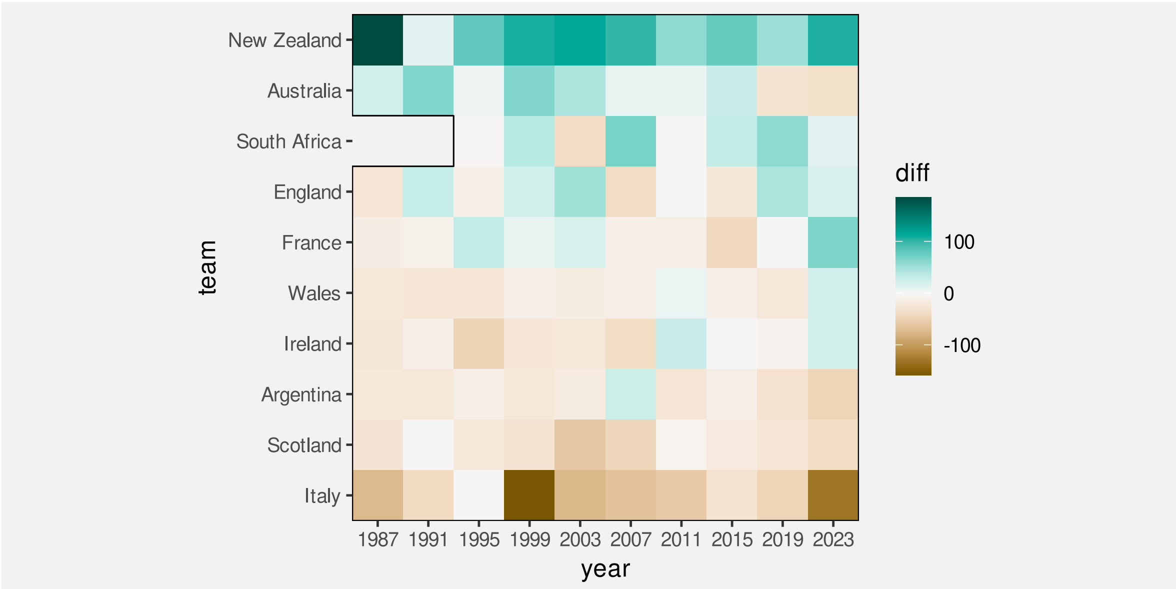


Figure 6.21: A heatmap of the total points differentials for Tier One nations at individual Rugby World Cups. South Africa was excluded from the 1987 and 1991 tournament due to its apartheid policies.

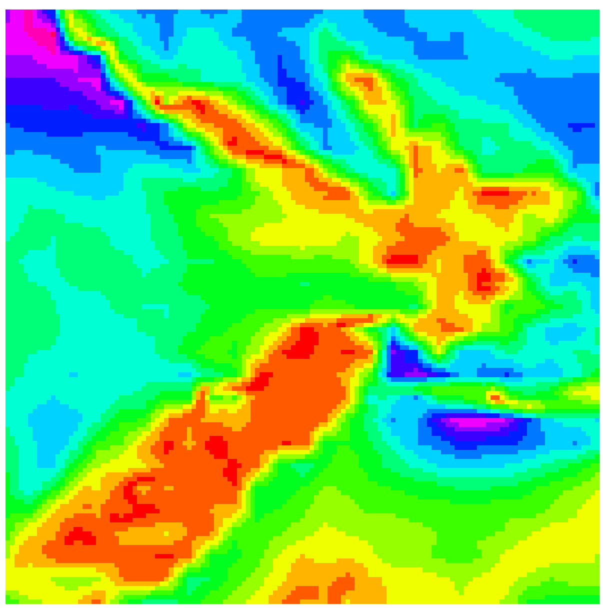


Figure 6.22: A snapshot of wind strength in the vicinity of New Zealand on September 14 2024. This uses the same data as Figure 2.5, but does not show wind direction. The image uses a rainbow colour palette (from red through green and blue to violet; see Figure 6.23).

Figure 6.23 shows the spectrum of colours for rainbow palettes.[81] This shows that a rainbow palette is mostly different in terms of hue and we have already established that hue is not a good visual feature for encoding quantitative data values (Section 3.4 and Section 6.1).

At the same time, The middle row of Figure 6.23 shows that the rainbow palette also varies in terms of luminance. We know that luminance can be an effective encoding for ordinal data values (Section 6.2), but the luminance changes in a rainbow palette are not monotonic. There are regions of almost constant luminance plus regions where luminance decreases followed by regions where luminance increases.[82] This makes a rainbow palette a poor encoding even for showing the order of numeric data values.

Finally, the changes in luminance and chroma within the rainbow spectrum produce bands of colours (red, yellow, green, cyan, blue, magenta). In other words, the spectrum appears discrete rather than continuous. This can result in a decoding that *adds* structure (discrete levels) to numeric data values that are in reality continuous.[83] For example, Figure 6.22 suggests three main regions–red-yellow, green-cyan, and blue-magenta—but the underlying data values vary more smoothly than that.

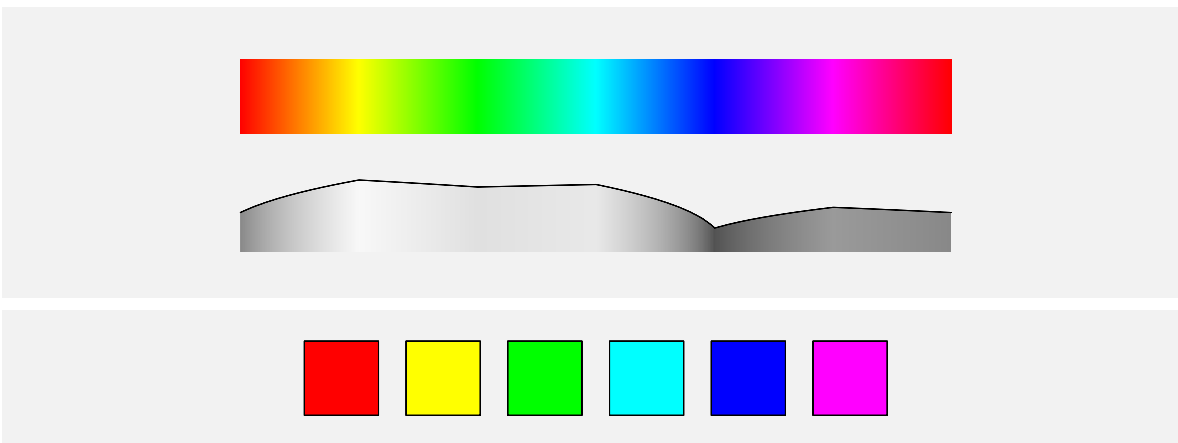


Figure 6.23: The top row shows the spectrum of colours from which a classic *rainbow* colour palette is taken. The middle row shows the changing luminance of the colours in the rainbow spectrum. The luminance of the blue colours is considerably less than the luminance of the yellows and greens and even the reds. The bottom row shows an example rainbow palette of 6 colours taken from the rainbow spectrum.

6.13 Case study: Viridis

One of the arguments for using a rainbow colour palette is that viewers like that it is colourful. We also saw in Section 6.9 that varying hue in addition to luminance can help to increase the capacity of an ordinal colour scale.[^multi-hue]

```
vary in hue as well as luminance and/or chroma (multi-hue palettes) lead
to faster and more accurate decoding than palettes with a constant
hue (single-hue palettes).  @rainbow-not-all-bad makes a similar argument.
```

The **viridis** colour palette, shown in Figure 6.24 (on the right), is an attempt to fix the problems of the rainbow palette, while still retaining a significant range of hues. Compared to Figure 6.22, it is much easier to decode regions of stronger winds and regions of weaker winds from the darker and lighter regions of Figure 6.24.

At the same time, the image on the left of Figure 6.24 shows a colour palette that only varies in terms of luminance and we can see that it is easier to see some details in the more colourful viridis map, such as the variations in low wind strength (the yellows) over New Zealand.

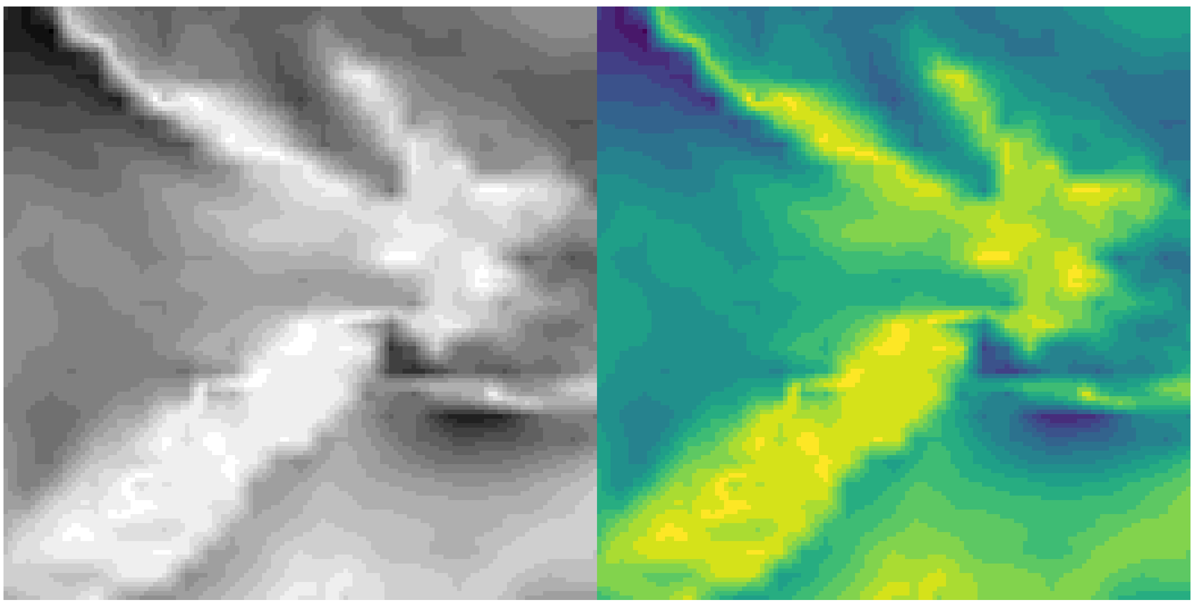


Figure 6.24: A snapshot of wind strength in the vicinity of New Zealand on September 14 2024. This uses the same data as Figure 2.5, but does not show wind direction. The image on the left encodes wind strength as shades of grey (darker is stronger) and the image on the right uses a viridis colour palette (Figure 6.25).

Figure 6.25 shows the spectrum for viridis colour palettes.[84] We can see that there is still a large range of hues, but the change in luminance is now monotonic, which at least allows decoding of ordinal data values. In addition, the range of hues has been selected to be appropriate for viewers with red-green colour blindness (Section 6.7).[85]

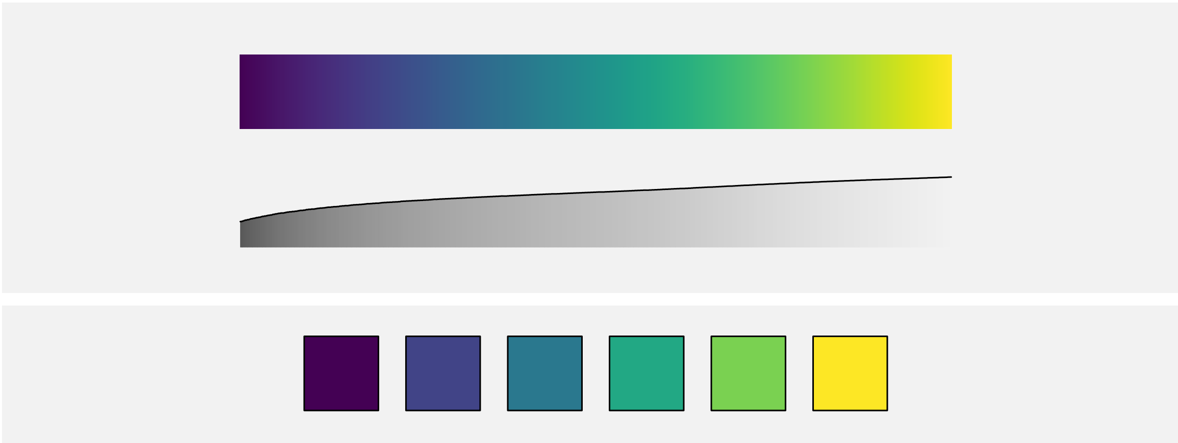


Figure 6.25: The top row shows the spectrum of colours from which a viridis colour palette is taken. The middle row shows the changing luminance of the colours in the viridis spectrum. The luminance change is monotonic. The bottom row shows an example viridis palette of 6 colours taken from the viridis spectrum.

6.14 Summary

Colour is really three visual features: **hue**, **chroma**, and **luminance**.

Hue is excellent for encoding **nominal** data values, though it has a limited capacity.

Chroma and **luminance** can be used to encode **ordinal** data values, but they have even lower capacity.

However, there are several complications:

- The decoding of data values from hue, chroma, and luminance is affected by surrounding colours and the size of the data symbol.
- Approximately 10% of viewers are unable to differentiate between red and green hues with similar chroma and luminance.
- The range of possible chroma values that can be decoded depends on both hue and luminance.

  The range of possible luminance values that can be decoded depends on both hue and chroma.

  The range of possible colours that can be produced is constrained by the technological limits of standard computer monitors.

Selecting which colours should be used to encode data values is difficult to get right and a good solution often involves varying all of hue, chroma, and luminance at once.

Consequently, it is usually a good idea to make use of pre-existing colour palettes that have been carefully designed to avoid most problems.

Notes

[66] The three-dimensional experience of colour arises from the fact that there are three different sorts of cones in the retina (see Section 2.1), each sensitive to a different range of light wavelengths: The L cones respond to long wavelengths (reds); the M cones respond to medium wavelengths (greens); and the S cones respond to short wavelengths (blues). The neural connections that combine the messages from the retina are also divided into three pathways: light-dark, which represents the sum of L, M, and S cones; red-green, which represents the difference between L and M cones; and blue-yellow, which represents the difference between S cones and the sum of L and M cones. For a more detailed description, see Ware (2021).

[67] Schloss, Leggon, and Lessard (2021) and Schloss et al. (2023) are examples of studies that have demonstrated a dark-is-more bias.

[68] The stance taken in this book, that chroma and luminance are only appropriate for qualitative (including ordinal) data, is more rigid than in many other places. Chroma and luminance are often included as an appropriate choice for encoding *quantitative* data values as well, albeit a poor one. For example, William S. Cleveland and McGill (1984) includes chroma (as *saturation*) and Munzner (2014) includes both luminance and chroma (as saturation) in their rankings of the accuracy of different visual features.

[69] This relative perception of colours is useful for being able to adapt to a wide range of lighting conditions. See Ware (2021) for a more detailed discussion.

[70] Stone (2012) describes "spreading", which means a blending or averaging of colours over small areas *by the visual system*: "In designing colors for digital visualization systems, one of the most critical factors is the interaction between size and color appearance."

[71] Figure 6.9 shows the extent of the *optimal colour solid*, which is essentially the colours that it is possible to produce by reflecting a light source off a surface that has perfect reflectance. This colour solid was determined assuming specific illumination conditions (D65), and assuming a standard human observer.

The colours are shown in CIE $L^*u^*v^*$ space (*aka* HCL space), which is (roughly) *perceptually uniform* so that perceptual differences between pairs of colours that are the same distance apart within the space are perceived as having the same visual difference.

The colours shown are produced using sRGB specifications, so many of the colours, especially at high chroma, are not realistic.

[72] Figure 6.10 shows the gamut of sRGB colours in CIE $L^*u^*v^*$ space (*aka* HCL space).

[73] In colour theory, hues that are close together on a colour wheel are referred to as *analogous* and hues that are opposite each other are *complementary*.

[74] This is more commonly known as *red-green colour blindness*. Blue-yellow colour blindness is also possible, but much rarer. There are also four distinct categories of red-green colour blindness, though the effect on the colour

perceived is similar in all cases. The images used in this book simulate deuteranomaly (the most common form of CVD). The approximate figure of 10% is from Ware (2021).

[75] This idea is described and demonstrated in Bartolomeo et al. (2025).

[76] The ColorBrewer tools are described in Harrower and Brewer (2003).

Okabe and Ito established the Color Universal Design organization, but the associated web site is no longer available. Okabe and Ito (2008) provides a general discussion of colour-blindness, including the palette used in Figure 6.20. An earlier piece of work that produced a palette of just 4 colours is described in Ichihara et al. (2008).

Zeileis and Murrell (2023) provides an overview of a large number of different palettes, within the context of producing data visualisations in R.

[77] This colour palette is based on the *BrBG* diverging palette from ColorBrewer.

[78] In contrast to a *diverging* colour palette, a colour palette for nominal data is often called a *qualitative* palette and a palette for ordinal or qualitative data is called a *sequential* palette.

[79] The resulting image is often referred to as a *pseudo-colour* image.

[80] The problems with rainbow palettes have been well-known for decades and attempts have been made to stamp out the use of rainbow colour palettes, but they remain popular in some fields of research (Borland and Taylor Ii 2007; Gołębiowska and Çöltekin 2022).

[81] The rainbow palette is typically a fully-*saturated*, maximum-*value* spectrum of hues—the most colourful and brightest colours that a typical computer monitor can produce.

Saturation and value are essentially less accurate versions of chroma and luminance. For example, a fully-saturated, maximum-value yellow has a higher chroma and luminance than a fully-saturated, maximum-value blue.

The rainbow spectrum is basically a path along the boundary of the colours in Figure 6.10, visiting the most extreme points for different hues. This path clearly is not going to have a simple monotonic change in either luminance or chroma.

[82] A rainbow palette is a colour encoding that is non-linear in an unhelpful way (Section 6.8).

The representation of luminance in Figure 6.23 also shows an example of a 3d illusion (Section 2.10). The curved line combined with the gradients of grey tempt us to see a curved wall of constant height that is receding from us.

[83] Heer and Stone (2012) discuss that viewers tend to group colours by discrete colour names and that those colour groups help to explain how colour palettes are decoded.

[84] There is no official publication that explains the development of the viridis colour spectrum, but there is a video on YouTube that is very good (Smith and Walt 2015).

[85] Nuñez (2018) take things a step further with a *cividis* colour palette that provides a CVD-safe version of viridis that also has a wider and *linear* variation in luminance (in Figure 6.25, we can see that the changes in luminance are non-linear and do not span the full range of luminance).

7 Congruent Mappings

In previous chapters, we have seen that, part of the reason why some well-known plots, like bar plots, are effective is because they map data values to the basic visual features of a data symbol. For example, encoding the number of youth offenders as the **length** of a bar in a bar plot (Figure 5.1). We have also seen that basic visual features like **length** are effective because we can accurately decode data values from those visual features.

The purpose of this chapter is to show that, in order to understand the effectiveness of a data visualisation, we need to consider more than just the accuracy or capacity of our mappings from data values to basic visual features.

7.1 Implicit decodings

Figure 7.1 shows two lines, one longer than the other. We can decode information from these lines without any explicit encoding of data values. For example, we can decode that one line is twice as long as the other without knowing anything about what "twice" represents in terms of data values. In other words, there is an **implicit decoding** from the lengths of the lines to data values (Figure 7.2).[86]

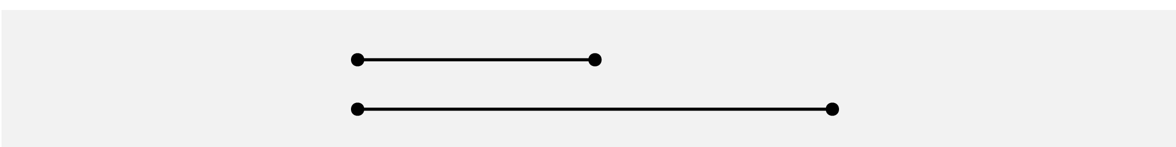


Figure 7.1: We can decode the fact that one line is twice the length of another line without there having to be any explicit encoding of data values to the lengths of the two lines.

When an implicit decoding exists, we can produce a more effective mapping by building upon the implicit decoding. We can explicitly encode data values as visual features in such a way that the decoding is **congruent** with the implicit decoding, in the sense that the implicit decoding leads back to the same data values that we explicitly encoded (Figure 7.3).[87]

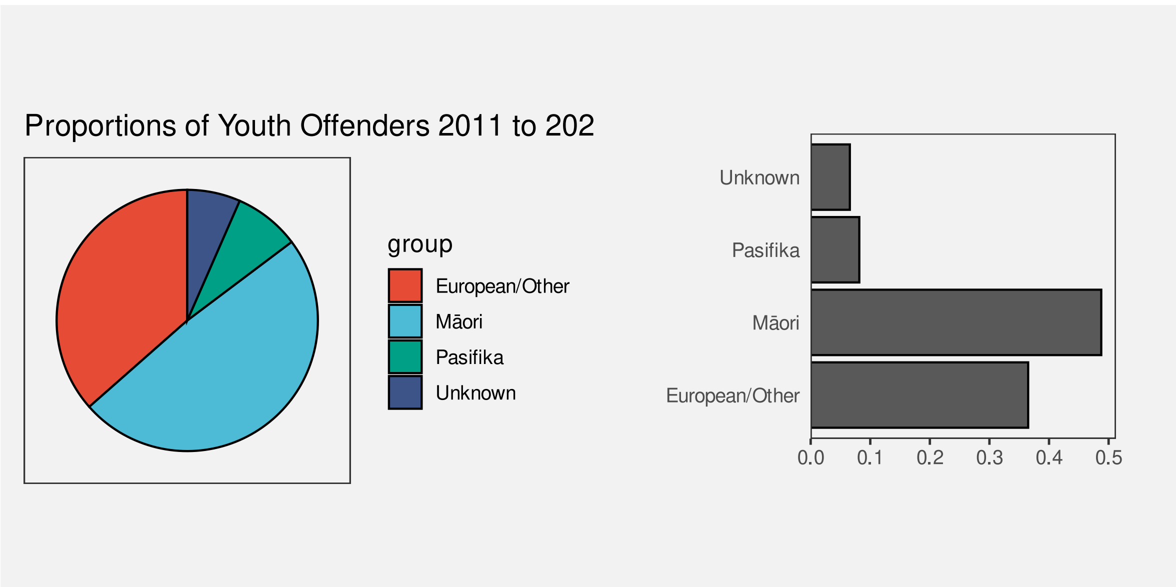
Figure 7.2: Some visual features have implicit encodings, which means that we can decode information from the visual features without having explicitly encoded any data values.

Figure 7.3: If we explicitly encode data values as visual features in a way that is consistent with an implicit decoding, we can more effectively decode data values from visual features.

7.2 Part-to-whole

Figure 7.4 shows a pie chart of the proportions of total youth offenders between 2011 and 2021 in different ethnic groups and a bar plot of the same data. We learned in Section 3.5 that encoding **quantitative** data values, like proportions, as **lengths**, like in the bar plot, is more accurate than encoding the proportions as **angles** or **areas**, like in the pie chart.

However, the superior representation of the fact that Māori offenders make up approximately 50% of the total is the pie chart rather than the bar plot. Although it is easy to determine from the length of the bar and the scale on the x-axis in the bar plot that the bar for Māori is just below 0.5, it is even easier, without any guides at all, to see that the light blue wedge in the pie chart is just under half a circle. It is also easy to see that the red wedge is close to a third of a circle.[88]



(a) A pie chart.

(b) A bar plot.

Figure 7.4: Data visualisations of the proportion of offenders from different ethnic groups.

The reason why the pie chart in Figure 7.4 is more effective than a bar plot for conveying simple proportions like a quarter or a third or a half is because the wedges in a pie chart have an implicit decoding to fractions of a whole. A semi-circle has a strong visual association with *half* of a circle and thirds and quarters are similarly evocative.

> One reason why a pie chart can be effective is because segments of a pie vividly convey simple fractions of a whole.

Figure 7.5 shows that if we place a bar representing a proportion bar within a reference rectangle that corresponds to a proportion of 1, then we can also create a part-to-whole congruence with bars, though the effect is not as strong as it is for pie wedges.

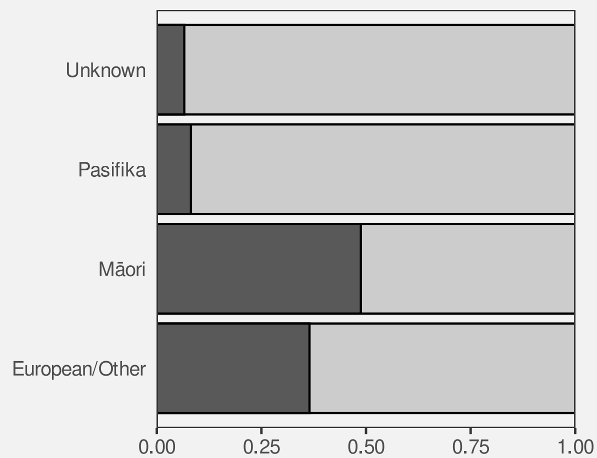


Figure 7.5: A bar plot of the proportion of offenders from different ethnic groups with reference rectangles to enhance the part-to-whole relationship.

7.3 More is more

We saw in Section 3.3 that visual features like **length** and **area** are appropriate for conveying quantitative values. For example, in Figure 7.6, the lengths of the bars are appropriate for conveying the total number of offenders in different ethnic groups.

Part of the reason why **length** and **area** work so well is because a **longer** bar implicitly conveys the idea of *more*.[89] By contrast, the **position** of data points in a dot plot of the same data, as shown in Figure 7.6, do not inherently convey the same sense of *more*. In a bar plot, the data symbols that correspond to larger data values are actually larger data symbols; the presentation of larger data values is more abstract in a dot plot.

Another way that **length** can convey information more naturally than **position** is visualising negative values. For example Figure 7.7 shows the total points differential (points scored minus points conceded) for Tier One nations at Rugby World Cups (see Table 4.1). The *direction* of the bars very clearly indicates the change from positive points differential to negative points differential.

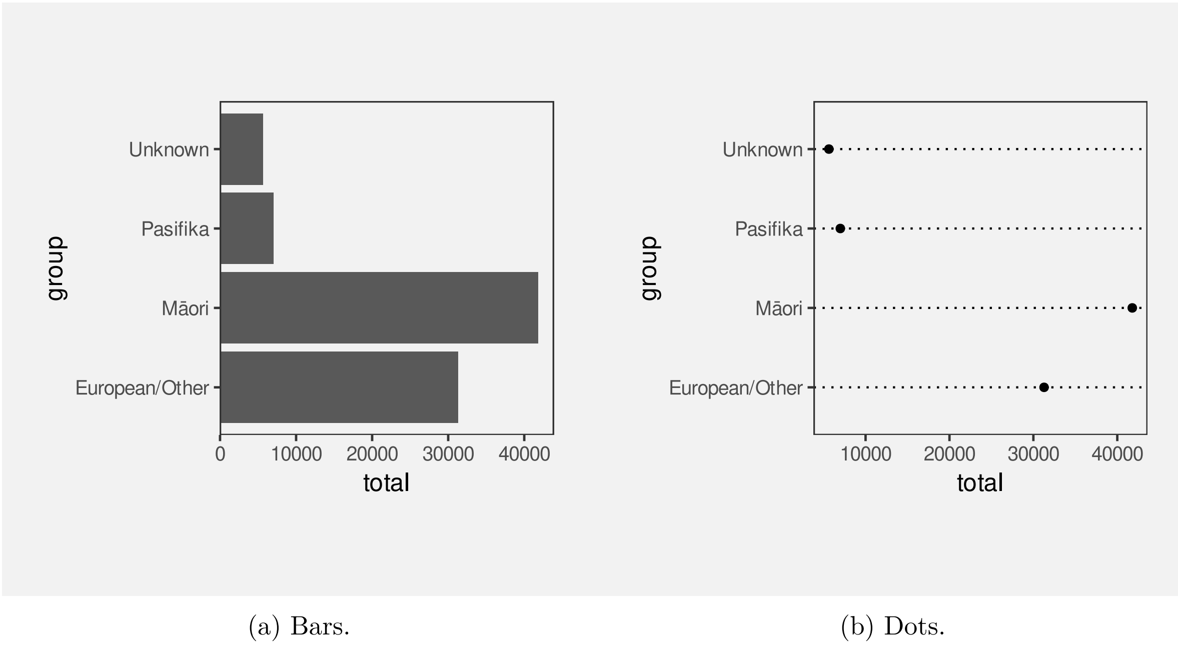


Figure 7.6: A bar plot and a dot plot of the total number of offenders for different ethnic groups.

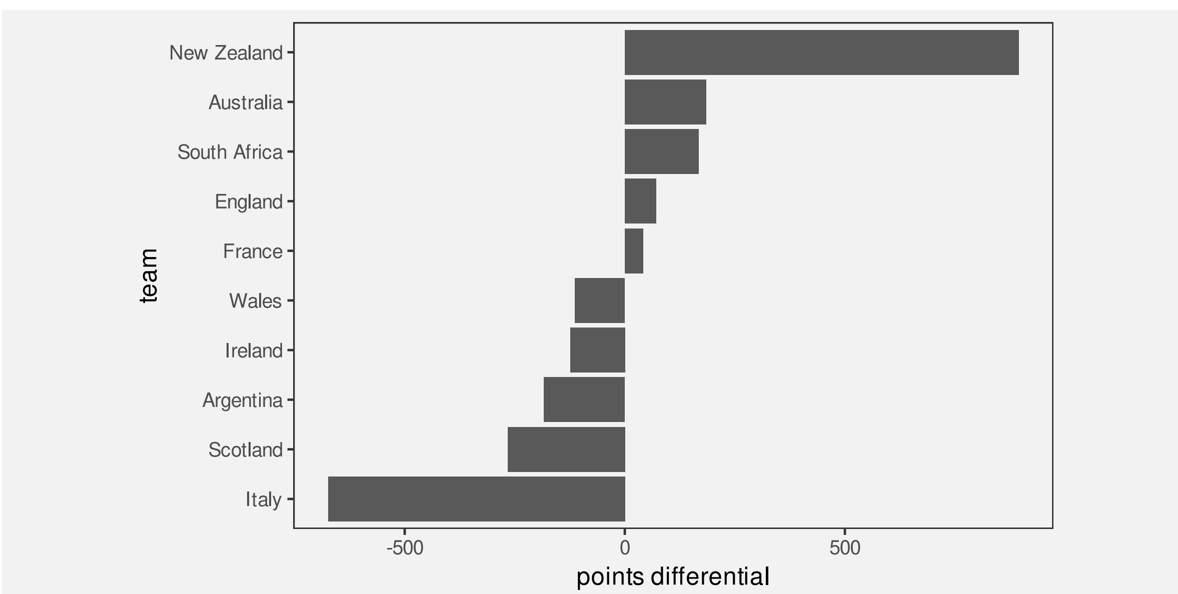


Figure 7.7: A bar plot of the total points differential for Tier One nations at Rugby World Cups.

By comparison, the change from positive to negative points differential is much more abstract in a dot plot of the same data (Figure 7.8).

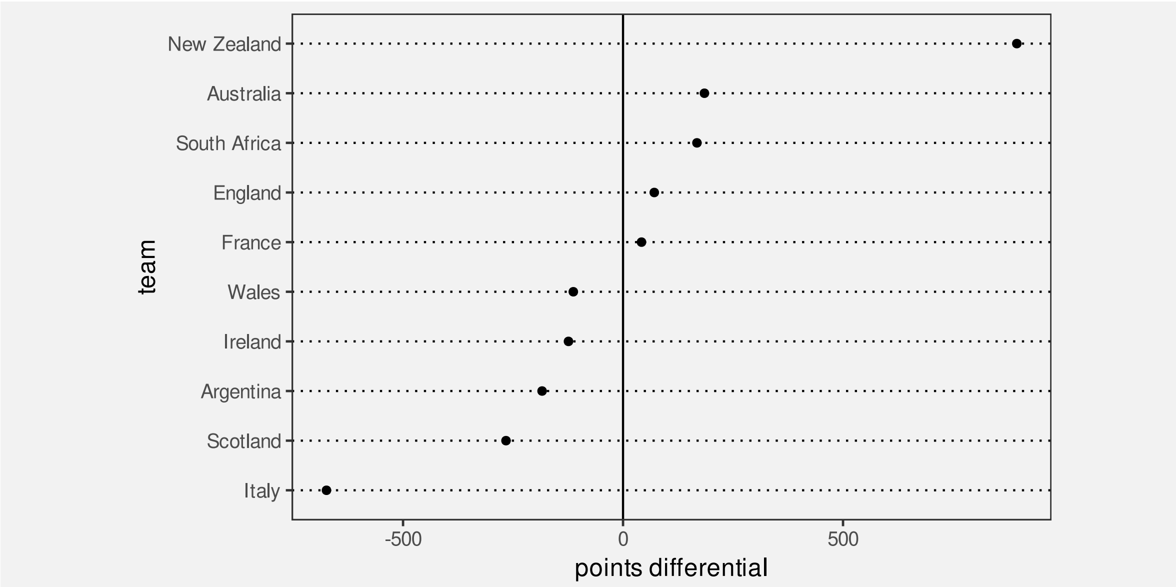


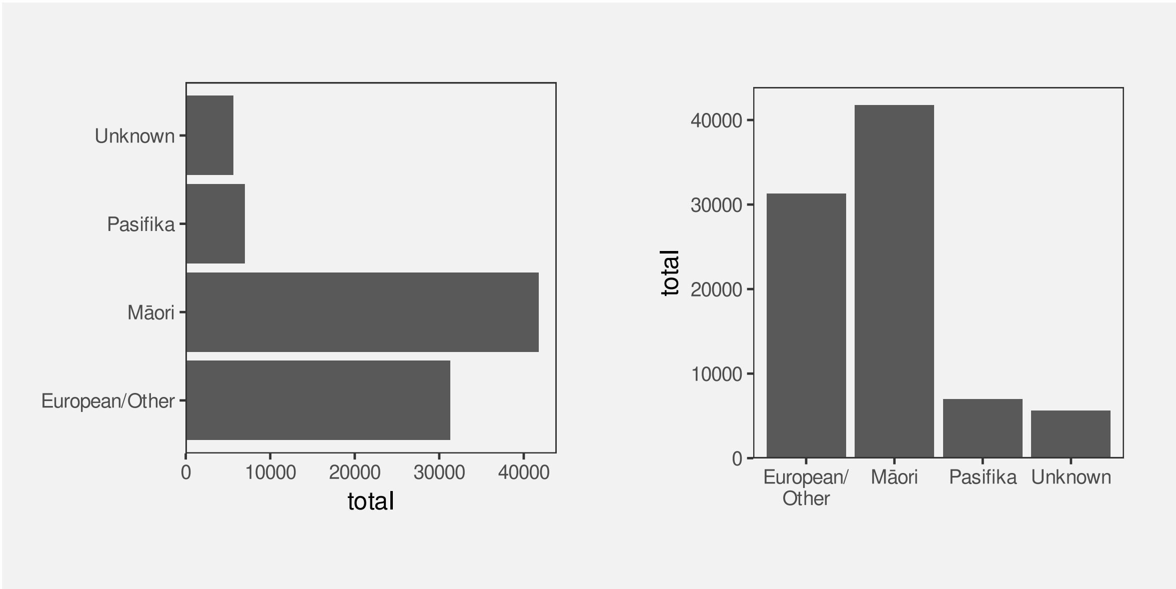
Figure 7.8: A dot plot of the total points differential for Tier One nations at Rugby World Cups.

> One reason why bar plots are effective is because larger data values are represented by larger data symbols.

A more subtle effect can be seen in Figure 7.9, which shows two different bar plots of the total number of offenders in different ethnic groups. In this case, the vertical bars are more natural for conveying count data like this because of the natural correspondence with stacks of items.[90]

7.4 Same is same

In Section 2.5 we saw that the visual system is very sensitive to *differences*, particularly when everything else remains the *same*. This means that mapping *changes* in data values to *changes* in data symbols will be effective because the visual system will notice the differences in the data symbols. More specifically, if a **visual feature** of a data symbol changes then the visual system will notice that change. Furthermore, if a visual feature of a data symbol changes while all other visual features of the data symbol remain the *same*, the visual system will notice the change more easily (Section 2.5). Put another way, visual differences are implicitly decoded



(a) Horizontal bars. (b) Vertical bars.

Figure 7.9: Data visualisations of the number of offenders from different ethnic groups.

as changes in the data values *and* visual constancy is implicitly decoded as no change in the data values.

For example, in a bar plot like Figure 3.2, the data symbols are bars and only the **lengths** of the bars and the vertical **positions** of the bars are different. The "widths" of the bars and the **colours** of the bars remain the same.

> Another reason why bar plots are effective is because the bars differ in length, but are otherwise the same.

7.5 Dissonant mappings

The fact that **length** and **area** implicitly convey amounts means that care must be taken *not* to confuse that association.[91] For example, Figure 7.10 shows the number of times each team entered the opposition's final third during a match at the 2023 Women's World Cup.[92] The final third of the pitch is shown as darker green regions, broken into five different zones, with a number of entries for each zone and each team represented by a line with an arrow.

In Figure 7.10, longer lines correspond to a larger number of entries. However, the bars start at the halfway line on the pitch rather than at the final third of the pitch. This means that the zero entries by South Africa into the second-from-right zone is encoded as a non-zero-length line. Apart from making comparisons of the line lengths very inaccurate, this creates a confusing visual effect because the viewer is expected to decode a non-zero length to a value of zero.

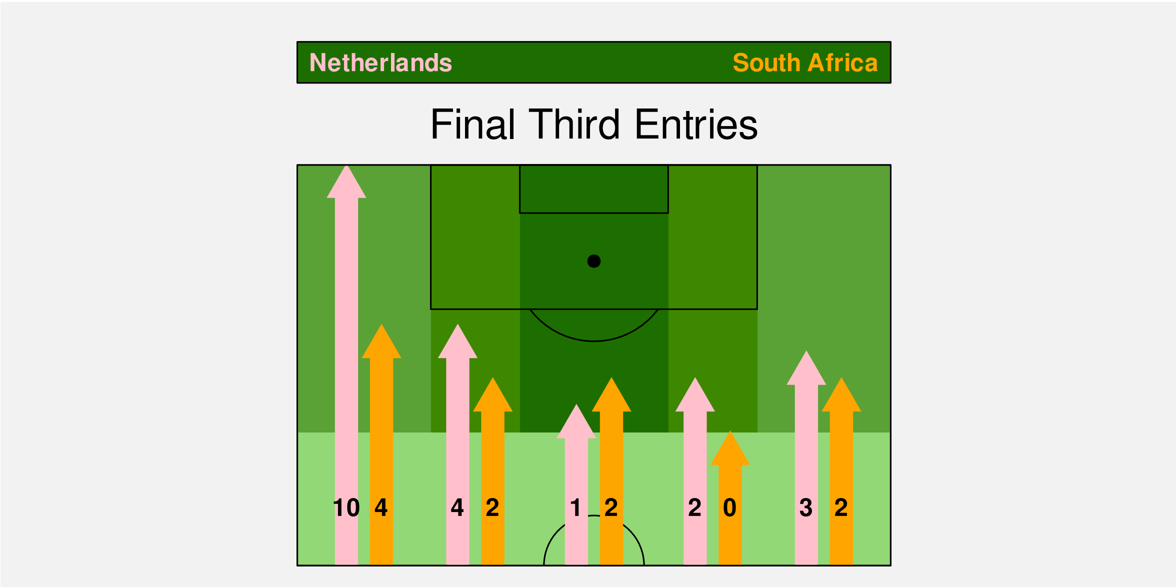


Figure 7.10: The number of times each team entered the opposition's final third in the Women's World Cup match between the Netherlands and South Africa.

For any example of **congruence**, where a visual feature implicitly decodes to a property of the data, there is an opportunity for **dissonance** if the visual feature is used inappropriately. We can improve the effectiveness of a data visualisation by taking advantage of visual congruence and we can reduce the effectiveness of a data visualisation if we create visual dissonance.[93]

If we fail to make our explicit encodings congruent with any implicit encodings—if we create a **dissonant** mapping—we can make it harder and more confusing to decode a data visualisation because there is more than one possible decoding (Figure 7.11).

We saw in Section 7.4 that we want to map differences in data values to differences in visual features *and* keep everything else the same. Failing to follow that approach will make it harder to perceive changes in the data values. However, a worse error, and another source of visual dissonance arises if we change visual features *without* any underlying change in the data values.[94]

Figure 7.12 shows a data visualisation of the relative number of people aged over 100 for females versus males.[95] There are numerous problems with this data visualisation, but here we will just focus on the horizontal **position** of the cake images. The cakes for females are positioned to the left of the cakes for males, which is acceptable because that difference in

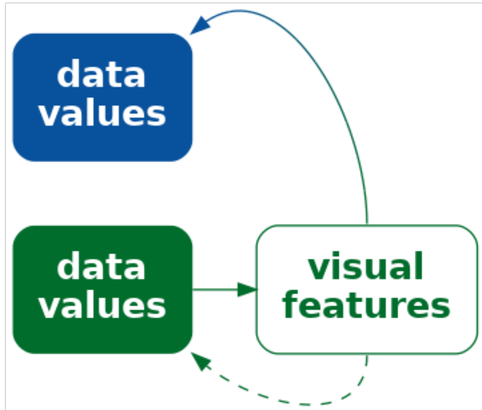


Figure 7.11: If we explicitly map data values to visual features, but the visual features have an implicit mapping to different data values, we can cause confusion and make it harder to effectively decode data values from visual features.

position reflects a difference in the data—males versus females. However, within each gender, the cakes are also staggered a small amount to the left then to the right. This visual difference in the cake positions does not correspond to any difference in the data values. We are tempted to seek meaning from the the left-to-right staggering of the cakes when no meaning exists.

> A data visualisation is *not* effective if it contains mappings that are dissonant because that can cause distraction and slower and/or incorrect **decoding** of data values from visual features.

7.6 Case study: Time's arrow

Figure 7.13 shows the results of a 2025 poll that asked what was the best thing that the Trump administration had done, comparing results in February to results in September. In terms of mappings, this bar plot makes good use of (vertical) **position** and (horizontal) **length** to encode different policy issues, the difference in dates, and the percent of respondents who voted for each policy. However, when we look at the top two bars, there is a temptation to read the pairs of bars in order from top to bottom.[96] If we do not attend carefully to the legend, we may perceive a *decrease* in the (relative) popularity of the immigration policy and an *increase* in the popularity of DOGE. This is a subtle example of visual dissonance.

Figure 7.14 shows another version of the bar plot, with the bars running vertically. In this plot, the natural reading from left to right leads to the correct interpretation: the (relative)

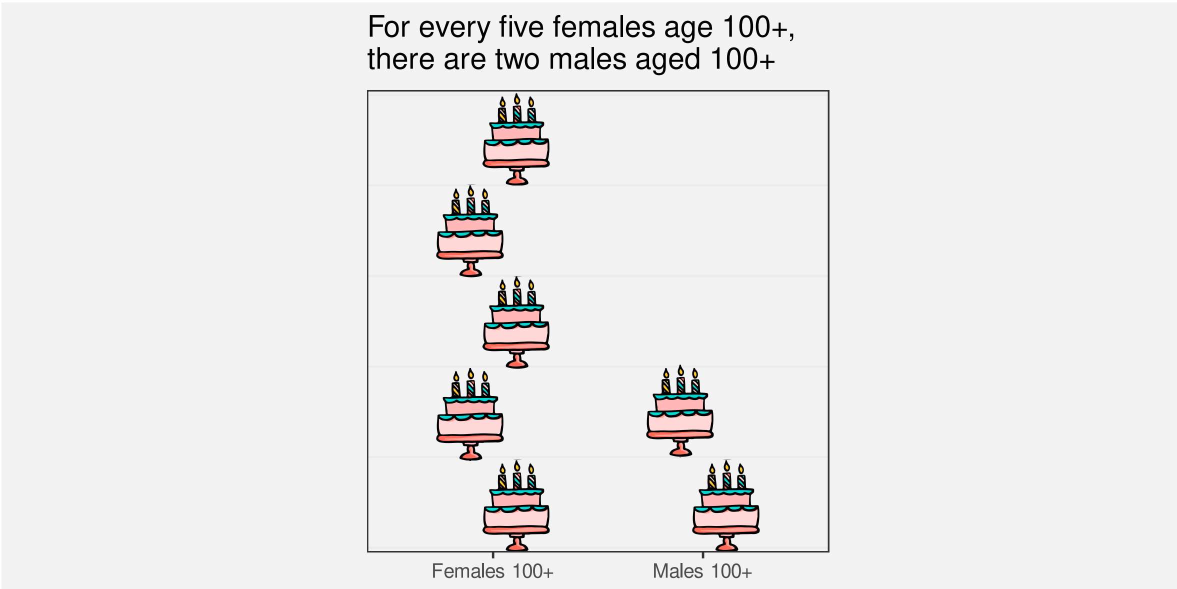


Figure 7.12: A data visualisation of the number of females aged over 100 relative to the number of makes aged over 100.

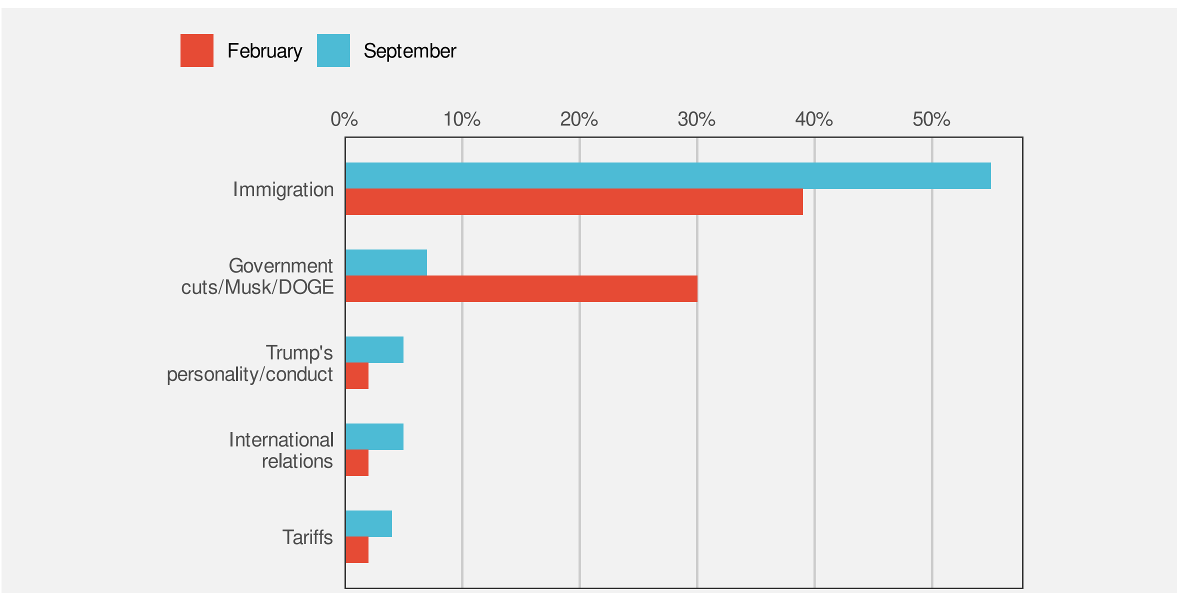


Figure 7.13: Bar plot of most popular Trump policies over time.

popularity of DOGE has declined while the popularity of the immigration policy has increased. This is an example of visual congruence.

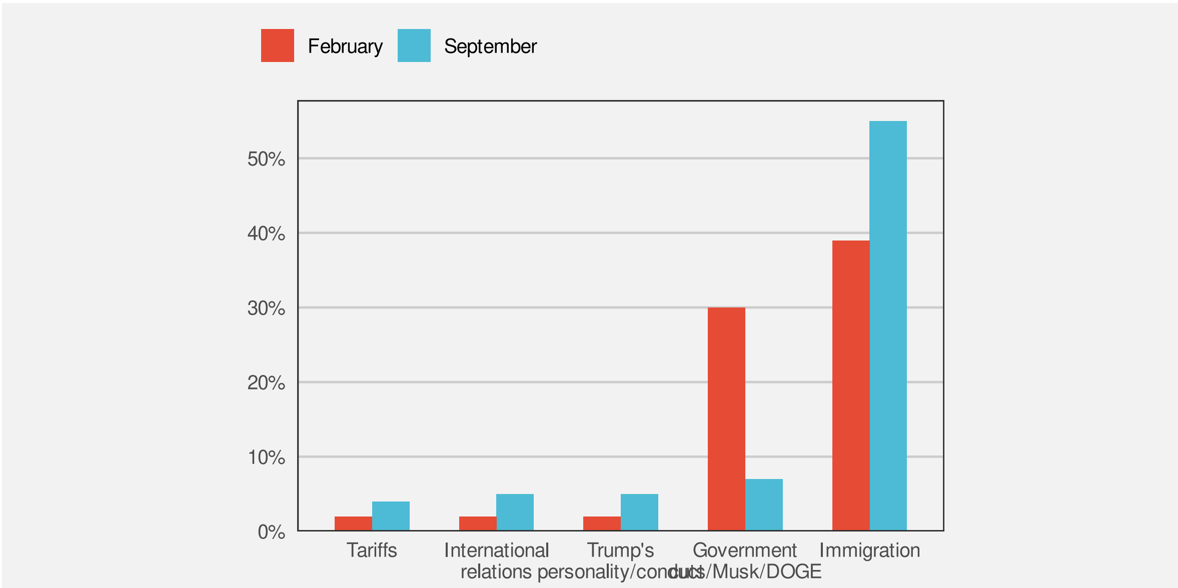


Figure 7.14: Bar plot of most popular Trump policies over time.

7.7 Case study: Jittering

We have data on the the number of points that were scored in every World Cup match by Tier One nations (the top 10 rugby-playing nations). Table 4.1 shows a sample of these data.

Figure 7.15 shows a dot plot of the number of points scored by teams from different hemispheres at Rugby World Cup matches. In this data visualisation, the data symbols are dots, the number of points scored is encoded as the horizontal **position** of the dots, and the hemisphere is encoded as the vertical **position** of the dots. We saw a similar visualisation of these data in Figure 4.5, which demonstrated the problem of overplotting of points that share the same data value.

In Figure 7.15, to ameliorate the overlapping of data points, a random amount of “jitter” has been added the vertical position of the dots. The filled interior of each dot is also semitransparent so that we can see where some points still overlap.

The dot plot in Figure 7.15 is effective for **decoding** to the number of points scored because each data value is encoded as the horizontal position of a single data symbol. Similarly, we can easily decode the hemisphere from each point. The proximity of the data symbols from the same hemisphere means that it is easy to perceive two “rows” of points (Section 2.7).

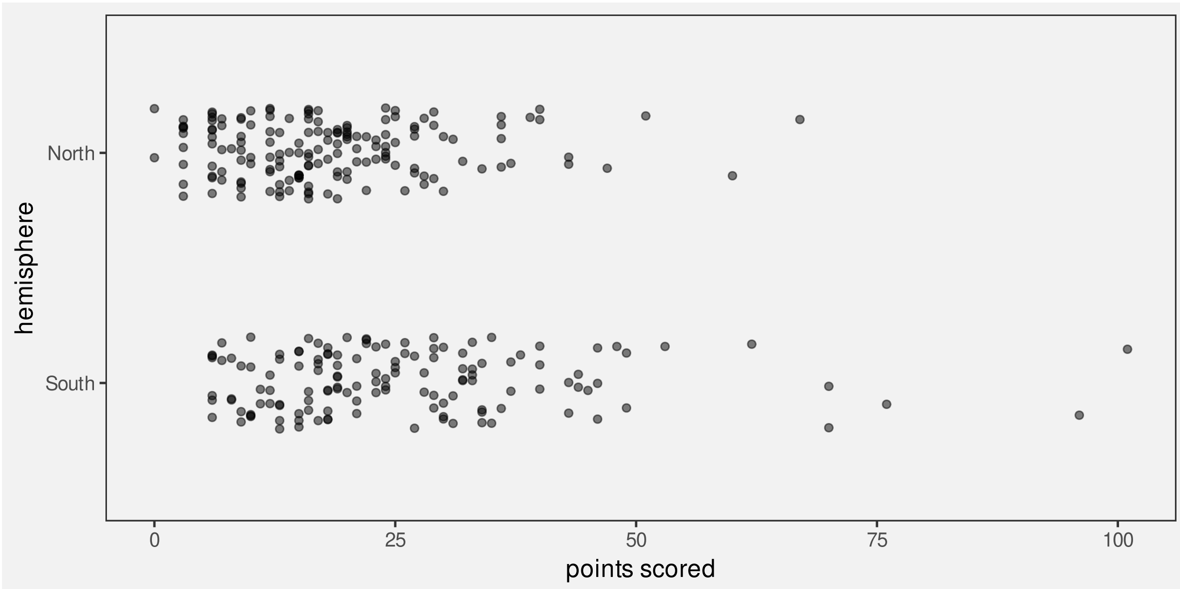


Figure 7.15: The number of points scored in Rugby World Cup games by teams from the northern and southern hemispheres.

On the other hand, the jittering means that there are visual differences between the vertical positions of the dots, within the same hemisphere, that do not correspond to any differences in data values. For example, there are two dots representing matches where a team from the Northern hemisphere scored zero points. The data values for these two dots are identical (zero points scored and northern hemisphere), but they are visually different. The randomness of the jittering may help to dilute any temptation to seek meaning from the artificial visual difference, but the addition of jittering is a potential source of confusion.

> Jittering data symbols is effective because it reduces the amount of overlap amongst data symbols, but jittering may also reduce the effectiveness of a data visualisation because it introduces visual dissonance.

7.8 Summary

The effectiveness of a data visualisation may depend on more than just the accuracy and capacity of visual features.

A **congruent** mapping is one where data values are *explicitly* encoded in a way that is consistent with an *implicit* decoding of the visual feature.

A data visualisation will be more effective if it is visually congruent, for example, data symbols are larger for larger data values or data symbols only change if the data values change.

A **dissonant** mapping is one where data values are *explicitly* encoded in a way that is *inconsistent* with an *implicit* decoding.

A data visualisations will be less effective if it is visually dissonant.

Notes

[86] These implicit mappings are part of the reason why different visual features are appropriate for encoding quantitative verus qualitative data values. The length of a data symbol is appropriate for encoding quantitative data values because we implicitly decode amounts from length.

[87] Tversky, Morrison, and Betrancourt (2002) describe a *congruence principle*: "the structure and content of the external representation should correspond to the desired structure and content of the internal representation," where the "external representation" is a data visualisation and the "internal representation" corresponds to an implicit decoding.

Bertini, Correll, and Franconeri (2020) discuss measures of effectiveness beyond accuracy and capacity and refers to the congruence principle above as well as the idea of *congnitive fit*: "performance on a task will be enhanced when there is a cognitive fit (match) between the information emphasized in the representation type and that required by the task type" (Vessey 1991). The latter was focused on a comparison of plots versus tables (plots for spatial comparisons and tables for more complex cognitive tasks involving symbolic reasoning), but it is analogous to the idea of congruent mappings.

[88] This defence of pie charts has existed for a very long time (Eells 1926a) and continues to receive empirical support in more recent times (Simkin and Hastie 1987).

[89] Kosslyn's Principle of Compatibility states that "A message is easiest to understand if its form is compatible with its meaning", plus explicitly "More Is More" (Kosslyn 2006).

Wilke (2019) has a similar Principle of Proportional Ink, which he attributes to Bergstrom and West (2017). A corollary is that bars in a bar plot should start at zero.

[90] Several authors make an argument for vertical bars being preferred because they correspond naturally to accumulation and growth (e.g., Few 2004).

[91] This is related to visual illusions (Section 2.10). There are decodings that the visual system performs whether we want it to or not, so we want to avoid conflicting with what the visual system does automatically.

[92] This data visualisation is an adaptation of a graphic shown during the live television coverage of the match.

[93] The idea of visual dissonance is just a logical inversion of the idea of visual congruence. This idea is not named in any literature that I could find, though *semantic clashes* (Bertini 2024) perhaps comes closest.

[94] Kosslyn's *Principle of Informative Changes* states that "People expect changes in properties to carry information" (Kosslyn 2006).

[95] This plot is based on a data visualisation from the New Zealand Listener October 14-20 2023.

[96] The idea for this data visualisation, and whether to question the vertical ordering, comes from a PolicyViz Newsletter. The original data source was a Washington Post poll).

8 Mapping Summaries

In Chapter 3 we described a data visualisation as a mapping that **encodes** data values as the visual features of a data symbol (Figure 3.1). For example, a bar plot, like Figure 3.2, maps the total number of youth offenders to the **lengths** of bars.

We also emphasised the importance of the **decoding** from visual features to data values. For example, the effectiveness of the bar plot in Figure 3.2 relies on our ability to accurately decode the lengths of the bars, particularly relative to each other.

In the simple scenario of a bar plot, the data values are counts (Table 3.1), which are encoded as the lengths of bars, which in turn allows us to decode lengths to retrieve the counts (Figure 3.6). Each **raw data value** (count) is mapped to the length of a bar.

In this chapter, we explore some data visualisations that are effective because they encode **data summaries** instead of raw data values.[97]

8.1 Box plots

Figure 8.1 shows box plots of the number of points scored by Tier One nations at Rugby World Cup matches (Table 4.1), with a separate box plot for teams from different hemispheres.[98] This is an effective data visualisation for comparing the distributions of points scored between hemispheres because we are able to easily see differences between the median values of the countries (the thick vertical lines) and differences between the interquartile ranges of the countries (the horizontal rectangles). For example, we can see that the median points scored by southern hemisphere teams is larger than the median for northern hemisphere teams and the interquartile range of points scored is also wider for southern hemisphere teams.

Box plots are quite different from bar plots and dot plots because the data symbol is a much more complicated shape: there is a vertical line to represent the median, a rectangle to represent the interquartile range (lower quartile to upper quartile), and horizontal lines to represent the extremes of the data values, from the the quartiles to the minimum and maximum values. Points are also included for “outliers” if data values lie further than 1.5 times the interquartile range beyond the quartiles.

Another important difference with a box plot is that it maps **data summaries**, rather than raw data values, to visual features of the data symbol (Figure 8.2). For example, the median

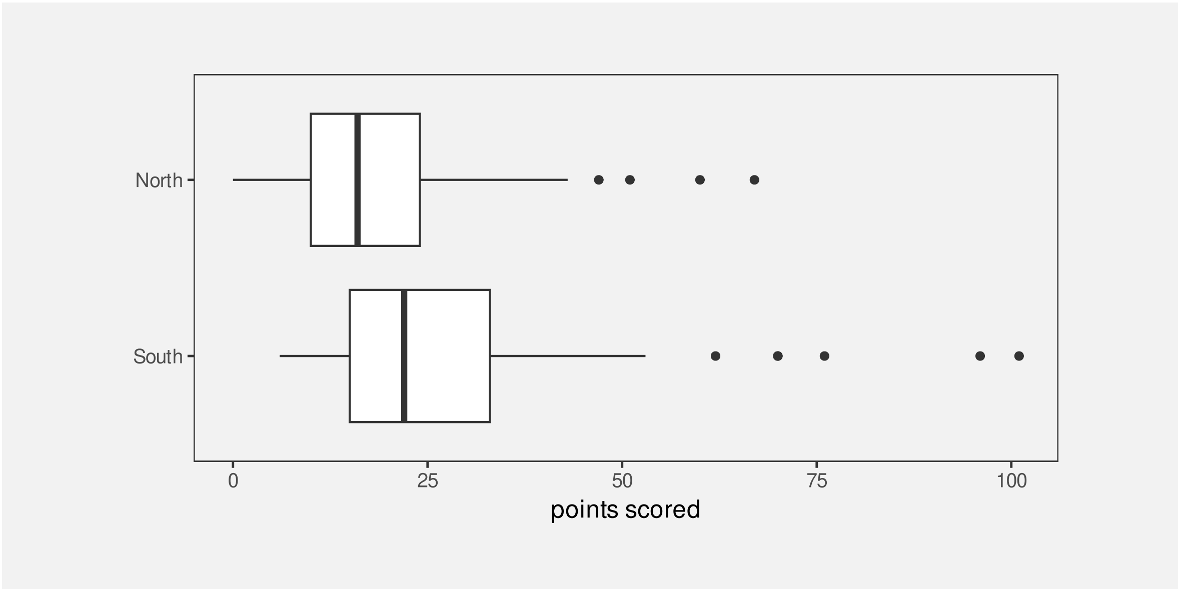


Figure 8.1: Box plots of the number of points scored in all Rugby World Cup games by Tier One nations.

values are mapped to the horizontal **positions** of thick vertical lines and the interquartile ranges are mapped to the **lengths** of horizontal rectangles.

To be explicit, the data values that are being **encoded** as visual features in Figure 8.1 are *not* the raw data values in Table 4.1; the values that are being encoded to visual features in Figure 8.1 are the **data summaries** shown in Table 8.1. The data summaries in Table 8.1 are the values that we can **decode** from the boxplots in Figure 8.1.[99]

Table 8.1: A table of the five-number summaries for the number of points scored by Tier One nations in Rugby World Cup matches, along with the country name. These data summaries are the values that are mapped to visual features to produce the box plots in Figure 8.1.

	minimum	lower quartile	median	upper quartile	maximum
South	6	15	22	33	101
North	0	10	16	24	67

Figure 8.1 is effective for allowing us to compare summaries like the medians between different hemispheres because, although the box plot data symbol is relatively complicated, a box plot still just encodes the data summaries to simple visual features. We know from Chapter 3 that **position** and **length** are effective for **decoding** the data summaries from the visual features (Figure 8.2).

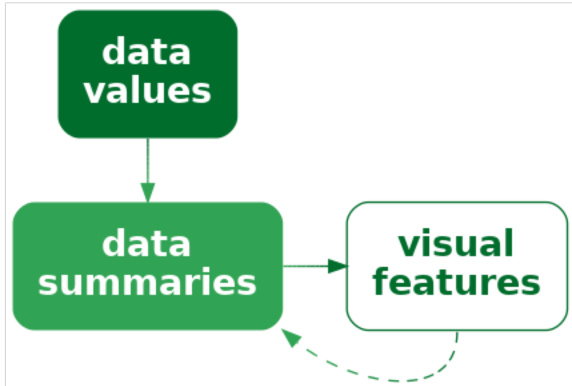


Figure 8.2: A data visualisation may encode data summaries, rather than the raw data, as visual features. This is effective for decoding the data summaries, but it is not effective for decoding the raw data.

Notice how, even though Table 8.1 contains very few values, it is much easier to perceive the differences between the two rows of values using the data visualisation in Figure 8.1 rather than the table of values in Table 8.1. Once again, we can see the power of mapping values to basic visual features. The only differences from previous sections is that we are mapping **data summaries** rather than raw data values and we are mapping to the visual features of a more complicated data symbol.

> A box plot is effective at conveying information about the distribution of data values because it maps **data summaries** about the distribution, like the median and interquartile range, to basic visual features.

8.2 Visual summaries

Figure 8.3 shows a dot plot of the number of points scored by countries from the different hemispheres in Rugby World Cup matches.[100] This data visualisation uses the same data as Figure 8.1, but it maps raw data values rather than data summaries and it uses much simpler data symbols (data points). This data visualisation encodes the number of points scored maps as the horizontal **position** of a data point and it encodes the hemisphere as the vertical **position** of a data point. There is an overplotting problem (Section 4.4), which is solved in this case by offsetting data symbols vertically for repeated data values.

Because this data visualisation encodes raw data values as visual features, it is possible to decode raw data values from the visual features (Figure 3.6). For example, it is possible to see that the lowest score (zero points) for a northern hemisphere team ocurred twice. It is also possible to see that there are scores that have never occurred in any matches: 1, 2, 4, and 5. These features are not visible in the boxplots in Figure 8.1.

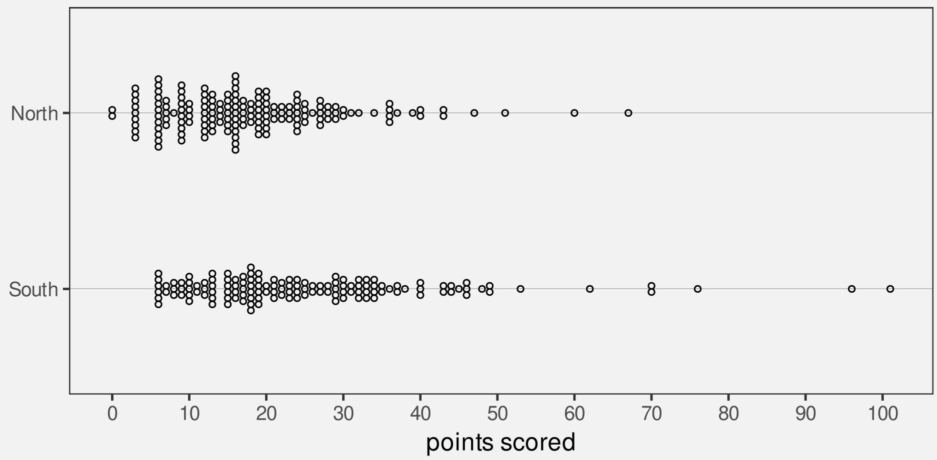


Figure 8.3: Stacked dotplots of the number of points scored by Tier One nations at Rugby World Cup matches.

The fact that we can decode raw data values from a dot plot is not news. That is easy because we have encoded raw data values as accurate visual features (see Section 3.7). However, Figure 8.3 also demonstrates that, in addition to decoding raw data values from visual features, our visual system allows us to decode **data summaries** from visual features.

For example, we can see that the southern hemisphere teams score slightly more points on average than the northern hemisphere teams; the data symbols for southern-hemisphere teams appear shifted to the right of the data symbols for northern-hemisphere teams. In other words, we can perceive **visual summaries**; we can perceive **data summaries** from collections of visual features. For example, we can visually average the **positions** of the data points.[101]

In Figure 8.3, we have encoded raw data values as visual features, which has enabled us to decode raw data values from individual visual features. In addition, **visual summaries** of the visual features enable us to decode **data summaries** from collections of visual features (Figure 8.4).

> A dot plot is effective not just for perceiving raw data values, but also for perceving data summaries like clusters and outliers, and central tendency and spread.

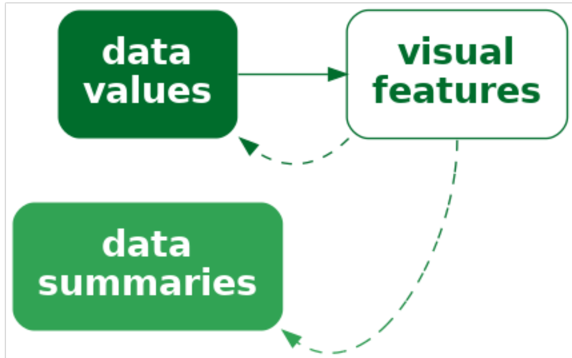


Figure 8.4: A data visualisation that encodes data values as visual features allows us to decode data values, but it may *also* allow us to decode summaries of the data, such as measures of central tendency or variability.

8.3 Case study: Histograms

Figure 8.5 shows a histogram of the number of points scored by Tier One nations at the Rugby World Cups (Table 4.1). A histogram is similar to a box plot in two ways: a histogram is a data visualisation that can be used to convey the distribution of a variable; and a histogram is a data visualisation that maps **data summaries**, rather than raw values, to visual features (Figure 8.2).

However, a histogram differs from a box plot in two main ways: the data symbol for a histogram is simpler than a box plot, because it consists of just rectangles or bars; and the data summaries for a histogram are more complicated than those for a box plot.

The raw data values for the histogram in Figure 8.5 are the points scored in each game by each country (Table 4.1). To get the data summaries for the histogram, the raw data values are binned; we break the range of the data into equal-sized intervals and count how many data values lie in each interval. The centre of each interval is then encoded as the horizontal **position** of a bar and the count of data values in each interval is encoded as the **length** of a bar.

To be explicit, the values that are being mapped to visual features in Figure 8.5 are not raw data values, but data summaries—intervals and counts—as shown in Table 8.2.[102]

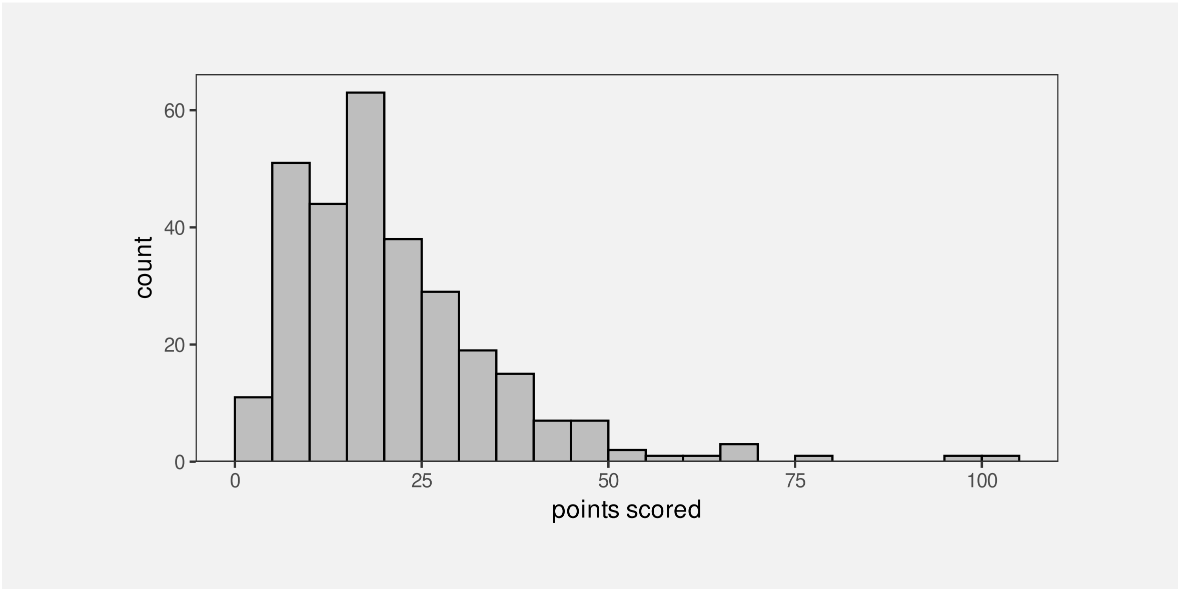


Figure 8.5: A histogram of the points scored per game by Tier One nations at Rugby World Cups.

Table 8.2: A table of intervals that cover the range of the number of points scored by Tier One nations in Rugby World Cup matches, along with the count of the number of points scored that fall within each interval. These data summaries are the values that are mapped to visual features to produce the histogram in Figure 8.1. Not all intervals are shown for reasons of space.

interval	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60	60-65
count	11	51	44	63	38	29	19	15	7	7	2	1	1

As with a box plot, encoding data summaries as accurate visual features, in this case **position** and **length**, means that we can easily decode data summaries from the visual features. For a histogram, this means that we can easily compare the lengths of the bars, which means we can easily compare the counts of data values within each interval. For example, the longest bars represent the intervals that contain a relatively large count of data values, which tells us about the mode of the data (or the number of modes in the data). In Figure 8.5 we can see that the largest number of data values occur around 15 to 20 points and we can see that the number of data values in each interval drops more rapidly above the highest bar than below the highest bar. The latter tells us about the spread and/or skewness of the data.

> A histogram is effective at conveying information about the distribution of data values because it maps data summaries to basic visual features and those data summaries can be compared to provide information about the distribution of the data.

8.4 Decoding summaries

When we encode a data summary as a visual feature, like the position of the median line of a boxplot (Figure 8.1), we know from previous chapters how effective the decoding is going to be. For example, because the median is encoded as position, we should be able to decode the median very accurately (Section 3.5).

However, if we rely on a visual summary to decode a data summary, for example, comparing the average points scored by teams from different hemispheres based on a collection of data points for each hemisphere (Figure 8.3), we are no longer dealing with just the decoding of a single visual feature. How well can we decode the average from the positions of several data symbols compared to decoding a single data value from a single data symbol? How effective is the decoding from a visual summary to a data summary (Figure 8.4)?

One issue that arises is that there are many possible visual summaries. For example, we might be interested in the average data value, or the variability of data values, whether there are unusual data values (outliers), or whether there are groups of data values (clusters).[103]

Furthermore, we can produce each sort of visual summary for each possible visual feature. For example, we can decode the average position of data symbols, or the average luminance of data symbols, or the average area of data symbols. We can decode unusual positions, unusual colours, or unusual angles.

Although it is well-established that our visual system is capable of generating visual summaries, less is known about the accuracy of those summaries, partly because the range of possible visual summaries is so large. However, we know that estimating averages from the positions of multiple data symbols is quite accurate and estimating averages from the luminance and/or chroma of multiple data symbols can also be effective.[104]

8.5 Dangers of summaries

The benefit of a data visualisation that encodes **data summaries** as visual features is that it allows us to decode data summaries very easily. For example, although we can perceive a general idea of central tendency from the dots in Figure 8.3 (see Section 8.2), it is much easier and more accurate to perceive median values from the thick vertical lines in Figure 8.1.

The flipside is that a data visualisation that encodes **data summaries** as visual features is *not* effective for decoding raw data values. For example, we cannot see from the box plot in Figure 8.1 that the scores 1, 2, 4, and 5 never occurred, while this is clear from Figure 8.3 because the dot plot does map the raw data values to data symbols.

> Box plots and histograms are *not* effective at conveying raw data values because they do not map raw data values to data symbols.

Another point to consider with a data visualisation that is based on data summaries is whether the data summaries provide reasonable summaries of the data.[105] For example, Figure 8.6 shows box plots of the number of points *conceded* by each team in Rugby World Cup games. The box plot for southern hemisphere teams suggests a unimodal distribution with a slight right skew.

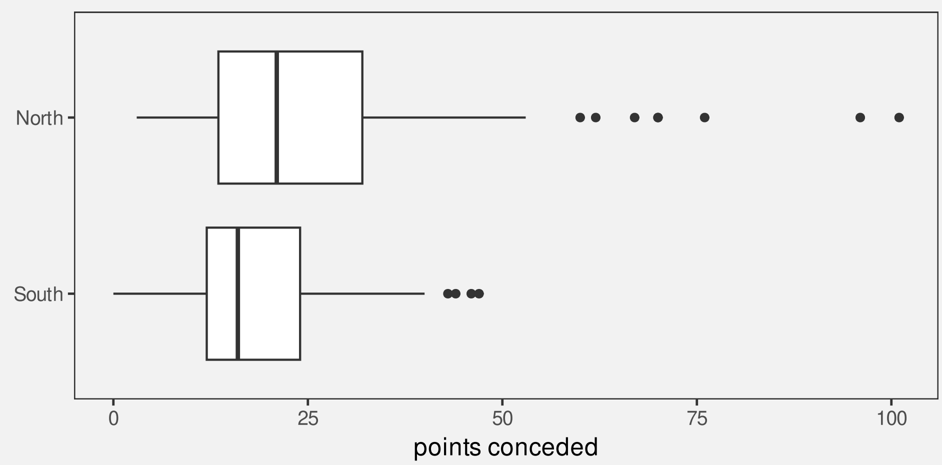


Figure 8.6: Box plots of the number of points conceded in all Rugby World Cup games by Tier One nations.

However, the dot plots in Figure 8.7 tell a different story. This data visualisation shows some evidence that the number of points conceded by southern-hemisphere teams may be bimodal, with some teams only conceding very few points (0, 3, or 6) in a number of games. The data summaries for a box plot assume a unimodal distribution, so a box plot will only ever tell a unimodal story, and that is only reasonable if the data are unimodal.

> A data visualisation that is based on data summaries can only be effective if the data summaries are sensible summaries of the data.

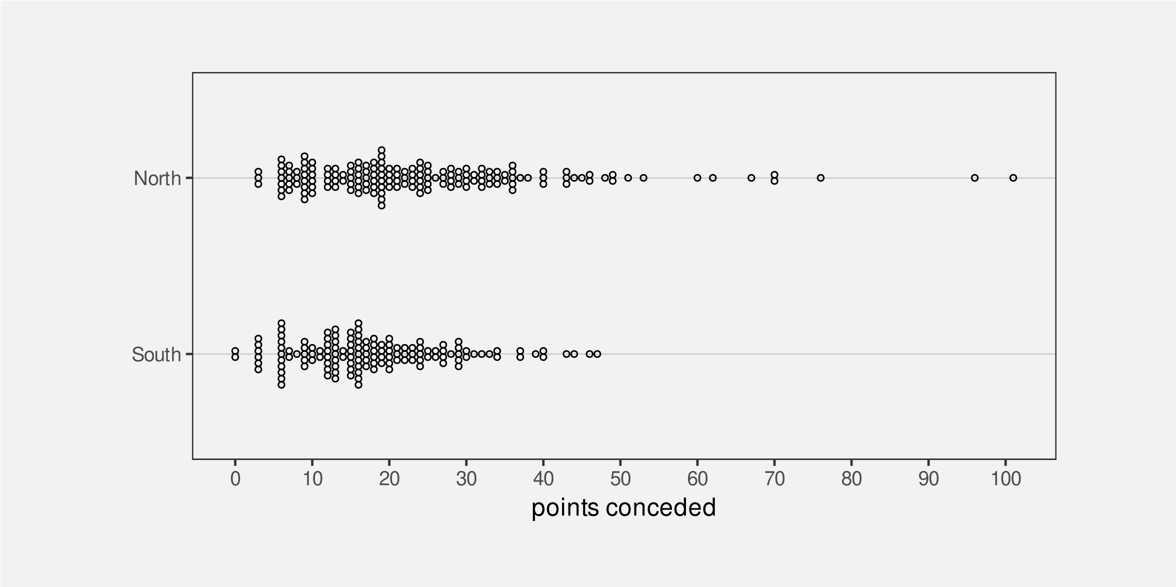


Figure 8.7: Stacked dotplots of the number of points conceded by Tier One nations at Rugby World Cup matches.

> A box plot is *not* effective if a five-number summary is not a sensible summary of the data values.

A further complication arises with histograms because we have to *choose* the intervals that are used to calculate the data summaries and that choice can have a large influence on the data summaries. For example, Figure 8.8 shows a histogram of the same data that we used for Figure 8.5, but with a greater number of narrower intervals. This visualisation tells a slightly different story than Figure 8.5. The most common score is now 6 points (rather than between 15 and 20 points) and we see features like gaps at 1, 2, 4, and 5 points and a stronger suggestion of bimodality. There are also point totals that appear strangely low, like 11 and 14.

Unlike a box plot, where the data summaries are essentially fixed calculations,[106] for a histogram we get to choose the data summaries to some extent, which means that we get to choose the story that the data visualisation tells. Figure 8.5 tells a simpler story, but Figure 8.8 tells a story that includes some important details.

When using a histogram for exploring a data set, it is a good idea to try out more than one choice of intervals in order to avoid missing an important story. When using a histogram to present a story about a data set, there is a responsibility to present a story that does not exclude important details.

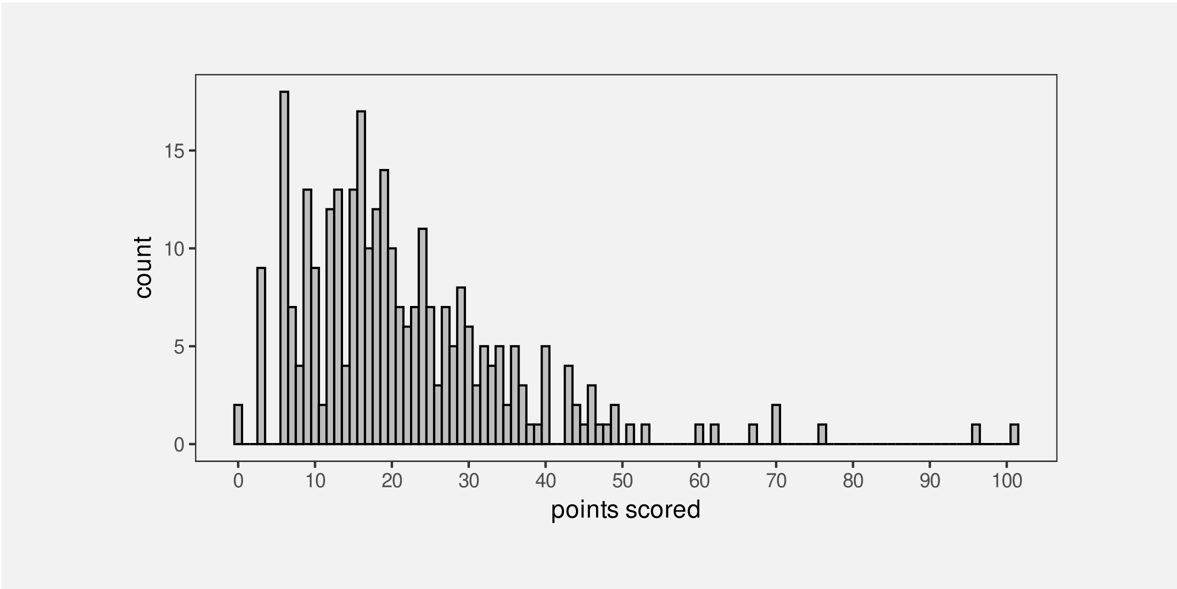


Figure 8.8: A histogram of the points scored per game by Tier One nations at Rugby World Cups. This histogram uses the same data as Figure 8.5, but bins the data using narrower intervals.

8.6 Case study: How charts lie

In this book, we assume that the purpose of a data visualisation is to faithfully and honestly encode data values and the goal is to produce a data visualisation that provides the most effective decoding (Section 1.3). Nevertheless, even less honest representations of data can also be thought of in terms of mappings, particularly mappings of data summaries.

For example, Figure 8.9 shows a bar plot of the predicted impact on the top tax rate in the United States from President George W. Bush to President Barack Obama.[107] One problem with this data visualisation is that it does not encode raw data values—the actual predicted top tax rates are 35 and 39.6. Instead, a "data summary" of the tax rate *minus 34* (the values 1 and 5.6) are encoded as the lengths of the two bars. We are unable to decode the raw data from this bar plot—we can only decode the data values that have been encoded—and this leads to an extreme exaggeration of the difference between the two top tax rates.

A data visualisation that encodes a data summary that is a distortion of the data or an inappropriate subset of the data is encoding unhelpful information to visual features. It is only then possible to decode unhelpful information from the data visualisation (Figure 8.10).

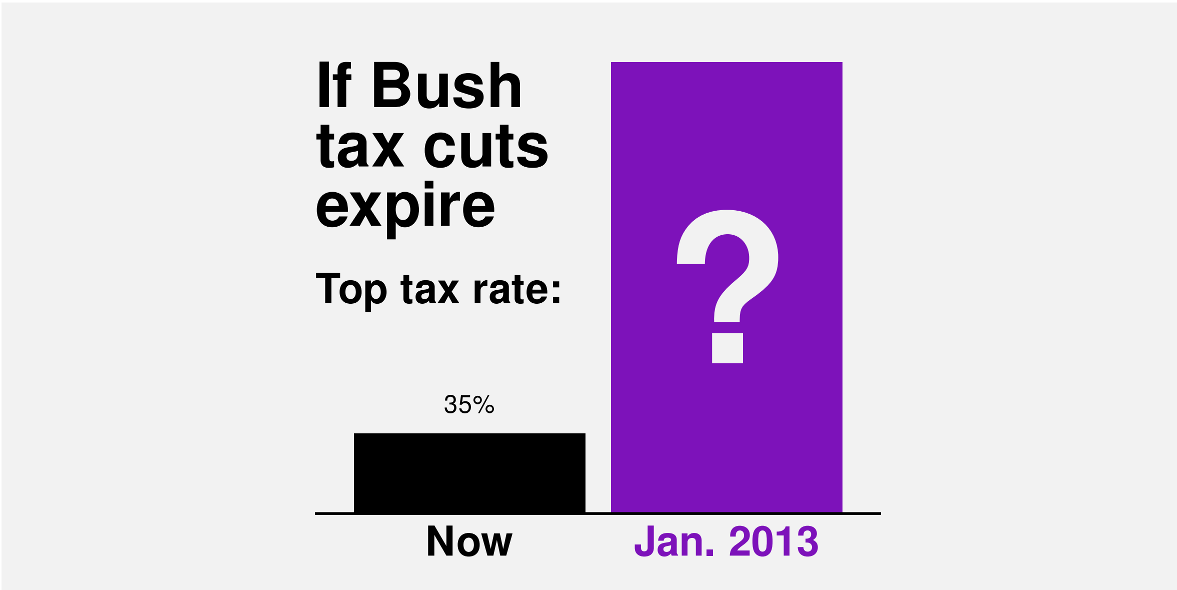


Figure 8.9: A bar plot of the increase in the top tax rate in the United States of America from President George W. Bush to President Barack Obama.

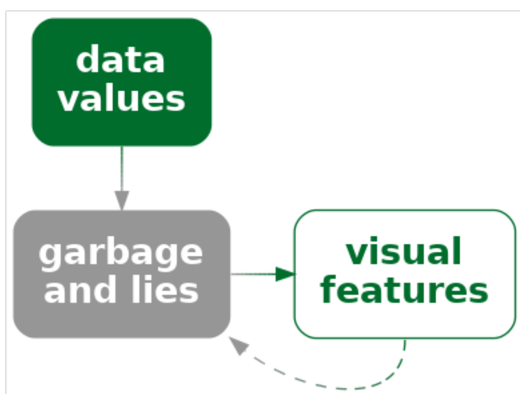


Figure 8.10: A data visualisation may map an unrepresentative or “cooked” version of the data, rather than the raw data, to visual features. Visual decoding leads back to the lies rather than the real data.

8.7 Visual downgrades

We have seen that a downside to mapping data summaries, rather than raw data values, is that we lose the ability to decode raw data values from data symbols. However, we pay this price in order to improve our ability to decode data summaries from the data symbols. For example, a boxplot makes it easier to decode the median of a set of data values.

By mapping data summaries, we deliberately lose the ability to decode raw data values in order to gain the ability to decode data summaries.

Another way that we may deliberately lose information is by selecting a visual feature that does not allow complete decoding of the original data values.

For example, Figure 8.11 shows the crime rate for 2021 in each police district of New Zealand. The data values for this map are shown in Table 8.3.

Table 8.3: The average crime rate over the period 2011 to 2021 for each police district of New Zealand.

district	rate
Auckland City	0.32
Bay of Plenty	0.62
Canterbury	0.38
Central	0.56
Counties/Manukau	0.39
Eastern	0.66
Northland	0.61
Southern	0.49
Tasman	0.62
Waikato	0.46
Waitematā	0.28
Wellington	0.33

The data symbols in Figure 8.11 are circles with the geographic location of the (centre of the) police district mapped to the position of the circles and the crime rate mapped to the (fill) colour, particularly the luminance, of the circles. Darker circles represent higher crime rates.

We know that encoding the crime rate as luminance is not an appropriate encoding for quantitative data values (Section 3.3). The best we can decode is ordinal data values from the luminance of the circles (Section 3.4).

However, we may accept this loss of information in order to gain the ability to decode the geographic position of the police districts. For example, although we cannot decode exact crime rates, we are able to see that the higher crime rates correspond to more rural areas; the

Figure 8.11: A map of New Zealand with .

lighter circles appear in the urban regions that include New Zealand's largest cities: Auckland, Wellington, and Christchurch.

In Figure 8.11, we can only decode to **data summaries** of the crime rate because we an only decode the **ranking** of the crime rates, not individual numeric crime rates and not numeric differences between crime rates (Figure 8.12).

On the other hand, this loss of information is acceptable if the decoding still allows us to make useful comparisons. Put more strongly, the numeric comparisons that we lose access to may not be the most important decodings, so the loss may not be that important.[108]

For comparison, Figure 8.13 shows a simple bar plot of the same crime rates. In this data visualisation, although we can decode and compare the numeric crime rates much more accurately, the rural/urban comparison is much more difficult.

8.8 Case study: Heatmaps

Heatmaps are a common way of representing three-dimensional data where two of the data variables are either qualitative or have been measured on a regular grid and the third variable is **quantitative**. The first two variables naturally map to horizontal and vertical **position** to produce an array of rectangles, while the third variable maps to **colour**.

We saw an example of a heatmap in Figure 1.2 and Figure 8.14 shows another example that is a variation on Figure 6.22 and Figure 6.24.

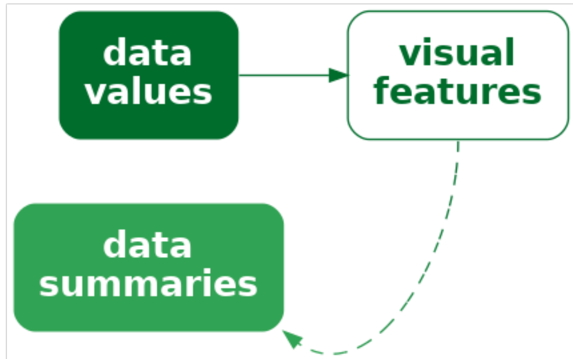


Figure 8.12: A data visualisation may encode data values as a visual feature that does not allow decoding back to the original data values. Instead, it is only possible to decode summaries of the data.

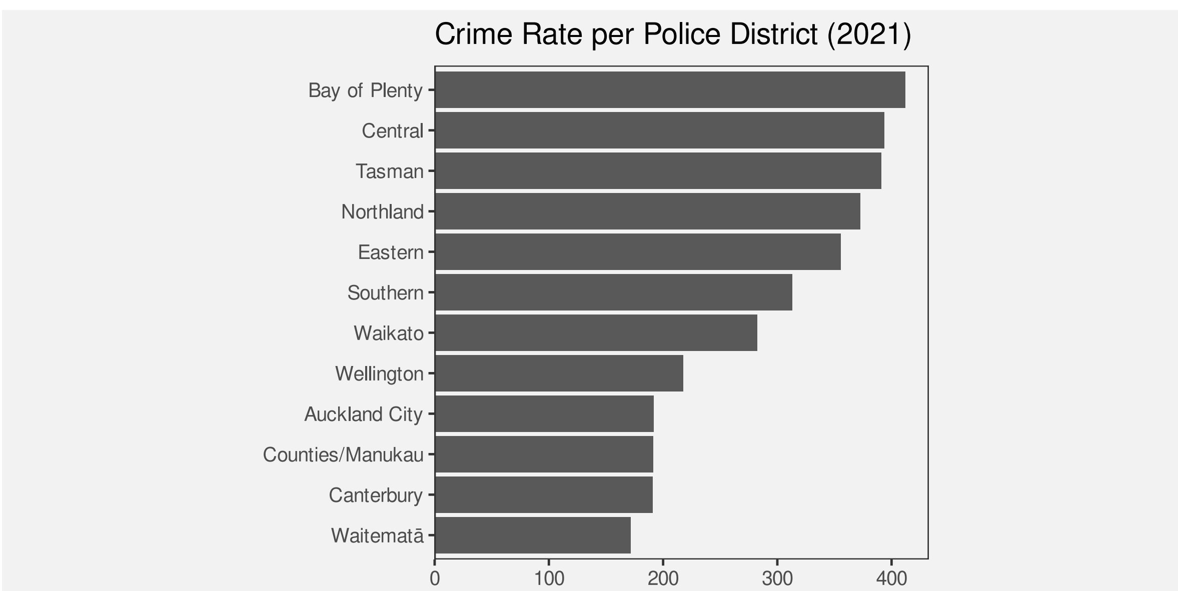


Figure 8.13: A bar plot of the crime rate for 2021 in each police district of New Zealand.

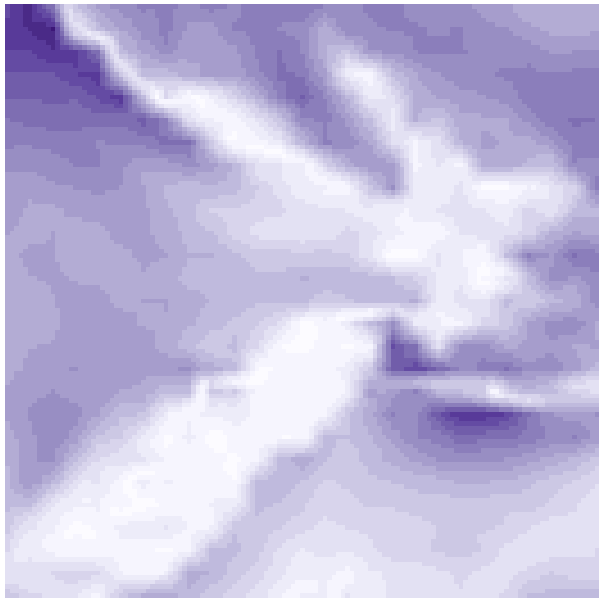


Figure 8.14: A snapshot of wind strength in the vicinity of New Zealand on September 14 2024. This uses the same data as Figure 2.5, but does not show wind direction.

The use of colour to encode quantitative data values is not optimal for accuracy (Section 3.3). However, as we saw in Section 8.7, as long as the colour scheme has monotonic luminance and/or chroma changes, it is still possible to decode ordinal values.

Furthermore, we are able to decode useful visual summaries from the multiple rectangle data symbols on a heat map (Section 8.2). For example, we can identify regions of darker or lighter colours, not just individual dark or light rectangles.[109]

On the downside, because each rectangle in a heatmap is surrounded by other rectangles of differing colours, it is difficult to accurately decode the colours of individual rectangles (Section 6.3).

In the case of a heatmap, ordinal comparisons are typically what is of most importance. The aim is *not* to be able to decode individual quantitative data values and this moderates the damage that is done by relative perception of the heatmap rectangles.

8.9 Cognitive load

Consider the following sentence about changes in global malnutrition rates:[110]

"rising food production reduced the malnutrition rate from 2 in 3 people in 1950 to 1 in 11 by 2019"

In order to comprehend the change in malnutrition rate, we are required to compare "2 in 3" to "1 in 11". This requires mental effort to calculate the ratios or fractions and then compare them. It is easier to compare the rates if we hold the denominator constant, comparing "2 in 3" to "2 in 22". It is easier still if we calculate the fractions, comparing 0.667 to 0.091. And it is easier still to compare those fractions visually (Figure 8.15).

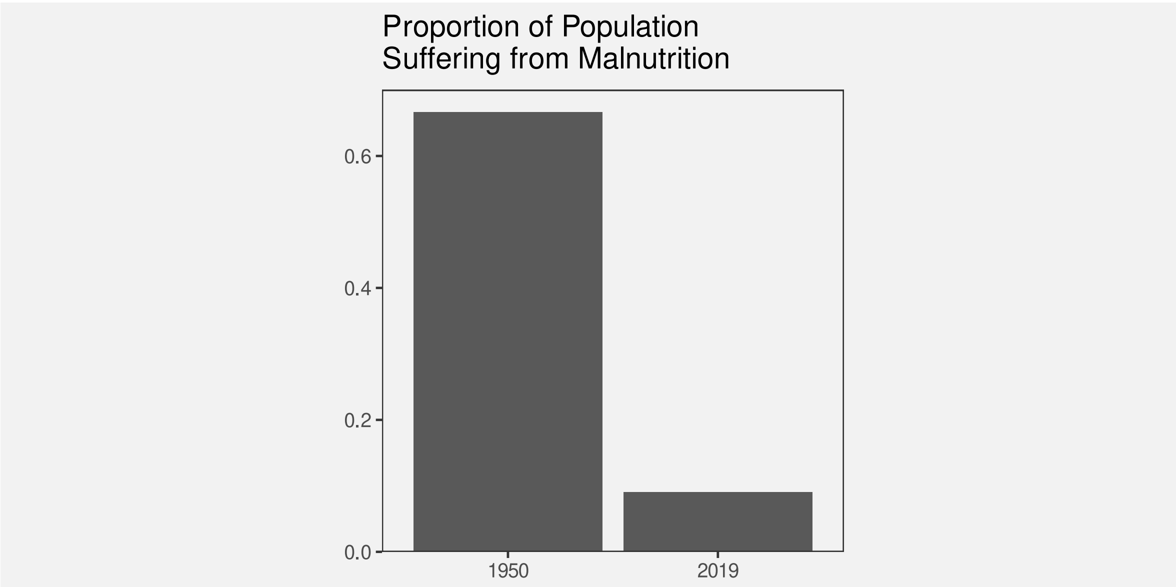


Figure 8.15: A bar plot comparing two fractions.

The point is that we can reduce the cognitive load that is required of the viewer if we make the mental calculations that are required easier. We can reduce the cognitive load further if we present the results of the calculation directly and even further if we encode the results of the calculation as simple visual features.

In other words, we can calculate **data summaries** or we can force the viewer to perform **visual summaries**.

Conversely, we can add to the cognitive load if we make the visual calculations harder. For example, Figure 8.16 shows a bar plot of the original "2 in 3" versus "1 in 11" data values. We now need to perceive the proportion of blue bars relative to red bars and compare those proportions, which is clearly more work than just comparing the two bars in Figure 8.15.

We ideally want to present data in a way that imposes the least cognitive load on the viewer. In terms of Section 2.9, we want to present the viewer with simple **visual tasks**.[111]

The best data visualisation will be different for different questions of interest, but if the question of interest involves comparing data summaries, then one way that we can make the viewer's life easier is by calculating summaries of the data and encoding data summaries as visual features, rather than encoding raw data values and asking the viewer to perform visual summaries.

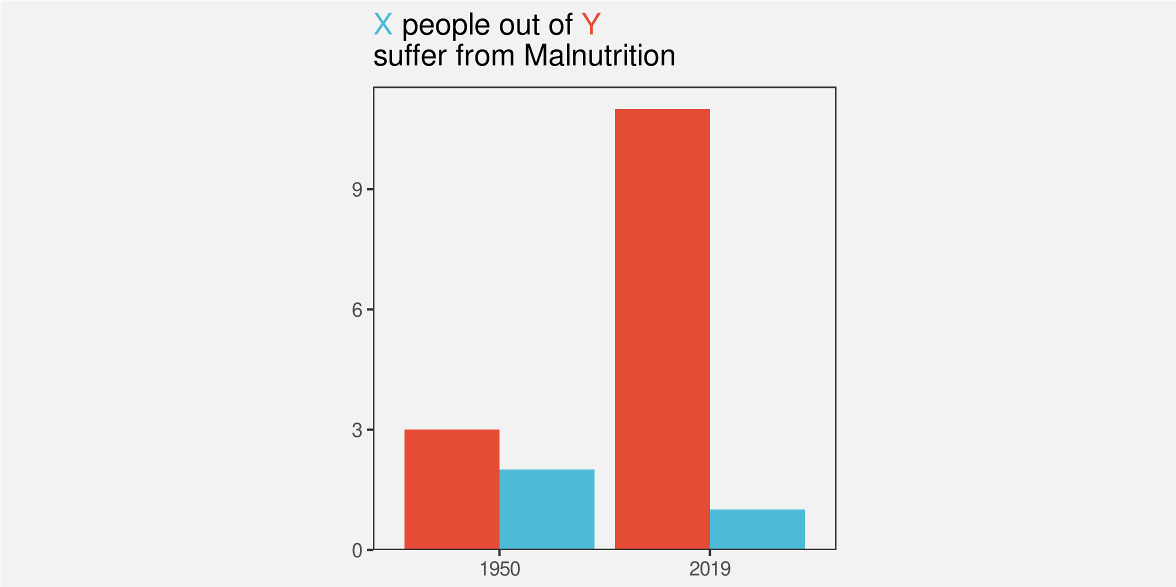


Figure 8.16: A bar plot comparing two fractions.

> It is possible in some cases to rely on visual summaries, but we need to be careful not to set a visual task that requires too much mental effort.

8.10 Summary

Data summaries, such as measures of central tendency, measures of variability, and tables of counts, provide information about the distribution of a set of data values.

Some data visualisations, like box plots and histograms, are effective because they map **data summaries** to visual features. This makes it easy to decode and compare data summaries.

Care must be taken to use data summaries that appropriately summarise the data values.

When we map raw data values to visual features, like in a dot plot, the encoding to individual visual features allows us to decode and compare raw data values, but, in addition, **visual summaries** of *collections* of visual features allow us to decode and compare data summaries.

A box plot that maps data summaries to visual features is more effective for perceiving data summaries than a dot plot that relies on visual summaries. However, a dot plot is more effective for perceiving raw data values.

Notes

[97] In this book, we use *data summary* to cover a broad range of transformations of the data. These are transformations that occur prior to the mapping of data values to visual features.

There are several more sophisticated break downs of the pipeline from data values to visual representations. For example, the Information Visualization Reference Model (Card, Mackinlay, and Shneiderman 1999) includes the transformation of data into a standard format (*data transformations*) prior to mapping data values to visual features and also includes the subsequent transformation and scaling of visual features to positions and shapes on the screen or page (*view transformations*).

Wu and Chang (2024) differentiates between *data transformations* and *design-specific transformations*, such as the calculation of summaries for a box plot.

The idea of a *data summary* corresponds well with the idea of a *statistic* in the *Grammar of Graphics* (Wilkinson 2005), although in this book, because we only consider static data visualisations, the connection with the raw data is lost. The *Grammar of Graphics* also includes transformations in the form of *scales* and *coordinates*, which roughly correspond to the *view transformations* of the Information Visualization Reference Model.

[98] John W. Tukey is credited as the inventor of the box plot (Tukey 1977).

[99] Ziemkiewicz and Kosara (2009) refer to this as the *readability* of *aggregations*.

[100] The general idea of arranging overlapping dots so that they are all visible, not just jittering them (Section 7.7), goes back to Wilkinson (1999) and Tukey (1977). The vertically-centred arrangement used here is also known as a *beeswarm plot* (Rosanbalm 2014).

[101] The ability to decode summary statistics from multiple visual objects is known as *ensemble perception*. Cui and Liu (2021) and Whitney and Yamanashi Leib (2018) provide modern overviews.

A related concept is *subitizing*, which is the ability to rapidly identify the number of visual objects, though that only works for very small numbers of objects (Ware 2021).

[102] A bar plot can also be thought of in terms of mapping data summaries rather than raw data. The data values that are encoded as the lengths of bars in a bar plot are tables of counts of raw data values; the raw data values are individual observations of a single qualitative variable, such as the ethnicity of a single youth offender.

See this way, there are raw data values that cannot be decoded from a bar plot. For example, information about an individual offender cannot be decoded from a bar plot because all similar individuals are represented together by a single bar data symbol.

[103] In the visual perception literature, the strict definition of *ensemble* encoding is limited to *statistical moments* (means and variances) (Whitney and Yamanashi Leib 2018), but the data visualisation literature takes a more relaxed approach (Szafir et al. 2016).

[104] Gleicher et al. (2013) show that the average position of data points in a scatterplot can be accurately determined and that this is robust to a number of variations in context, such as the number of data points and the presence of distractors.

Albers, Correll, and Gleicher (2014) show that the average luminance of rectangles in a colour field can be determined reasonably well. This study is also a good demonstration of both the multitude of visual summaries that can be studied and the variability of performance across different combinations of visual summaries and visual features.

[105] The classic demonstration of the idea that numerical summaries may not satisfactorily summarise the raw data is Anscombe's quartet (Anscombe 1973). Matejka and Fitzmaurice (2017) provide a more modern update, including Alberto Cairo's *datasaurus* (Cairo 2016a).

[106] It is not true that there is only one way to generate the data summaries for a box plot. McGill, Tukey, and Larsen (1978) describe several early variants and *letter-value plots* are an example of a more modern adaptation (Hofmann, Wickham, and Kafadar 2017).

[107] This is a reproduction of a chart from Alberto Cairo's "How Charts Lie" (p12-13), which itself was a reproduction of a chart shown by Fox News.

[108] Bertini, Correll, and Franconeri (2020) argue that ordinal comparisons are more commonly useful in practice than quantitative comparisons.

"1. Graphics are for the qualitative/descriptive—conceivably the semiquantitative—never for the carefully quantitative."

"2. Graphics are for comparison-comparison of one kind or another - not for access to individual amounts."

Tukey (1993)

[109] Correll et al. (2012) provide evidence for superior ensemble perception of "high" regions within a time series from "colour fields" (heatmaps) compared to traditional time series line plots.

[110] (Smil 2022, 46)

[111] The 10 data visualisation tasks in Amar, Eagan, and Stasko (2005) all correspond to some sort of numerical summary of the data. We can choose to perform the numerical summary and then visualise the result, or we can choose to present the raw data and ask the viewer to perform a visual summary. We should choose whichever will be easier for the viewer.

9 Combining Mappings

Most data visualisations involve more than one data variable. For example, the bar plot in Figure 9.1 involves data on the **gender** of offenders and data on the **count** of offenders for each gender.

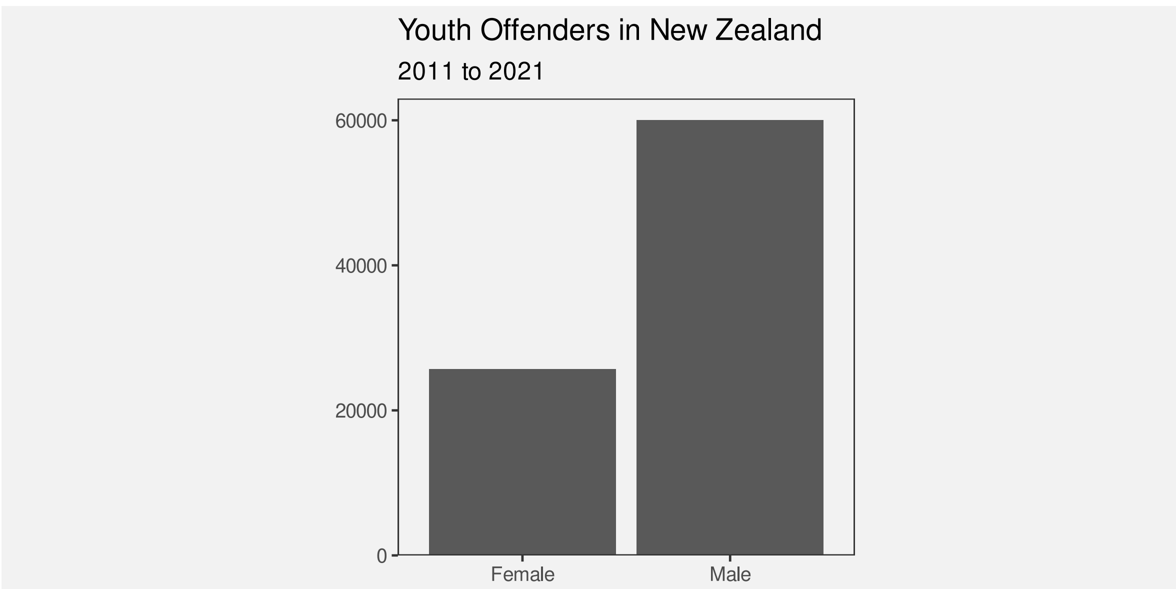


Figure 9.1: A bar plot of the total number of male offenders and the total number of female offenders in New Zealand from 2011 to 2021.

In Figure 9.1, each data value is mapped to a **visual feature** of a data symbol (Figure 3.1). For example, the gender is encoded as the horizontal **position** of the bars and the count of offenders is encoded as the **length** of the bars.

We have so far only looked at mappings from data values to visual features in isolation. For example, we know that the mapping from gender to position is effective because position is appropriate for qualitative data and position has sufficient capacity to differentiate between two genders. We also know that mapping the count of offenders to length is effective because length is appropriate for quantitative data and length provides an accurate decoding.

In this chapter, we begin to consider what happens when mappings are used in combination. For a start, we will just extend our thinking to data visualisations that involve mapping **two** data variables to **two** visual features, like gender and count to position and length. Furthermore, we will restrict ourselves to mappings where each pair of data values are encoded as a single data symbol (Figure 9.2). For example, in Figure 9.1, each combination of gender and count produces a single bar.

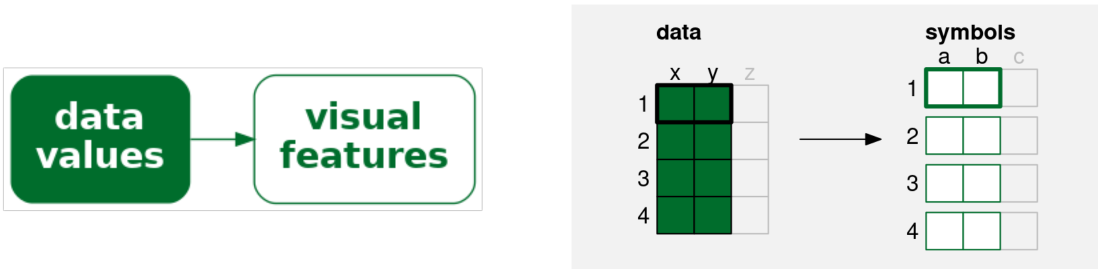


Figure 9.2: A simple data visualisation maps **two** data values to **two** visual features to create a data symbol. Each pair of data values (`x` and `y`) is mapped to two visual features (`a` and `b`) of a **single** data symbol.

9.1 Independent visual features

A bar plot involves encoding a qualitative variable as bar **position** and a quantitative variable as bar **length** (e.g., Figure 3.2). We know that each visual feature is effective on its own—we can decode from positions to qualitative values and we can decode from lengths to quantitative values—but another reason why the bar plot is effective overall is because the two visual features act independently.[112] Our perception of the horizontal **length** of the bars is not affected by the vertical **position** of the bars.[113]

Figure 9.3 demonstrates that we can also add **colour** to the mix without creating any interactions between visual features. The fact that the bottom two bars are blue and red does not affect our ability to perceive that they have different lengths.

> One reason why a bar plot is effective is because data values are encoded as visual features that can be decoded independently.

9.2 Scatter plots

Table 9.1 shows data gathered on the performance of all twenty teams at the 2023 Rugby World Cup. Most measures are positive, for example, more **points** is better, but there are also a couple of negative measures: more **tackles** may just mean that the team was weak and

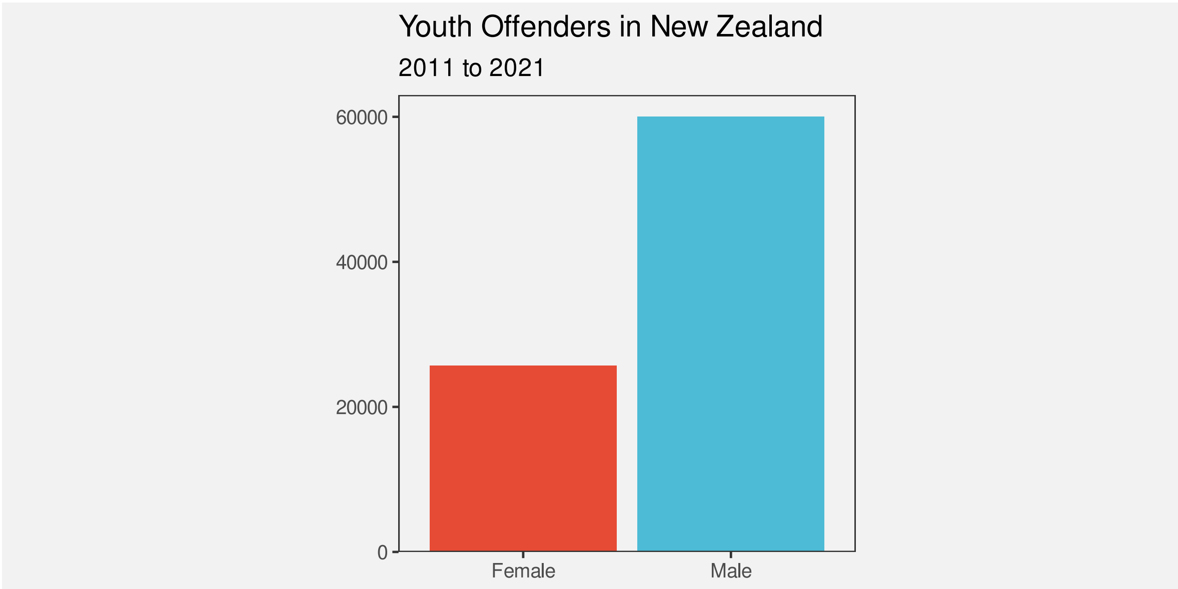


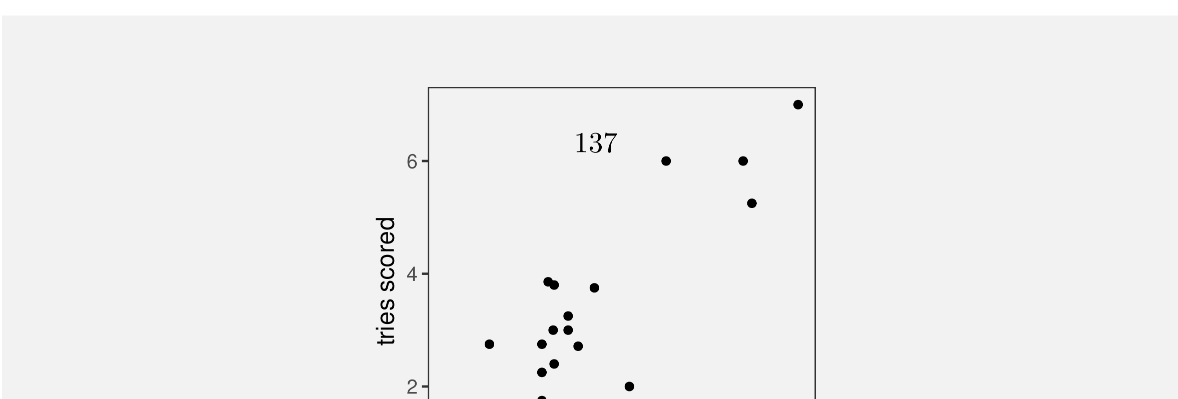
Figure 9.3: A bar plot of the total number of male offenders and the total number of female offenders. The data symbols in this plot come from mapping gender to the vertical positions and colours of the bars and mapping the total number of offenders to the lengths of the bars.

was always being attacked; and more disciplinary **cards** (either **y**ellow or **r**ed) is definitely a bad sign.

Table 9.1: Performance measures for teams at the 2023 Rugby World Cup. Each measure is a per-game average because some teams played more games than others. There are 20 teams in total, with only the first 6 shown here.

country	sphere	ycards	rcards	breaks	tackles	points	converts	offloads	tries	runs
Namibia	South	1.0	0.5	2.5	102.0	9.2	0.5	2.2	0.8	92.0
Romania	North	1.2	0.0	2.8	142.5	8.0	0.8	1.5	1.0	81.0
Chile	South	1.2	0.0	3.8	132.2	6.8	0.5	5.2	1.0	102.2
Samoa	South	1.2	0.2	3.8	109.5	23.0	2.0	9.2	2.8	102.5
Australia	South	0.5	0.0	5.2	108.2	22.5	1.8	7.8	2.8	110.8
Georgia	North	0.5	0.0	5.2	149.8	16.0	1.0	8.0	1.8	114.0

Figure 9.4 shows a scatter plot of the number of "clean breaks" per game versus the number of "tries scored" per game for each team. A clean break means that a team has broken through the opposition defence and a "try scored" means that the team has earned 5 points. A clean break implies an opportunity to score a try, but does not guarantee that a try will be scored.



in the scatter plot in Figure 9.4: the **quantitative** number of clean breaks is encoded as the horizontal **position** of the data points and the **quantitative** number of tries scored is encoded as the vertical **position** of the data points.

We know from Chapter 3 that we should be able to accurately decode the original data values from the positions of the data points. For example, we can easily see that one team scored a little over 7.5 clean breaks per game and we can see that two teams scored about 6 tries per game.

However, there is something else going on in the scatter plot. The two explicit encodings—horizontal and vertical position—**interact** to create an additional **emergent** visual feature: **position in space** (Figure 9.5).[114] For example, we can perceive not only horizontal and vertical distances between data points, but also euclidean distances between data points ("as the crow flies").

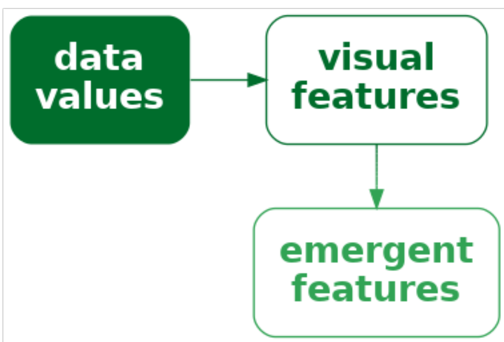


Figure 9.5: If we encode data values as visual features and the visual features interact, the result can be additional visual features. It is possible for the additional visual feature to be dominant relative to the original visual features or even to obscure the original visual features.

> One reason why a scatter plot is effective at conveying relationships between variables is because the mappings of data values to visual features—horizontal position and vertical position—interact to produce an additional visual feature: position in space.

Furthermore, our visual system allows us to decode **visual summaries** (see Section 8.2) from **position in space** (see Figure 9.6).[115] For example, in Figure 9.4 we can easily see a positive **correlation** between the number of clean breaks and the number of tries scored. The points lie roughly along a line from bottom-left to top-right.

We can also easily see **clusters** of data points; the four data points in the top-right corner of the plot appear separate from the other data points because they share a similar position in space (Section 2.7).

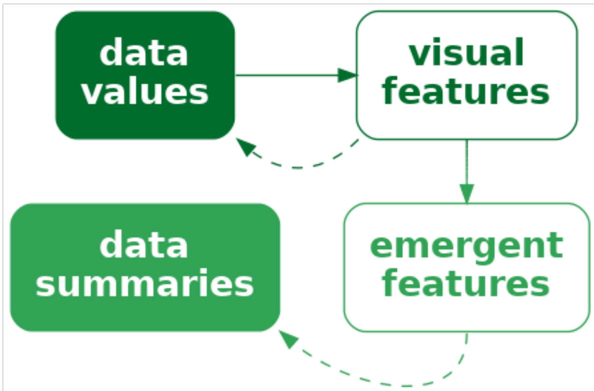


Figure 9.6: If we encode data values as visual features and the visual features interact, the result can be emergent visual features. The original visual features allow us to decode raw data values and the emergent features allow us to decode data summaries.

> One reason why a scatter plot is effective at conveying relationships between variables is because we are able to decode visual summaries from a collection of positions in space.

Although a scatter plot is similar to a bar plot because they both involve two encodings—position and length for a bar plot and horizontal position and vertical position for a scatter plot—there are additional decodings possible from the data symbols in a scatterplot because the additional visual feature of position in space emerges from the combination of horizontal and vertical position.

9.3 Visual features that interact

We have seen that some combinations of visual features, such as **position** and **length** (Figure 9.1) and **position** and **hue** (Figure 9.3) do not interact, while other combinations of features, such as horizontal and vertical **position** (Figure 9.4) do interact. What about other combinations of visual features? Figure 9.7 shows all pairwise combinations of visual features that we have considered (Figure 3.5).[116]

The top row of Figure 9.7 shows that **position** can be combined independently with every other visual feature, except itself (Section 9.2), even when we vary both visual features at once.

For other combinations, we have an array of data symbols, with one visual feature changing down rows and another visual feature changing across columns. If the visual features are independent, we should be able to decode one visual feature from the rows and one visual feature from the columns. For example, for the combination of **hue** and **pattern**, we can clearly see columns of the same hue and rows of the same pattern.

If the visual features interact, we should not be able decode separate visual features. For example, for the combination of **chroma** and **luminance**, all we can see is an array of different shades; there are no clear rows of different chroma or columns of different luminance.

There are also combinations that are not fully independent and have weak interactions between visual features. For example, the decoding of hue, chroma, and luminance is somewhat affected by the area of the data symbol. Stone (2012) provides a clearer demonstration of this effect. Another weak interaction is between hue and luminance; this still allows for diverging colour palettes (Section 6.11).

One clear message from Figure 9.7 is that all visual features interact with themselves. If we try to encode two separate data variables as the same visual feature, it will be difficult to decode the individual data variables. We can also see that combinations of length, angle, area, and pattern produce interations, as do combinations of area, hue, chroma, and luminance. However, combinations of one of hue, chroma, or luminance with any of position, length, angle, area, or pattern are independent.

9.4 Case study: Mosaic plots

Table 9.2 shows a table of counts for offences committed in New Zealand in 2021. For each offence, we know the sex of the offender and what action was taken against the offender.

Table 9.2: Counts of offences committed in New Zealand in 2021 by the sex of the offender and the action taken against the offender.

	Female	Male
Court Action	13950	39526
Non-Court Action	8717	19663
Not Proceeded With	381	933

Figure 9.8 shows a mosaic plot of the data in Table 9.2.[117] This is an effective data visualisation for making several comparisons: the overall proportion of male versus female offenders; within

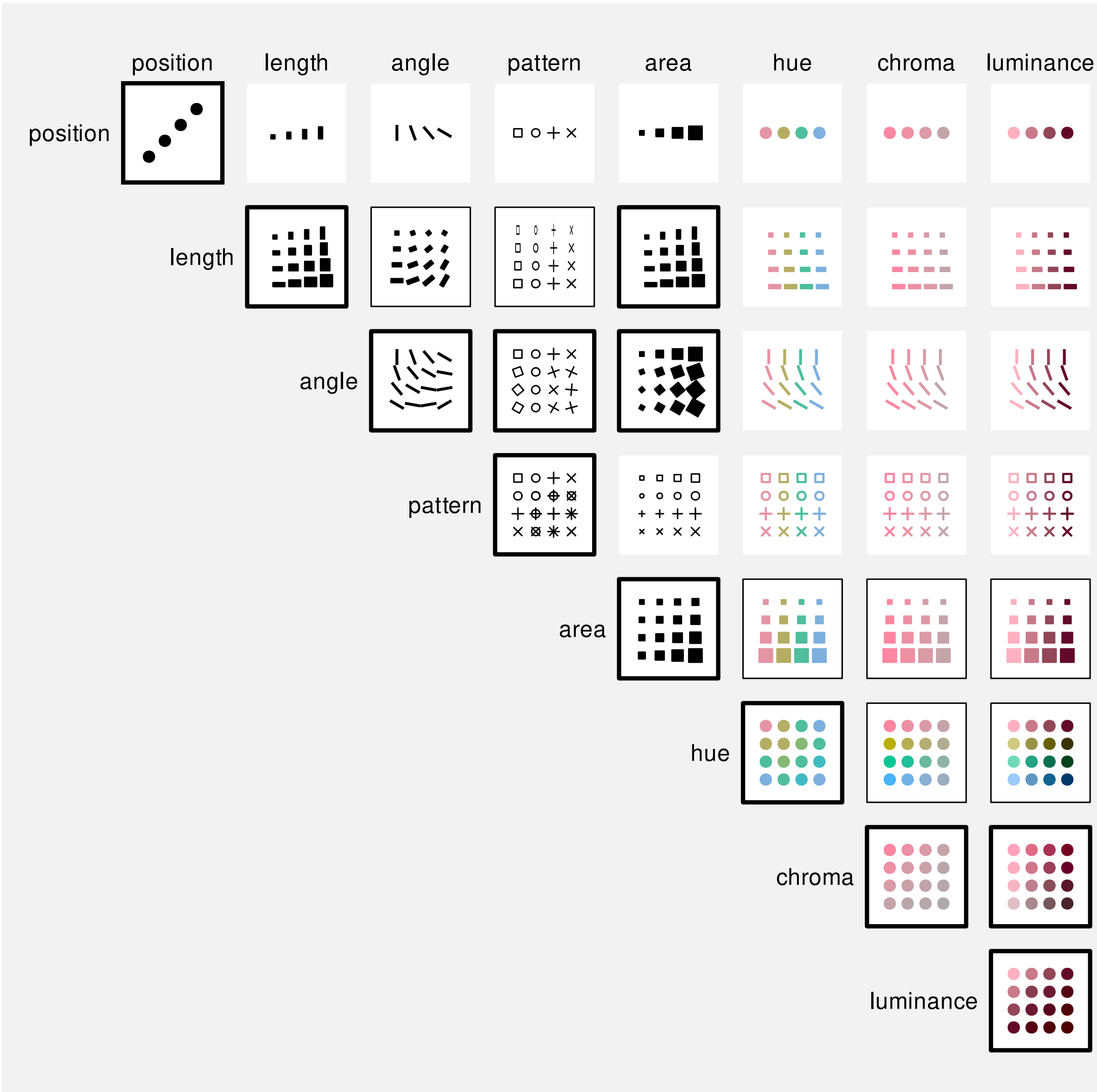


Figure 9.7: All pairwise combinations of basic visual features. Combinations that strongly interact have a thick black border, combinations that are clearly independent have no border, and combinations where there is weak interaction have a thin black border.

each sex, the proportion of different actions against offenders; between sexes, the distribution of different actions; and the overall proportions of combinations of sex and action against offenders.

For example, we can see that there are more male offenders than female offenders, court action is more common than non-court action for both males and female offenders, court action is more common for male offenders than for female offenders, and the most common offences involve male offenders and result in court action.

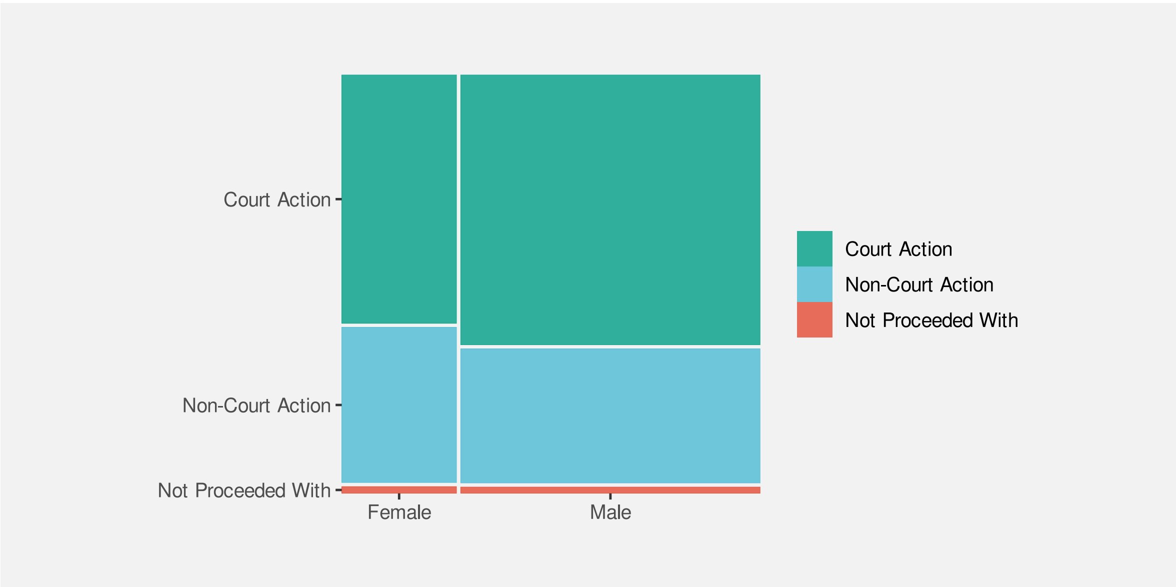


Figure 9.8: A spine plot

A mosaic plot is another example of a data visualisation that maps **data summaries** to data symbols rather than raw data. The data symbols are rectangles, the proportions of male versus female offenders are encoded as the widths of the rectangles, and the proportions of different actions taken, within males and females, are encoded as the heights of the rectangles (see Table 9.3).

> One reason why a mosaic plot is effective is because it maps proportions and conditional proportions, rather than raw counts, to the visual features of rectangles.

A mosaic plot is also an example of a data visualisation that maps to visual features that **interact**. The explicit mappings involve encoding proportions as the widths and heights of the rectangles and we know from Section 3.3 that this will allow accurate decoding from lengths to proportions.

Table 9.3: The data summaries calculated from Table 9.2 that are encoded as the widths and heights of the rectangles in Figure 9.8.

(a) Widths

	Female	Male
Court Action	0.28	0.72
Non-Court Action	0.28	0.72
Not Proceeded With	0.28	0.72

(b) Heights

	Female	Male
Court Action	0.61	0.66
Non-Court Action	0.38	0.33
Not Proceeded With	0.02	0.02

> One reason why a mosaic plot is effective is because it maps proportions to lengths.

However, the interaction of these mappings produces an **emergent feature**, which is the **area** of the rectangles (Figure 9.5). Importantly, these areas correspond to additional data summaries, in this case, the overall proportion of offences in each combination of sex and action taken.

> One reason why a mosaic plot is effective is because the mappings to length and height interact to produce area, which represents overall proportions.

One weakness of mosaic plots is that area is not the most accurate visual feature (Section 3.5), so we are not able to decode overall proportions from the rectangle areas as accurately as we are able to decode the proportions that are represented by the separate widths and heights of the rectangles.

Another weakness is that the lengths and heights of the rectangles are unaligned (Section 5.2). This compromises the accuracy of comparisons of widths between categories or the comparisons of heights between categories. This weakness is greater if there are more categories and/or larger differences between categories than we see in Figure 9.8. For example, Figure 9.9 shows a mosaic plot of the proportions of offenders in different age groups and, within age groups, the proportions of actions against the offender at a finer level of detail than in Figure 9.8. If we attempt to compare the proportion of offences that result in “Formal Warnings” in younger age groups versus older age groups, we are hampered by the fact that the pink rectangles do not have a common vertical baseline. As a side note, we are also hampered in that case by the distance and distractors between the rectangles (Section 5.1); the ordering of the variables and the ordering of categories can have an impact on the effectiveness of a mosaic plot just like ordering the bars of a bar plot.

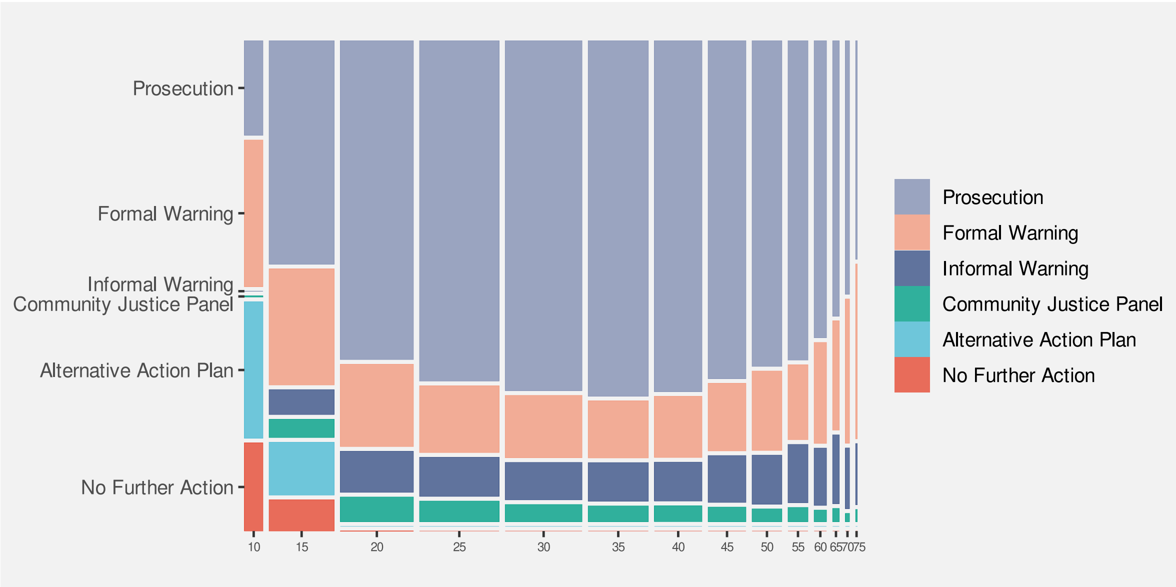


Figure 9.9: A spine plot with problems

These weaknesses are exacerbated for mosaic plots of more than two variables. For example, the mosaic plot in Figure 9.10 shows the proportions of youth versus adult offenders, then, within age levels, the proportions of male versus female offenders, then, within combinations of age levels and sex, the proportions of different actions taken against the offender. Even though there are only very few levels for each variable, our ability to compare heights and widths of rectangles is impaired by inconsistent baselines, distance between rectangles, and distractors.

A final weakness with mosaic plots is that the **shape** of the rectangles changes as well as the **area**. In Section 7.5 we saw that changes in visual features should only reflect changes in the data. Figure 9.11 shows that we can create different rectangles with the same area, but different shapes. There can be rectangles within a mosaic plot with the same area, which represents the same overall probability data value, but with different shapes. The different shapes are a visual signal of differences in the data when in reality no difference exists.

9.5 Case study: Interaction plots

Figure 9.12 shows an interaction plot of the number of offenders in different ethnic groups, comparing 2011 to 2021. The raw data are shown in Table 9.4.

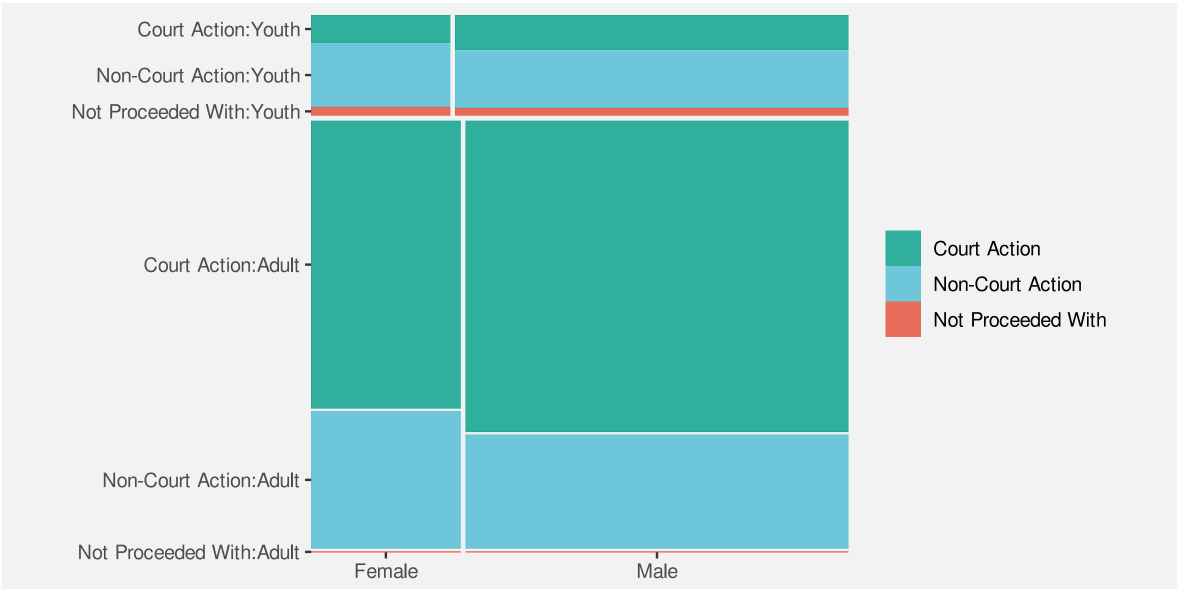


Figure 9.10: A mosaic plot of three variables

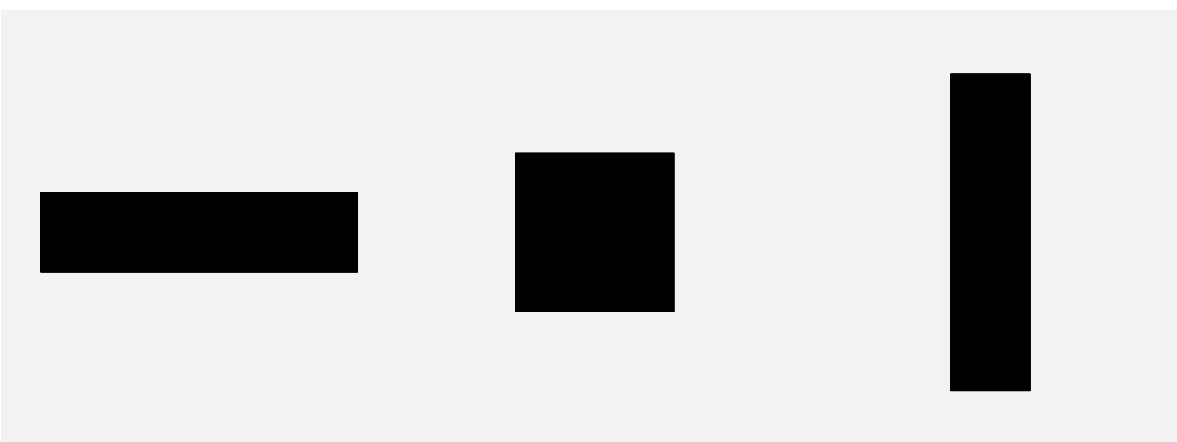


Figure 9.11: Three different rectangles that all have the same area.

Table 9.4: The number of offenders in different ethnic groups in 2011 and in 2021.

group	year	count
Māori	2011	5957
Pasifika	2011	1092
European/Other	2011	5775
Unknown	2011	194
Māori	2021	2869
Pasifika	2021	328
European/Other	2021	1833
Unknown	2021	1589

The purpose of the interaction plot is to show the *change* in the number of offenders between 2011 and 2021 for each of the ethnic groups. If the lines of the interaction plot were parallel we would conclude that the same change has happened to all ethnic groups; the fact that the lines are not parallel indicates that there have been different changes in some ethnic groups compared to other ethnic groups.[118]

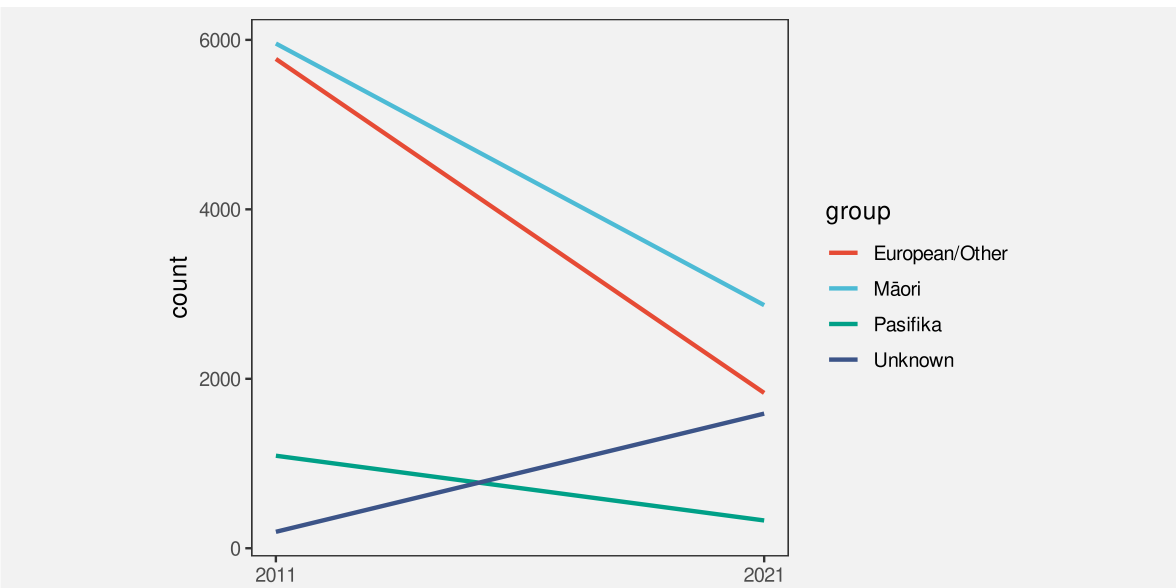


Figure 9.12: An interaction plot showing the change in the number of offenders in different ethnic groups between 2011 and 2021.

The data symbols in Figure 9.12 are straight line segments. The year is encoded as the horizontal start and end **position** of the lines and the number of offenders is encoded as the vertical start and end **position** of the lines. The ethnic group is encoded as the **colour** of the lines. These mappings are effective for decoding the raw data values in Table 9.4 from the end points and the colours of the lines.

In addition, the explicit mappings generate an **emergent feature**: the **angle** of the lines. Furthermore, the angle of the lines allow us to decode a **data summary**: the *change* in the number of offenders between 2011 and 2021 (see Figure 9.6).

> An interaction plot is effective because it allows us to decode changes in the data values from the angles of the lines.

It is more difficult to decode and compare the *change* in the number of offenders for each ethnic group in a bar plot like the one shown in Figure 9.13. Partly this is because we have to decode the lengths of two bars for each ethnic group rather than the slope of a single line.[119]

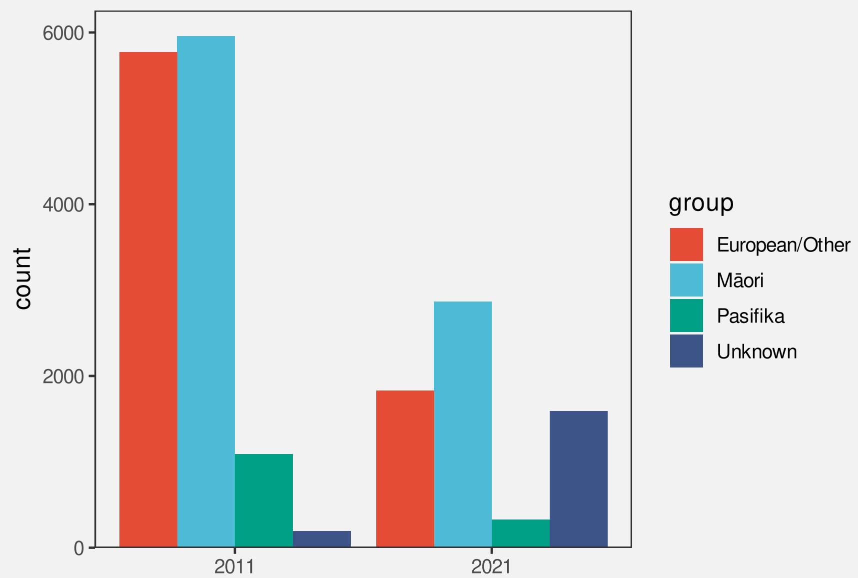


Figure 9.13: A bar plot showing the number of offenders in different ethnic groups in 2011 and in 2021.

9.6 Decoding combinations of visual features

Independent features reduce to the effectiveness of the individual features.

Interactions less clear. Interaction of width and height produce known feature, but interaction of x and y produce new feature. What do we know about position in space?

Although there is some experimental support for reasonable accuracy of correlation from scatter plot (with some bias - specifically underestimation), there is not much experimental evidence for which is best for perceiving correlations Eg scatter plot vs parallel coords etc [120]

9.7 Dangers of combining mappings

Figure 9.14 shows a plot of changes in health spending by successive New Zealand governments.[121] In this data visualisation, the average change in health spending over a government's entire term is encoded as the **height** of a bar, with the duration of that government's term encoded as the **width** of the bar. Because height and width are visual features that interact, the result is an **area** for each government's term. However, that area has no sensible interpretation; what does a percent multiplied by a number of years mean? The areas of the bars do not correspond to any property of the data values.

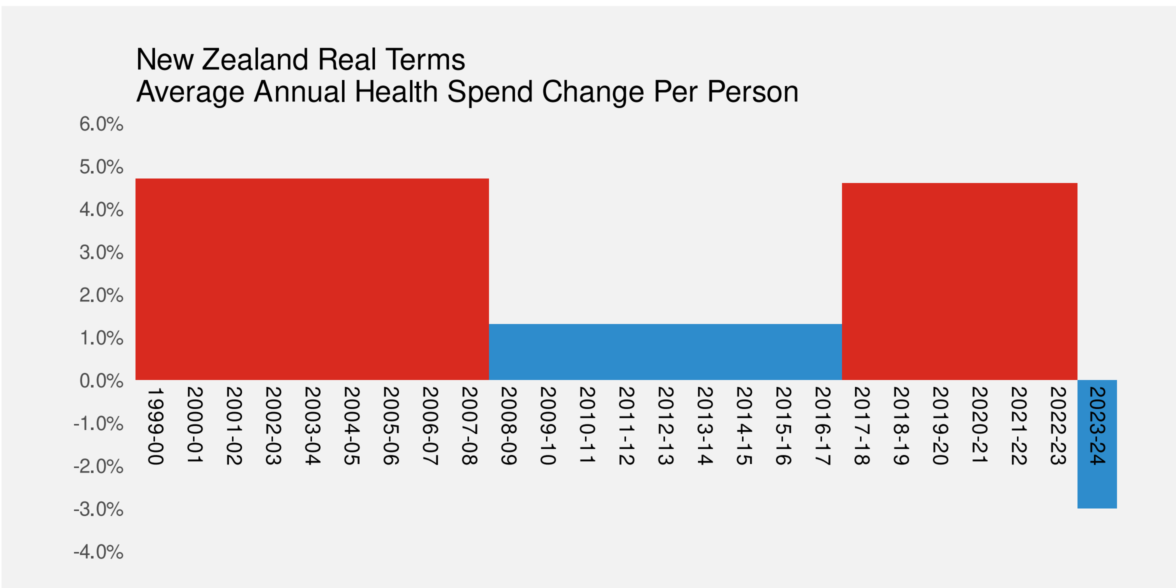


Figure 9.14: A plot showing the differences in health spending for successive New Zealand governments using filled bars.

The interaction of visual features can be a boon when the interaction produces an emergent feature that can be decoded to properties of the data (as in a scatter plot, or mosaic plot, or interaction plot). However, it is possible to create a poor data visualisation if we produce an emergent feature that has no correspondence with data values (Figure 9.15).[122]

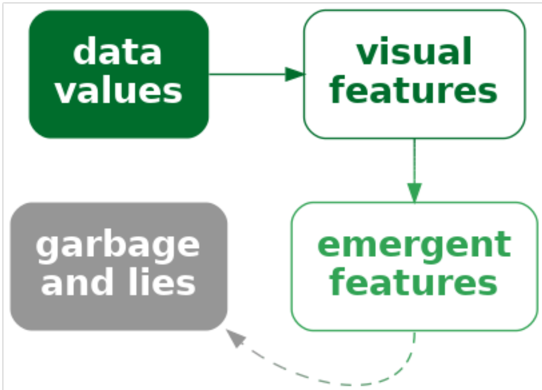


Figure 9.15: If we encode data values as visual features and the visual features interact, the result can be emergent visual features. However, this is only appropriate if the emergent features correspond to some property of the data. Otherwise, the decoding may lead to unhelpful and misleading interpretations.

> A data visualisation can be misleading if we encode data values using a combination of visual features that have a strong interaction, but the values that we decode from the resulting emergent feature do not correspond to any meaningful summary of the data.

Figure 9.16 shows a different representation of the data. In this case, the average change in health spending is encoded as the (vertical) **position** of a line and the duration is encoded as the **length** of the line. These visual features do not interact, so we no longer have a misleading emergent visual feature.

9.8 Case study: Redundant mappings

Figure 9.17 shows a bar plot of various performance metrics for the football player Lamine Yamal from July 2024.[123] The percentile values show how Yamal ranked against other similar players on each metric. He is a relatively good football player.

Figure 9.17 is a typical bar plot in that it encodes qualitative data values as the (vertical) position of the bars and quantitative data values as the lengths of the bars. However, there is also a third encoding in Figure 9.17: the percentile values are also encoded as the colour of the bars.[124]

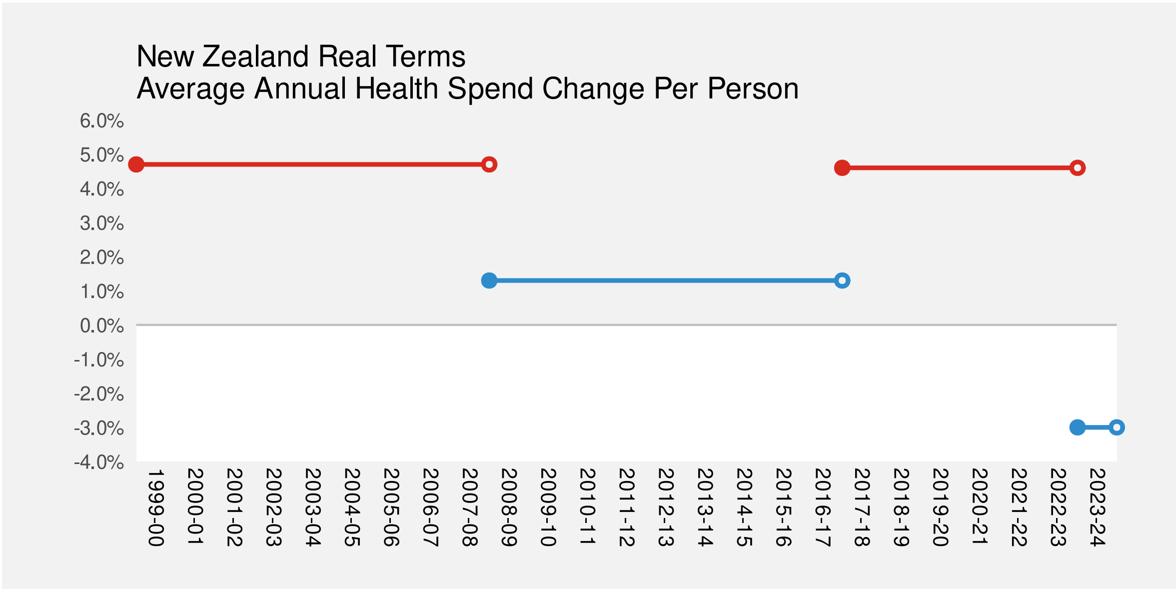


Figure 9.16: A plot showing the differences in health spending for successive New Zealand governments using horizontal lines.

Statistic	Per 90	Percentile
Non-Penalty Goals	0.16	29
npxG: Non-Penalty xG	0.22	57
Shots Total	2.56	68
Assists	0.23	64
xAG: Exp. Assisted Goals	0.27	84
npxG + xAG	0.49	74
Shot-Creating Actions	4.44	71

Figure 9.17: FBREF statistics for Lamine Yamal from July 2024.

The additional encoding of percentile is **redundant** in the sense that we can already decode the perentile from the length of the bars. However, having the additional encoding means that we have two ways to decode the percentile: from the length and from the colour (Figure 9.18).

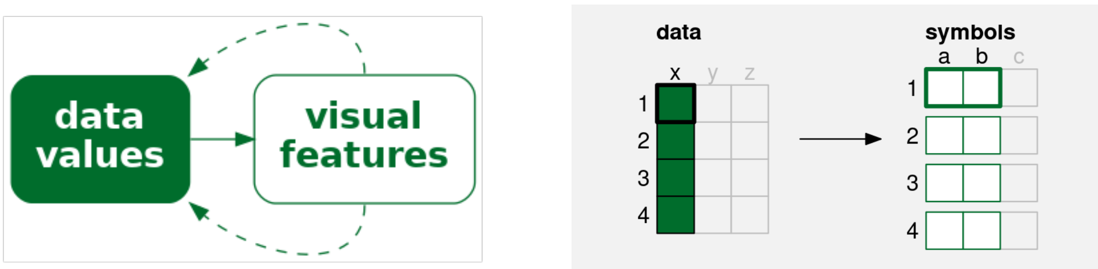


Figure 9.18: A redundant encoding maps a single data value to multiple visual features. Each data value (`x`) is mapped to not just one visual feature (`a`), but also a second visual feature (`b`), of a single data symbol. If we encode data values using multiple visual features, then multiple decodings are possible.

The effectiveness of a redundant encoding partially depends upon the independence of the two visual features. The features are only redundant if we are able to decode data values from the two features separately, as we can with colour and length.

One useful application of redundant encoding is to accommodate viewers with CVD (Section 6.7). Having a secondary decoding alongside colour allows for the fact that the colour decoding may fail for some viewers. For example, Figure 9.19 shows what Figure 9.17 might look like to a viewer with severe red-green colour blindness. The distinction between red and green bars is gone and that in turn confuses the decoding of the bar luminance (changes in luminance are not monotonic with changes in length). However, thanks to the redundant encoding, the simple comparison between bar lengths remains.

One argument against using a redundant encoding is that we are introducing additional visual complexity. Each individual visual feature may not be as effective as it would be on its own because it is not the *only* visual change (Section 2.5). For example, Figure 9.20 shows a variation on Figure 9.17 with the redundant encoding removed—all of the bars are just the same dark grey. It is arguably easier to decode the lengths of the bars in Figure 9.17 because the bars are less complex visual objects. This is even more true when comparing with the confusing changes in luminance of the bars in Figure 9.19.[125]

Statistic	Per 90	Percentile
Non-Penalty Goals	0.16	29
npxG: Non-Penalty xG	0.22	57
Shots Total	2.56	68
Assists	0.23	64
xAG: Exp. Assisted Goals	0.27	84
npxG + xAG	0.49	74
Shot-Creating Actions	4.44	71

Figure 9.19: FBREF statistics for Lamine Yamal from July 2024. This is a version of Figure 9.17 with the colours adjusted to simulate severe red-green colour blindness.

Statistic	Per 90	Percentile
Non-Penalty Goals	0.16	29
npxG: Non-Penalty xG	0.22	57
Shots Total	2.56	68
Assists	0.23	64
xAG: Exp. Assisted Goals	0.27	84
npxG + xAG	0.49	74
Shot-Creating Actions	4.44	71

Figure 9.20: FBREF statistics for Lamine Yamal from July 2024. This is a version of Figure 9.17 with the redundant colour encoding removed.

9.9 Summary

Almost all data visualisations involve *combinations* of mappings. More than one set of data values are encoded as more than one visual feature of data symbols.

The mappings involved in a bar plot—quantitative data values encoded as lengths and qualitative values encoded as position—are effective because we are able to perceive some combinations of visual features, such as length, position, and colour **independently**, which means that we can effectively decode both position and length from a bar plot.

A scatter plot is effective for perceiving relationships between variables because the mappings of quantitative data values to both horizontal and vertical positions **interact** to produce **position in space** and our visual system is capable of producing useful **visual summaries** from position in space, such as correlation.

Independence between visual features is useful when we want to decode separate data values. Interactions between visual features is useful when we want to produce an emergent feature that allows us to decode data summaries.

Conversely, independence between visual features is of no help if we want to produce an emergent feature. Furthermore, interaction between visual features is unhelpful, or even misleading, if we cannot decode any meaningful information from the emergent feature.

Notes

[112] Independent visual features are called *separable*, in contrast to *integral* features that interact. An early use of these terms in the context of data visualisation (or at least *statistical graphics*) is Carswell and Wickens (1988).

A further distinction is made for *configural* visual features, which are combinations that interact, but still allow decoding of the individual visual features (Carswell and Wickens 1990).

These ideas can be traced back to the much less readable Garner (1974).

[113] To be honest, there is an interaction between position and length because lengths that are further apart from each other vertically are harder to compare (Section 5.1). However, compared to visual features that strongly interact, length and position are relatively independent.

[114] The use of *emergent* visual features in this book, to describe the result of *integral* (or *configural*) visual features interacting, corresponds loosely to various usages of *emergent* in the literature.

MacEachren (1995) uses *emergent* in this way, but, for example, Cragin and Pomerantz (2015) and Palmer (1999) use *emergent* more to describe visual features that arise from combinations of separate visual objects (data symbols).

"The arrangement of several dots in a line give rise to emergent properties, such as length, orientation, and curvature, that are different from the properties of the dots that compose it." (Palmer 1999, fig. 2.1.5)

115 The ability to decode correlations from scatter plots is another example of *ensemble* perception (Szafir et al. 2016; Rensink 2017).

116 Munzner (2014) provides some examples of combinations of *visual channels* in Figure 5.10. Ware (2021) gives a broader selection of combinations of *display attributes* in Figure 5.24.

117 This two-dimensional mosaic plot could also be called a spine plot. Mosaic plot is the more general term that includes plots of more than two qualitative variables.

Mosaic plots also provide a simple visual check for independence between factors. For example, in Figure 9.8 we can see that the action taken is reasonably independent of the sex of the offender because the heights of the rectangles are similar for both Male and Female offenders (the conditional proportions are roughly the same).

118 An interaction plot is typically used in a more experimental setting where the x-axis might be different treatments and the groups might be different patient groups and the y-axis might be a measure of response to treatment. The outcome is then whether the treatment has had the same effect for different patient groups (or not).

119 Pinker1990GraphComprehension (p. 111) discusses the problem of extracting different sorts of information from different data symbols, in particular bars versus lines.

120 Examples of studies that demonstrate a tendency to underestimate correlation coefficient are Rensink and Baldridge (2010) and Cui, Kini, and Liu (2024).

121 Figure 9.14 is based on Figure 2 from Huskinson (2024).

122 The idea that combining visual features that interact is more appropriate for decoding data summaries (and vice versa) is similar to the *proximity compatibility principle* (Wickens and Carswell 1995).

Shah and Hoeffner (2002a) use the following nice quote from earlier work (Carswell and Wickens 1987):

"Integrated, object-like displays (e.g., a line graph) are better for integrative tasks, whereas more separable formats (e.g., bar graphs) are better for less integrative or synthetic tasks such as point reading."

123 Figure 9.17 is based on an image produced by the FBREF web site for football statistics and history.

124 In fact, because the colours are from a diverging colour palette (Section 6.11), there are arguably four encodings: whether the percentile is below 50 or above 50 is encoded as a red or green hue and the amount above or below 50 is encoded as the luminance/chroma of the bar.

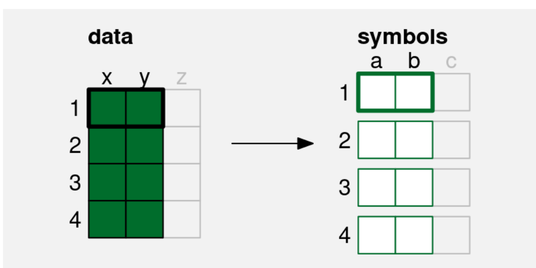
125 Redundant encodings are not universally supported in the literature either. For example, VanderPlas and Hofmann (2017) provide evidence that additional cues improve plot perception and Borkin et al. (2016) report that "redundancy helps effectively communicate the message". However, Franconeri et al. (2021b) are less in favour: "redundant encoding should be avoided in most cases, except when used to make visualizations accessible for viewers with color-vision impairments."

10 Visual Shapes

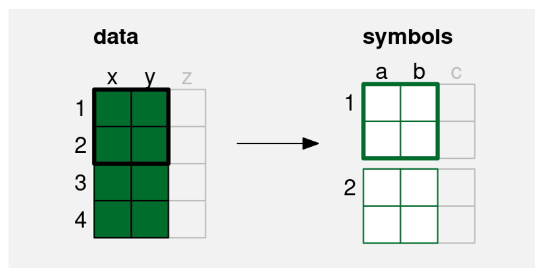
TODO refer back to Section 4.3 and mention that shapes are one solution, with all of the drawbacks of not being able to decode individual data values. Footnote surjective/injective?

All of the data visualisations that we have considered so far have mapped a **single data value** from one or more variables to a **single data symbol** (Figure 10.1a). For example, a bar plot like Figure 3.2 maps one category and one count to one bar. A scatter plot like Figure 9.4 maps one data value from a quantitative variable and one data value from another quantitative variable to one data point.

In this chapter, we consider data visualisations that map **multiple data values** from one or more variables to a **single data symbol** (Figure 10.1b). In this chapter, we begin to consider more complex data symbols.



(a)



(b)

Figure 10.1: (a) A simple data symbol, like a bar in a bar plot or a data point in a scatter plot involves mapping **one** data value from each of two variables to two visual features of **one** data symbol; a single pair of data values produces a single data symbol (this is a reproduction of the diagram in Figure 9.2). (b) A more complex data symbol is produced if we map **multiple** data values from each of two variables to two visual features of a **single** data symbol. Multiple data values from `x` and `y` are mapped to the visual features, `a` and `b`, of a single data symbol. Multiple pairs of values produce a single data symbol.

10.1 Line plots

Figure 10.2 shows a line plot of the number of offenders per year in different ethnic groups from 2011 to 2021 (see Table 5.1). This data visualisation is very effective for perceiving changes over time in the number of offenders within each group as well as comparing the changes over time between groups.

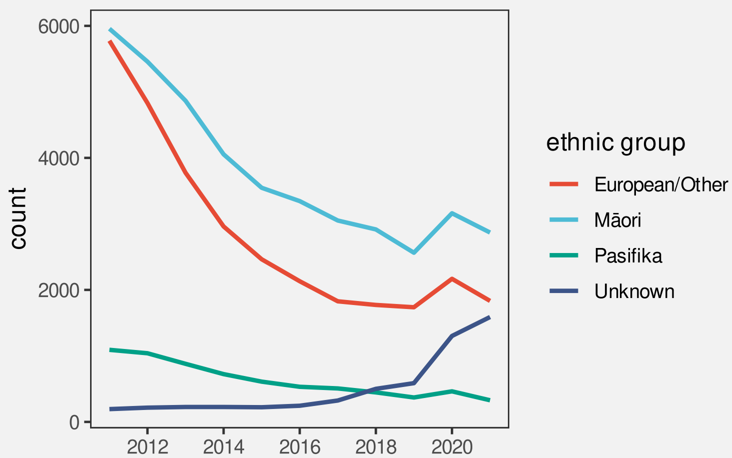


Figure 10.2: A line plot of the number of offenders per year in different ethnic groups from 2011 to 2021.

There are some familiar mappings in this data visualisation: year is mapped to horizontal **position**, the number of offenders is mapped to vertical **position**, and the ethnic group is mapped to **colour**. However, the **data symbols** in this data visualisation, the coloured lines, are different from any previous examples that we have seen. Although there are 44 rows of data, there are only 4 data symbols. Multiple rows of data are used to create each data symbol. Each coloured line in Figure 10.2 represents 11 pairs of year and count values.

This contrasts with visualisations like bar plots that use simpler data symbols. For example, the bar plot of the same data in Figure 5.7 has 44 bars.

Figure 10.2 is effective at conveying changes over time because the coloured lines create **visual shapes** (Figure 10.3) and those visual shapes convey information about collections of data values. For example, we can perceive slopes and curves and peaks and valleys in each of the coloured lines.[126]

In terms of the simple visual processing model from Section 2.2, these more complicated shapes can convey more sophisticated information, at the expense of more cognitive effort, and with a

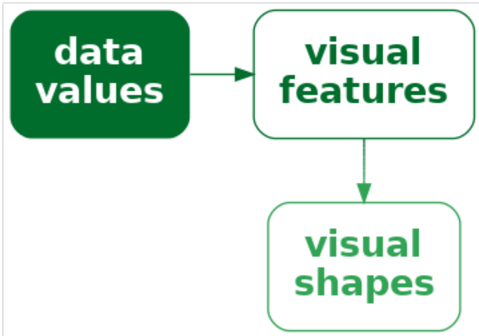


Figure 10.3: If we map multiple data values to multiple visual features of a single data symbol, like in a line plot, we create a smaller number of more complex data symbols, **visual shapes**, that contain a larger amount of information.

limit on how many shapes we can process simultaneously (see Figure 2.4). In other words, the more complex data symbols allow us to decode **data summaries** (Figure 10.4). For example, in Figure 10.2 we are able to perceive overall trends in the number of offenders, periods of more rapid change versus periods of slower change, and local maxima and minima.[127]

This ability to perceive data summaries, such as peaks and slopes, when we use lines as data symbols is even more apparent if we compare Figure 10.2 with Figure 5.7. When we have bars as data symbols, these data summaries are much less apparent.

In terms of Chapter 7, the use of lines for data symbols is also **congruent** with trends in the data; we naturally associate slopes with increases and decreases in data values.[128]

> One reason why line plots are effective is because they encode multiple data values into a complex data symbol that creates a visual shape, from which we are able to decode data summaries.

On the other hand, the more complex data symbols in a line plot, that contain multiple data values, are less effective at conveying the *individual* raw data values. The coloured lines in Figure 10.2 are not simple shapes from which it is easy to perceive basic **visual features**. This means that, although we may be able to decode data summaries, it is not easy to decode the raw data values (Figure 10.4). For example, compare the ease with which we can determine the number of European/Other offenders in 2013 from Figure 10.2 versus Figure 5.7.

This deficiency is significant because the main mappings in a line plot are from **quantitative** data values to (horizontal and vertical) **position**. We know from Chapter 3 that these should be very effective mappings that allow us to accurately decode the raw data values. However,

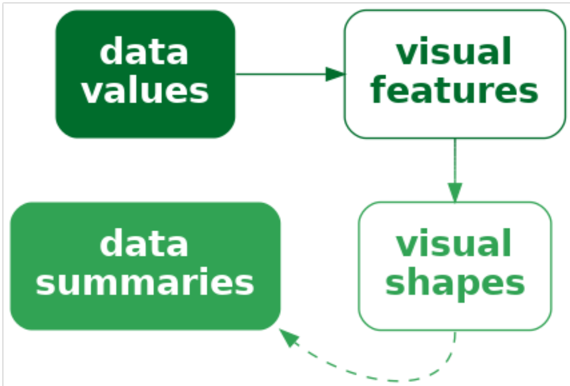


Figure 10.4: If we map data values to a **visual shape** that contains a large amount of information, then we may be able to decode more sophisticated **data summaries**.

in a line plot, we are not mapping a single data value to a basic visual feature of a single, simple shape. Instead, a single data value is being mapped, along with many other values, to a basic visual feature of a complicated shape. In effect, the individual data values lose their individual identity.[129]

TODO Refer back to Section 3.1 where we had bijective (but only use that terminology in footnote), so we could decode individual data values.

> One reason why a line plot is *less* effective for decoding raw data values is because each raw data value is not mapped to its own data symbol.

The creation of complex data symbols from multiple data values bears some similarity to the creation of emergent features by combining visual features (Section 9.2). In both cases, we have *explicit* encodings from data values to basic visual features, but we also produce an *implicit* additional visual effect. The difference here is that with complex data symbols, we are collapsing multiple data values into a data symbol, so we run a higher risk of losing the ability to perceive individual data values.[130]

There is also a similarity between complex data symbols and visual summaries (Section 8.2). In both cases, we gain the ability to decode data summaries from collections of data values. Again, the difference is that with visual summaries, we are summarising many simple data symbols that each represent individual data values, whereas with a complex data symbol we decode the data summary from a single data symbol that obscures the individual data values.

10.2 Case study: Scatter plots with lines

Figure 10.5 shows a data visualisation of the number of offenders per year in different ethnic groups (Table 5.1). This is very similar to Figure 10.2, but we have added a data point at every horizontal and vertical **position**.

Like Figure 10.2, we can decode overall trends from the line data symbols, but we can also decode individual data values more easily because now every individual data value is also mapped to its own data symbol.

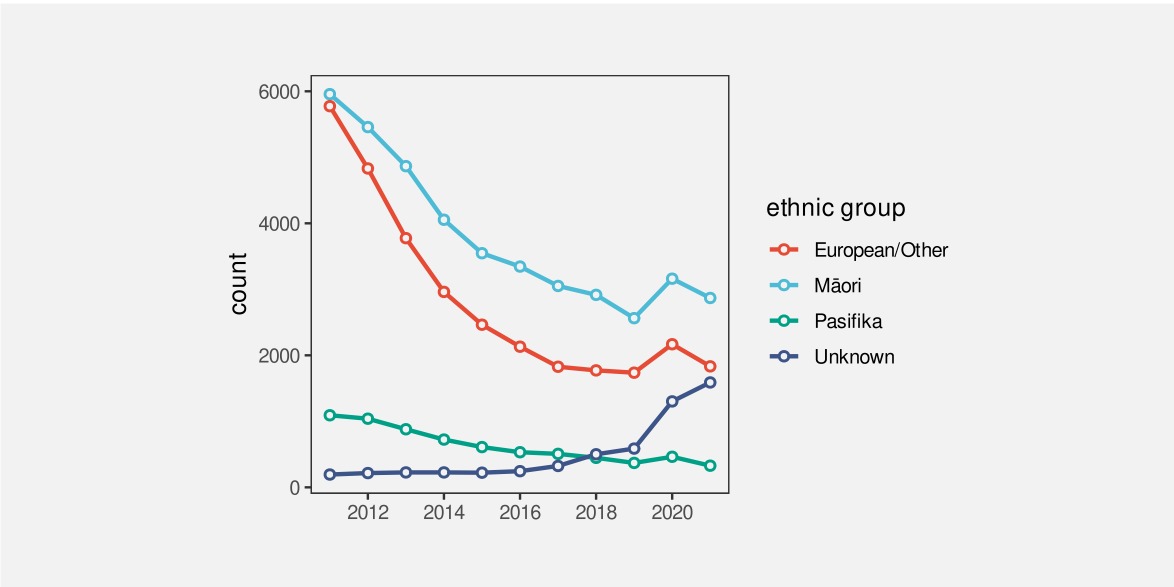


Figure 10.5: A plot of the number of offenders per year in different ethnic groups from 2011 to 2021 with both lines and points.

10.3 Case study: Density plots

Figure 10.6 shows a density plot of the number of points scored by Tier One nations at the Rugby World Cups (Table 4.1). As with a histogram (Figure 8.5), this data visualisation is effective for conveying the distribution of data values. For example, we can perceive that there is evidence of a single mode and right skew.

Also like a histogram, a density plot actually maps **data summaries**, density estimates rather than raw data values, to visual features (see Table 10.1). In the case of a histogram, the data summary values are counts within bins (see Table 8.2) and each data summary produces a separate bar. However, a density plot encodes all of the data summary values to a single data symbol, a line, to create a **visual shape**. The density estimates are mapped to the horizontal

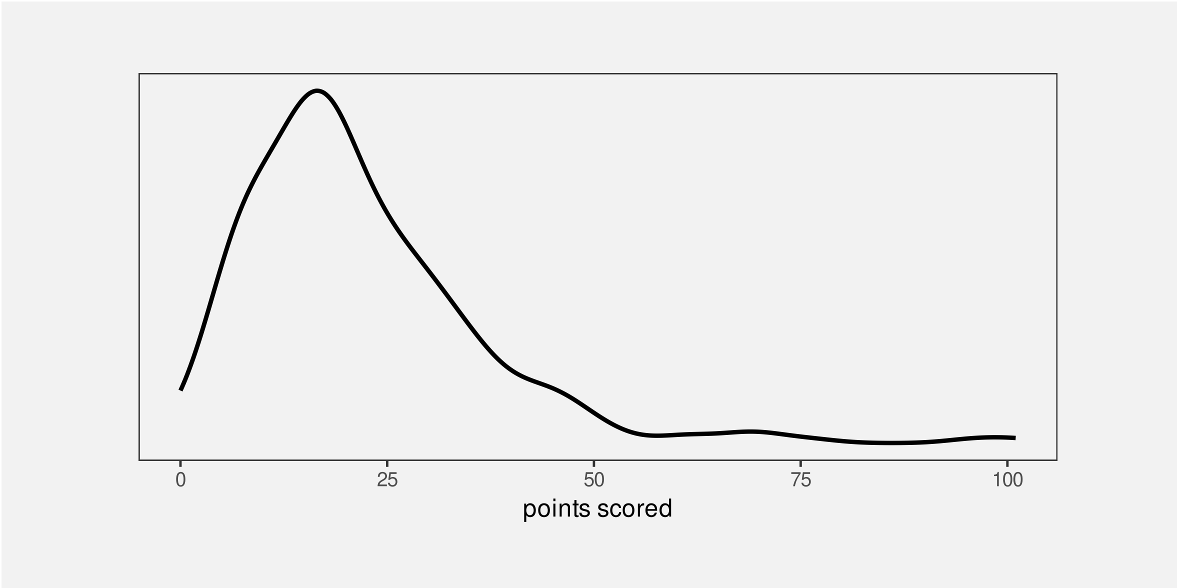


Figure 10.6: A density plot of the points scored per game by Tier One nations at Rugby World Cups.

and vertical **positions** of a single line data symbol. The visual shape allows us to perceive summaries of the density estimates. For example, whether there is a single maximum density and how the density changes to either side of that peak.

Table 10.1: A table of density estimates at a sample of values over the range of the number of points scored by Tier One nations in Rugby World Cup matches. These data summaries are the values that are mapped to visual features to produce the density plot in Figure 10.6. There are 512 rows of values, but only the first 6 are shown here.

scored	density
0.00	0.0055
0.20	0.0059
0.40	0.0063
0.59	0.0067
0.79	0.0071
0.99	0.0075

Figure 10.7 shows the path from raw data values to data summaries (density estimates), which are encoded as the visual features (horizontal and vertical position) of a visual shape (a line) in a density plot and the summary information that we can decode from the visual shape. We

cannot decode the raw data values from a density plot, nor can we very easily decode the data summaries (the density estimates). The only real purpose of the data symbol in a density plot is to allow us to decode **summaries** of the data summaries, which are visual features of the density estimates (like the number of peaks and where the peaks occur).

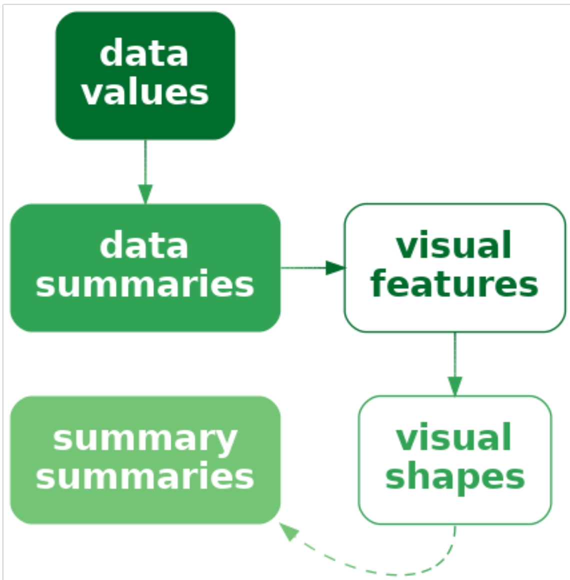


Figure 10.7: A density plot maps data summaries to visual features to create a visual shape from which we can decode **summaries** of the data summaries.

> A density plot is effective because it conveys high-level features of density estimates of the raw data.

Another commonality between density plots and histograms is that we get to choose how the data summaries are calculated from the raw data values. In the case of a density plot, we must choose a "kernel" and a "bandwidth", both of which affect the smoothness of the density estimates. This means that we may need to explore more than one kernel and bandwidth (see Section 8.5).

A final point about density plots, especially in comparison to histograms is that it is more effective to present multiple density curves, rather than multiple histograms. This is demonstrated in Figure 10.8 and Figure 10.9. Multiple histograms overlap each other, which makes it difficult to perceive the separate histograms and this is not satisfactorily solved by adding semitransparency. By comparison, multiple density curves create very little overlap, and even though the orange curve passes "under" the blue curve, we effortlessly perceive the orange curve as a single, coherent shape (see Section 2.7).

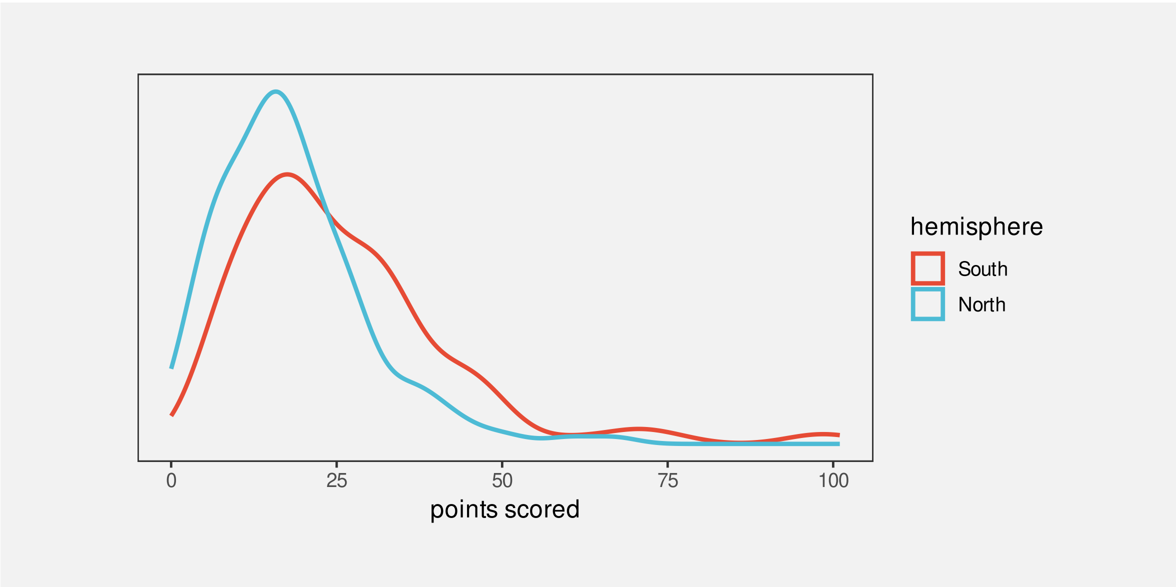


Figure 10.8: Density plots of the points scored per game by Tier One nations at Rugby World Cups, broken down by hemisphere.

10.4 Aspect ratio

How aspect ratio affects ability to detect features in lines

Also use of white space around points in scatter plot ?

Cite Cleveland (1994) for white space margin affecting strength of relation?

10.5 Dangers of visual shapes

Need a caveat that more complex shapes does NOT necessarily mean useful mappings back to data summaries - could just end up with a mess (Figure 10.10).

Lines imply continuity (at least) if not linear interpolation so do example where connecting with lines not such a great idea. Refer back to Chapter 7 (congruence).

Example from "Bars and lines: A study of graphic communication" Zacks and Tversky (1999) (Fig 2) of inappropriate line plot: line connecting males and females => some subjects made trend interpretation, the worst being "The more male a person is, the taller he/she is"

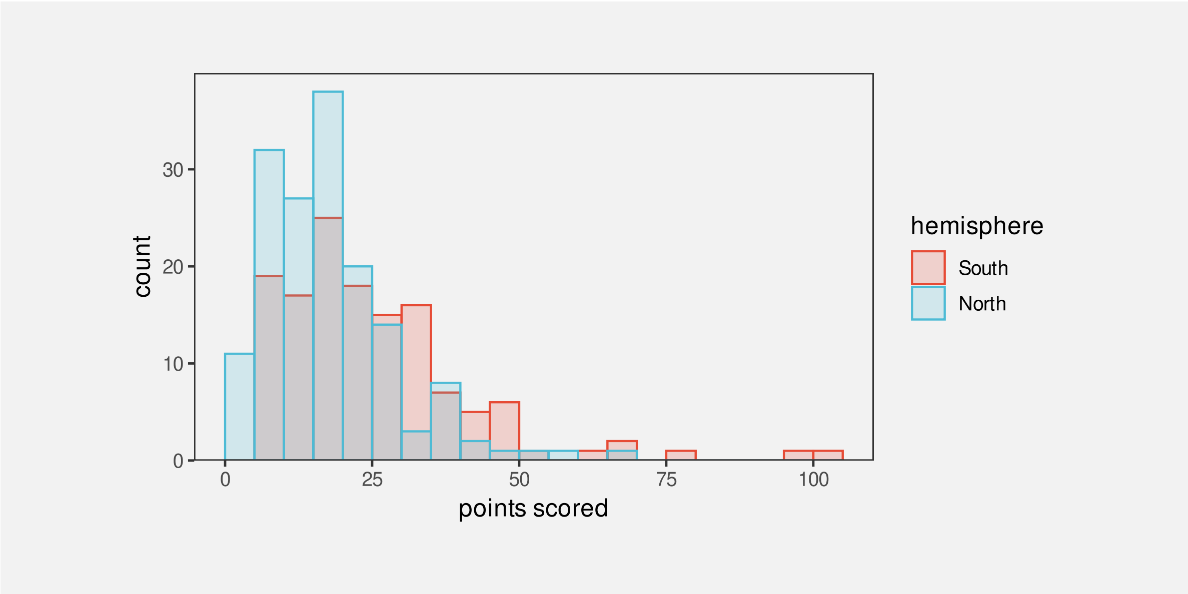


Figure 10.9: Histograms of the points scored per game by Tier One nations at Rugby World Cups, broken down by hemisphere.

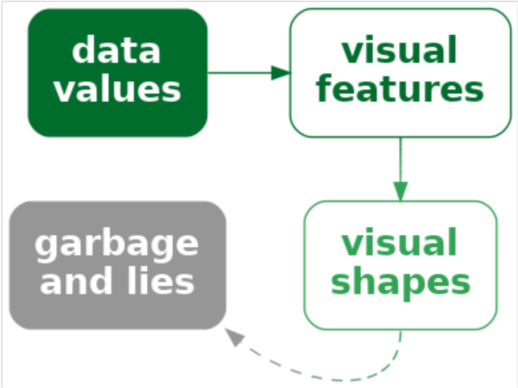


Figure 10.10: Mapping data values to the visual features of a complex shape may not be effective if the complex shape decodes to unhelpful or misleading information.

10.6 Case study: Violin plots

And other variations on boxplots

First point out that a boxplot is really a visual shape: multiple data summaries generate a single “box” data symbol. Although it is still very easy to decode individual data summaries from the box shape.

- Case study: SinaPlots

A SinaPlot is a nice combination of decode for raw data and visual shape for data summary.[131] Figure 10.11 is actually a combination of a SinaPlot and a violin plot.

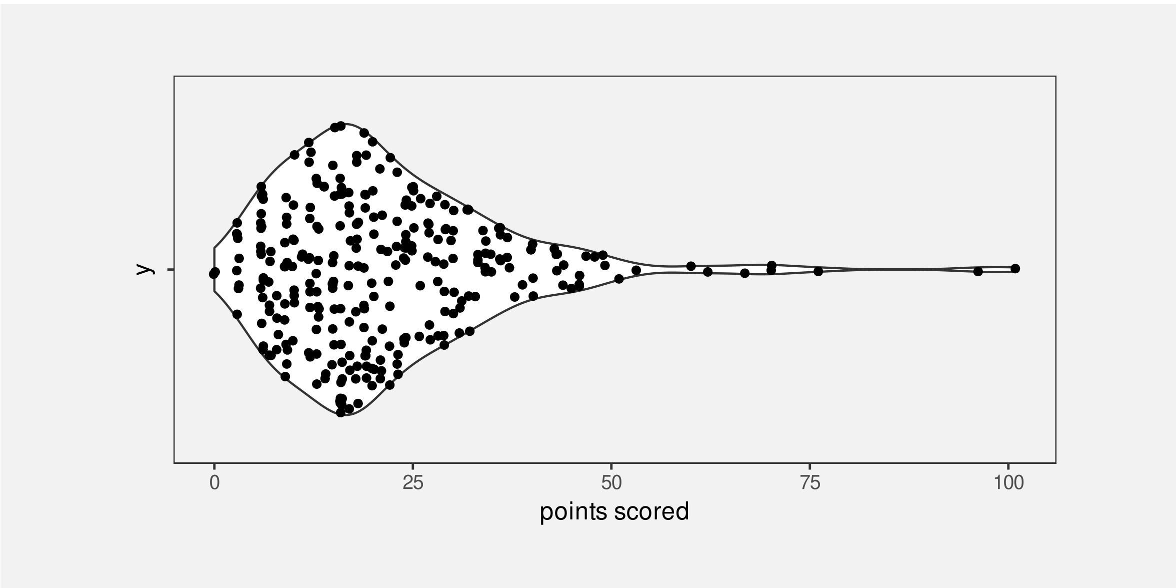


Figure 10.11: A SinaPlot of the points scored per game by Tier One nations at Rugby World Cups.

10.7 Case study: loess curves

Another example of data -> summary -> features -> shapes -> summaries

10.8 Case study: Line of best fit

Nice example of a complex calculation, the results of which are what gets visualised. Also example of performing the calculation numerically rather than asking the viewer to perform it visually. This was in summaries.qmd, but it cannot be there because it requires scatter plots, which come in the next section. Add this after mention of visualising models?

Add footnote to Hadley's more complex model visualisation paper https://vita.had.co.nz/papers/model-vis.pdf https://dl.acm.org/doi/10.1002/sam.11271

model fits like lm straight line and ref Hadley's data-space vs model-space "data models" to mirror "data statistics" (revisit box plots as VERY simple models? also talk about failing assumptions of these models, like unimodal for boxplot)

footnote cite to Tukey for visualising phenomena rather than raw data? "Data-Based Graphics: Visual Display in the Decades to Come" Tukey (1990)

10.9 Non-linear mappings

Refer back to Section 4.7

NOTE that there ARE some situations where non-linear position. scales are ok, e.g., log-log. Length is still dodgy. You DO still get ordinal info, even if cannot decode sizes of differences. Non-linearity might actually be ok on non-position scales, e.g., a colour scale for numeric data, where differences are smaller at one end; all you are after, if you are using colour, is ordinal, so changing colours more swiftly at one end of the scale is ok (?) At least some 'colorspace' functions allow non-linear palettes. Revisit in qualitative.Rmd to point out that non-linear ordinal scales are ok, e.g., jet colour scale

Nice example of log-log scales in The Signal and the Noise (Nate Silver) See 787resources/Examples/2025-08-signal-noise-log-log

Log log scales a good example of asking what is it you want to see? You cannot extract raw data or a compare raw data points. But all you want is to see the data summary of correlation so no harm no foul.

Can get useful *shapes* out of non-linear scales

cite the paper (cook? hofmann?) that looked at identification of curves vs straight lines in log-log plots? but only in visual-shapes chapter when call back to non-linear scales in position chapter

Nice kiwi-biased example on this thread https://stats.stackexchange.com/questions/27951/when-are-log-scales-appropriate But I think the point still holds that you LOSE the ability to judge size of differences - you can only decode things like order and groups (?)

The ratio plot might be more interpretable if the axis is labelled as "number of times larger" ?

10.10 Case study: log-log plots

Involves non-linear scales, but the point is the shape that you (hope to) get.

Could also do q-q plots?

10.11 Summary

Mapping multiple data values to a single data symbol produces a **visual shape**, like a line on a line plot.

Data summaries, such as trends over time, can can be decoded from a visual shape. However, decoding raw data values from a visual shape may be harder, cmopared to decoding raw data values from a simple data symbol like a bar.

Notes

126 "VISUAL PATTERN RECOGNITION" Few (2006) Makes this same point [p 10] Demonstration of difference between density plot (mostly see a shape) and histogram (mostly see bar heights).

127 "Understanding Charts and Graphs" Kosslyn (1989) [somewhere after p 205] Lines are objects "For example, if one wants a reader to see an interaction, a line graph is better than a bar graph: not only is a line a single perceptual unit, and hence there will be less to hold in short-term memory, but we are good at detecting slope differences-which convey the relevant information (see Pinker, 1988)."

128 According to Zacks and Tversky 1999 people use "trend words" (increasing/decreasing) versus "discrete words" (higher/lower than) AND are faster to judge absolute values from bars but angle from line graphs.

129 citation for surjective, injective, bijective

Refer back to Section 4.3

Adam has a nice example in his thesis Figure 4.5, a simpler version of which is a line or bar plot that contains a "total" line or bar - the visual objects do not map neatly, and disjointly to subsets of the data - there is overlap. This makes mapping back to the data harder or at least more ambiguous ?

130 Also link to "configural displays" (David Peebles (2008)) and "display proximity" (Barnett & Wickens, 1988; Carswell & Wickens, 1987; Wickens & Andre, 1990) "According to the proximity compatibility principle (Wickens & Carswell, 1995), representations with high display proximity facilitate high mental proximity tasks requiring the integration of information from multiple sources. The relationship between display and mental proximity

has been revealed in several studies (e.g., Barnett & Wickens, 1988; Bennett & Flach, 1992; Bennett, Toms, & Woods, 1993; Carswell & Wickens, 1987; Wickens & Andre, 1990; Wickens & Carswell, 1995) and it is argued that the advantage comes when emergent features (Pomerantz & Pristach, 1989) of the representation indicate additional information about the domain." He also mentions line plots as being in between separable bar plots and configural radar plots.

[131] Sidiropoulos et al. (2018)

11 Multiple Mappings

TODO a side-by-side bar plot and even a stacked bar plot both use nested position!!! Also stacked dot plots like fig-rwc-dotplot.

Figure 11.1 shows a data visualisation of the performance measures for teams in the 2023 Rugby World Cup (Table 9.1). This visualisation is different from anything we have seen previously because it involves multiple mappings. The data symbol in this case is a simple data point, but we have mapped data values to *five* different visual features of each data point. The number of clean breaks is encoded as the horizontal **position** of the data points and the number of tries is encoded as the vertical **position** of the data points. The number of points scored is encoded as the **size** of the points, the hemisphere is encoded as the **shape** of the points, and the number of runs is encoded as the **colour** of the points.

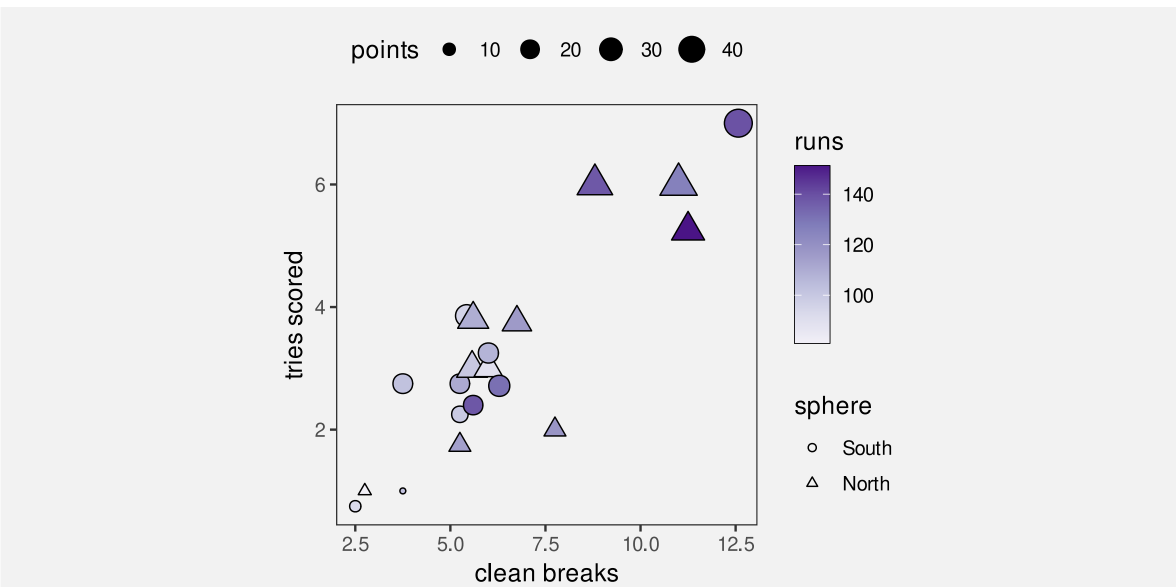


Figure 11.1: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup.

Figure 11.1 is effective in some ways because some of the mappings involve effective visual features. For example, we can easily decode the number of clean breaks and the number of tries from horizontal and vertical positions. We can also decode a correlation between clean breaks and tries from the emergent feature of **position in space** (Section 9.2).

However, other mappings are less effective. For example, it is less easy to accurately decode number of points scored from size and even harder to decode the number of runs from colour. In addition, we are hampered by the fact that many different visual features are changing at once, so it is harder to perceive individual differences (see Section 2.5).

One goal of visualising multiple variables like we have in Figure 11.1 is to discover or display multivariate relationships between the variables. For example, we can see that there are four teams in the top-right corner of the plot that are all relatively large and relatively dark in colour, which suggests that there is a group of teams that run more, make more clean breaks, score more tries, and score more points compared to the other teams.

This chapter looks at some of the different approaches to visualising several variables at once. How do we perceive multivariate relationships? And can we still perceive individual data values at the same time?

11.1 Parallel coordinates plots

One issue with Figure 11.1 is that it attempts to map each data variable to a separate visual feature. Once we have used both horizontal and vertical **position**, we have to start using less accurate and less appropriate visual features, like **area** and **colour** for quantitative data values (see Chapter 3). If we attempted to map all of the variables in Table 9.1, we could actually run out of visual features to use (Figure 3.5).

An alternative approach is to map more of the data variables to the same visual feature. For example, Figure 11.2 shows a parallel coordinates plot of the data from Figure 11.1. In this visualisation, all of the quantitative variables have been encoded as **position**, with just hemisphere (qualitative) encoded as colour.[132]

A parallel coordinates plot maps data summaries rather than raw data values (Chapter 8). For example, all data values are standardised to a range between 0 and 1.[133] Table 11.1 shows some of the values that are actually mapped to visual features in Figure 11.2. Each variable is encoded as a different horizontal **position** and the values within each variable are encoded as vertical **position**. In this way, all quantitative data values are encoded as **position**.

Table 11.1: The data summaries that are mapped to visual features in Figure 11.2.

country	sphere	variable	value
Namibia	South	runs	0.16
Romania	North	runs	0.00
Chile	South	runs	0.30
Samoa	South	runs	0.31
Australia	South	runs	0.42
Georgia	North	runs	0.47

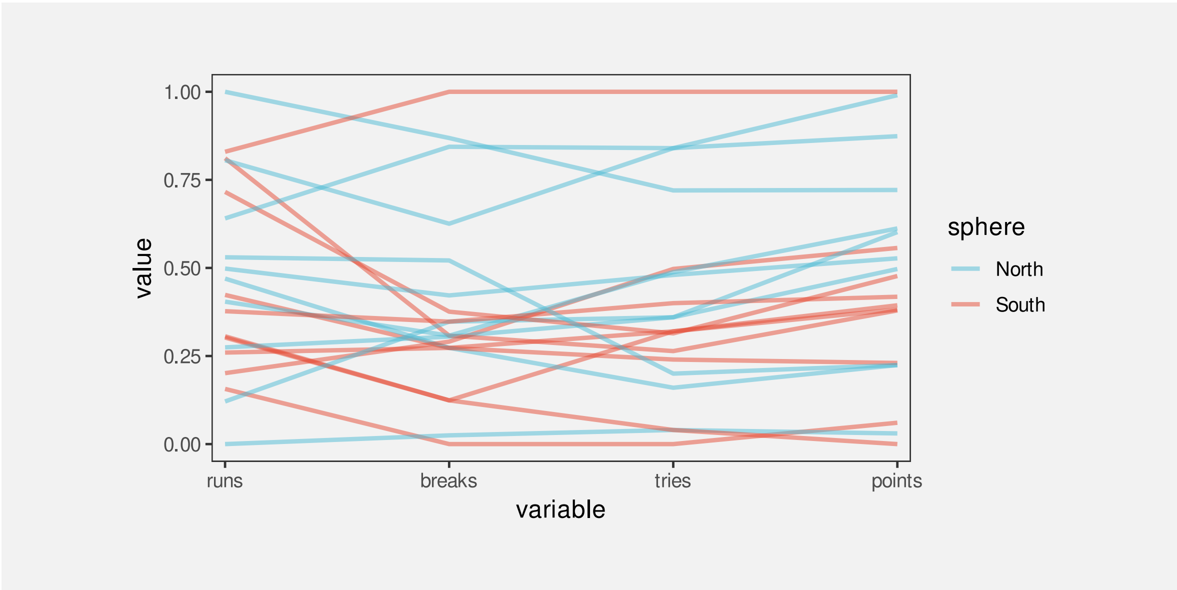


Figure 11.2: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup.

In addition, the hemisphere is encoded as **colour** and a separate line data symbol is drawn for each country. This means that a visual shape is drawn for each country (Figure 11.3; Chapter 10).

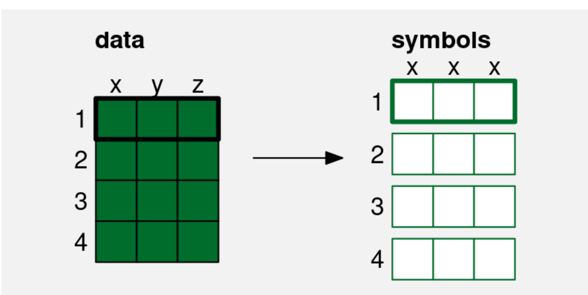


Figure 11.3: In a parallel coordinates plot, a single row of data—one data value from each of several variables—is mapped to a visual shape (a line).

On one hand, a parallel coordinates plot appears to makes sense because it encodes multiple data values to **position**, which is an accurate visual feature for quantitative data. However, this advantage is immediately squandered because the values that are encoded are not the raw data values (Figure 8.2), so a parallel coordinates plot is not effective for decoding raw data values from the positions of the lines. Furthermore, the data symbols are lines, so each data symbol encodes multiple data values (Section 10.1), which makes it difficult to decode even individual data summary values from the lines.

The effectiveness of a parallel coordinates plot comes from the visual shapes produced by the lines. For example, in Figure 11.2, we are able to identify a group of four lines at the top of the plot that are separate from the other lines. This grouping across at least three variables (clean breaks, tries, and points) is easier to see compared to Figure 11.1. We can also more clearly see that there are two teams that have similar numbers of runs to the top four, but then are mixed in with the other teams on the other variables. Furthermore, the fact that many of the lines are reasonably horizontal rather than criss-crossing each other wildly suggests that there is an overall correlation between all four quantitative variables, not just in the top four teams.

> A parallel coordinates plot can be effective at identifying high-dimensional groups or correlations because it maps multiple data values from different variables to a single visual shape.

11.2 Case study: Bubble plots

An example of using area/size for third variable.

Provide some better options.

11.3 Small multiples[134]

One difficulty with Figure 11.2 is that many of the lines overlap each other, which makes it difficult to separately perceive individual visual shapes. Figure 11.4 shows an alternative approach to visualising the same data, called a profile plot. Similar to a parallel coordinates plot, a visual shape is drawn for each country, but the shape in this case is a filled in polygon, effectively the lines in Figure 11.2 with the area below them (down to zero) filled in.

Again, this sort of data visualisation can be effective because of the overall shapes. For example, we can see a group of four shapes that are larger than the others. We can also see that Argentina and Fiji appear very similar, which was not apparent in previous plots.

Figure 11.4 uses the same data summaries as Figure 11.2 (Table 11.1), and similar mappings: each variable is encoded as a different horizontal **position** and the values within each variable are encoded as vertical **position** and hemisphere is mapped to **colour**. The data symbol is also similar: multiple data values are mapped to a single data symbol, in this case a polygon.

The difference in Figure 11.4 is that, rather than having all of the data symbols on the same plot, each data symbol is drawn in its own **position in space**. This makes it easy to view each distinct data symbol.

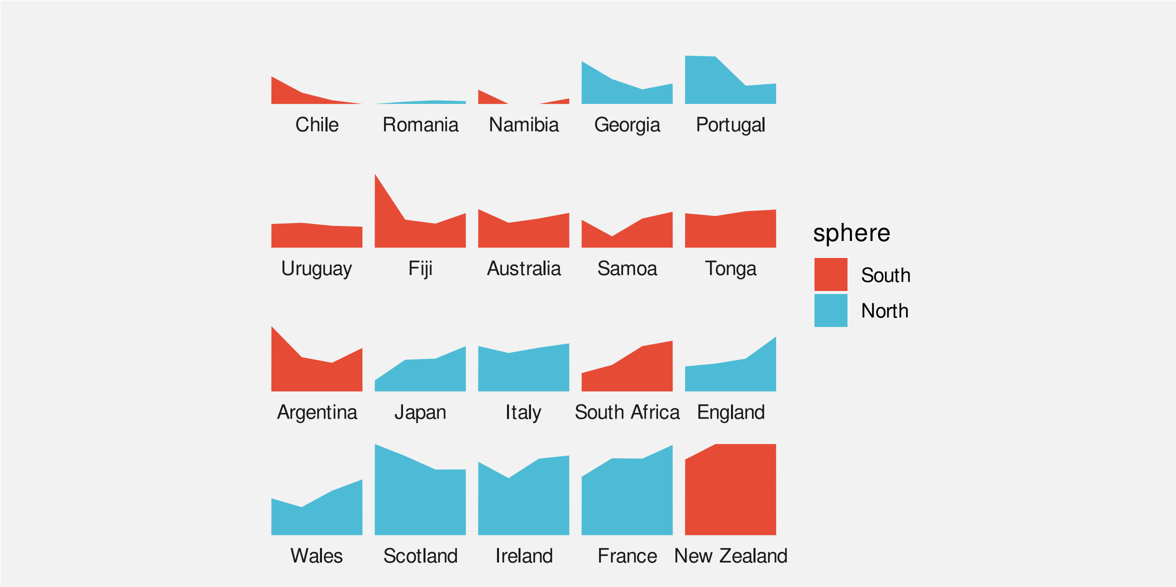


Figure 11.4: A data visualisation of the performance measures for teams in the 2023 Rugby World Cup.

The cluster of four larger shapes in Figure 11.4 is also easier to see because the countries have been ordered by the number of points scored (the right-hand height of the shape). As we have seen before, ordering data symbols can make a big difference (Section 5.1).

11.4 Facetting

TODO refer back to enclosure in visual-model.Rmd

Figure 11.5 shows a line plot of the number of youth offenders per year in different Police Districts of New Zealand. This visualisation makes use of several effective mappings. For example, we encode year as horizontal **position** and we encode count as vertical **position**. However, there is also a weaker mapping: the Police District. We encode districts as **colour**, but because there are 12 different districts, we are exceeding the capacity of colour (Section 3.6). It is not easy to differentiate all of the different colours (not helped by overplotting of the lines).

A better mapping for Police District would be **position**, which has a much higher capacity, but position has already been taken by year and count.

Figure 11.6 shows one solution: a facetted plot.[135] Each district has been encoded as a **position in space** and we draw a separate plot for each district. This makes it very easy to distinguish the individual districts.

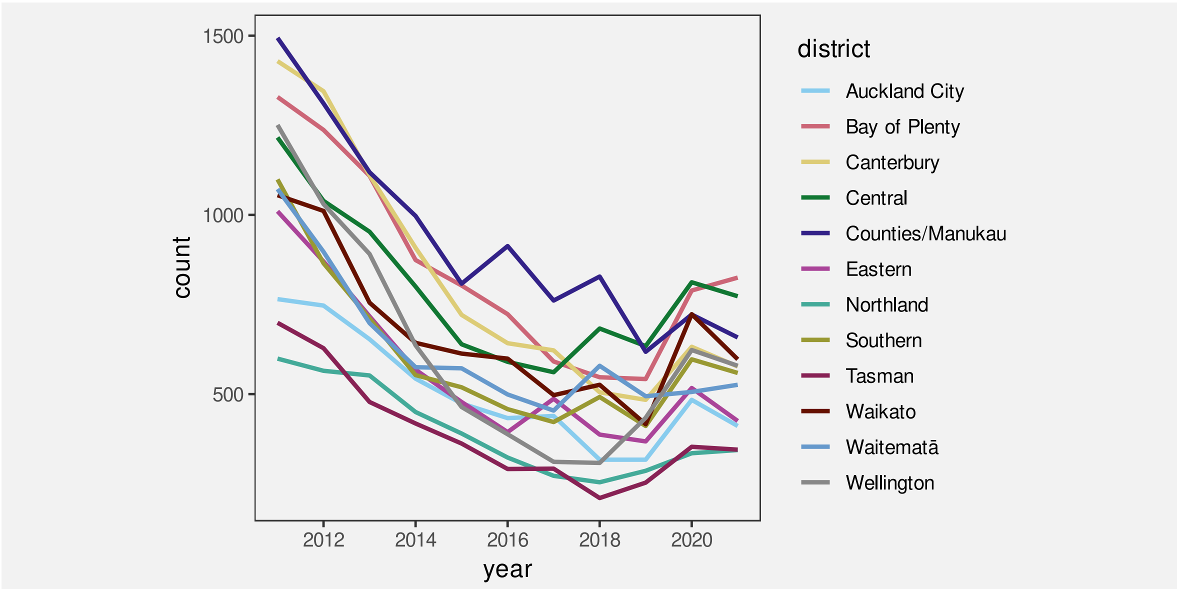


Figure 11.5: A data visualisation of the number of offenders per year in different Police Districts of New Zealand, from 2011 to 2021.

One downside is that comparisons between districts are more difficult because counts and/or years are **unaligned** (Section 4.2). In this case, we have mitigated that problem by drawing a grey line for all districts in each plot. Similar to grid lines, this makes it easier to compare the line for each district to other districts (Section 5.4).

We have again ordered the panels so that districts are near to other districts with a similar starting count of offenders (Section 5.1).

11.5 Case study: Scatterplot matrices

Figure 11.7 shows a scatter plot matrix of the performance measures for teams in the 2023 Rugby World Cup (Table 9.1).

This is another example of reusing **position** to encode data values. Each combination of variables is encoded as a separate scatter plot at a different position in space and, within each scatter plot, two variables are encoded as horizontal and vertical position.

Within each scatter plot, there is an emergent feature of position in space, which allows us to decode the correlation between different pairs of variables. Although this only allows us to see pairwise correlations between variables, we can also get a glimpse of higher-level correlations across several of the scatterplots.

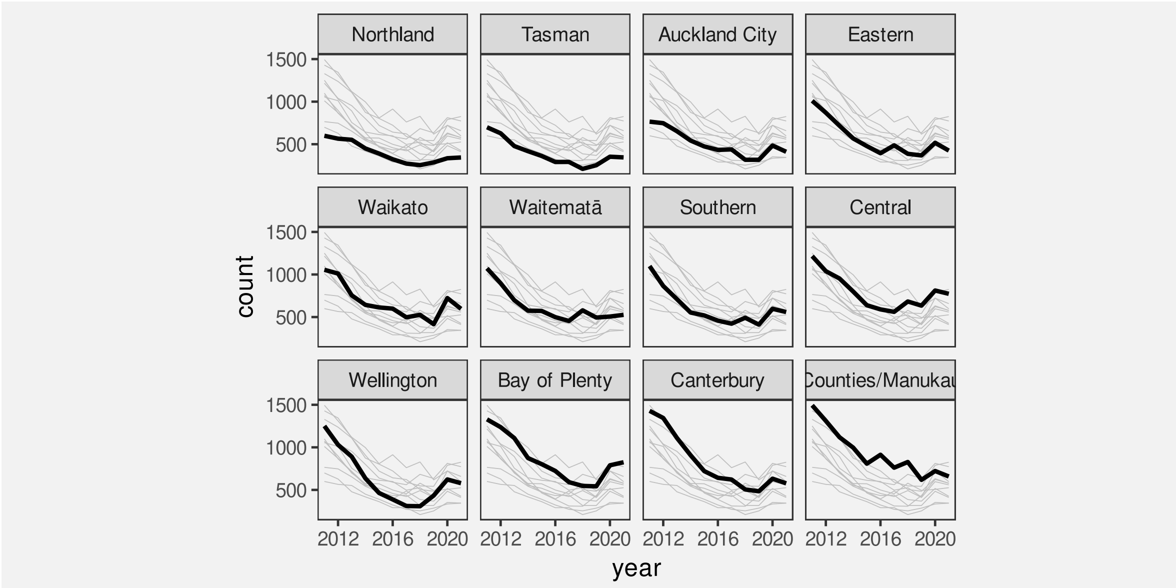


Figure 11.6: A data visualisation of the number of offenders per year in different Police Districts of New Zealand, from 2011 to 2021.

For example, looking down the diagonal of the matrix, we can see that: the number of runs is correlated loosely with the number of clean breaks; the number of clean breaks is more strongly correlated with the number of tries; and the number of tries is very strongly correlated with the number of points.

If we look down the first column of the matrix, we can see that: the number of runs is correlated loosely with the number of clean breaks, a little more loosely with the number of tries and a little more loosely still with the number of points.

11.6 Non-cartesian mappings

Parallel plots is actually an example of this, so can just refer back to Section 11.1

Refer back to Section 4.5

11.7 Case study: kiviat plots or radar plots or spider plots

(David Peebles (2008))

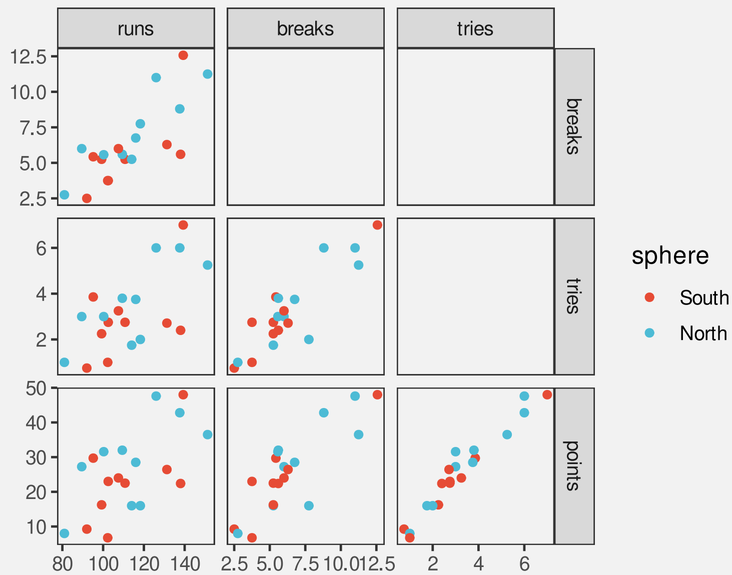


Figure 11.7: A scatter plot matrix of the quantitative performance measures for teams in the 2023 Rugby World Cup.

11.8 Case study: 2D density plots

Stat summary visualised

Compare with visual summary of density of points

Different pros for each approach

11.9 Dangers of multiple mappings

Figure 11.1 showed that the naive approach of just using a separate visual feature for each data variable leads to complex data visualisations that are difficult to decode. Although there are better options, like small multiple and facetting, there is still a limit to how much information can effectively be displayed at once within a static data visualisation.

This is one area where dynamic and interactive graphics can be useful because the viewer is able to rapidly switch between multiple views of the data, which means that each individual view can be simpler.[136]

11.10 Summary

When we need to visualise multiple variables at once, it is not effective to map each different variable to a different visual feature.

An alternative is to reuse the same visual feature for multiple different variables. In particular, small multiples, or facetting, allows us to use **position** for more than just two variables.

Notes

[132] "GoFish: A Grammar of More Graphics!" Pollock and Satyanarayan (2025) Cite this as a grammar that allows more explicit reuse of position (via "geometric operators"). NOTE that the stacking of bars in a stacked bar plot is also a simple version of the reuse of position.

[133] Acknowledge and point to different ways that the data values can be transformed (or summarised).

[134] Tufte's small multiples

[135] trellis multi-panels, ggplot2 facets, GoG ??

[136] Cite again the classic interactive lit ? Point to some tools that might help, like GGobi?

12 Visual Objects

Figure 12.1 shows a scatter plot of the number of clean breaks and the number of tries for countries at the 2023 Rugby World Cup (Table 9.1). This data visualisation is similar to Figure 9.4, but it adds an extra mapping. Country names are encoded as the **shape** of the data symbols. This is a reasonable mapping, despite the large number of different countries, because each different shape only appears once. We are able to distinguish all countries from each other and, thanks to the legend, we can identify individual countries.

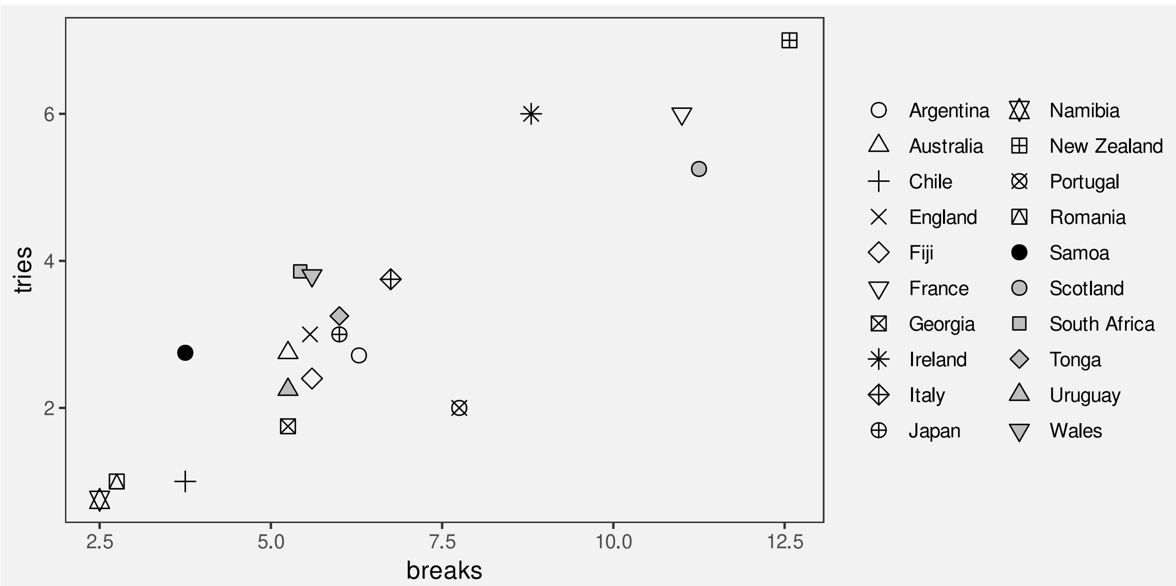


Figure 12.1: A scatter plot with a different shape per country

Figure 12.2 shows another scatter plot of the number of clean breaks and the number of tries for countries at the 2023 Rugby World Cup, but this time with flags as the data symbols. In one sense, this is just a variation on Figure 12.1 because the country names are encoded as the **shape** of a data symbol. Each data symbol is a shape that allows us to distinguish between the different countries.

The difference between the data symbols in Figure 12.1 and the data symbols on Figure 12.2 is that the flags are not only more complicated shapes, but they are familiar shapes. The flags convey differences, but they also convey the more sophisticated concept of the country names.

In the terms of Section 2.2, the flags are **visual objects** that might take longer to process, but convey more information.

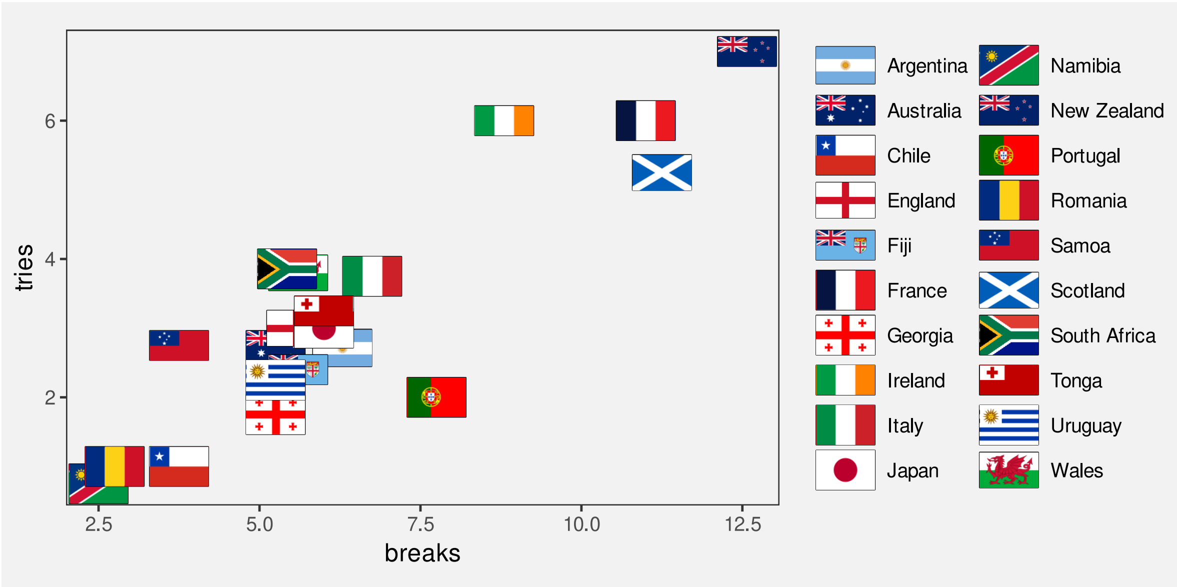


Figure 12.2: A scatter plot with flags as data symbols

12.1 Learned decodings

We **encode** data values as visual features so that the viewers of a data visualisation can **decode** the data values from the visual features (Figure 3.6).

In Section 7.1, we saw that some encodings are effective because there are implicit decodings. For example, a longer bar in a bar plot naturally corresponds to a larger quantity. Basic visual features, like length, are effective thanks to **pre-existing** decodings that are built-in features of our visual system.

Part of the power of **visual objects** also comes from pre-existing decodings. However, these decodings are **learned** rather than being built-in. In some cases, we are able to **decode** values from a visual feature based on prior experience and learning, rather than due to an implicit encoding (Figure 12.3).[137]

For example, Figure 12.4 shows three coloured circles. Without any explicit encoding, we can decode the red circle as "stop", the orange circle as "slow", and the green circle as "go". This is not a decoding that we are born with; it is a decoding that we learn through experience.

The flags in Figure 12.2 provide examples of learned decodings. We associate each flag with a particular country because we have seen these flags before and they have learned the decoding

Figure 12.3: Some visual features have learned encodings, which means that we can decode values from the visual features without having explicitly encoded values as visual features.

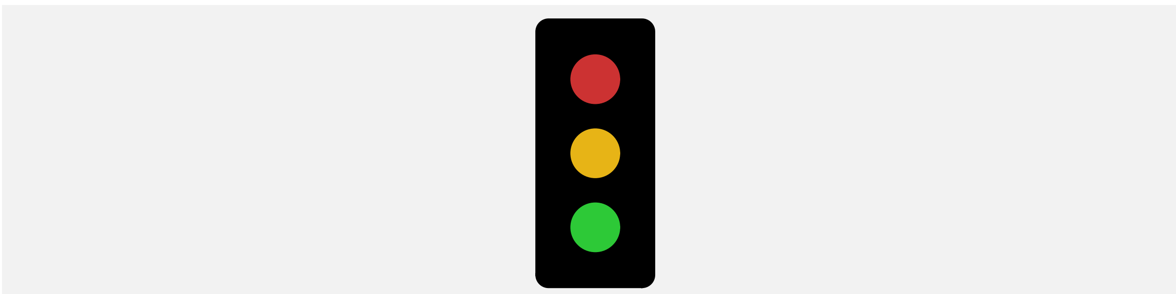


Figure 12.4: An example of visual symbols that have pre-existing, learned decodings.

from flag to country. Figure 12.5 demonstrates this point by repeating the scatter plot, but removing the legend. We are still able to identify not only that the data symbols represent different countries, but we can identify (at least some of) the country names as well.

Just to make the point completely clear, Figure 12.6 shows Figure 12.1 without a legend. Although we can still perceive that the data symbols are different from each other, there is nothing about the shape of the data symbols in Figure 12.6 that conveys the country names. We can clearly see that Figure 12.5 contains more information than Figure 12.6.

> A plot that uses visual objects as data symbols can be effective because the visual objects can convey information based on learned decodings.

Figure 12.5 also demonstrates a weakness of **visual objects**. The flags work well for identifying the country names *only if* we are familiar with all of the different flags. For example, it may only be Australians and New Zealanders who can reliably identify the difference between their two flags.

> A plot that uses visual objects as data symbols may *not* be effective if the learned decodings have not been learned by all viewers.

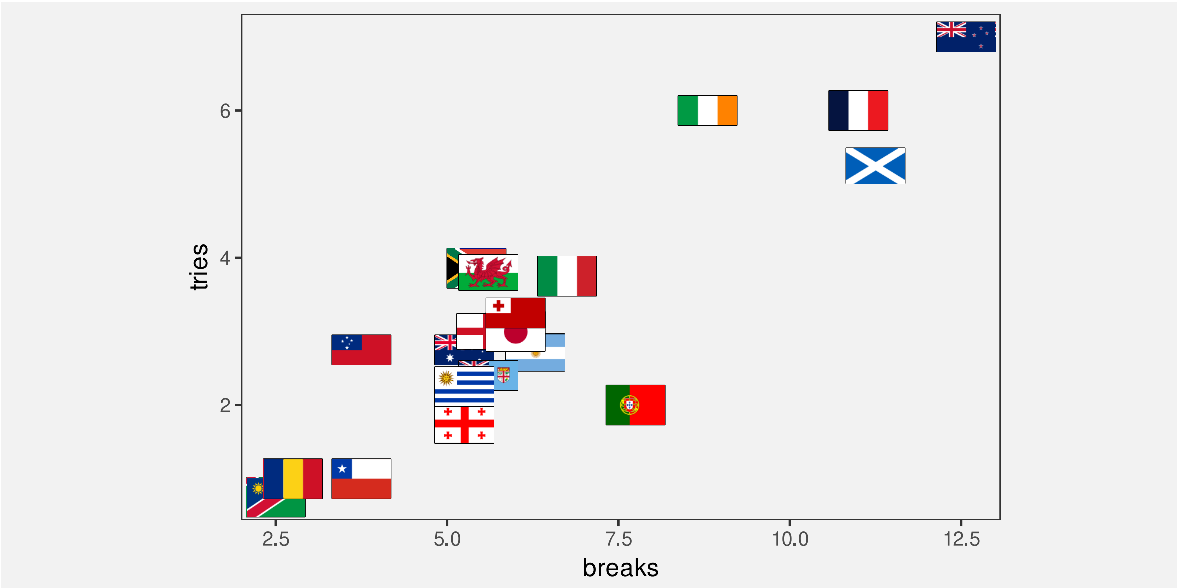


Figure 12.5: A scatter plot with flags as data symbols

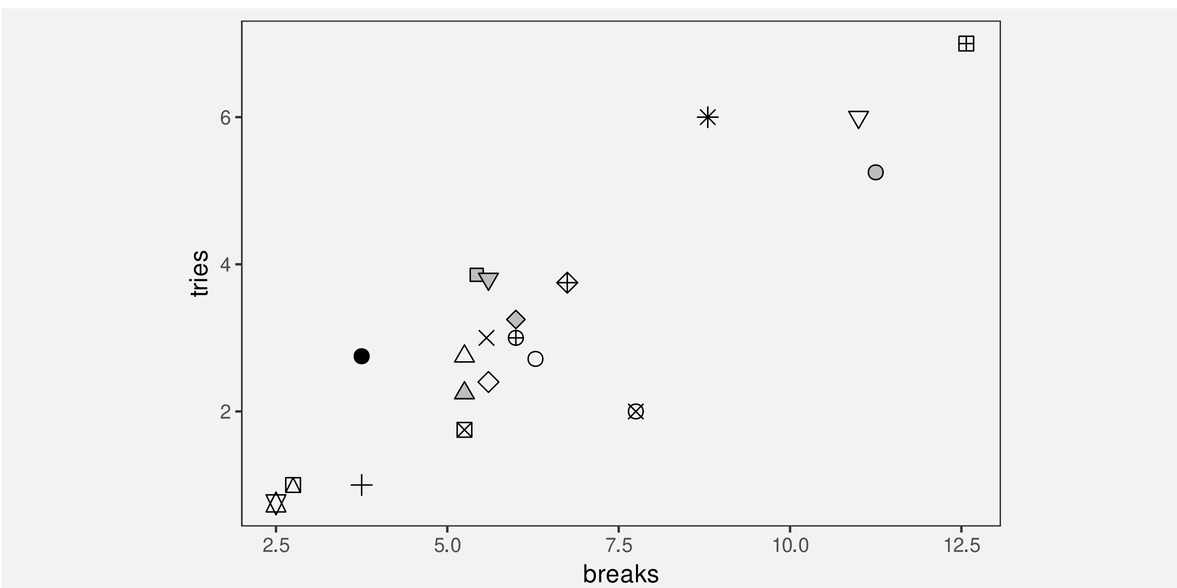


Figure 12.6: A scatter plot with a different shape per country

TODO Refer back to Section 4.1. Visual shapes are one way to decode to absolute data values (rather than just relative)

12.2 Case study: Choropleth maps

Figure 12.7 shows a choropleth map of the crime rate for 2021 in each police district of New Zealand.[138] The data values for this map are shown in Table 8.3.

The data symbol in Figure 12.7 is a polygon representing the geographic boundary of each police district. This implicitly entails an encoding of a police district as the **position** of a polygon. Crime rate is encoded as the fill colour of each polygon.

The use of polygons as data symbols is effective because the polygons form familiar shapes with familiar spatial arrangements (familiar to a New Zealand audience at least). The polygons are **visual objects** that convey the concepts of districts within New Zealand. The spatial arrangements are also a form of visual **congruence**; the arrangements mirror the real-world spatial relationships between the regions (see Chapter 7).

Figure 12.7: A choropleth map of the average crime rate over the period 2011 to 2021 for each police district.

> A choropleth map is effective because it maps geographic regions to familiar visual objects that resemble the real world regions.

A weakness of Figure 12.7 is that it encodes the **quantitative** crime rate as colour (chroma and/or luminance). This means that we can only decode ordinal comparisons between regions (see Section 6.2). For example, we can see that the highest crime rate occurs in the district to the east of the North Island, but it is difficult to perceive how much higher that crime rate is compared to the next highest crime rate.

> A choropleth map is *only* effective for making ordinal comparisons between geographic regions because it encodes quantitative values as colour.

Figure 12.8 shows a bar plot of the same data, which demonstrates the advantages and disadvantages of the choropleth map in Figure 12.7. In Figure 12.8, it is much easier to compare the crime rate values because crime rates are encoded as bar **length**. However, it is much harder to associate the different regions with their spatial arrangements because all we have are the region names. Each police district is encoded as the vertical position of a bar, with no correspondence to the spatial arrangement of the regions in the real world.

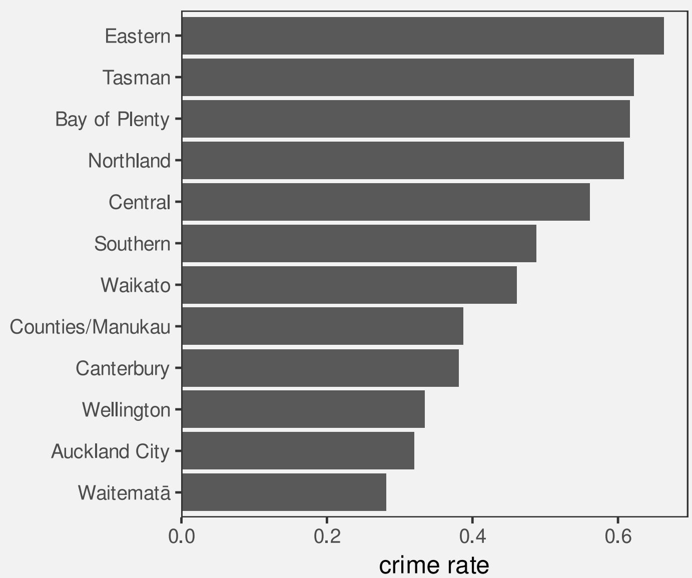
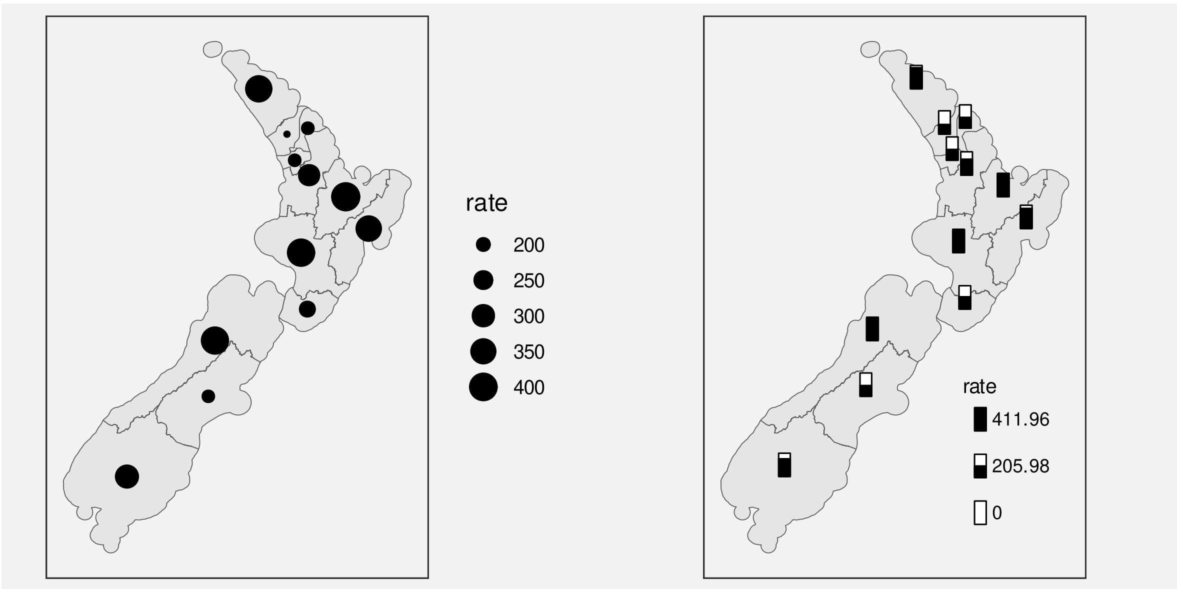


Figure 12.8

Another potential weakness with choropleth maps is the fact that regions differ in terms of **area**. We know from Section 7.3 that larger data symbols are associated with larger data values. The larger polygons accurately reflect the fact that some police districts are geographically larger than others, but there is potential for larger polygons to be erroneously decoded as larger crime rates.

Two alternative approaches are shown in Figure 12.9. These both retain the region shapes of the map, but map crime rates to the **area** of circles or the **length** of bars. These data symbols have a spatial arrangement that corresponds to the real-world arrangement of police districts and they provide a better decoding of crime rates from data symbols compared to the colours in the choropleth map (although area is still not the most accurate visual feature and the lengths are unaligned; see Section 3.5).



(a) Dots as data symbols.

(b) Bars as data symbols.

Figure 12.9: Alternative map-based representations of the average crime rate over the period 2011 to 2021 for each police district.

12.3 Chart junk and clutter

Figure 12.10 shows a version of Figure 12.7 with two cartoon villians added in the top-left and bottom-right corners.[139] These sorts of annotations are often, derisively, referred to as **chart junk**, with the suggestion that they add no value and make it harder to extract the important information.[140]

The cartoon villians are an example of **visual objects**. They (crudely) convey the concepts of crime and criminals. However, they are *not* **data symbols** because there is no mapping from the raw data to the cartoon villians. Their position is unrelated to the data, their colour is unrelated to the data, and their size is unrelated to the data. The best that can be said is that they relate to the **metadata**, the general topic of the data visualisation. Annotations like this are closer in purpose to **labels** (see Section 13.3).

In Section 2.5 we learned that we should keep things simple and only present visual changes that correspond to changes in the data. In Section 2.6 we learned that our attention is drawn

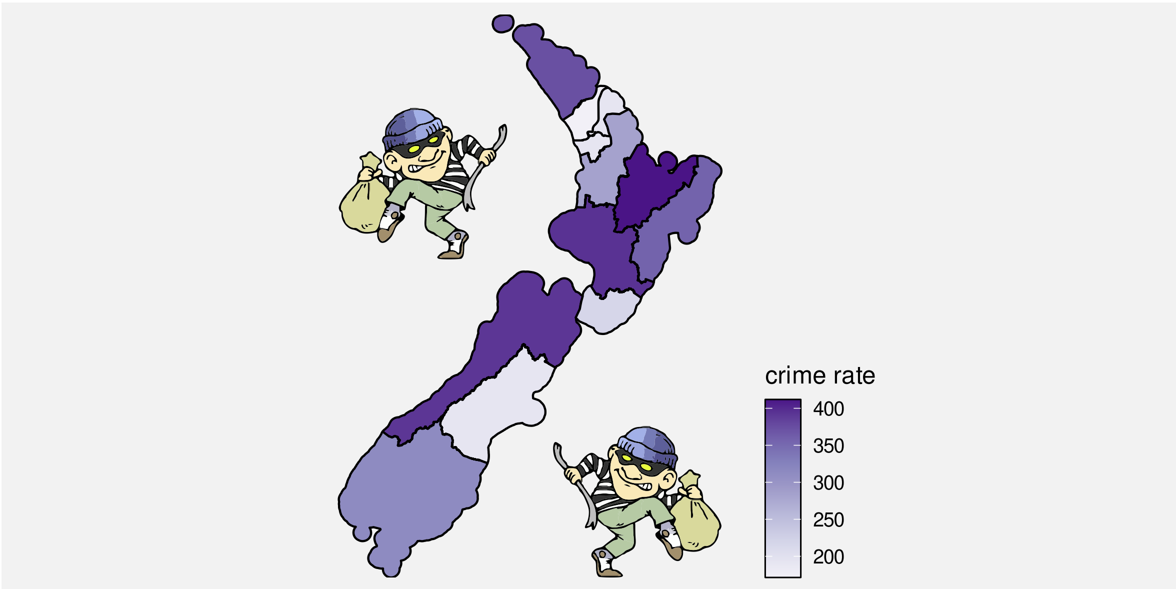


Figure 12.10: A choropleth map of the average crime rate over the period 2011 to 2021 for each police district (like Figure 12.7), with cartoon villian images added.

to the larger and brighter items within an image. The cartoon villians are relatively large, complex, and colourful and bear no connection to the raw data, so there is a good chance that they will distract from the visual features that do convey data values.

> Chart junk is *not* effective because it is impossible to decode data values from chart junk (no data values have been encoded) and because chart junk distracts from the visual elements that do represent data values.

However, it is important to point out that not all **visual objects** are chart junk. For example, the data symbols in Figure 12.10, the map regions, are also **visual objects**, but they differ from the cartoon villians because they are related to the data values.

Furthermore, **visual objects** that have no connection to the raw data may still have some positive effects. For example, the cartoon villians are not devoid of information; they convey the concept of crime more dramatically than the simple text label “crime rate”. The more complex and interesting images may attract attention to the data visualisation as a whole and there is some evidence that such annotations improve the memorability of a data visualisation.[141]

> Chart junk can be effective because it has a greater emotional impact and is more memorable.

Overall, visual objects that are not related to data values are *not* effective for decoding data values, so any other benefits have to be weighed against that cost.[142]

12.4 Case study: Pictograms

Figure 12.11 shows a pictogram of the number of pet dogs and pet cats globally.[143] The data symbols in this plot are images, or **icons**, of a cat and a dog, with the number of pets encoded as the height of each icon.[144] Icons are an example of **visual objects**; they are effective at conveying categories because they visually resemble physical objects, in this case, cats and dogs.

Unfortunately, because the cat and dog images are not simple shapes, it is harder to accurately perceive the heights of the images, so it is harder to decode the number of cats and dogs. Even worse, we actually perceive the size, or **area**, of the images. The dog image is wider than the cat image so the number of dogs is visually over-emphasised.

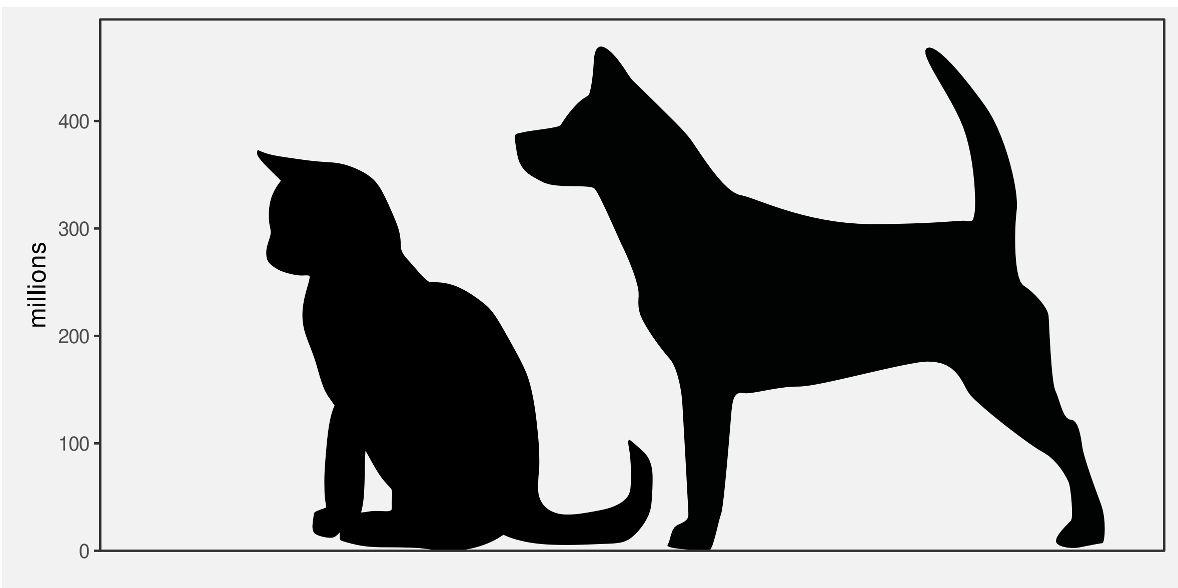
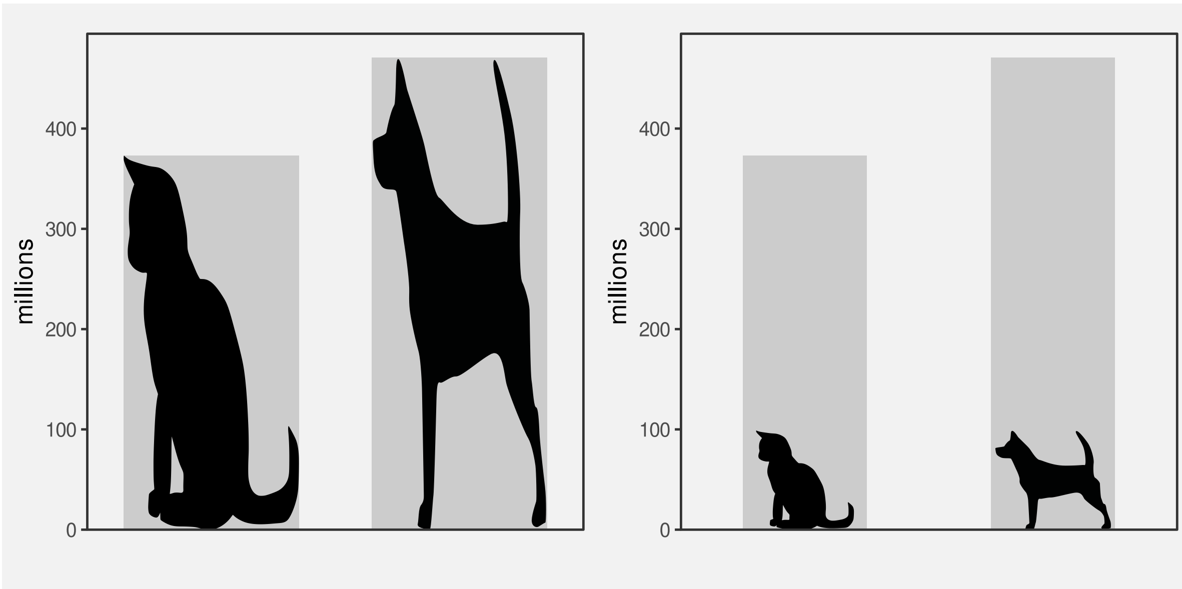


Figure 12.11: A data visualisation of the number of pet cats and pet dogs worldwide.

Figure 12.12 shows two alternative uses of icons that attempt to resolve some of the problems in Figure 12.11. In Figure 12.12a, the images are distorted so that they have the same width and so that only the height of the images changes. There is also a bar to help with the perception of **length**. While the bar makes it much easier to decode the number of pets, the distortion is not effective because it hinders the perception of the icon.

Figure 12.12b uses a bar to represent the number of pets and just has a small version of the icon at the base of each bar. We know that the bar is effective for decoding the number of pets and the small icons are effective as labels for the bars.



(a) distorted icons

(b) icon labels

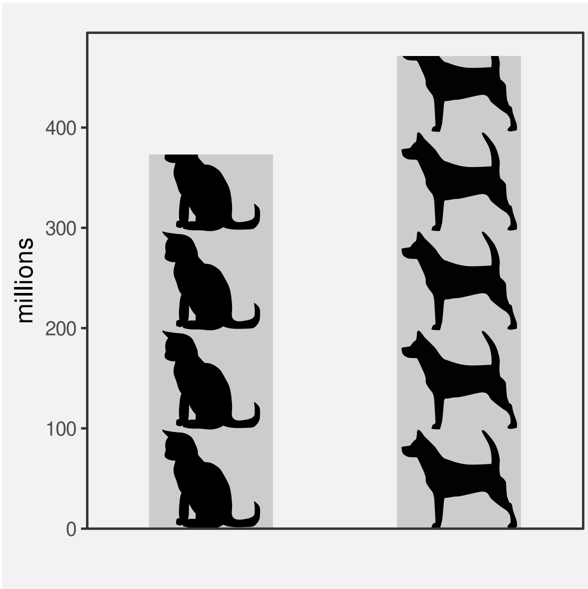
Figure 12.12: Two alternative uses of icons.

Figure 12.13 shows two more alternatives to Figure 12.11. Figure 12.13b is just a standard bar plot. We know that this is effective for decoding the number of pets.

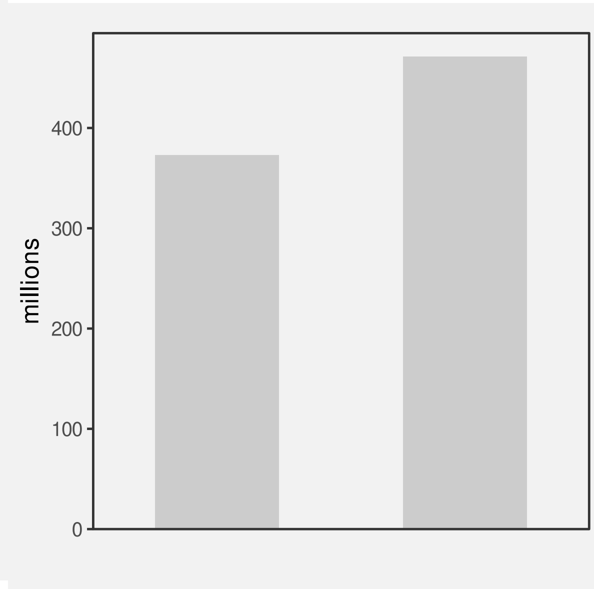
Figure 12.13a uses repetitions of the icons to represent the number of pets, still within a bar. In this case, each icon represents 100 million pets. The ability to count icons is effective for decoding the overall counts.[145]

12.5 Case study: Chernoff faces

Figure 12.14 visualises the 2023 Rugby World Cup data (Table 9.1) using Chernoff faces.[146] The data symbol in this visualisation is a little unusual; each country is represented by a combination of circles, curves, and lines that together produce a **visual object** that resembles a face. The mappings in this visualisation are also a little unusual. The number of points scored is encoded as the **curvature** of the smile, the number of tries is encoded as the **angle** of the eye brows, and the number of clean breaks is encoded as the horizontal **position** of the eyes.



(a) Icons are repeated to represent the number of pets.



(b) A basic bar plot

Figure 12.13: Two more alternative data visualisations of the pet data.

```
Warning: The `scale_name` argument of `continuous_scale()` is deprecated as of ggplot2
3.5.0.
```

The mappings in a Chernoff face are *not* very accurate (see Section 3.5; **curvature** is ranked *below* colour in terms of accuracy),[147] so Cheroff faces are *not* very effective for decoding raw data values.

However, the familiarity of Chernoff faces means that we are easily able to perceive common combinations of the facial features. For example, we can see several sad, worried faces at the top-left of Figure 12.14 and several happy, perhaps even smug, faces at the bottom-right. We can perceive countries that share similar combinations of performance measures. In other words, we can perceive multi-dimensional correlations and multi-dimensional groupings in the data.

In this case, the happier faces have been arranged to correspond to the more successful teams and the teams have been ordered from least points scored to most points scored. The resulting proximity helps with perceiving groups of similar faces.

> A Chernoff face is effective at conveying multi-dimensional relationships because the data symbol is a familiar visual object from which we can easily perceive simple human emotions.

Figure 12.14: A Chernoff faces plot of the ...

12.6 Dangers of learned mappings

Figure 12.15 provides a demonstration of the Stroop Effect.[148] The task is to name the colour of each word. This task is difficult to do because there are two decodings involved and the two decodings are inconsistent. Each set of letters has a learned decoding based on the **shape** of the letters—the word that the letters spell out—but there is also a learned decoding based on the actual **colour** of the letters—the colour names that we learn as children.

Figure 12.15: The Stroop effect. Try to name the *colour* of each word.

One issue with relying on learned decodings is that we may learn more than one association for the same visual feature. For example, the colour red is commonly associated with hot (versus blue for cold), but also with stop (versus green for go), plus also passion, or even anger.[149] Furthermore, many associations are specific to a particular culture. For example, red is associated with luck and prosperity in Chinese culture, among other things.[150]

Relying on a learned decoding can cause confusion if there are multiple possible decodings. It may not be clear which data value should be decoded from the visual feature (Figure 12.16).

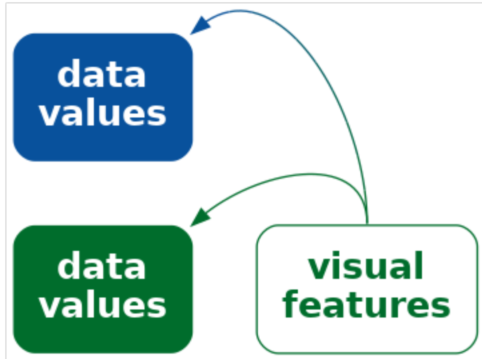


Figure 12.16: A visual feature may have more than one learned decoding, which means that we may decode more than one value from the visual feature.

The problem is much worse if the learned decoding(s) conflict with an explicit encoding of data values. For example, in Figure 12.2, if we use the incorrect flag for a country, for example, by swapping the Australian and New Zealand flags, we not only introduce confusion, but a misleading decoding. This is similar to the problem of dissonant mappings (Section 7.5 and Figure 7.11).

Suppose that we made a similar change to Figure 12.1, swapping the open triangle for Australia and the square-with-a-plus symbol for New Zealand. That would not cause any problem because there is no learned decoding to conflict with. We only get the problem if we swap flags in Figure 12.2 because of the conflict with learned decodings.

> Employing a learned decoding incorrectly can at best weaken and at worst destroy the effectiveness of a data visualisation.

12.7 Dangers of visual objects

One issue with more complex data visualisations is that their effectiveness depends on a larger amount of learning and experience. If the viewer does not have the knowledge required to decode a visual object, then the data visualisation will fail. We saw an example of this in Figure 12.5, where the familiarity of different country flags will vary with nationality and age, for example.

An extreme version of this problem is the use of not just complex or unusual data symbols, but entirely novel data visualisations.[151]

Andrews curve of RWCperGame statistics (analogue of Figure 11.2). **Cite Andrews article.** Can see the three distinct groups of countries more easily in this than in parallel plot, but hard to understand exactly what you are seeing. Plus NO chance of recovering original data values (another example of visualising data summary).

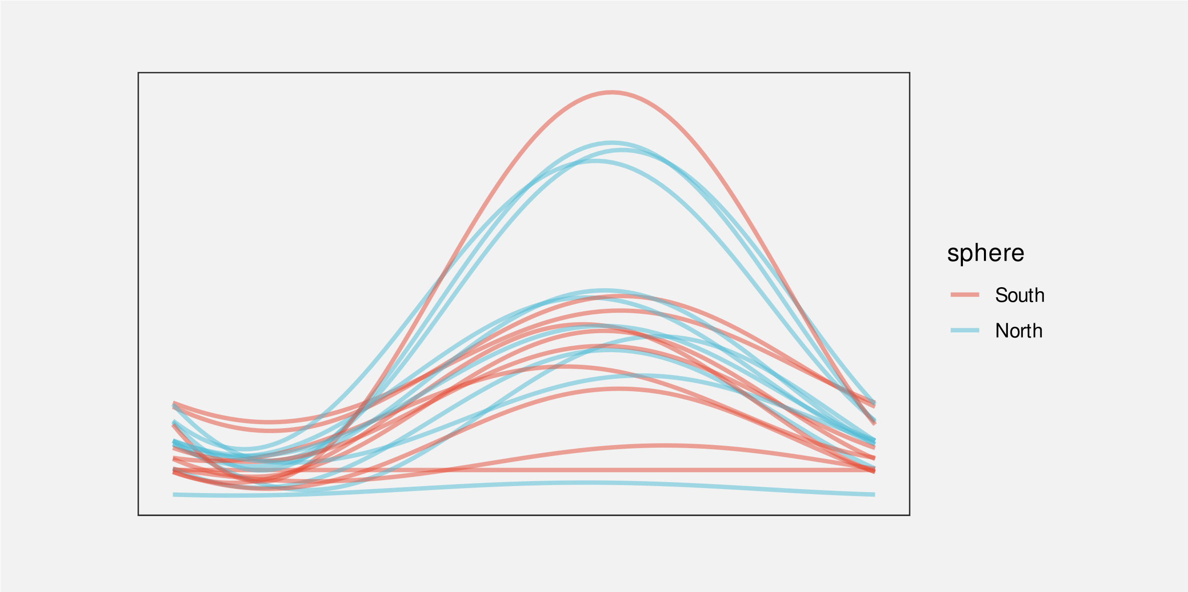


Figure 12.17: An Andrews plot of the performance measures for teams in the 2023 Rugby World Cup.

Another example of this problem are the hurrican “cones of uncertainty” that Cairo highlights in Chapter 5 of How Charts Lie (footnote!).

Mention that little is known about the accuracy of visual objects (though they must have good capacity because they are more abstract, right?)[152]:

Refer back to Section 4.6 as example of plot type that experts in specific fields learn to read.

12.8 Summary

If we encode data values as familiar visual objects, it becomes possible to decode more complex concepts based on prior learning and experience.

This may obviate the need for legends.

Notes

[137] "Mapping Color to Meaning in Colormap Data Visualizations" Schloss et al (2018) They call this an "inferred mapping"

"Affective Congruence in Visualization Design: Influences on Reading Categorical Maps" Anderson and Robinson (2021)

[138] Add citations to MacEachren and Bertin and Dan Carr ?

[139] Attribution for the original public domain image.

[140] Mention Tufte (data-ink ratio), Stephen Few? see DataVis/README for links to the debate.

- "Declutter and Focus: Empirically Evaluating Design Guidelines for Effective Data Communication" Ajani et al (2021) says that "The visualization practitioner community prescribes two popular guidelines for creating clear and efficient visualizations: declutter and focus."

[141] Links to studies that try to show benefits of chart junk. Rahman et al (2024) Mention affect (emotion) study?

"Chart and Graphs" Karsten (1925)

This is the source of the "insult to man's intelligence" remark about pie charts, though note the full quote:

"In a sense, it might be construed as an insult to a man's intelligence to show him a pie-chart, but the insult is not often resented. For if your main object be to get a story across, you are justified in taking that means which will encounter least resistance, and in making your story as simple as possible. For publicity purposes, the pie-chart is therefore almost invariably better than the bar." [p98]

[142] There is a really nice summary of when chart junk might be acceptable somewhere in DataVis/README

[143] reference to Statista https://www.statista.com/statistics/1044386/dog-and-cat-pet-population-worldwide/

[144] different terminology: icons, glyphs, pictograms Vehement opposition to these from some quarters

[145] cite the study that reckons using icons can help with perceiving counts. More effective when each icon represents 1 (and so there are no fractional icons) ?

[146] Chernoff (1973)

[147] Link to a resource that has a comprehensive list of visual channels and includes curvature.

[148] citation please!

[149] Cite "Sometimes more is more: iterative participatory design of infographics for engagement of community members with varying levels of health literacy" Arcia et al (2016) where people interpret icons in multiple ways, e.g. specifically "pineapple" rather than generic "fruit".

Also cite something about cultural differences in perception. e.g., "More of what? Dissociating effects of conceptual and numeric mappings on interpreting colormap data visualizations" Soto et al (2023)

"Review of Graph Comprehension Research: Implications for Instruction" Shah and Hoeffner (2002) Give example of ambiguous colour association "green means profitable for financial managers but infected for health care workers (Brockmann, 1991)"

[150] "A Study on the Metaphor of "Red" in Chinese Culture" Qiang (2011)

[151] Kosslyn's Principle of appropriate knowledge ?

Could mention Pinker's (1990) "graph schema" (prior knowledge about a graph type).

Shah and Hoeffner (2002b) provides several examples of young students misinterpreting the message from line plots, which suggests that we need to *learn* how to make effective use of even basic data visualisation formats.

"Bars and lines: A study of graphic communication" Zacks and Tversky (1999) speculates on the development of graphical conventions.

[152] in general, there needs to be more explicit acknowledgement of the lack of experimental results for complex data vis scenarios. c.f., "Testing Statistical Charts: What Makes a Good Graph?" Vanderplas, Cook, and Hofmann (2020) p 71.

13 Text

Figure 13.1 is a version of Figure 1.1 with all of the text removed. Although we can see that some values, or rather three sets of values, haved decreased, we cannot tell what the values are or by how much they have decreased. If there was ever any doubt, Figure 13.1 makes it clear that text is doing a lot of important work in a data visualisation.

TODO Refer back to Section 4.1. Text is one way to decode to absolute data values (rather than just relative)

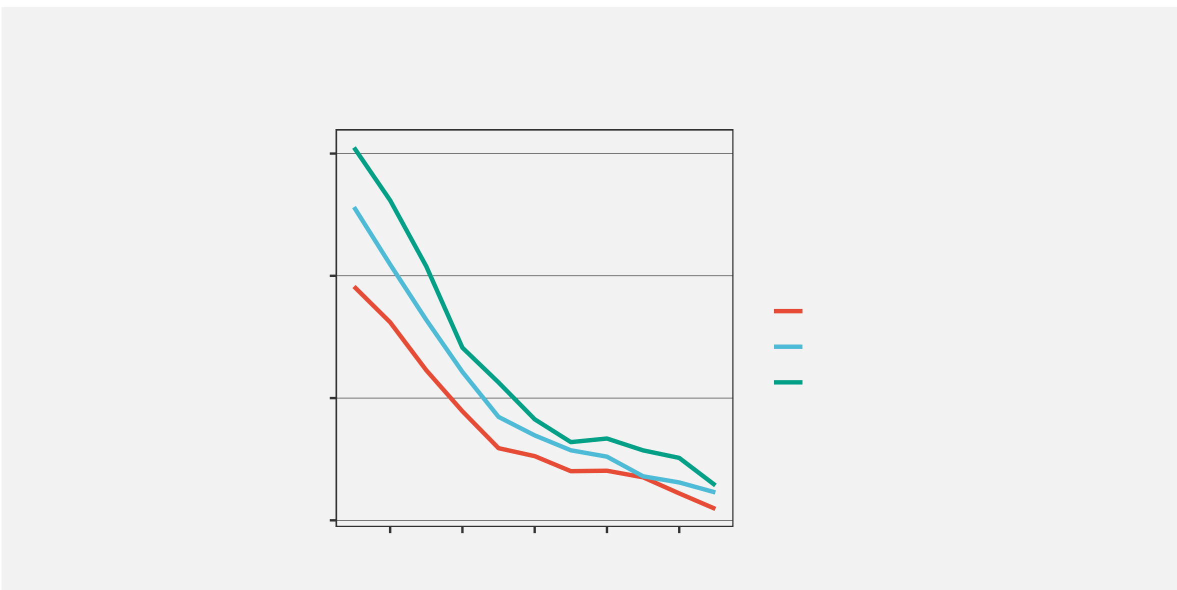
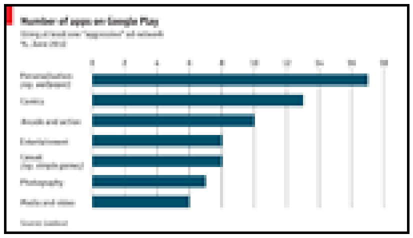


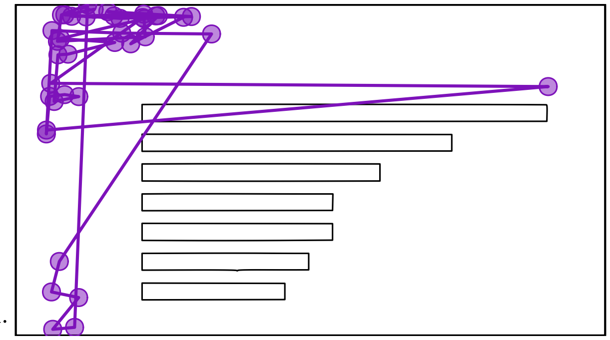
Figure 13.1: A line plot with all text removed. Without text, it is impossible to know what data values are represented in this plot.

Eye-tracking studies also show that people spend a lot of time looking at the text within a data visualisation. For example, in Figure 13.2, there are more fixations in the regions that contain text than anywhere else. There is also evidence that text elements are more memorable than the other elements of a data visualisation.[153]

This chapter looks at the role that text plays in making data visualisations work.[154]



(a) A thumbnail of the original data visualisation.



(b) Fixations and saccades from eye tracking data.

Figure 13.2: Eye tracking data from the MASSVIS project that shows fixations (circles) and saccades (lines between circles) for one person viewing an image for 10 seconds.

13.1 Data symbols

Figure 13.3 is a version of Figure 3.2 with white text added to each bar showing the exact number of offenders in each ethnic group. This is an example of using text as a **data symbol**.

Just as data values are mapped to the visual features of the bar data symbols in Figure 13.3, data values are also mapped to the visual features of the text data symbols. For example, the ethnic groups are encoded as the vertical **position** of the bars *and* as the vertical **position** of the text. The data values for the number of offenders in each ethnic group are encoded as the **length** of the bars *and* as the horizontal **position** of the text. Furthermore, the data values for the number of offenders in each ethnic group are encoded as the **shape** of the text—the letters and digits that make up the text.

Just as we can decode data values from the visual features of the bars in Figure 13.3, we can decode data values from the visual features of the text. For example, we can see just from the horizontal position of the text, without reading the text values, that some counts are (much) larger than others.

> Encoding data values as the position, colour, and size of text data symbols is effective for the same reasons that these visual features are effective for other types of data symbols.

We see examples of the use of simple visual features in text in almost any piece of writing. For example, section headings are typically larger and bolder than the main text in a report and the Executive Summary in this book makes use of colour and angle to create visual connections between associated terms within a bullet-point list.

However, the unique power of text comes from **decoding** the **shape** of the text. Text is the ultimate example of a **learned mapping** (Section 12.1). We all learn to read text (and

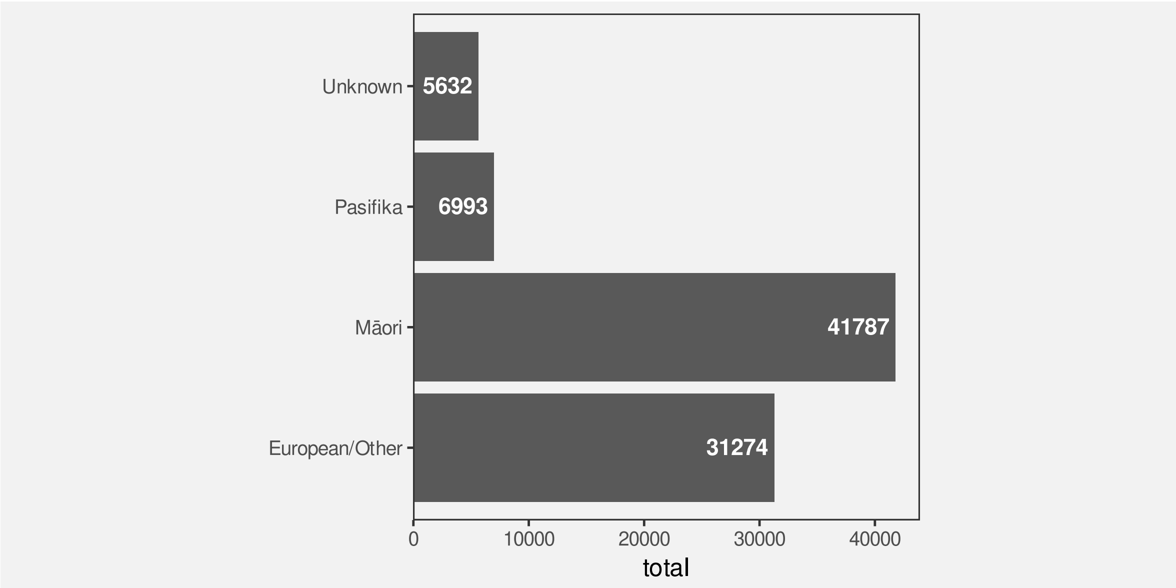


Figure 13.3: A bar plot of the total number of offenders for different ethnic groups. The data symbols in this plot are the horizontal bars *and* the text values at the end of each bar.

numbers), which means that, when we are presented with a text shape, there is a pre-existing decoding to the semantic content of the text (Figure 12.3). For example, in Figure 13.3, we are able to decode the data value `41787` (the count for Māori) from the text "41787". Furthermore, this decoding is extremely accurate and very flexible.

Figure 13.4 shows another example of text data symbols

In terms of the criteria that we discussed in Chapter 3, the value of text data symbols is that they allow us to decode data values very **accurately** (Figure 2.1). Text is also very flexible because it is appropriate for both **quantitative** and **qualitative** data values. Finally, text has a very large **capacity**; we are able to distinguish between many different words and numbers.

> Text data symbols are effective because they can be used to represent any sort of data and they allow a very accurate decoding of data values.

Compared to the bar data symbols in Figure 13.3, the text data symbols are much more complex shapes. In terms of the simple visual processing model from Section 2.2, the text data symbols are similar to **visual objects**. This means that text can convey more sophisticated information, at the expense of more cognitive effort, and with a limit on how many shapes we can process simultaneously (see Figure 2.4).[155]

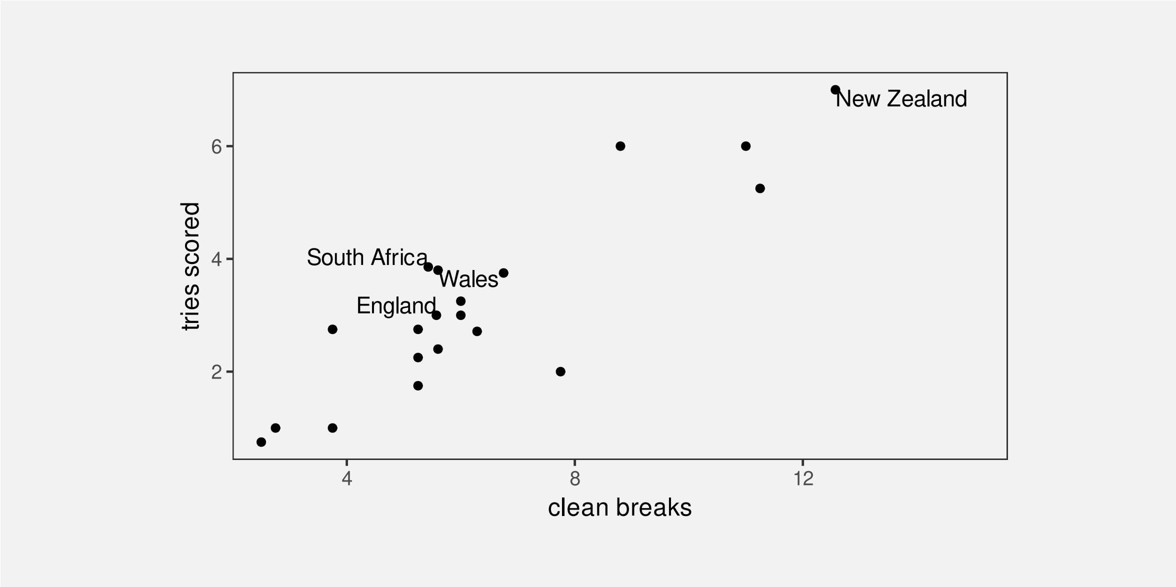


Figure 13.4: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

One weakness of text data symbols is that comparing values (rather than just decoding a single value) requires more cognitive effort compared to comparing simple visual features (like the lengths of bars). For example, in Figure 3.2, it is quick and easy to approximate the ratio of Pasifika to Māori or European/Other to Māori from the lengths of the bars, but those comparisons are slower and take more conscious effort using the text data symbols in Figure 13.3.

> It is harder to compare data values that are mapped to text because it requires more effort to decode data values and to perform mental calculations.

In addition, our visual system cannot generate **visual summaries** of large amounts of text (Section 8.2). This is why a table of numbers, like Table 1.1, is ineffective for perceiving groups of values or trends in values.

> Large numbers of text data symbols are *not* effective because the visual system cannot decode visual summaries from the text.

Why is a picture is worth a thousand words? Because 1000 words is 1000 fixations, identifying 1000 visual objects, and then interpreting their combined meaning.

13.2 Guides

Figure 13.5 is a version of Figure 1.1 with the guides highlighted in purple. There are tick marks and tick labels on the x-axis and y-axis, there are horizontal grid lines, and there are legend elements to show the mapping from the different ages to different colours.

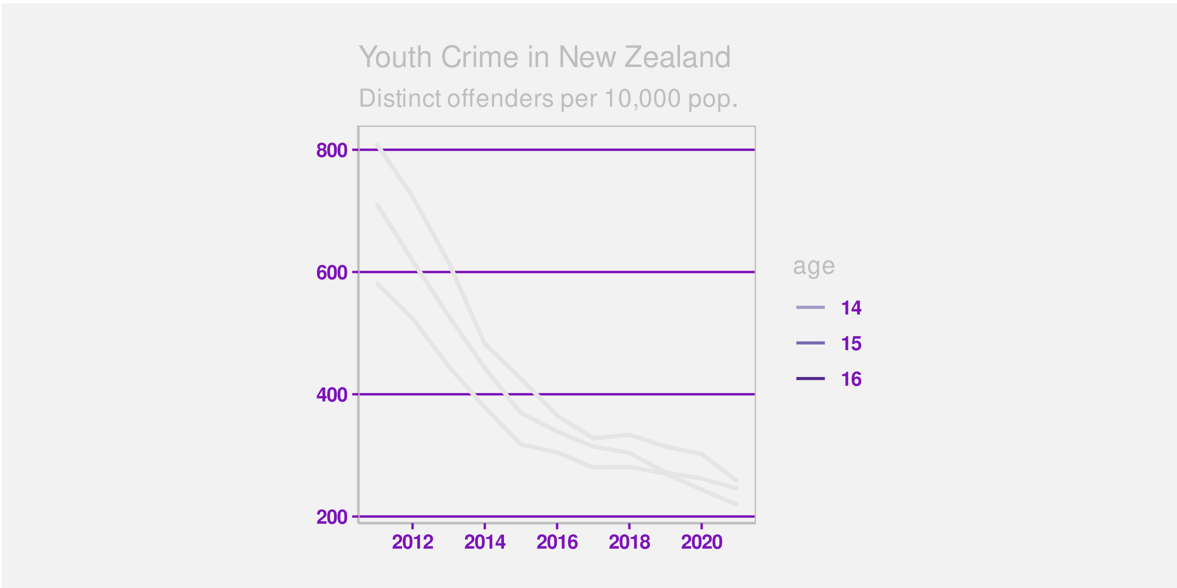


Figure 13.5: A line plot of the crime rate amongst youth offenders aged 14 to 16 from 2011 to 2021. The guides in this plot have been highlighted in purple.

The first thing to note about the guides in Figure 13.5 is that they are all composed of **data symbols**, with **data values** mapped to **visual features** of the data symbols (Chapter 3). This is perhaps easiest to see with the grid lines. Each grid line is a data symbol where a data value, a rate of offending, has been encoded as the vertical **position** of the line. Similarly, each tick mark on the y-axis is a line data symbol where a rate of offending has been encoded as the vertical **position** of the line and each tick mark on the x-axis is a line data symbol where a year has been encoded as the horizontal **position** of the line. Finally, each of the tick labels on the y-axis is a text data symbol where a rate of offending has been encoded as the vertical **position** of the text *and* as the **shape** of the text. Similar mappings apply to the x-axis tick labels. The legend is also made up of line data symbols and text data symbols, this time with age data values encoded as the **colour** of lines and the **shape** of text.

The **decoding** of the **shape** of the **text** data symbols within the guides shows the critical role that text plays within guides. If we look again at Figure 13.1, there are coloured lines on the plot and there are coloured lines on the legend. The fact that these lines have different colours indicates that there are three different groups, but the colours alone do not tell us what the groups are. We can decode different values from the colours of the lines, but we cannot decode exact data values. In the original Figure 1.1, the shapes of the text labels on the legend allow us to decode the original data values—the three ages 14, 15, and 16.

TODO text is typically the **only** data symbol within a data visualisation that can be decoded to an **individual**, **absolute** data value.
Most other decoding is relative (e.g., Section 4.1, Section 5.4, Section 6.3) or to a data summary (e.g., Section 10.1)

The proximity of the text labels in the legend to the coloured lines in the legend means that we associate each label with each coloured line (Section 2.7). So the text labels in the legend are the essential data symbols in Figure 1.1 because only they allow us to decode from the colours of the lines to the three age groups.

The text labels on axes perform a similar service. For example, the numbers on the axes in Figure 1.1 are the only data symbols that provide exact decoding from positions to data values (years or rates). The lines within the plot, by themselves, only show that values are decreasing. It is only the relative position of the lines within the plot to the tick labels on the axes (and the grid lines) that allows us to decode from the lines within the plot to (approximate) data values.

Text data symbols within guides are effective because guides contain only a few data values that need to be decoded precisely. This is the situation where text excels.

> Guides are effective because they encode known data values as text data symbols as well as the data symbols used in the main plot.

13.3 Labels

TODO Refer back to Figure 4.9 from text. The labels are the confusing part on the log plot

Figure 13.6 shows a version of Figure 1.1 with the labels highlighted in purple. In this data visualisation, the only labels are the overall plot title and the legend title. It is also common to see axis titles.

The difference between labels and the other components of a data visualisation, the data symbols and the guides, is that labels relate to **metadata** rather than raw data values. We can still consider this a mapping, just one that involves encoding metadata rather than data values (Figure 13.7).

Labels contain information like descriptions of variables and units of measurement and references to data sources. These are all important pieces of information, but they are more complex, more abstract, and far fewer in number than data values. Fortunately, text is very capable of expressing complex information and works very well in small amounts, so it is ideal for this role.

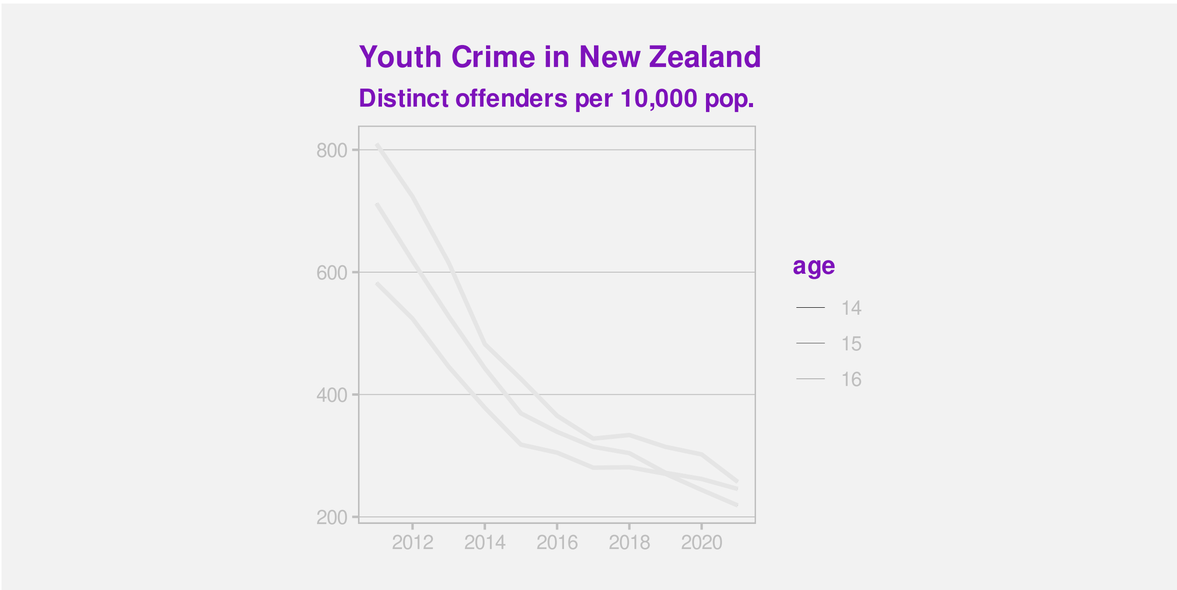


Figure 13.6: A line plot of the crime rate amongst youth offenders aged 14 to 16 from 2011 to 2021. The labels in this plot have been highlighted in purple.

Figure 13.7: Labels can be viewed as an encoding of metadata as text. We can encode complex information using text and we can decode text with extremely high fidelity.

To emphasise that point, imagine how impossible it would be to encode the information in the title of Figure 13.6, or just the legend title, only using simple geometric data symbols like bars and points. Using pictograms, which have learned decodings like text, would also be very difficult because it is much harder to compose pictograms in the way that text can be composed into phrases and sentences.

> Text is the only way to encode the type of complex and abstract information that is required for titles and axis labels.

Another use of text in a data visualisation is to label specific points of interest. For example, Figure 13.8 shows a variation of Figure 9.4, with text added to help explain the result of the Rugby World Cup final to disillusioned New Zealanders. This demonstrates further the value of text as the only way to encode complex and nuanced information.[156]

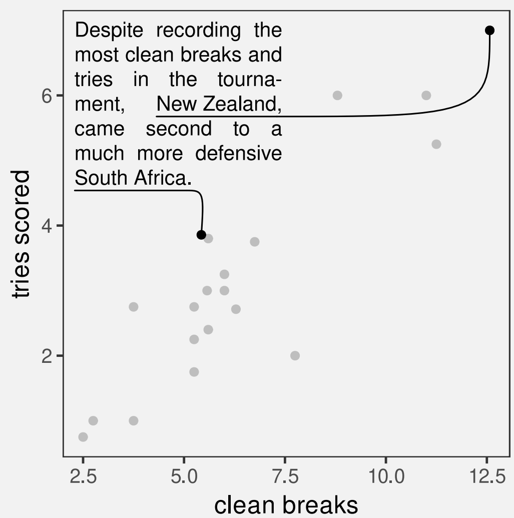


Figure 13.8: The number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams in the 2023 Rugby World Cup.

Figure 13.8 also demonstrates that text is an effective way of directing attention (Section 2.6). Text labels can explicitly describe data values or, more usefully, data summaries that the data visualisation is designed to show.

REFER to Section 2.6.

13.4 Case study: Word clouds

Figure 13.9 shows a word cloud data visualisation based on the text of the Wikipedia page on the Rugby World Cup.[157] This visualisation shows the frequency with which different words appear on the Wikipedia page. The data values that are being visualised are shown in Table 13.1.

Figure 13.9: A word cloud of the frequency of words in the Wikipedia page for the Rugby World Cup.

This data visualisation is unusual because it consists entirely of text **data symbols**. The mappings in Figure 13.9 include encoding each word as the **shape** of a text data symbol and encoding frequency as the **size** of a text data symbol. There is no need for a guide for the mapping from words to text shape because there is a learned decoding (Figure 12.3); we can decode each text data symbol into a word data value because that decoding has already been learned.

There is no guide for the mapping from word frequency to text size, so we cannot decode exact word frequencies. Table 13.1 shows the six most frequent words. However, the visualisation is still effective at conveying the *relative* frequencies. The larger words occurred more frequently. For example, we can see the names of the teams that were most successful at the tournament, like New Zealand and South Africa, are much larger than less successful teams.

Table 13.1: A table of the number of times different words occur in the Wikipedia page for the Rugby World Cup (for words that appear more than once). There are 287 different words, but only the first 6 are shown here.

word	freq
rugby	72
world	66
tournament	56
cup	54
new	41
zealand	38

The encoding of frequencies as the **size** of the words does not provide the best **accuracy** for decoding frequencies (Section 3.5) and this made worse by the fact that longer words become even larger. This means that we cannot identify the exact frequency of any one word, we cannot identify the quantitative difference in frequency between pairs of words, and we cannot easily perceive the distribution of frequencies across words. A word cloud is only effective for approximate ordinal comparisons of word frequencies (Chapter 3).

Frequency is also mapped to **position in space**, with more frequent words placed towards the centre of the word cloud. This is an example of redundant mapping; more frequent words are both larger *and* more central (Section 9.8).

> A word cloud is effective for conveying words because it maps words to text data symbols. It is also partially effective for conveying relative frequencies of words because it maps word frequency to both size and position in space.

Another benefit of encoding frequency as position in space is the efficient use of space. For example, Figure 13.10 shows the same data presented as a bar plot with the frequency of each word mapped to a separate bar. Although the accuracy is improved for comparing frequencies, the bar plot does not have as much room for the words themselves.

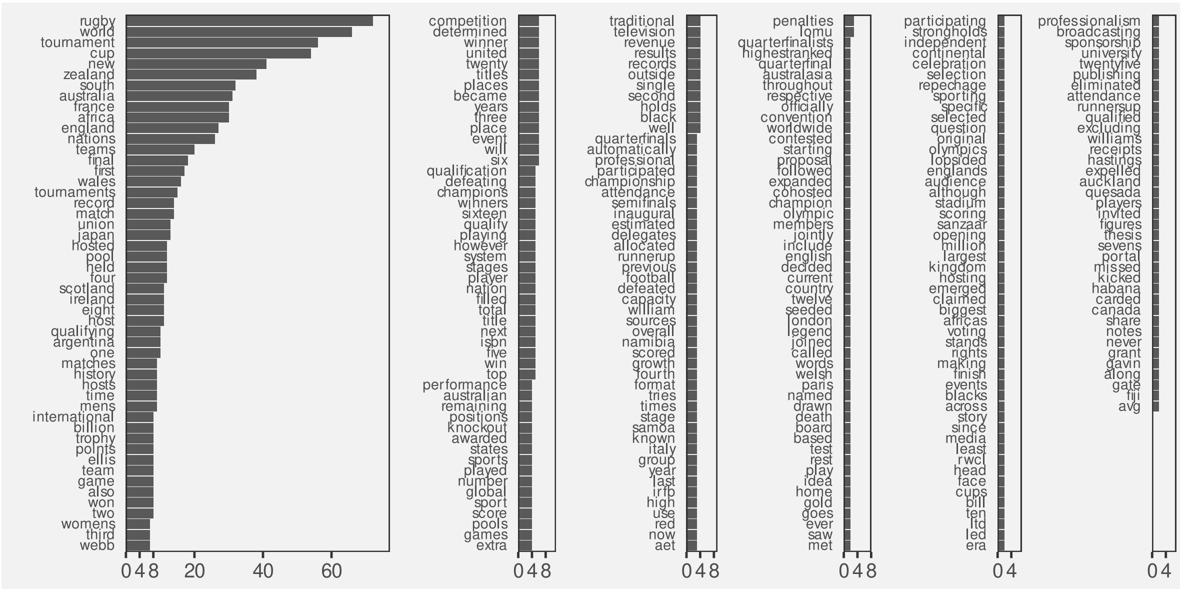


Figure 13.10: A bar plot of the frequency of different words in the Wikipedia page for the Rugby World Cup.

One weakness of the word cloud is that the **angle** of words changes with no corresponding change in the data values. This is an example of a **dissonant** mapping, which can cause confusion (Section 7.5). The viewer will notice the differences in angle and attempt to decode that difference to a difference in data values that does not exist.

> The different angles of text in a word cloud is not effective because it does not encode any differences in the data.

13.5 Case study: Tables

Table 13.2 shows a subset of the 2023 Rugby World Cup data set (Table 9.1), just including the top eight nations in the tournament. We established in Section 13.1 that visualising data values in a table provides an excellent basis for accurately decoding individual data values, but imposes a larger cognitive burden for comparing values, and is a very poor for extracting visual summaries, such as trends.

For example, it is very difficult to determine from Table 13.2 whether there is any relationship between the values in the four numeric columns.

Table 13.2: Performance measures for the top 8 teams at the 2023 Rugby World Cup. Each measure is a per-game average because some teams played more games than others.

country	sphere	runs	breaks	tries	points
South Africa	South	95.14	5.429	3.857	29.71
England	North	100.3	5.571	3	31.57
Fiji	South	138	5.6	2.4	22.4
Wales	North	109.4	5.6	3.8	32
Argentina	South	131.3	6.286	2.714	26.43
Ireland	North	137.6	8.8	6	42.8
France	North	126	11	6	47.6
New Zealand	South	139.3	12.57	7	48

However, there are standard guidelines for formatting tables of values that can improve the readability of a table.[158] Table 13.3 shows the same data as Table 13.2, but with all of the numeric columns right-aligned, a consistent number of decimal places in each column, fewer decimal places than the data actually possess, and the rows have been ordered from the highest number of points per game to the lowest.

It is easier to see some structure in Table 13.2. For example, the number of tries appears to decrease as well as the number of points. We can also see some evidence of the number of breaks decreasing as well, though it is weaker.

There are some familiar explanations for the improvement in the table display. The reduction in detail, with fewer decimal places, means that there is less complexity in the visual field. The consistent number of decimal places means that less is changing, so differences are easier to detect (Section 2.5). Also, the consistent baseline provided by right-alignment means that the text **length** of the numbers (the number of digits in each number) provides a crude encoding of the size of the number, particularly in the column showing the number of clean breaks. However, it is worth noting that some of the **accuracy** of the text has been sacrificed in order to make some of these gains.

Table 13.3: Performance measures for the top 8 teams at the 2023 Rugby World Cup. This is a version of Table 13.2 with some standard table design guidelines applied.

country	sphere	runs	breaks	tries	points
New Zealand	South	139	12.6	7.0	48.0
France	North	126	11.0	6.0	47.6
Ireland	North	138	8.8	6.0	42.8
Wales	North	109	5.6	3.8	32.0
England	North	100	5.6	3.0	31.6
South Africa	South	95	5.4	3.9	29.7

Table 13.3: Performance measures for the top 8 teams at the 2023 Rugby World Cup. This is a version of Table 13.2 with some standard table design guidelines applied.

country	sphere	runs	breaks	tries	points
Argentina	South	131	6.3	2.7	26.4
Fiji	South	138	5.6	2.4	22.4

For comparison, Figure 13.11 shows a scatter plot of the data from Table 13.2. The data symbol is a circle, with the number of tries mapped to horizontal **position**, the number of points mapped to vertical **position**, the number of clean breaks mapped to **size**, and the number of runs mapped to **colour** (shade). The relationships between tries and points and between breaks and both tries and points are still easier to perceive in this plot than in Table 13.3, but Table 13.3, even with its rounding, still provides better accuracy, particularly for breaks and runs.

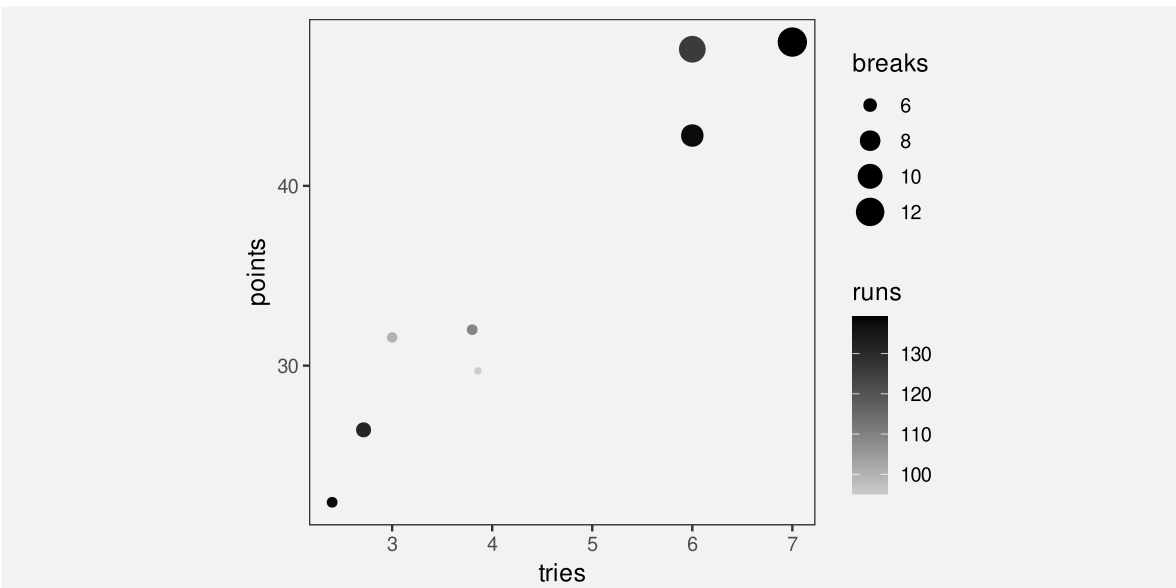


Figure 13.11: A scatter plot of the performance measures for the top 8 teams at the 2023 Rugby World Cup.

13.6 Case study: Stem-and-leaf plots

Figure Figure 13.12 shows a histogram of the number of tries scored by teams in the 2023 Rugby World Cup. The data symbols in Figure 13.12 are bars. The raw data values are the average number of tries per game for each team, but the lengths of the bars encode **data**

summaries: the number of teams with an average number of tries within each range: 0 to 1, 1 to 2, and so on.

Because the borders of the bars are not drawn, we get an emergent shape from the combined bars (similar to a density plot; Section 10.3).

This histogram allows us to see features of the data summaries, such as a suggestion that there is a small subset of teams that score a higher number of tries than the other teams (a bimodal distribution; though in this case we have a very small amount of data). However, there is no way to decode the raw data values from this histogram.

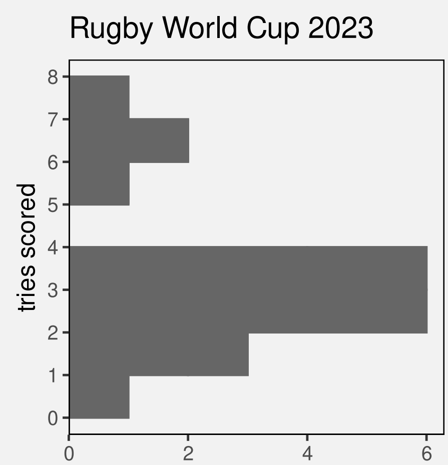


Figure 13.12: A histogram of the number of tries scored. The bars are horizontal for easy comparison with the stem-and-leaf plot.

Figure Figure 13.13 shows a stem-and-leaf diagram of the same data (with tries scored rounded to one decimal place). The data symbol in this case is a text digit. Every team is encoded as a digit from 0 to 9, which represents the first decimal place for the average number of tries scored by each team. For example, New Zealand scored 7.0 tries per game on average and that is encoded as a `0` beside the `7` on the y-axis. The number of tries is also encoded as a combination of vertical position (the whole number of tries) and horizontal position (teams with the same whole number of tries are stacked horizontally, but also ordered by the first decimal place value).

The number of teams within each range (0 to 1, 1 to 2, etc) is encoded by the number of digits. Another way to look at it is that the data symbol is a piece of text consisting of several digits and the number of teams is encoded as the **length** of the piece of text.

As with the histogram in Figure 13.12, there are emergent shapes with two distinct humps visible, one taller than the other. However, in Figure 13.13 we are able to decode the raw data values as well. Furthermore, because the data symbols are text digits, we are able to decode individual data values very accurately.

In summary, Figure 13.13 combines standard decodings (**length**) with **learned decodings** (text shape) to provide access to both high-level features of data summaries *and* raw data values.

Figure 13.13: A stem-and-leaf plot of the number of tries scored.

Figure 13.14 shows a variation on Figure 13.13 with the hemisphere encoded as the colours of the digits. This provides another demonstration that we can use simple visual features like colour to encode data values with text data symbols.

Figure 13.15 shows another variation on Figure 13.13, this time with the hemisphere encoded as the **weight** of the digits. This demonstrates the idea that there are text-specific visual features that can be used to encode data values. Other examples are font family (e.g., Arial versus Roman) and font slant (italic versus upright).[159]

Stem-and-leaf plots might be seen as an archaic form of data visualisation, but we can see that they are effective in many ways.[160]

Unfortunately, despite all of their advantages, stem-and-leaf plots have a limited usefulness because they can only accommodate very small data sets. There is a similarity between a stem-and-leaf plot and the stacked icon plot in Figure 12.13a. However, whereas an icon can easily be used to represent some multiple of objects, that option is not available for text digits.

Figure 13.14: A stem-and-leaf plot of the number of tries scored with hemisphere encoded as colour.

Figure 13.15: A stem-and-leaf plot of the number of tries scored with hemisphere encoded as font weight.

13.7 Case study: Direct Labelling

some sort of ggrepel version of the flags example in visual-objects?

relate visual objects (other than text) to direct labelling?

"Review of Graph Comprehension Research: Implications for Instruction" Shah and Hoeffner (2002) [p 56] Some quotes from discussion of usefulness of colour, but in relation to the use of legends: "reduce the difficulty viewers face in keeping track of graphic referents because of the demands imposed on working memory" [p 57] Cites Kosslyn (1994) for "label graph features directly with their referents"

"A Model of the Perceptual and Conceptual Processes in Graph Comprehension" Carpenter and Shah (1998) Eye tracking study "viewers must continuously reexamine the labels to refresh their memory"

13.8 Dangers of text

Figure 13.16 shows a data visualisation of the number of gun deaths in Florida between 1990 and 2012, covering the enactment of Florida's "stand your ground" law that loosened the constrainst on an individual's rights to use deadly force.[161] The title of the plot points out that gun deaths *decreased* after the law was enacted, which is perhaps surprising given that use of deadly force was less constrained after the introduction of the law.

However, if we look closely at the plot itself, we can see that the y-axis scale is inverted. A downward trend is actually an *increase* in gun deaths, so the data shows that gun deaths actually increased after the law was enacted.

This example demonstrates that the text within a data visualisation is not only given a lot of attention, but it can have a dominant effect on the message that the viewer takes away.[162]

Figure 13.16 is an example of a deliberately misleading data visualisation, which we obviously want to avoid. However, the broader point is that we need to be at least as careful with our choice of text content as we are with our selection of mappings from data values to visual features and we have a responsibility to work hard to make sure that the message from data symbols and from text labels is completely consistent.

13.9 Alt text

Is a data visualisation effective if we cannot see it? We saw in Section 6.7 that a data visualisation may not be effective if a colour encoding does not take into account red-green colour blindness. But what about a low-vision or blind audience?

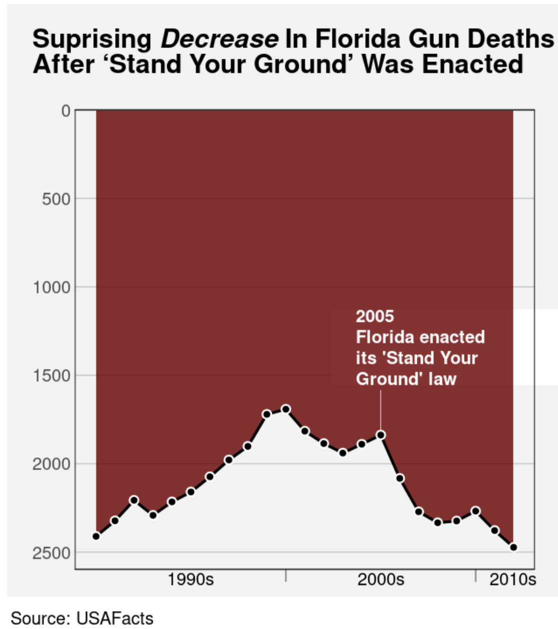


Figure 13.16: The title influences our perception; the message from the data symbols is actually quite different.

Although data visualisation is based on the visual system, it is still possible to make a data visualisation effective for a low-vision or blind audience by providing a text description of the data visualisation. Alt text is a text description of a data visualisation that will be recognised and read out by a screen reader.

The alt text for a data visualisation will ideally describe what variables are being displayed and how they are encoded (what sort of plot is it), data summaries like trends, clusters, outliers, and correlations, and include the main message of the visualisation, including causal explanations and wider implications.[163]

Another option sometimes employed is to provide tables of data in addition to the data visualisation, though this will not always be possible or practical.

All of the images in this document have alt text (or at least they should!).

13.10 Summary

Data values can be encoded as the visual features of text, such as position and colour, just like for other data symbols.

The shape of text—the characters used in text—is particularly important because we have a learned decoding of text. We can read data values from text.

We can use text to encode any type of data, we can represent a very large number of different categories, and we can decode individual data values from text extremely accurately.

On the other hand, decoding a large number of data values from text is very slow and we cannot generate visual summaries from text.

Guides are really just collections of data symbols, with specific data values encoded as the visual features to demonstrate how data values have been encoded.

The text elements within guides are essential because they provide the only precise decoding of exact data values. This is what allows the decoding of the main data symbols within a data visualisation.

Overall, text is very valuable in a data visualisation for accurately encoding a small number of data values, or to express complex information. But text is not appropriate for representing a large number of raw data values.[164]

Notes

[153] "a memorable visualization is often also an effective one." ("Beyond Memorability: Visualization Recognition and Recall" Borkin et al (2016)) Only a thumbnail version of the original image is shown in Figure 13.2 in order to comply with terms of use of the MASSVIS data set.

[154] Provide citation to Brath book at least. Perhaps cite my text article, however that gets to see the light of day.

[155] Talk about there being separate verbal processing areas in the brain. Text has to come in as visual, so undergoes visual processing, but text is also processed separately for its language value. Cite Ware p315-316

[156] Russell Ackoff's "From Data to Wisdom" Text is very flexible - it can convey data, information, knowledge, understanding, and wisdom. Most other graphical elements are much more limited.

[157] (Wikipedia 2024)

[158] For example, Schwabish (2020).

Ehrenberg (1977) advice on tables

"UNDERSTANDING GRAPHS AND TABLES" Wainer (1992) (Wainer 1992)

[159] (Brath and Banissi 2014) at least

[160] Quote from John Tukey about stem-and-leaf plots:

If we are going to make a mark, it may as well be a meaningful one. The simplest - and most useful - meaningful mark is a digit.

Attributed to “Some Graphic and Semigraphic Displays” in T. A. Bancroft, ed. Statistical Papers in Honor of George W. Snedecor (Ames, Iowa, 1972), p. 296.

[161] Based on a Business Insider data visualisation; data from USA Facts.

[162] “Trust and Recall of Information across Varying Degrees of Title-Visualization Misalignment” Kong et al (2019) Greater impact that can even override message in data. “Towards Understanding How Readers Integrate Charts and Captions: A Case Study with Line Charts” Kim et al (2021) Studied what happens when caption and text are incoherent; viewer goes with the prominent visual feature. Main message is to make the text and chart coherent.

[163] “Accessible Visualization via Natural Language Descriptions: A Four-Level Model of Semantic Content” Lundgard and Satyanarayan (2022)

[164] Acknowledge Ware 4th Ed: G8.18 (p327) “When a large number of data points must be represented in a visualization, use sybmols instead of words or pictorial icons.” G8.19 (p327) “Use words directly on the chart where the number of symbolic objects in each category is relatively few and where space is available.”

14 Mapping Structure

Figure 14.1 shows a line plot of the rate of youth crime in New Zealand for different ethnic groups and a pie chart showing the proportions of different types of crime.[165] There is a problem with this figure because the line plot and the pie chart share the same colour palette even though they are visualising different variables (ethnicity and crime type). For example, the ethnic group `European/Other` is encoded as the orange colour of a line *and* the crime type `Theft` is also encoded as an orange colour, in this case of a wedge.

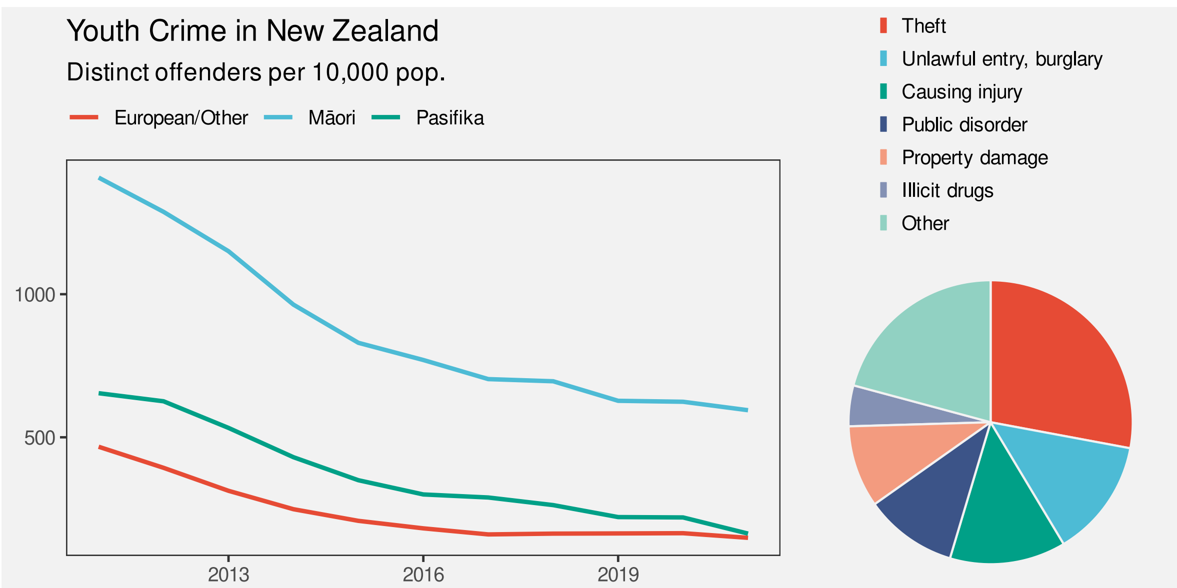


Figure 14.1

This creates a confusing data visualisation because the same visual feature possesses multiple conflicting decodings (Figure 14.2). There is nothing wrong with the encodings in the line plot by itself or with the encodings in the pie chart by itself, but the overall combination of elements within Figure 14.1 is ineffective.

In this chapter we look at the overall arrangement and coordination of the elements of a data visualisation.[166]

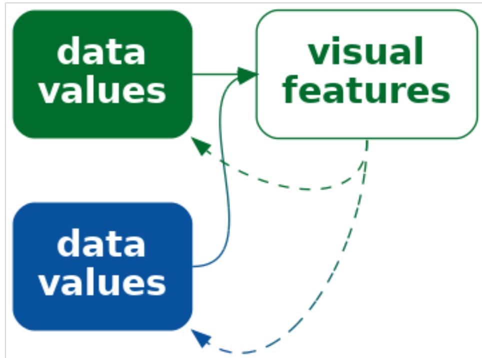


Figure 14.2: If we encode more than one data value to the same visual feature, it is unclear which is the correct decoding.

14.1 Mapping organisation

Figure 14.3 shows a scatter plot of the number of clean breaks and the number of tries for countries at the 2023 Rugby World Cup (Table 9.1). This data visualisation is similar to Figure 9.4, but with an additional encoding of the hemisphere for each country as the colour of the data points.

The legend (guide) in Figure 14.3 demonstrates the scale for the colour encoding: which colour is used to encode each hemisphere. In Section 13.2, we described how the proximity of the labels "South" and "North" to the coloured points in the legend allow us to associate the colours with the different hemispheres. The different values of hemisphere are encoded as the vertical positions of the points and the text labels; both the orange point and the label "South" have the same vertical position.

What about the horizontal position of the points and the text in the legend? These do not encode hemisphere values because they are the same for both North and South. What about the position of the legend title, "hemisphere"?

All of the elements of the legend are positioned together and to the right of the main plot. Their proximity to each other (and separation from the main plot) mean that we see the legend as a separate group (Section 2.7). This creates a visual organisation for the plot, which helps the viewer to navigate and comprehend the different elements of the plot (could do with a CITE here). This is analogous to reading a body of text, which is split into paragraphs and sections with headings to help the reader navigate the text.

One way to express this is that the **position** of the elements of the legend **encode** the **organisation** of the overall data visualisation. Our visual system **decodes** the legend as a separate element based on the proximity of the text and points in the legend.

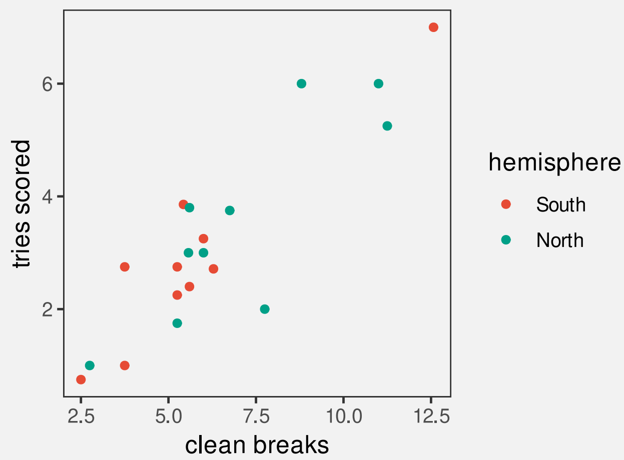


Figure 14.3: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

> One way in which a data visualisation can be effective is if it provides information about how to read the plot as well as information about the data itself.

Another way to say it is that each element of the data visualisation in Figure 14.3 belongs to one of two groups: the main plot or the legend. There is a **qualitative** value, `plot` or `legend`, associated with each element of the data visualisation. These qualitative values are encoded as the **position** of the elements within the data visualisation: on the left or on the right.

Figure 14.5 shows a variation of Figure 14.3, with exactly the same plot elements, but in a different arrangement. There is greater separation of the legend from the main plot. The legend has also been moved so that its top is aligned with the top of the main plot. Similarly, the axis titles have been moved so that they align with the top and right sides of the plot, and the y-axis label has been rotated to horizontal. More subtly, the right edge of the y-axis label is aligned with the right edge of the y-axis tick labels and the points in the legend are aligned with the left edge of the legend title.

All of these changes are aimed at making the data visualisation more orderly (Section 2.8). The legend is more obviously a separate element from the main plot and the axis titles more obviously connect to the main plot. A more orderly arrangement of the visual elements will provide fewer distractions and lead to a more effective data visualisation because the important

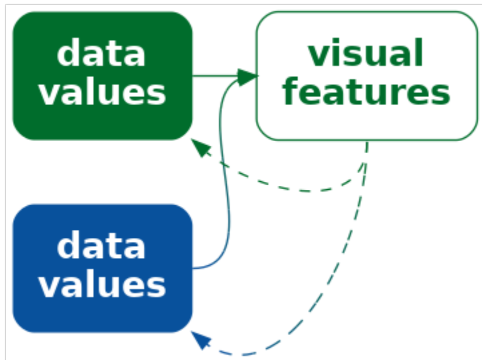


Figure 14.4: We can encode the structure of a data visualisation as position so that elements that are part of the same structure are close to each other.

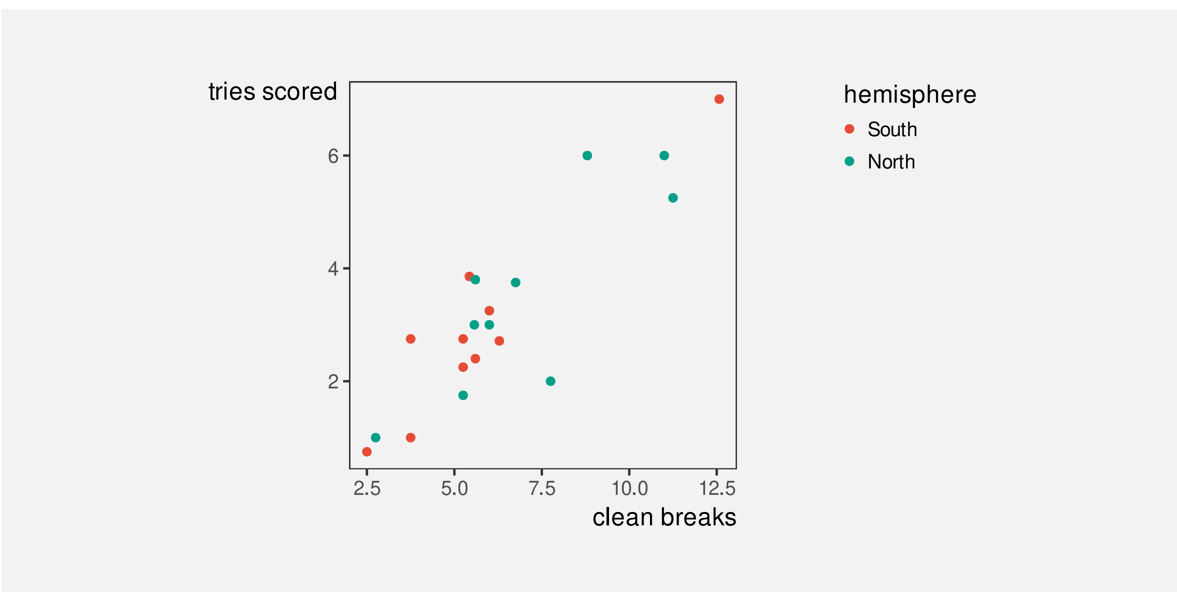


Figure 14.5: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

differences in the data will be more readily perceived (Section 2.5).[167]

The strong alignment and regular patterns within facetted plots like Figure 11.6 are another effective example of keeping everything else the same in order to champion the changes in the data. Conversely, another weakness of placing data symbols on maps, like in Figure 12.9b, is the distracting complexity that results.

14.2 Consistent mappings

TODO Refer back to congruence and dissonance in congruence.Rmd and differences and similarities in visual-model.Rmd

Figure 14.6 shows a variation on Figure 14.3 with a modified legend. In this plot, the hemisphere of a country is encoded using orange and green within the main plot, but then hemisphere is encoded using light blue and brown in the legend. This is clearly an ineffective data visualisation, but it highlights the importance of maintaining consistent mappings between different elements of a plot.[168]

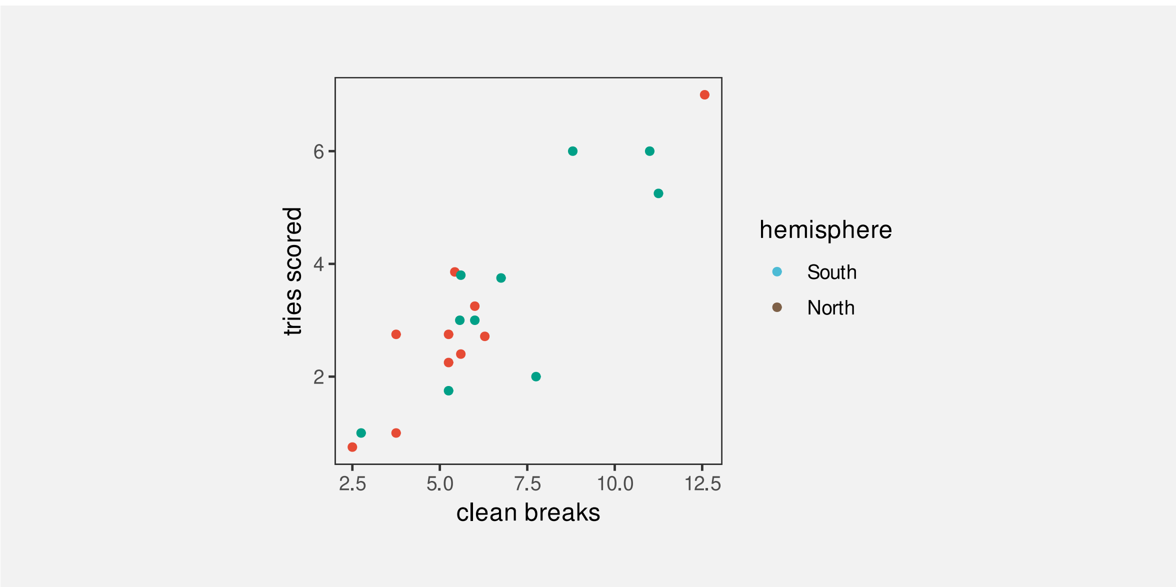


Figure 14.6: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

Figure 14.7 demonstrates that we can harness consistent mappings to improve upon the original Figure 14.3. In this plot, the encoding of hemisphere as the colour of points is not only consistent between the main plot and the legend, but it has been extended to the colour of legend text. The visual grouping of elements that relate to the same hemisphere is now stronger within the legend and across the entire data visualisation (Section 2.7).

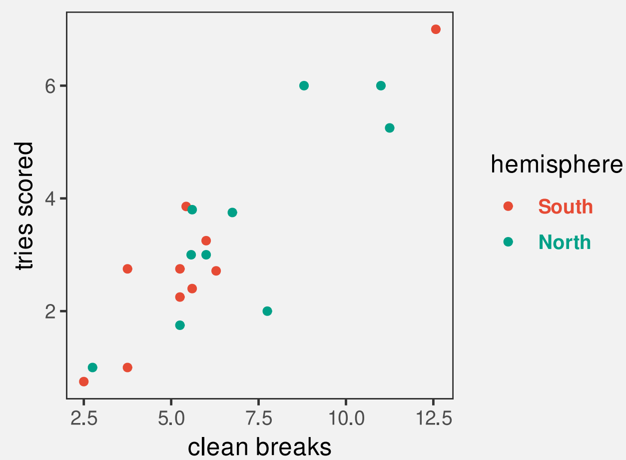


Figure 14.7: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

14.3 Dangers of inconsistent mappings

Figure 14.8 shows a bar plot of the impact of a proposed tax bill during the second Trump administration.[169] Each bar represents a change in income for a particular income bracket. Lower incomes are to the left and higher incomes are to the right. There are labels to describe each income bracket and there are labels that describe each change in income. Where the change in income is large, the label for change in income is written within the bar (at the top of the bar), but for smaller changes in income, the bar is too small to fit the label, so the label is placed just above the bar. For the two leftmost bars, the change in income is actually negative and the change in income label is placed *below* the bar. The labels that describe the income brackets are mostly below the bars, but for the two leftmost bars the labels for the income brackets are placed above the bars.

In summary, there are two labels for each bar, one below the bar and one above (or at the top of) the bar. This consistent positioning of the labels encourages the viewer to decode two groups of labels: one group above the bars and one group below the bars. This can mislead us into comparing, for example, “-1K” for the leftmost bar with “4.3M” for the rightmost bar, when the correct comparison is between “-1K” (for the lowest income bracket) and “389.3K” (for the highest income bracket).[170]

Just as it makes no sense to change the encoding of data values within a data visualisation (Figure 14.6) or across multiple data visualisations within the same document (Figure 14.1), it makes no sense to change the encoding of structure within a data visualisation.

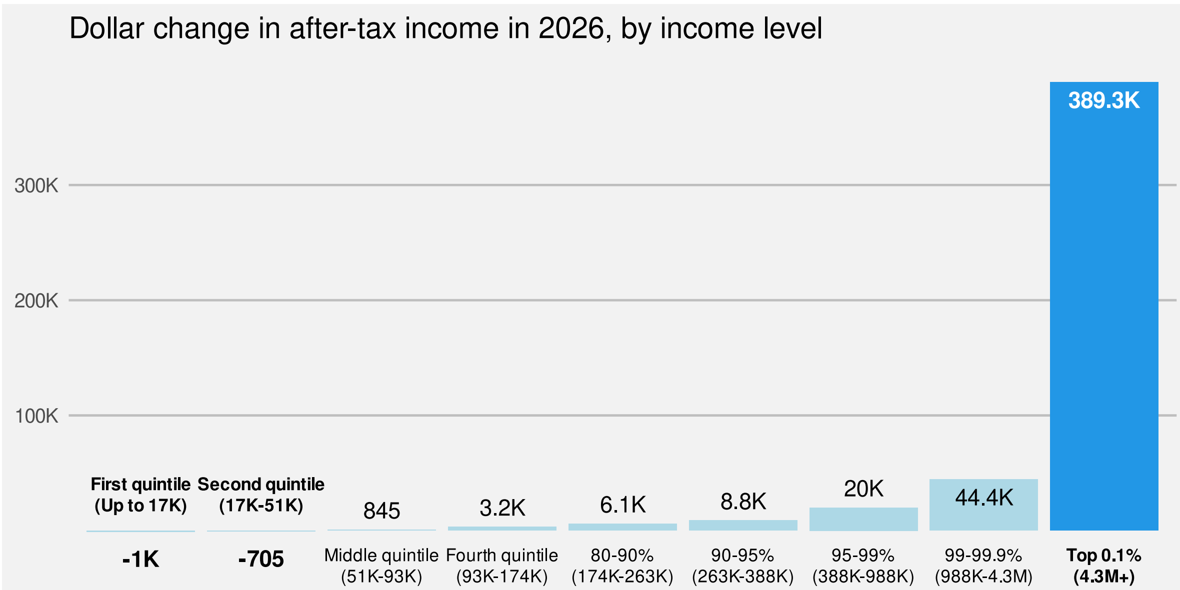


Figure 14.8: A bar plot of the distributional effects of Trump’s “Big, Beautiful Bill”.[1]

14.4 Mapping importance

Figure 14.10 shows a variation on Figure 14.7 with many elements of the plot drawn in light grey. The effect of this change is to create two visual groups, one lighter and less saturated than the other. This emphasises the main data symbols in the main plot and in the legend, and de-emphasises the axes and labels. Our attention is drawn to the darker and more saturated elements first (Section 2.6), with the other elements in the background, which we can focus on if we need the additional information.[171]

By comparison, in Figure 14.7 there is a similar visual impact from all elements of the plot, so it is less clear where to look first. All of the elements of the data visualisation compete for our attention.

[1]Based on plot from “Popular Information” web site (Judd Legum) https://popular.info/p/the-ugly-truth-about-trumps-big-beautiful Data source: Penn Wharton Budget Model https://budgetmodel.wharton.upenn.edu/estimates/2025/5/16/distributional-effects-of-house-budget-reconciliation-as-of-thursday-may-15

This is another example of encoding (qualitative) non-data values as visual features (see Section 14.1). In this case, we are encoding the **importance** of different visual elements as visual features, in this case colour. We decode the differences in colour to a more important group of elements and a less important group of elements.

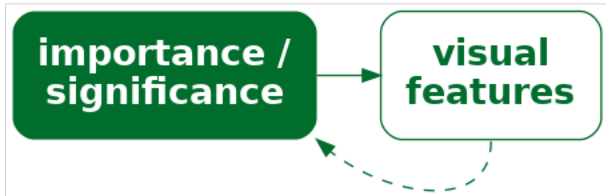


Figure 14.9: We can encode the importance of elements within a data visualisation as visual features so that elements that are more important are more visible.

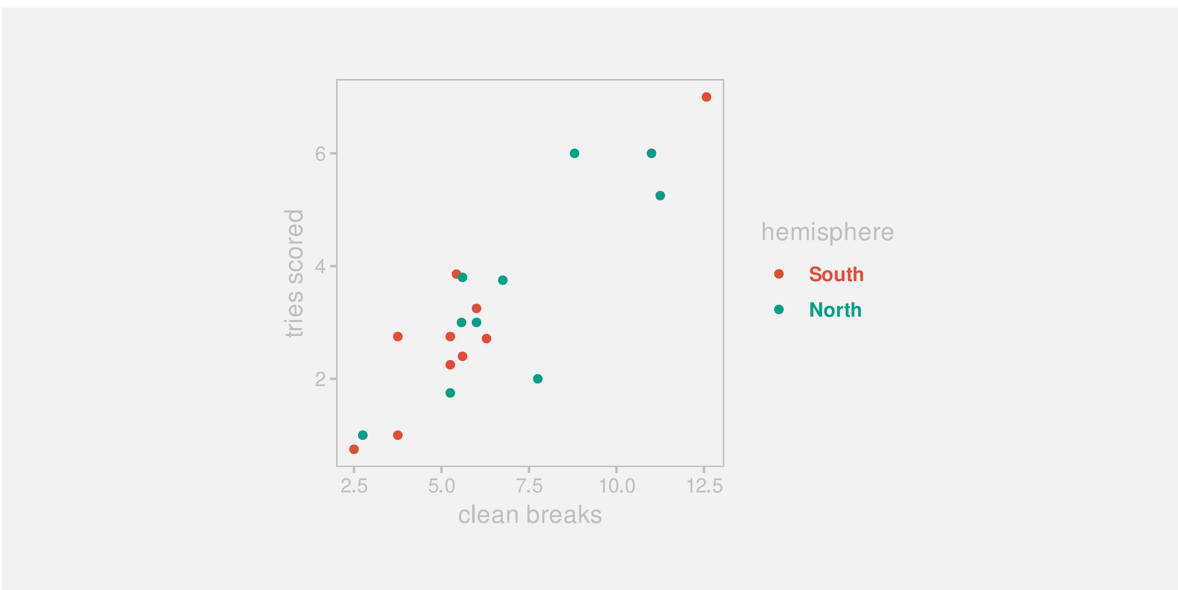


Figure 14.10: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

Figure 14.11 shows a similar application of importance. In this case, we are highlighting two data points (New Zealand and South Africa). The distinct colour and larger size of the two points encodes the importance of those points and draws attention to those points, with everything else as background context.

TODO an example of complementary colours providing greater contrast than analogous colours (Section 6.6).

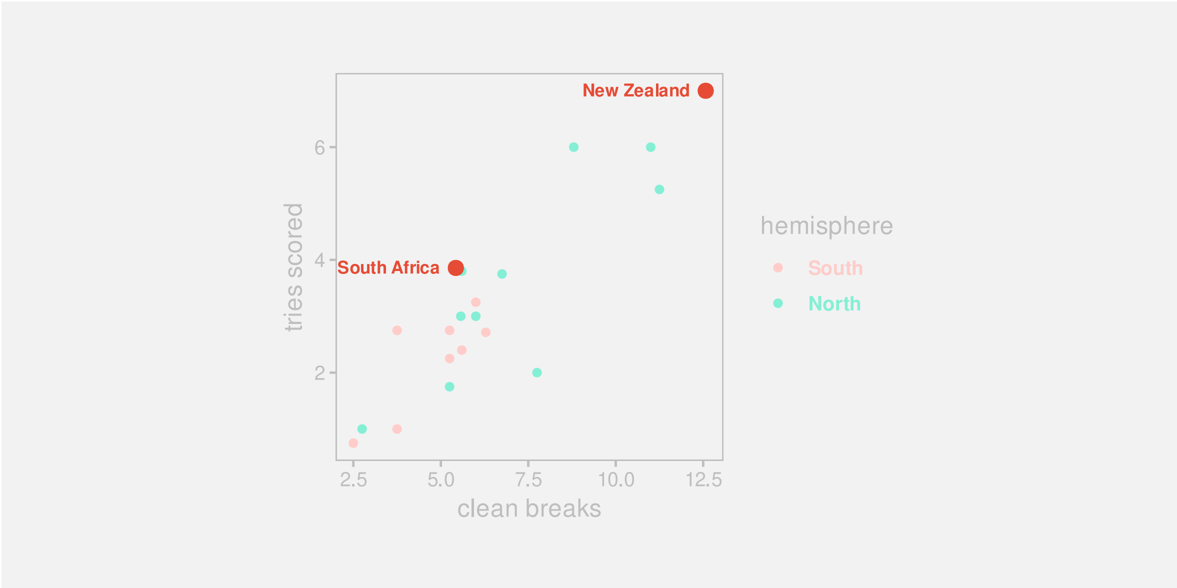


Figure 14.11: A scatter plot of the number of times a team breaks through the opposition defence and the number of tries that a team scores (both are per-game averages) for teams at the 2023 Rugby World Cup.

Section 2.5

14.5 Dangers of mapping importance

Distractions

Add examples of making the WRONG thing important (e.g., use thick lines on bar borders or bright colours on unimportant elements) ?

Refer back to chart junk and clutter (Section 12.3).

14.6 Summary

Just as we can map data values to visual features, to communicate information about the data, we can map non-data information to visual features, to communicate important aspects of the structure of a data visualisation.

The structure or organisation of a data visualisation is a qualitative value that can be effectively encoded as the position of different elements.

Communicating structure helps the viewer to navigate within a data visualisation.

Position is especially important

Notes

[165] Based on figures from the Youth Justice Indicators Report from 2021 .

[166] Acknowledge that many of these ideas align with the ideas of contrast, repetition, alignment, and proximity that are espoused in Naomi Robbins book.

[167] "Bridging Educational Theories of Cognitive Load to Visualization Design and Evaluation" Cabouat et al (2025) Shows that a more "readable" data vis requires less mental effort to extract information. May get more info from a closer reading of this paper.

[168] This particularly echoes the repetition part of CRAP design.

[169] Based on plot from "Popular Information" web site (Judd Legum) https://popular.info/p/the-ugly-truth-about-trumps-big-beautiful Data source: Penn Wharton Budget Model https://budgetmodel.wharton.upenn.edu/estimates/2025/5/16/dist effects-of-house-budget-reconciliation-as-of-thursday-may-15

[170] This is precisely the mistake that was made when this data visualisation was discussed on The Majority Report (~6:00).

[171] CRAP contrast A larger difference is better because it suggests a larger difference in importance. A small difference suggests a less important difference. Also, a small difference is more confusing, distracting, and unpleasant. Like a background hum (c.f. Robbins)

Review

Add review at the end that reinforces the threads like visual feature, visual shape, visual object and mappings

Possible encoding end points:

- visual feature
- visual summary
- emergent feature
- visual shape
- visual object
- text

Possible decodings from encoding end points:

- qual/quant data value
- data summary
- higher level information/knowledge/wisdom

Do two versions (at least):

- Set of questions for when you already have a plot and want to understand why it works or not

- Another set for if you want to create a data vis

Basic questions:

- What are you asking the visual system to do?
- Is the visual system good at that task?
- Are you asking the viewer to produce visual summaries when it would be easier to produce data summaries?

Present process by which a data vis can be broken down. What data symbols are being used? (simple or complex) What mappings are being used? What do we want to decode? (data or summaries) If mapping back to raw data, are the mappings accurate? What can be mapped back to? Are mappings congruent with data?

...

- Use heatmap as case study?
- do the side-by-side pie chart idea as an example of reasoning your way to a new visualisation.
- Move the discussion/revisit of tables from text section to here ?
- Another possible application of data vis knowledge: Analyse one of the weird boxplot variations that look a bit like a ziggurat?

List of questions to evaluate a data vis (maybe nest these a bit):

- What do you want the viewer to be able to decode?
- Do the mappings support that goal?
- What are the data variables
- What visual features are they mapped to
- What type of data
- Can the visual features represent the data
- Is there the capacity to represent all categories
- How accurate are the decodings
- Is there congruence or dissonance in the mapping
- Mapping data or data summary
- Do visual features interact
- Are there emergent features
- Is the data symbol a visual shape
- Is that appropriate
- A visual object

- Is that appropriate
- Learned mappings somehow (distinct from implicit mappings)
- Do all visual differences correspond to differences in the data?
- Are all differences in the data represented by visual differences?
- Does each data value map to its own data symbol If not, individual data values may be difficult to decode
- Do multiple data values map to a single data symbol If not, data summaries may be difficult to decode

And so on …

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